



Learning & Growing With Team

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MGT411-Money and Banking

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Final Lectures 23 - 45

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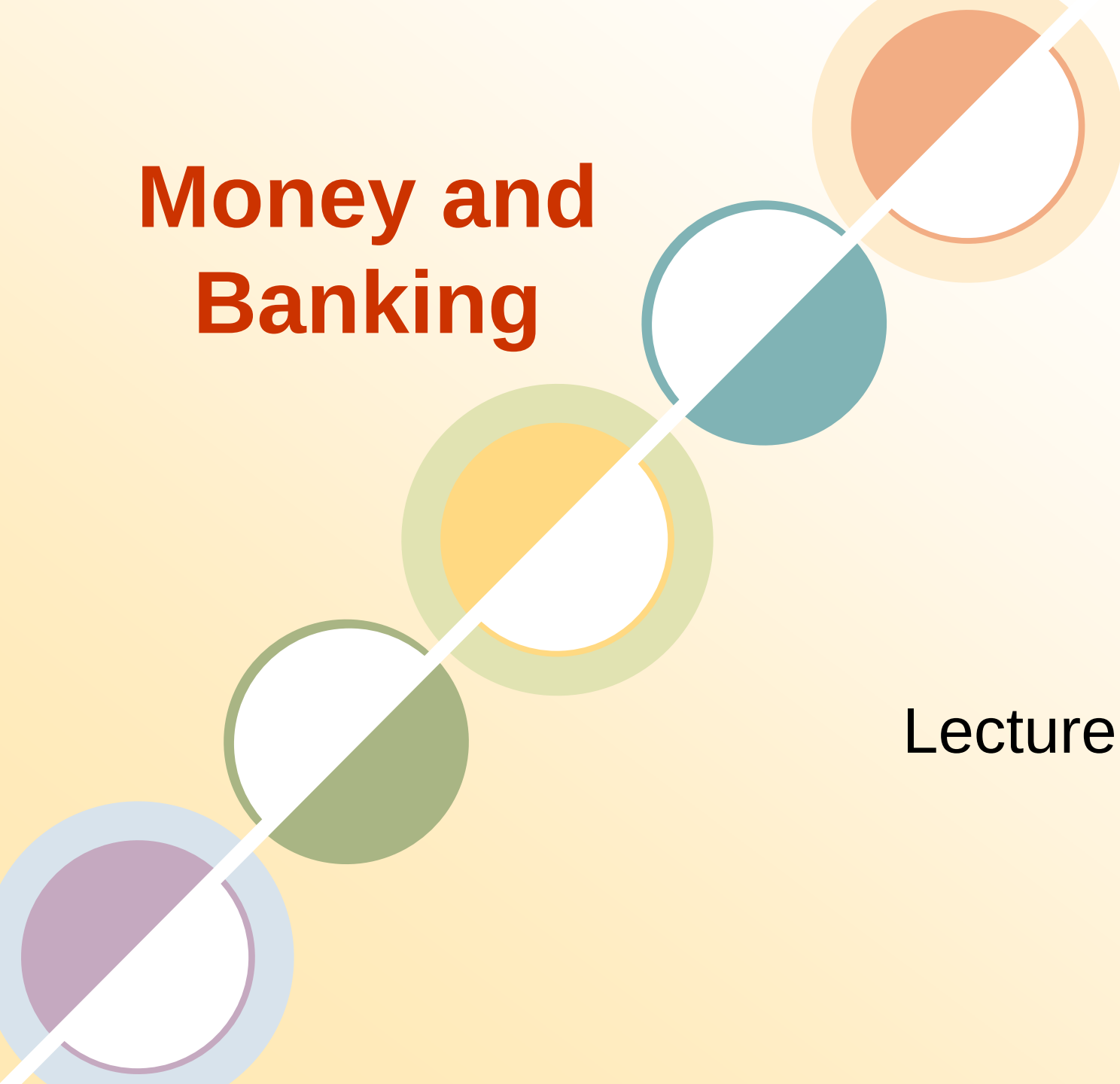
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Money and Banking

Lecture 23



Review of the Previous Lecture

- Role of Financial Intermediaries
 - Pool Savings
 - Safekeeping, accounting services and access to the payments system;
 - Liquidity;
 - Information Services
 - Risk diversification
- Information Asymmetry and Information Costs
 - Adverse Selection
 - Moral Hazards

Topics under Discussion

- Banking
 - Types of banks
 - Balance Sheet of Commercial Banks
 - Assets
 - Liabilities

Banking

- Banking is a combination of businesses designed to deliver the services
 - pool the savings of and making loans
 - Diversification
 - Access to the payments system
 - Accounting and record-keeping
- The intent of banks is to profit from each of these lines of business

Banking

- There are three basic types of depository institutions:
 - Commercial banks,
 - Savings institutions
 - Credit unions.

Banking

- Commercial banks
 - They accept deposits and use the proceeds to make consumer, commercial and real estate loans.

Banking

- Saving Institutions
 - Financial intermediaries to serve households and individuals
 - Provide mortgage and lending as well as saving deposit services

Banking

- Credit Unions
 - Nonprofit depository institutions that are owned by people with a common bond
 - These unions specialize in making small consumer loans

The Balance Sheet of Commercial Banks

- Balance Sheet Identity

$$\text{Total Bank Assets} = \text{Total Bank Liabilities} + \text{Bank Capital}$$

The Balance Sheet of Commercial Banks

- Banks obtain funds from individual depositors and business as well as by borrowing from other financial institutions and through the financial markets.
- They use these funds to make loans, purchase marketable securities and hold cash.

The Balance Sheet of Commercial Banks

- The difference between a bank's assets and liabilities is the bank's capital or Net Worth
- The bank's profits come both from service fees and the difference between interest earned and interest paid.

2005 **2004**
(Rupees '000)

ASSETS

Cash and balances with treasury banks
Balances with other banks
Lendings to financial institutions
Investments - net
Advances - net
Other assets - net
Operating fixed assets
Deferred tax assets - net

23,665,549

1,469,333

9,998,828

69,481,487

180,322,753

5,464,426

8,182,454

191,967

23,833,253

5,708,323

10,965,297

67,194,971

137,317,773

6,154,370

7,999,821

-

298,776,797

259,173,808

LIABILITIES

Bills payable
Borrowings from financial institutions
Deposits and other accounts
Sub-ordinated loan
Liabilities against assets subject to finance lease
Other liabilities
Deferred tax liabilities - net

8,536,674

27,377,502

229,345,178

1,598,080

-

8,611,600

-

7,566,684

7,590,864

221,069,158

1,598,720

-

6,525,999

269,499

275,469,034

244,620,924

NET ASSETS

23,307,763

14,552,884

Table 12.1 Balance Sheet of U.S. Commercial Banks, August 2004

Assets in billions of dollars (numbers in parentheses are percentage of total assets)			
Cash items (including reserves)		321.6	(4.1)
Securities		1,893.5	(23.9)
U.S. Government and agency	1,170.7	(14.8)	
State and local government and other	722.8	(9.1)	
Loans		5,044.6	(63.7)
Commercial and industrial	887.1	(11.2)	
Real estate (including mortgage)	2,411.6	(30.5)	
Consumer	672.3	(8.5)	
Interbank	364.7	(4.6)	
Other	708.9	(9.0)	
Other Assets		<u>655.1</u>	<u>(8.3)</u>
Total Commercial Bank Assets		7,914.8	
Liabilities in billions of dollars (numbers in parentheses are percentage of total liabilities)			
Checkable deposits		645.0	(8.9)
Nontransaction deposits		4,487.7	(62.3)
Savings deposits and time deposits	3,340.9	(46.3)	
Large, negotiable time deposits	1,146.8	(15.9)	
Borrowings		1,589.0	(22.0)
From banks in U.S.	459.1	(6.4)	
From nonbanks in U.S.	1,129.9	(15.7)	
Other liabilities		<u>487.4</u>	<u>(6.8)</u>
Total Commercial Bank Liabilities		7,209.1	
Bank Assets - Bank Liabilities = Bank Capital		705.7	

Assets: Uses of Funds

- Cash Items
 - reserves
 - cash items in process of collection
 - vault cash
- Securities
 - secondary reserves
- Loans

Assets: Uses of Funds

- Cash Items

- Reserves

- includes cash in the bank's vault and its deposits at the central bank
 - held to meet customers' withdrawal requests

- Cash items in the process of collections

- uncollected funds the bank expects to receive

- The balances of accounts that banks hold at other banks (correspondent banking)

- Because cash earns no interest, it has a high opportunity cost. So banks minimize the amount of cash holding

Assets: Uses of Funds



Securities:

- Stocks
 - T-Bills
 - Government and corporate bonds
 - Securities are sometimes called secondary reserves because they are highly liquid and can be sold quickly if the bank needs cash.
- 
- 

Assets: Uses of Funds

- Loans:
 - The primary asset of modern commercial banks;
 - business loans (commercial and industrial loans),
 - Real estate loans,
 - Consumer loans,
 - Inter-bank loans,
 - Loans for the purchase of other securities.

Assets: Uses of Funds

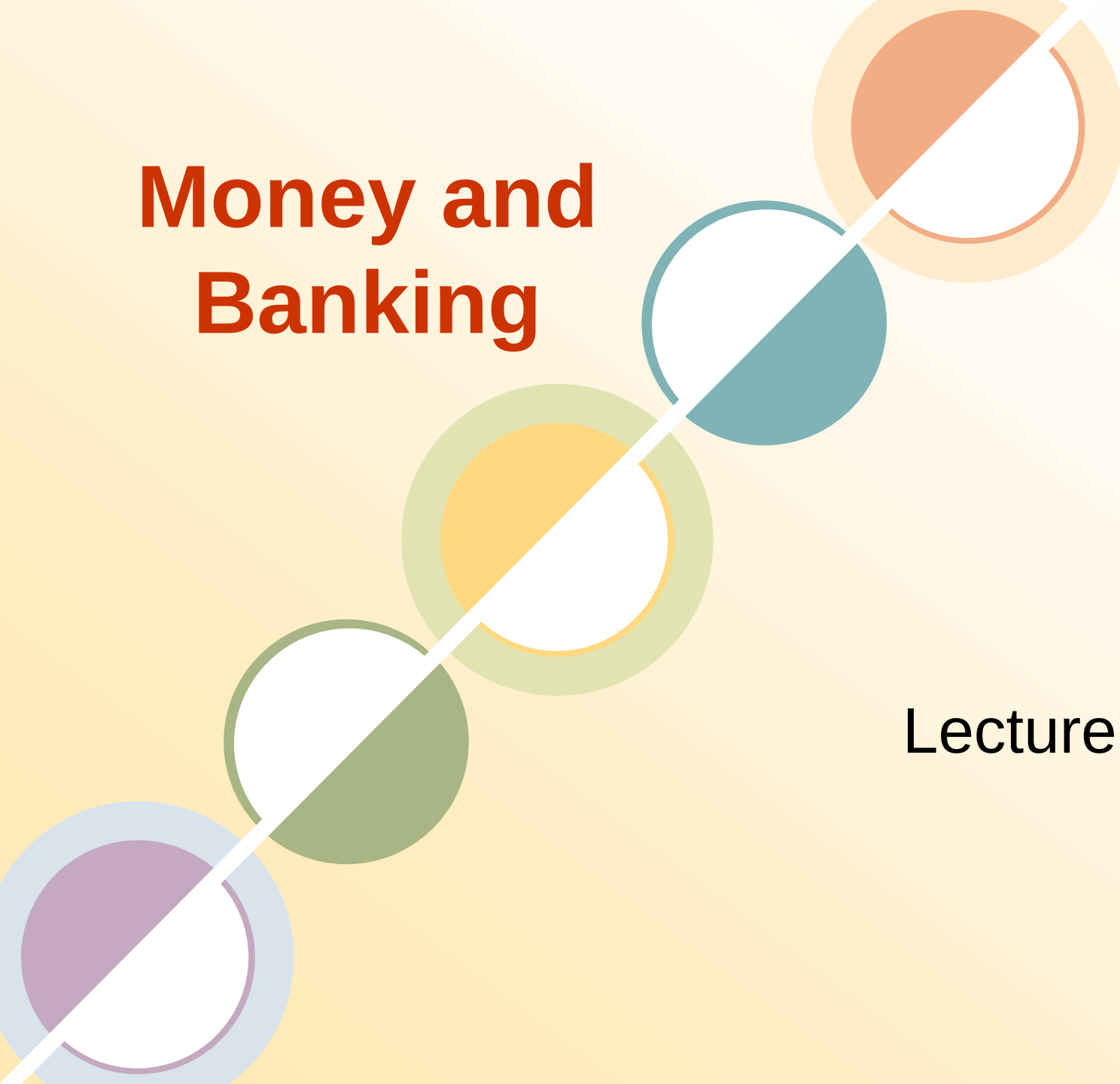
- The primary difference among the various types of depository institutions is in the composition of their loan portfolios
 - Commercial banks make loans primarily to business
 - Savings and loans provide mortgages to individuals
 - Credit unions specialize in consumer loans

Summary

- Banking
 - Types of banks
 - Balance Sheet of Commercial Banks
 - Assets
 - Liabilities

Money and Banking

Lecture 24



Review of The Previous Lecture

- Banking
 - Types of Banking Institutions
 - Commercial Banks
 - Savings Institutions
 - Credit Unions
 - Balance Sheet of Commercial Banks
 - Assets: uses of funds
 - Cash
 - Securities
 - Loans

Topics under Discussion

- Balance Sheet of Commercial Banks
 - Assets: uses of funds
 - Bank Capital and Profitability
- Off-Balance-Sheet Activities
- Bank Risk
 - Liquidity Risk
 - Credit Risk
 - Interest Rate Risk
 - Trading Risk
 - Other Risks

Liabilities: Sources of Funds

- Checkable Deposits
- Non-transactions Deposits
- Borrowings
 - discount loans
 - federal funds market.

Liabilities: Sources of Funds

- Checkable deposits:
 - A typical bank will offer 6 or more types of checking accounts.
 - In recent decades these deposits have declined because the accounts pay low interest rates

Liabilities: Sources of Funds

- Nontransactions Deposits:
 - These include savings and time deposits and account for nearly two-thirds of all commercial bank liabilities.
 - When you place your savings in a Certificate of Deposit (CD) at the bank, it is as if you are buying a bond issued by that bank
 - CDs can vary in terms of their value, the large ones can be bought and sold in financial markets

Liabilities: Sources of Funds

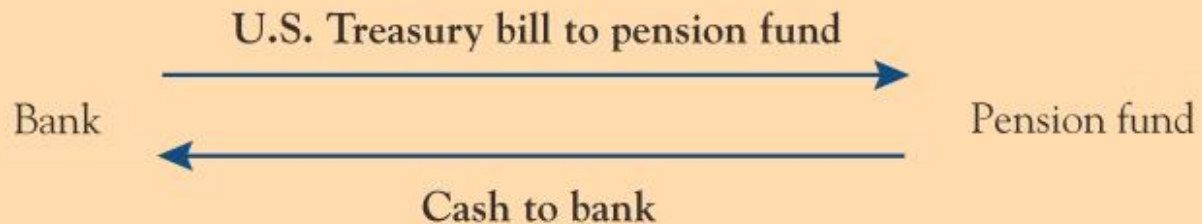
• Borrowings:

- Banks borrow from the central bank (discount loans)
- They can borrow from other banks with excessive reserves in the inter-bank money market.
- Banks can also borrow by using a repurchase agreement or repo, which is a short-term collateralized loan
 - A security is exchanged for cash, with the agreement that the parties will reverse the transaction on a specific future date (might be as soon as the next day)

Figure 12.3 Mechanics of an Overnight Repurchase Agreement

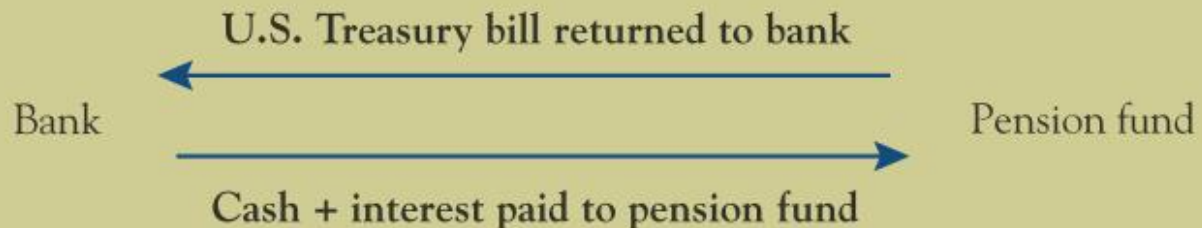
Day One

Bank sells U.S. Treasury bill to pension fund in exchange for cash



Day Two

Bank *repurchases* U.S. Treasury bill from the pension fund in exchange for cash plus interest



Bank Capital and Profitability

- The net worth of banks is called bank capital; it is the owners' stake in the bank
- Capital is the cushion that banks have against a sudden drop in the value of their assets or an unexpected withdrawal of liabilities
- An important component of bank capital is loan loss reserves, an amount the bank sets aside to cover potential losses from defaulted loans
 - It is reduced by the defaulted loans written-off

Bank Capital and Profitability

- There are several basic measures of bank profitability
- Return on Assets,

$$\text{ROA} = \frac{\text{Net profit after taxes}}{\text{Total bank assets}}$$

- It is a measure of how efficiently a particular bank uses its assets
- A manager can compare the performance of bank's various lines of businesses by looking at different units' ROA

Bank Capital and Profitability

- The bank's return to its owners is measured by the Return on Equity

$$\text{ROE} = \frac{\text{Net profit after taxes}}{\text{Bank capital}}$$

- ROA and ROE are related to leverage
- A measure of leverage is the ratio of bank assets to bank capital. Multiplying ROA by this ratio yields ROE

Bank Capital and Profitability

$$\text{ROA} \times \frac{\text{Bank Assets}}{\text{Bank Capital}}$$

$$= \frac{\text{Net profit after taxes}}{\text{Total bank assets}} \times \frac{\text{Bank Assets}}{\text{bank Capital}}$$

$$= \frac{\text{Net profit after taxes}}{\text{Bank Capital}} = \text{ROE}$$

- Return on equity tends to be higher for larger banks, suggesting the existence of economies of scale

Bank Capital and Profitability

- Net interest income is another measure of profitability;
- It is the difference between the interest the bank pays and what it receives
- It can also be expressed as a percentage of total assets to yield (net interest margin). It is the bank's interest rate spread
- Well run banks have high net interest income and a high net interest margin.
- If a bank's net interest margin is currently improving, its profitability is likely to improve in the future.

Table 12.2 Profitability of U.S. Commercial Banks

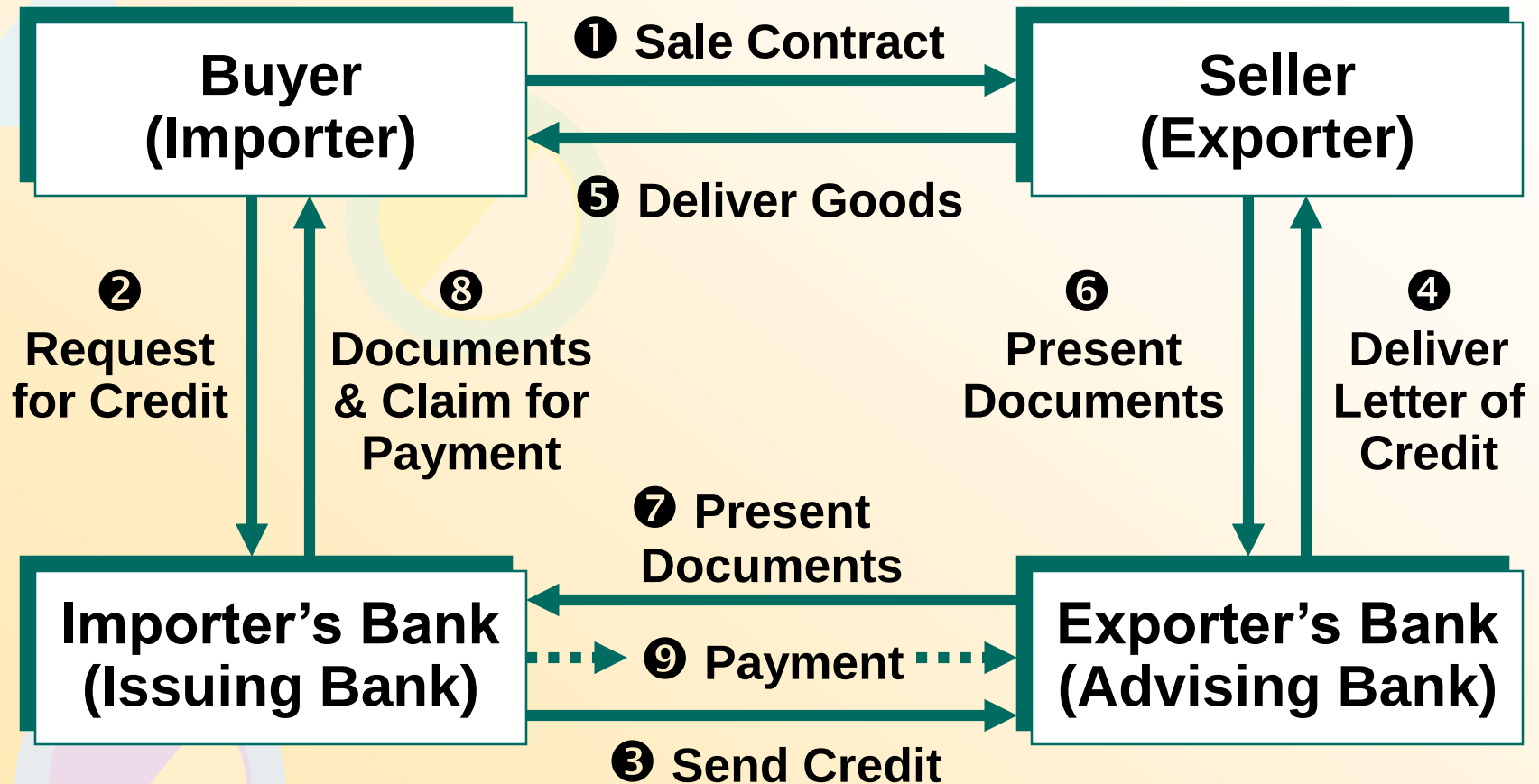
(in millions of dollars, except bottom four rows)

	1991	1996	2001
A. Interest income - Interest expense (Net interest income)	\$121,288	\$161,172	\$210,809
B. Other revenue*	58,482	92,515	153,734
C. Operating costs	124,233	159,241	218,706
D. Gross profit (A + B - C)	55,537	94,446	145,837
E. Loan losses (provisions)	34,128	15,483	41,008
F. Net operating profit (D - E)	21,409	78,963	104,829
G. Realized capital gains from sale of real estate	2,971	530	4,434
H. Net profits before taxes (F + G)	24,380	79,493	109,263
I. Assets	3,420,381	4,554,234	6,454,543
Net interest margin (A/I)	0.0355%	0.0354%	0.0327%
Return on assets (H/I) (ROA)	0.0071	0.0175	0.0169
Return on equity (ROE)	0.1258	0.2147	0.1860
Net interest income/Total income [A/(A + B)]	0.6747	0.6353	0.5782

Off-Balance-Sheet Activities

- Banks engage in these activities in order to generate fee income; these activities include providing trusted customers with lines of credit
- Letters of credit are another important off-balance-sheet activity; they guarantee that a customer will be able to make a promised payment.
- In so doing, the bank, in exchange for a fee, substitutes its own guarantee for that of the customer and enables a transaction to go forward

Off-Balance-Sheet Activities



Off-Balance-Sheet Activities

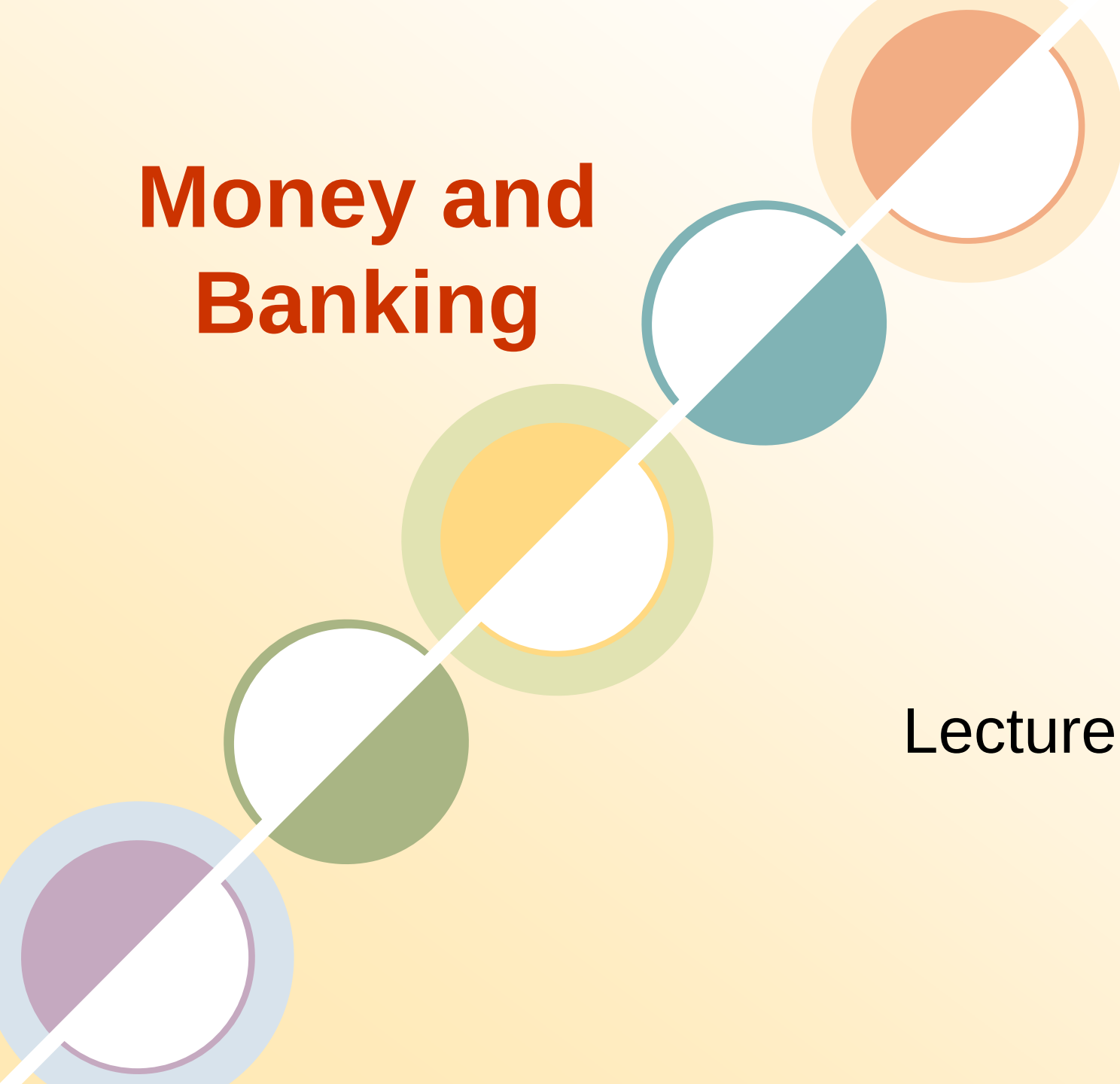
- A standby letter of credit is a form of insurance; the bank promises that it will repay the lender should the borrower default
- Off-balance-sheet activities create risk for financial institutions and so have come under increasing scrutiny in recent years

Summary

- Balance Sheet of Commercial Banks
 - Liabilities: Sources of Funds
 - Bank Capital and Profitability
- Off-Balance-Sheet Activities

Money and Banking

Lecture 25



Review of the Previous Lecture

- Balance Sheet of Commercial Banks
 - Assets: Uses of Funds
 - Liabilities: Sources of Funds
 - Bank Capital and Profitability
- Off-Balance-Sheet Activities

Topics under Discussion

- Bank Risk
 - Liquidity Risk
 - Credit Risk
 - Interest Rate Risk
 - Trading Risk
 - Other Risks

Bank Risk

- Banking is risky because depository institutions are highly leveraged and because what they do
- In all the lines of banking trades, the goal of every bank is to pay less for the deposits the bank receives than for the loan it makes and the securities it buys.

Liquidity Risk

- Liquidity risk is the risk of a sudden demand for funds and it can come from both sides of a bank's balance sheet (deposit withdrawal on one side and the funds needed for its off-balance sheet activities on the liabilities side)
- If a bank cannot meet customers' requests for immediate funds it runs the risk of failure; even with a positive net worth, illiquidity can drive it out of business

Liquidity Risk

Balance Sheet of a Bank Holding \$5 Million in Excess Reserves

Assets		Liabilities	
Reserves	\$15 million	Deposits	\$100 million
Loans	\$100 million	Borrowed funds	\$30 million
Securities	\$35 million	Bank capital	\$20 million

Liquidity Risk

- One way to manage liquidity risk is to hold sufficient excess reserves (beyond the required reserves mandated by the central bank) to accommodate customers' withdrawals.
- However, this is expensive (interest is foregone).

Liquidity Risk

- Two other ways to manage liquidity risk are:
 - adjusting assets
 - adjusting liabilities.

Liquidity Risk

Balance Sheet of a Bank Holding No Excess Reserves

Assets		Liabilities	
Reserves	\$10 million	Deposits	\$100 million
Loans	\$100 million	Borrowed funds	\$30 million
Securities	\$40 million	Bank capital	\$20 million

- If a customer makes a \$5 million withdrawal, the bank can't simply deduct it from reserves.
- Rather it will adjust another part of balance sheet

Liquidity Risk

- A bank can adjust its assets by
 - selling a portion of its securities portfolio,
 - or by selling some of its loans,
 - or by refusing to renew a customer loan that has come due

Liquidity Risk

Balance Sheet of a Bank Following a \$5 Million Withdrawal and Asset Adjustment

Withdrawal Is Met by Selling Securities

Assets		Liabilities	
Reserves	\$10 million	Deposits	\$95 million
Loans	\$100 million	Borrowed funds	\$30 million
Securities	\$35 million	Bank capital	\$20 million

Withdrawal Is Met by Reducing Loans

Assets		Liabilities	
Reserves	\$10 million	Deposits	\$95 million
Loans	\$95 million	Borrowed funds	\$30 million
Securities	\$40 million	Bank capital	\$20 million

Liquidity Risk

- Banks do not like to meet their deposit outflows by contracting the asset side of the balance sheet because doing so shrinks the size of the bank
- Banks can use liability management to obtain additional funds by
 - borrowing (from the central bank or from another bank) or
 - by attracting additional deposits (by issuing large CDs)

Liquidity Risk

Balance Sheet of a Bank Following a \$5 Million Withdrawal and Liability Adjustment

Withdrawal Is Met by Borrowing

Assets		Liabilities	
Reserves	\$10 million	Deposits	\$95 million
Loans	\$100 million	Borrowed funds	\$35 million
Securities	\$40 million	Bank capital	\$20 million

Withdrawal Is Met by Attracting Deposits

Assets		Liabilities	
Reserves	\$10 million	Deposits	\$100 million
Loans	\$100 million	Borrowed funds	\$30 million
Securities	\$40 million	Bank capital	\$20 million

Credit Risk

- This is the risk that loans will not be repaid and it can be managed through diversification and credit-risk analysis
- Diversification can be difficult for banks, especially those that focus on certain kinds of lending

Credit Risk

- Credit-risk analysis produces information that is very similar to the bond-rating systems and is done using a combination of statistical models and information specific to the loan applicant
- Lending is plagued by adverse selection and moral hazard, and financial institutions use a variety of methods to mitigate these problems

Credit Risk

- Screen loan application
- Monitor borrowers after they have received loan
- Collateral or high net-worth demand
- Developing long term relationships

Interest-Rate Risk

- The two sides of a bank's balance sheet often do not match up because liabilities tend to be short-term while assets tend to be long-term; this creates interest-rate risk
- In order to manage interest-rate risk, the bank must determine how sensitive its balance sheet (assets and liabilities) is to a change in interest rates;

Interest-Rate Risk

- If we think of bank's assets and liabilities as bonds, the change in interest rate will affect the value of these bonds, more importantly due to the term of bonds
- So if interest rate rises, the bank face the risk that the value of their assets may fall more than the value of their liabilities (reducing the bank's capital)

Interest-Rate Risk

- Suppose 20% of bank's assets fall into the category of assets sensitive to changes in the interest rate. While rest of 80% are not sensitive to changes in interest rate
 - If interest rate is stable 5%, then each \$100 yields \$5 in interest
- Now suppose 50% of bank's deposits (liabilities) are interest rate sensitive and 50% are not
 - Half of the liabilities are deposits that earn variable returns so costs vary with market rate

Interest-Rate Risk

- For making profit, interest rate on liabilities must be lower than the interest rate on assets.
 - The difference is the bank's margin!
- Assuming interest rate on liabilities is 3%, the net interest margin is $5 - 3 = 2\%$
- What happens as interest rate rises by 1%

An Example of Interest-Rate Risk

The impact of an interest-rate increase on bank profits (per \$100 of assets)

	Assets	Liabilities
Interest-rate sensitive	\$20	\$50
Not interest-rate sensitive	\$80	\$50
Initial interest rate	5%	3%
New interest rate on interest-rate-sensitive assets and liabilities	6%	4%
	Revenue from Assets	Cost of Liabilities
At initial interest rate	$(0.05 \times \$20) + (0.05 \times \$80) = \$5.00$	$(0.03 \times \$50) + (0.03 \times \$50) = \$3.00$
After interest-rate change	$(0.06 \times \$20) + (0.05 \times \$80) = \$5.20$	$(0.04 \times \$50) + (0.03 \times \$50) = \$3.50$

Profits at initial interest rate: $(\$5.00) - (\$3.00) = \$2.00$ per \$100 in assets

Profits after interest-rate change: $(\$5.20) - (\$3.50) = \$1.70$ per \$100 in assets

Gap Analysis

Gap between interest-rate-sensitive assets and interest-rate-sensitive liabilities:

$$\begin{aligned}
 & (\text{Interest-rate-sensitive assets of } \$20) \\
 & - (\text{Interest-rate-sensitive liabilities of } \$50) \\
 & = (\text{Gap of } - \$30)
 \end{aligned}$$

Interest-Rate Risk

- **When bank has more interest rate sensitive liabilities than does interest rate assets, an increase in interest rate will cut into the bank's profits**

Summary

- Bank Risk
 - Liquidity Risk
 - Credit Risk
 - Interest Rate Risk

Money and Banking

Lecture 26

Review of the Previous Lecture

- Bank Risk
 - Liquidity Risk
 - Credit Risk
 - Interest Rate Risk

Topic under Discussion

- Bank Risk
 - Interest Rate Risk (Cont.)
 - Trading Risk
 - Other Risks
- Globalization of Banking
- The Future of Banks
- Non-depository Institutions
 - Insurance Companies
 - Securities Firms
 - Finance Companies
 - Government Sponsored Enterprises

Interest-Rate Risk

- Gap analysis highlights the gap or difference between the yield on interest sensitive assets and the yield on interest-sensitive liabilities
- Multiplying the gap by the projected change in the interest rate yields the change in the bank's profit
- Gap analysis can be further refined to take account of differences in the maturity of assets and liabilities

An Example of Interest-Rate Risk

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$$\begin{aligned}
 & \text{(Interest-rate-sensitive assets of \$20)} \\
 & - \text{(Interest-rate-sensitive liabilities of \$50)} \\
 & = \text{(Gap of - \$30)}
 \end{aligned}$$

Interest-Rate Risk

- Banks can manage interest-rate risk by matching the interest-rate sensitivity of assets with the interest-rate sensitivity of liabilities,
 - Purchase short term securities to match variable rate deposits
 - Make long term loans at floating rates
 - but this approach increases credit risk

Trading Risk

- Banks today hire traders to actively buy and sell securities, loans, and derivatives using a portion of the bank's capital in the hope of making additional profits
- However, trading such instruments is risky (the price may go down instead of up); this is called trading risk or market risk

Trading Risk

- Managing trading risk is a major concern for today's banks, and bank risk managers place limits on the amount of risk any individual trader is allowed to assume
- Banks also need to hold more capital if there is more risk in their portfolio

Other Risks

- Banks that operate internationally will face
 - foreign exchange risk (the risk from unfavorable moves in the exchange rate)
 - sovereign risk (the risk from a government prohibiting the repayment of loans).
- Banks manage their foreign exchange risk by attracting deposits denominated in the same currency as the loans and by using foreign exchange futures and swaps to hedge the risk

Other Risks

- Banks manage sovereign risk by diversification, by refusing to do business in a particular country or set of countries, and by using derivatives to hedge the risk
- Banks also face operational risk, the risk that their computer system may fail or that their buildings may burn down
- To manage operational risk the bank must make sure that its computer systems and buildings are sufficiently robust to withstand potential disasters

Risks Banks Face and How They Manage Them

Type of Risk	Source of Risk	Recommended Responses
<i>Liquidity Risk</i>	Sudden withdrawals by depositors	1. Hold sufficient cash reserves to meet customer demand. 2. Manage assets—sell securities or loans (contracts the balance sheet) 3. Manage liabilities—attract more deposits (maintains the size of the balance sheet)
<i>Credit Risk</i>	Default by borrowers on their loans	1. Diversify to spread risk. 2. Use statistical models to screen for creditworthy borrowers. 3. Monitor to reduce moral hazard.
<i>Interest-Rate Risk</i>	Mismatch in maturity of assets and liabilities coupled with a change in interest rates	1. Closely match the maturity of both sides of the balance sheet. 2. Use derivatives such as interest-rate swaps.
<i>Trading (Market) Risk</i>	Trading losses in the bank's own account	Closely monitor traders using risk management tools, including value at risk.

The Globalization of Banking

- Toward the end of the 20th century, sharp rise in international trade increased the need for international financial services

The Globalization of Banking

- Banks can operate in other countries by
 1. opening a foreign branch,
 - Offer same services as in home country
 2. creating an International Banking Facility (IBF)
 - Accept deposits from and make loans to foreigners outside the country

The Globalization of Banking

- 3. creating an Edge Act subsidiary,
 - Engage in international banking transactions
- 4. purchasing a controlling interest in a foreign bank
- Foreign banks can take advantage of similar options.

The Globalization of Banking

- The growth of international banking has had an economic impact, increasing the competition in and efficiency of banking markets
 - A borrower from France, Brazil, Singapore or Pakistan can shop for loan virtually anywhere in the world, while a depositor seeking the highest return can do the same

The Globalization of Banking

- This phenomenon has made banking a tougher job
 - Profits are harder to come by as borrowers and depositors have more options
- But overall the improved efficiency of financial system has enhanced growth everywhere

The Globalization of Banking

- One of the most important aspects of international banking is the eurodollar market, in which dollar-denominated deposits in foreign banks are exchanged
- The eurodollar market was created in response to restrictions on the movement of international capital imposed at the end of World War II with the creation of Bretton Woods system

The Globalization of Banking

- Today, the eurodollar market in London is one of the biggest and most important financial markets in the world,
 - The interest rate at which banks lend each other eurodollars (the London Interbank Offered Rate or LIBOR) is the standard against which many private loan rates are measured

The Future of Banks

- Today's banks are bigger, fewer in number and more international than those of the past, and they offer more services
- Financial holding companies are a limited form of universal banks, firms that engage in non-financial as well as financial activities
 - Banking , Insurance and securities

The Future of Banks

- The owners and managers of these financial firms cite three reasons to create them:
 1. They are well diversified,
 2. They are large enough to take advantage of economies of scale,
 3. They hope to benefit from economies of scope
 - *offering many products under the same “brand” name can also reduce costs*

The Future of Banks

- Individual firms provide the same services as more traditional intermediaries do
 - Money market mutual funds provide liquidity
 - Mortgage brokers help in borrowing for home purchase
 - Leasing companies provide car, and consumer financing
 - Discount brokers provide low cost access to financial markets

The Future of Banks

- Thanks to recent technological advances, almost every service traditionally provided by financial intermediaries can now be produced independently, without the help of a large organization
- Moreover, the production of information to mitigate the problems of adverse selection and moral hazard has become a business in and of itself

The Future of Banks

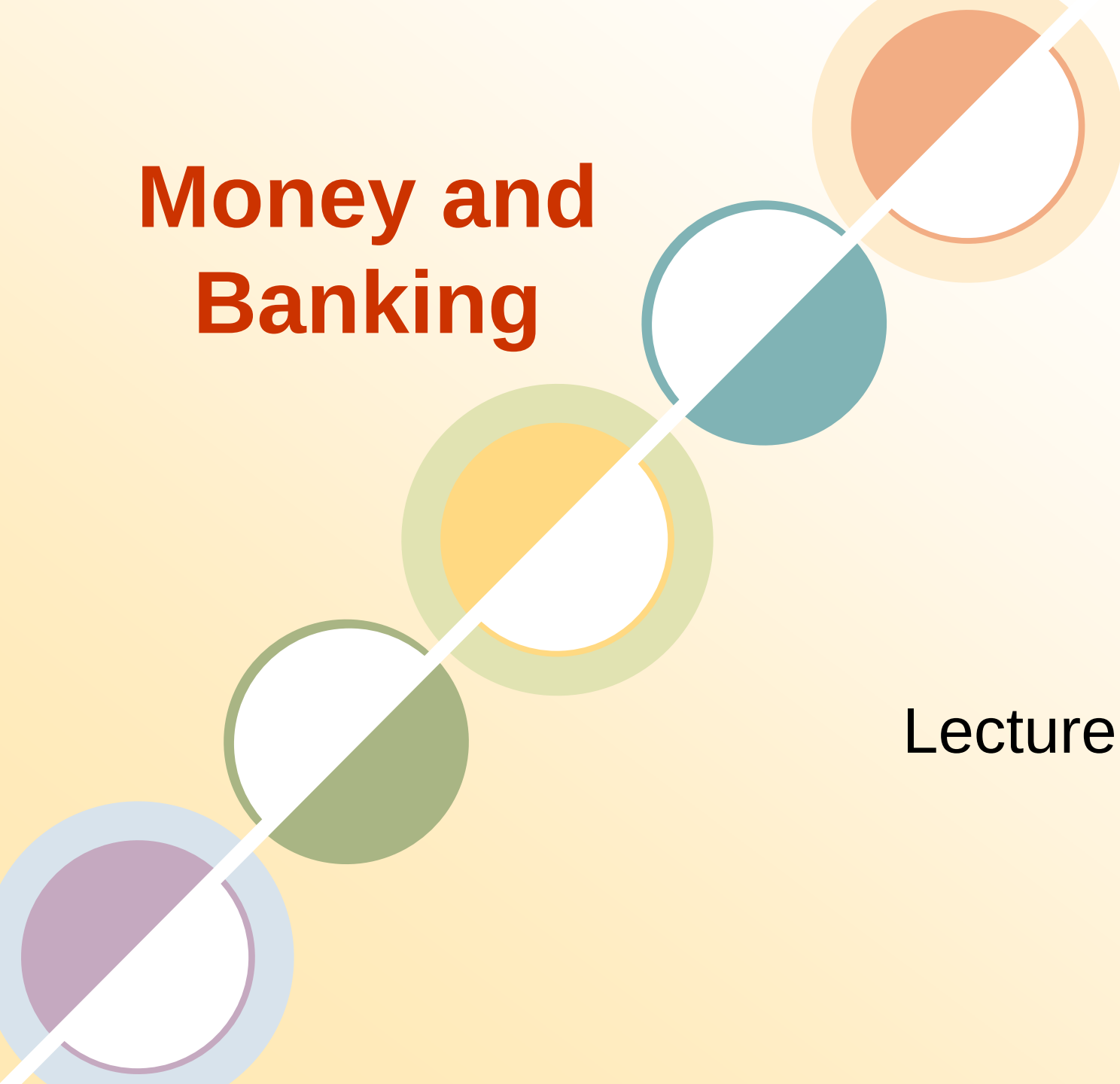
- As we survey the financial industry we can discern two opposite trends:
 - Large firms are working hard to provide one-stop shopping for financial services
 - Industry is splintering into a host of small firms, each of which serves a very specific purpose

Summary

- Globalization of Banking
- The Future of Banks
- Non-depository Institutions
 - Insurance Companies
 - Securities Firms
 - Finance Companies
 - Government Sponsored Enterprises

Money and Banking

Lecture 27



Review of the Previous Lecture

- Bank Risk
 - Liquidity Risk
 - Credit Risk
 - Interest Rate Risk
 - Trading Risk
 - Other Risks
- Globalization of Banking
- The Future of Banks

Topics under Discussion

- Non-depository Institutions
 - Insurance Companies
 - Securities Firms
 - Finance Companies
 - Government Sponsored Enterprises

Non-depository Institutions

- Insurance Companies
- Securities Firms
 - Brokerage firms
 - Investment banks
 - Mutual fund companies
- Finance Companies
- Government Sponsored Enterprises

Insurance Companies

- Insurance companies began hundreds of years ago with long sea voyages
- The most famous insurance company, Lloyd's of London, was established in 1688
- Besides insuring traditional assets like airplane and ships, it also insures singers' voices, pianists' fingers and even food critics' taste buds

Insurance Companies

- Underwriting process refers to the risk assessment and loss reimbursement guarantee by the individual risk experts of the relevant field joining together to form a syndicate.
- When an insurance contract is offered, these syndicates sign up for a certain portion of the risk in return for a portion of the risk premiums

Insurance Companies

- Insurance process
 - Insurance companies accept premiums in exchange for the promise of compensation if certain event occurs
 - A home owner pays premium in return for the promise that if the house burns down, the insurance company will pay to rebuild it
 - So for individuals, insurance is way for transferring the risk

Insurance Companies

- In terms of financial system as a whole, insurance companies:
 - Pool small policies and make large investments
 - Diversify risks across a large population
 - Screen and monitor policyholders to mitigate the problem of asymmetric information

Insurance Companies

- Two Types of Insurance Company:
 - life insurance
 - property and casualty insurance

Insurance Companies

- Type of Life insurance

- Term life insurance

- which makes a payment to the insured's beneficiaries upon the death of the insured
 - Group insurance is obtained through employers

Insurance Companies

- Whole life insurance
 - Combination of term life insurance and a savings account
 - A payment of a fixed premium over lifetime in return for a fixed benefit in case of death of policy holder
 - The cash value can be refunded if the policyholder decides to discontinue the policy
 - Over the years, the emphasis shifts from insurance to savings

Insurance Companies

- Property and casualty Insurance
 - Auto insurance is a combination of property insurance on the car and casualty insurance on the driver
 - The policyholder pays premium in exchange for protection

Insurance Companies

- Balance sheet
 - Liabilities
 - Promises to policyholders
 - Assets
 - Combination of bonds and stocks
 - Short term money market instruments (in case of property and casualty insurance)

Insurance Companies

- The Role of Insurance Companies:
 - Insurance companies pool risk to generate predictable payouts
 - Adverse selection and moral hazard create problems in the insurance market that are worse than those in the stock and bond markets
 - Cancer Patients
 - Fire Insurance

Insurance Companies

- To deal with this, insurance companies carefully screen applicants before issuing them policies
 - Medical Examination
 - Driving Records
- Policies may also include restrictive covenants in order to reduce moral hazard
 - Fire extinguishing system and training
 - careful

Insurance Companies

- The future of insurance must be considered in the light of advances in medical technology, particularly with regard to the decoding of the human genome.
- In the future, people with inherited tendencies toward certain diseases may not be able to get insurance

Securities Firms

- The broad class of securities firms include brokerages, investment banks, and mutual fund companies.
- In one way or another, these are all financial intermediaries
- The primary services of brokerage firms are accounting and the provision of access to secondary markets.

Securities Firms

- They also provide loans to customers who wish to purchase stock on margin, and they provide liquidity by offering check-writing privileges and by allowing investors to sell assets quickly
- All securities firms are very much in the business of producing information; but this is truly at the heart of the investment banking business

Summary

- Non-depository Institutions
 - Insurance Companies
 - Securities Firms
 - Brokerage Firms

Money and Banking

Lecture 28

Review of the Previous Lecture

- Non-depository Institutions
 - Insurance Companies
 - Securities Firms
 - Brokerage Firms

Topics under Discussion

- Securities Firms
 - Investment Banks
 - Mutual Funds
- Finance Companies
- Government Sponsored Enterprises
- Banking Crisis
 - Sources of Runs, Panics and Crisis
- Government Safety Net
 - Government: Lender of Last Resort

Securities Firms

- Investment banks are the conduits through which firms raise funds in the capital markets
- Through their underwriting services, investment banks issue new stocks and a variety of other debt instruments

Securities Firms

- In underwriting, the investment bank guarantees the price of a new issue and then sells it to investors at a higher price;
 - However, this is not without risk, since the selling price may not in fact be higher than the price guaranteed to the firm issuing the security

Securities Firms

- Information and reputation are central to the underwriting business;
 - underwriters collect information to determine the price of the new securities and then put their reputations on the line when they go out to sell the issues
- In addition to underwriting, investment banks provide advice to firms that wish to merge with or acquire other firms, for which advice they are paid a fee

Finance Companies

- Finance companies raise funds in the financial markets by issuing commercial paper and securities and use the funds to make loans to individuals and corporations
- These companies are largely concerned with reducing the transactions and information costs that are associated with intermediated finance

Finance Companies

- Most finance companies specialize in one of three loan types:
 - consumer loans,
 - business loans,
 - sales loans (for example, the financing for a consumer to purchase a large-ticket item like an appliance).
 - Some also provide commercial and home mortgages

Finance Companies

- Business finance companies provide loans to businesses, for equipment leasing
- Business finance companies also provide short-term liquidity to firms by offering
 - inventory loans (so that firms can keep the shelves stocked)
 - accounts receivable loans (which provide immediate resources against anticipated revenue streams)

Government-Sponsored Enterprises

- The government is directly involved in the financial intermediation system through loan guarantees and in the chartering of financial institutions to provide specific types of financing
 - Zarai Taraqati Bank Limited (ZTBL)
 - Small and Medium Enterprise (SME) Bank
 - House Building Finance Corporation (HBFC)
 - Khushhali Bank

Financial Intermediary	Primary Sources of Funds (Liabilities)	Primary Uses of Funds (Assets)	Services Provided
Depository Institution (Bank)	Checkable deposits Savings and time deposits Borrowing from other banks	Cash Loans Securities	• Pooling of small savings to provide large loans • Diversified, liquid deposit accounts • Access to payments system • Screening and monitoring of borrowers
Insurance Company	Expected claims	Corporate bonds Government bonds Stocks Mortgages	• Pooling of risk • Screening and monitoring of policy-holders

Summary of Financial Industry Structure

Financial Intermediary	Primary Sources of Funds (Liabilities)	Primary Uses of Funds (Assets)	Services Provided
Securities Firm	Short-term loans	Commercial paper Bonds	• Management of asset pools • Clearing and settling trades
Investment Bank			• Immediate sale of assets • Access to spectrum of assets, allowing diversification • Evaluation of firms wishing to issue securities • Research and advice for investors
Mutual-Fund Company	Shares sold to customers	Commercial paper Bonds Mortgages Stocks Real estate	• Pooling of small savings to provide access to large, diversified portfolios, which can be liquid

Summary of Financial Industry Structure

Financial Intermediary	Primary Sources of Funds (Liabilities)	Primary Uses of Funds (Assets)	Services Provided
Finance Company	Bonds Bank loans Commercial paper	Mortgages Consumer loans Business loans	• Screening and monitoring of borrowers
Government-Sponsored Enterprise	Commercial paper Bonds Loan guarantees	Mortgages Farm loans Student loans	• Access to financing for borrowers who cannot obtain it elsewhere

Banking Crisis

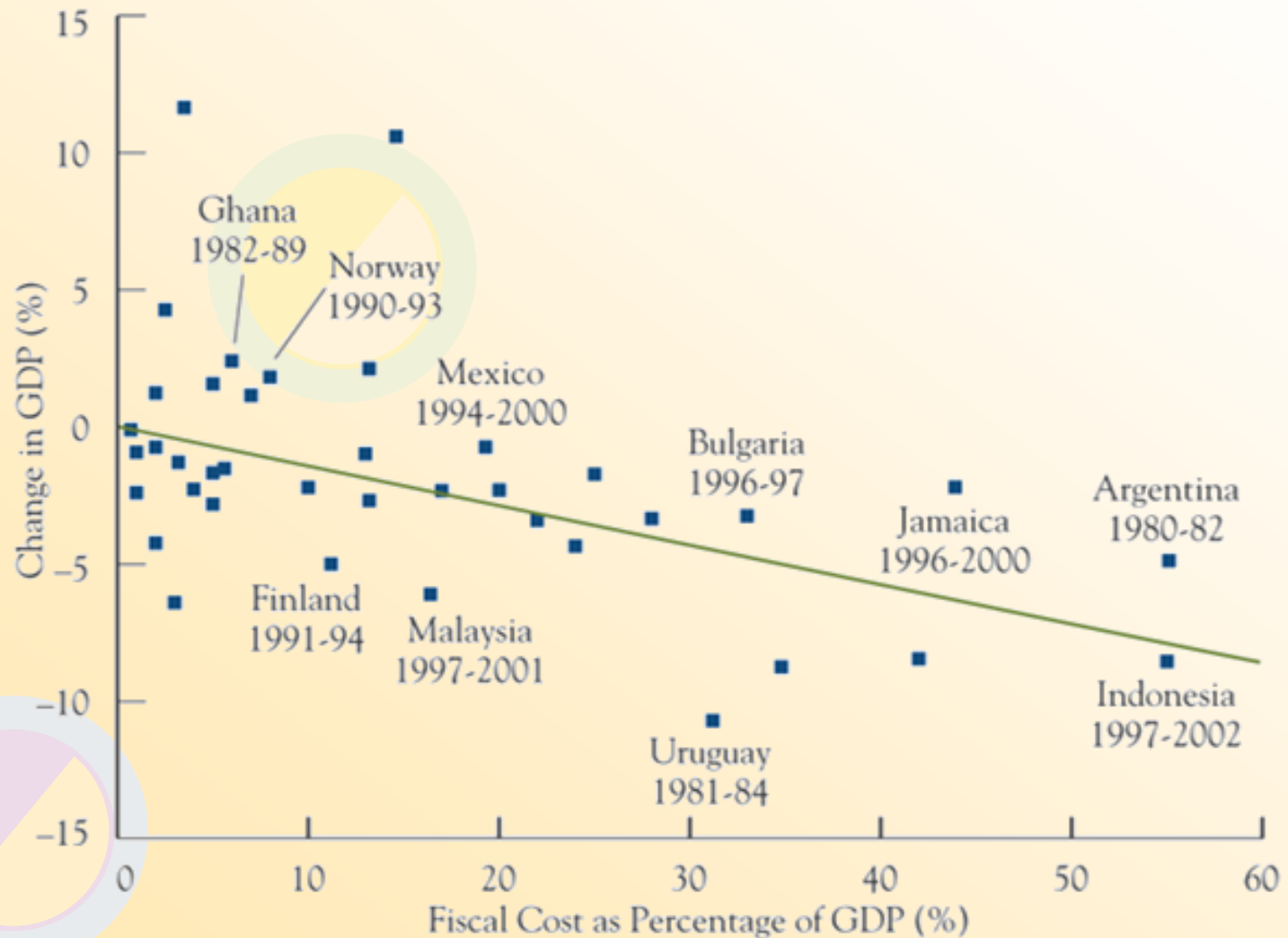
- Banking crises are not a new phenomena; the history of commercial banking over the last two centuries is replete with period of turmoil and failure.
- By their very nature, financial systems are fragile and vulnerable to crisis

Worst Banking Crises since 1980

Country	Crisis Dates	Estimated Cost of Resolution (as % of GDP)
Argentina*	1980–1982	55%
Indonesia*	1997–1998	55%
China	1990s	47%
Jamaica*	1994	44%
Chile*	1981–1983	42%
Thailand*	1997	35%
Macedonia	1993–1994	32%
Israel	1977–1983	30%
Turkey*	2000	30%
Uruguay	1981–1984	29%
Korea	1998	28%
Côte d'Ivoire	1988–1991	25%
Japan	1990s	24%
Uruguay	1981–1984	24%
Malaysia*	1997–1998	20%

South Asian
Crisis

Relationship between Size of a Financial Crisis and Change in GDP Growth



The Sources and Consequences of Runs, Panics, and Crises

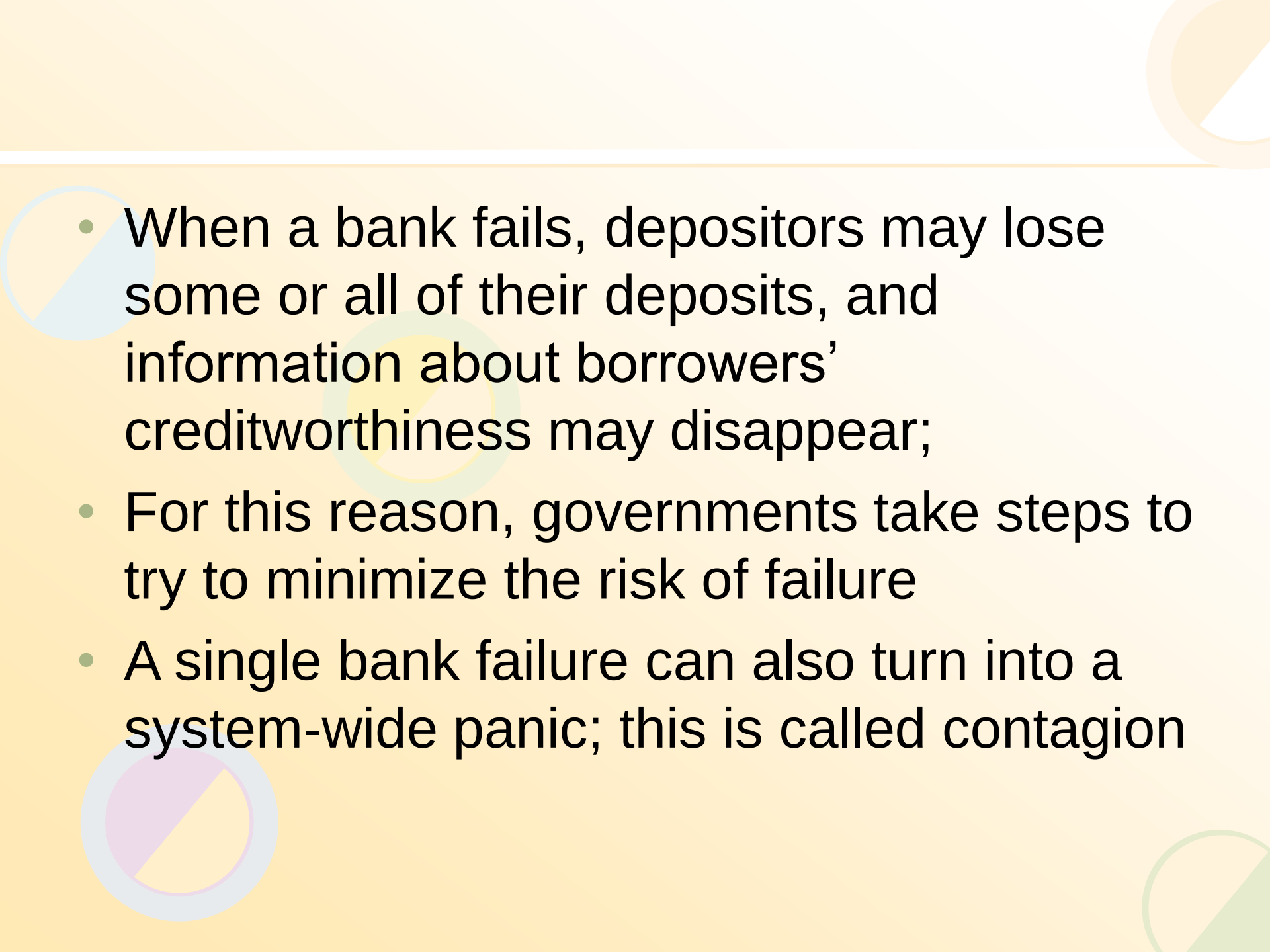
- In a market based economy, the opportunity to succeed is also the opportunity to fail!
- Banks serve some essential functions in the economy
 - Access to payment system
 - Screen and monitor borrowers to reduce information problems
- So if bank fails, we lose ability to make financial transactions. Collectively, the economy is endangered.

The Sources and Consequences of Runs, Panics, and Crises

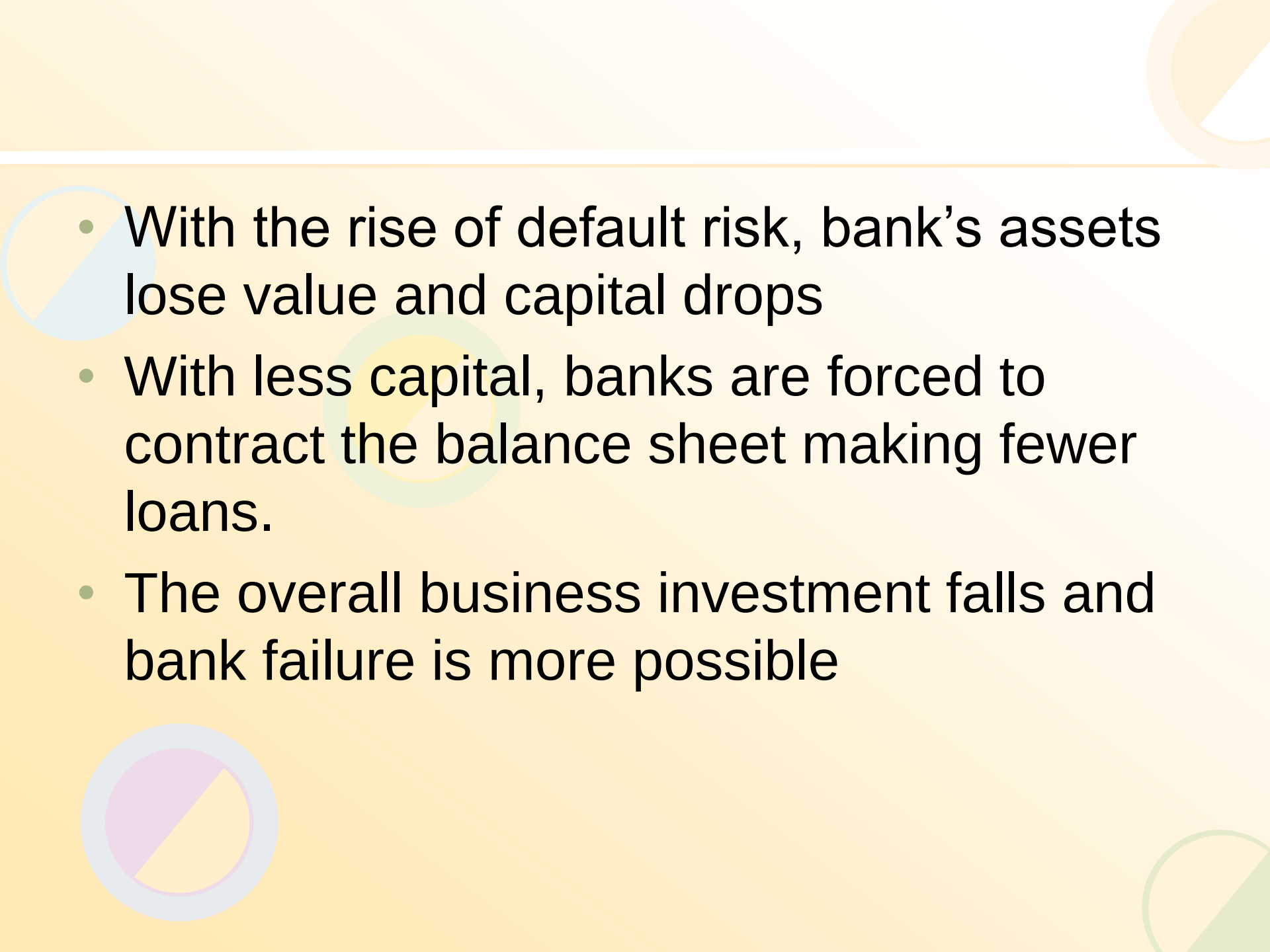
- Banks' fragility arises from the fact that they provide liquidity to depositors, allowing them to withdraw their balances on demand, on a first-come, first-served basis
- If bank can not meet this promise of withdrawal, because of insufficient funds, it will fail

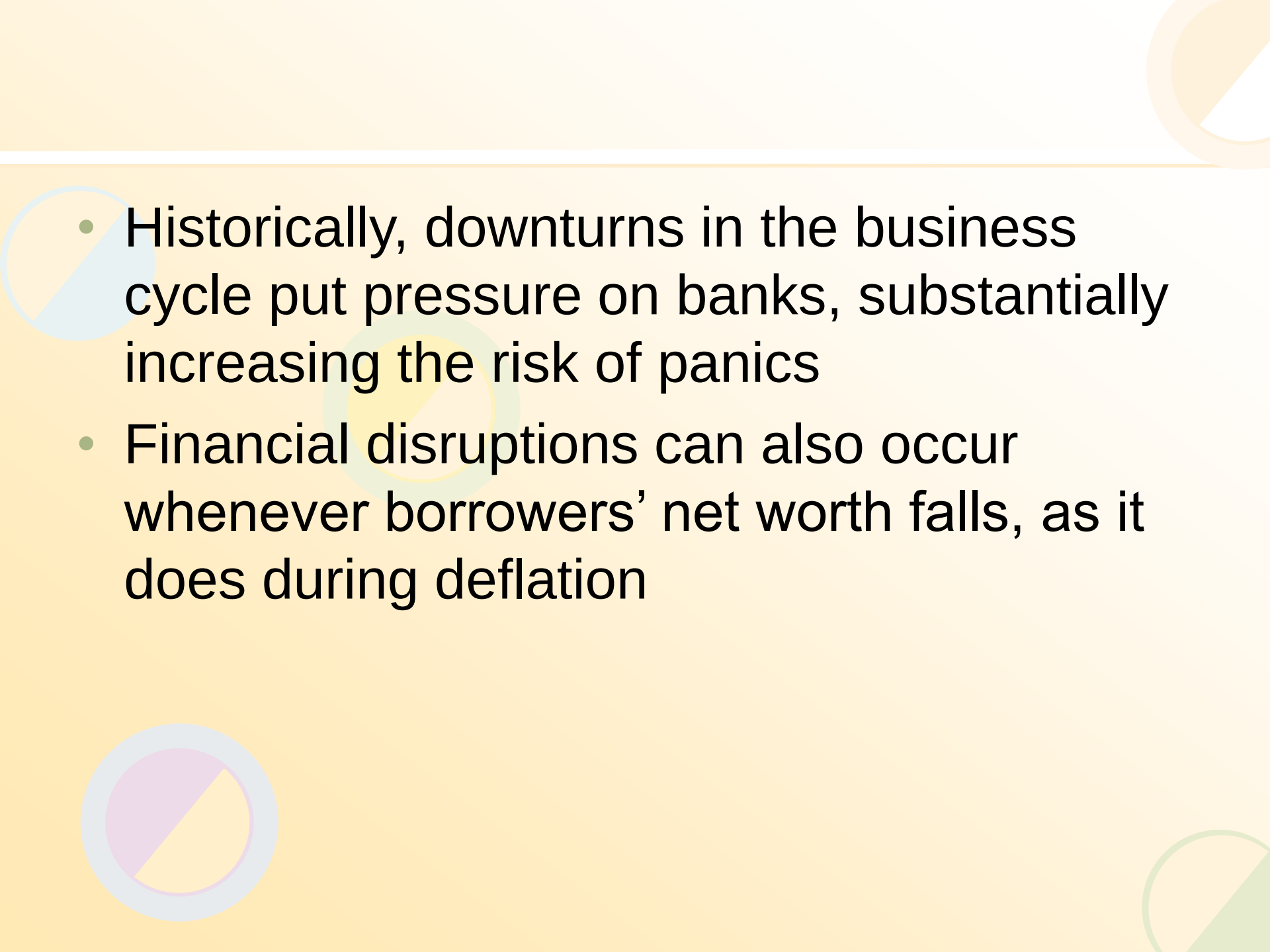
- Reports that a bank has become insolvent can spread fear that it will run out of cash and close its doors;
 - Depositors will rush to convert their balances into cash
 - such a run on a bank can cause it to fail

- What matters during a bank run is not whether a bank is solvent but whether it is liquid
 - Here solvency means that the value of the bank's assets exceeds its liabilities (positive net worth)
 - Liquidity refers to the sufficient reserves of the bank to meet withdrawal demands
- False rumors that a bank is insolvent can lead to a run which renders it illiquid

- 
- When a bank fails, depositors may lose some or all of their deposits, and information about borrowers' creditworthiness may disappear;
 - For this reason, governments take steps to try to minimize the risk of failure
 - A single bank failure can also turn into a system-wide panic; this is called contagion

- While banking panics and financial crises can result from false rumors, they can also occur for more concrete reasons;
 - anything that affects borrowers' ability to repay their loans or drives down the market price of securities has the potential to imperil the bank's finances
 - Recessions have a clear negative impact on bank's balance sheet
 - Low profitability of firm makes debt repayment much harder
 - People lose jobs and cant pay their loan

- 
- With the rise of default risk, bank's assets lose value and capital drops
 - With less capital, banks are forced to contract the balance sheet making fewer loans.
 - The overall business investment falls and bank failure is more possible

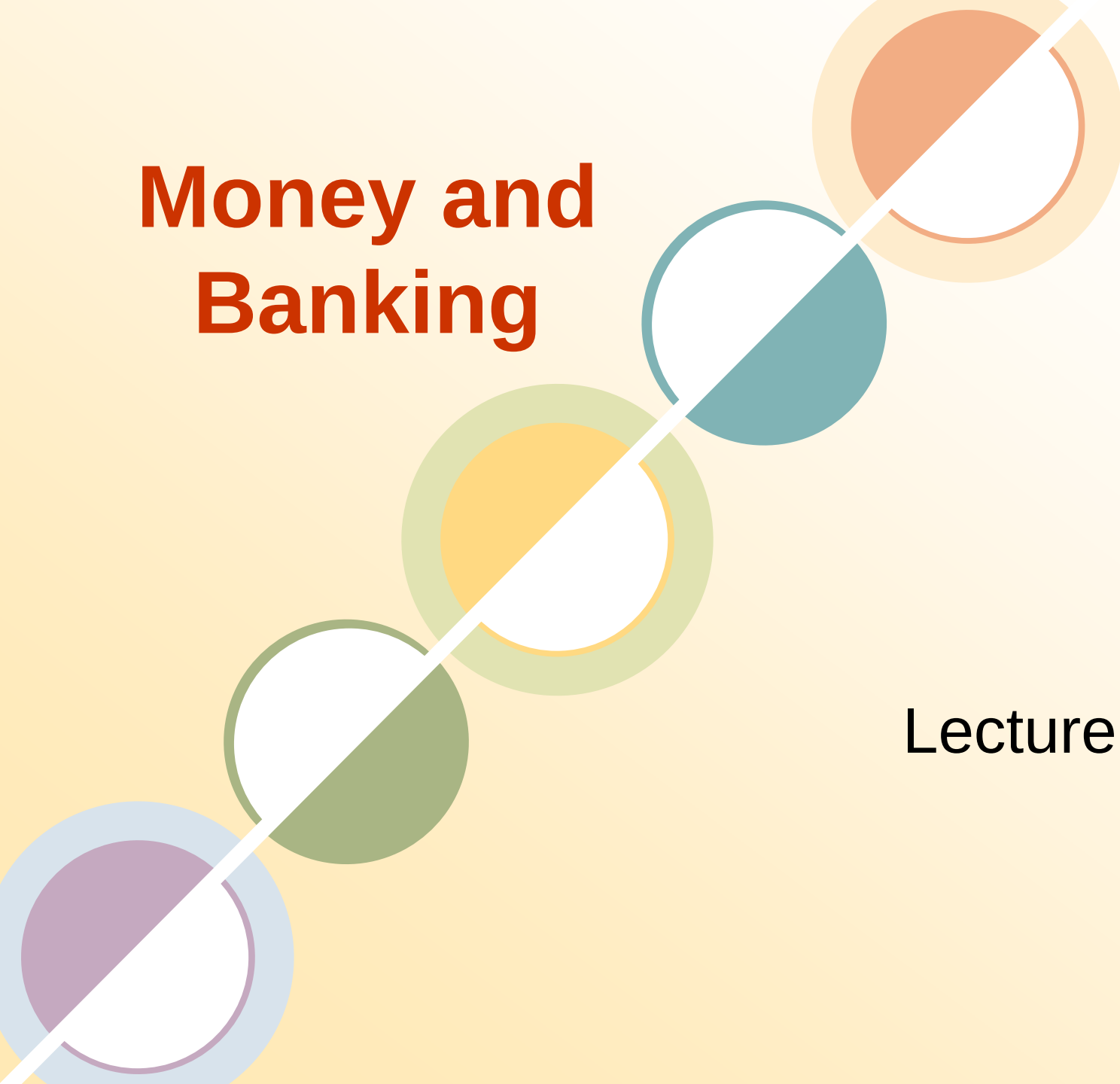
- 
- Historically, downturns in the business cycle put pressure on banks, substantially increasing the risk of panics
 - Financial disruptions can also occur whenever borrowers' net worth falls, as it does during deflation

Summary

- Non-depository Institution
 - Insurance Companies
 - Securities Firms
 - Finance Companies
 - Govt. Sponsored Enterprises
- Bank Crisis
 - Sources of Bank Runs, Panics and crisis`

Money and Banking

Lecture 29

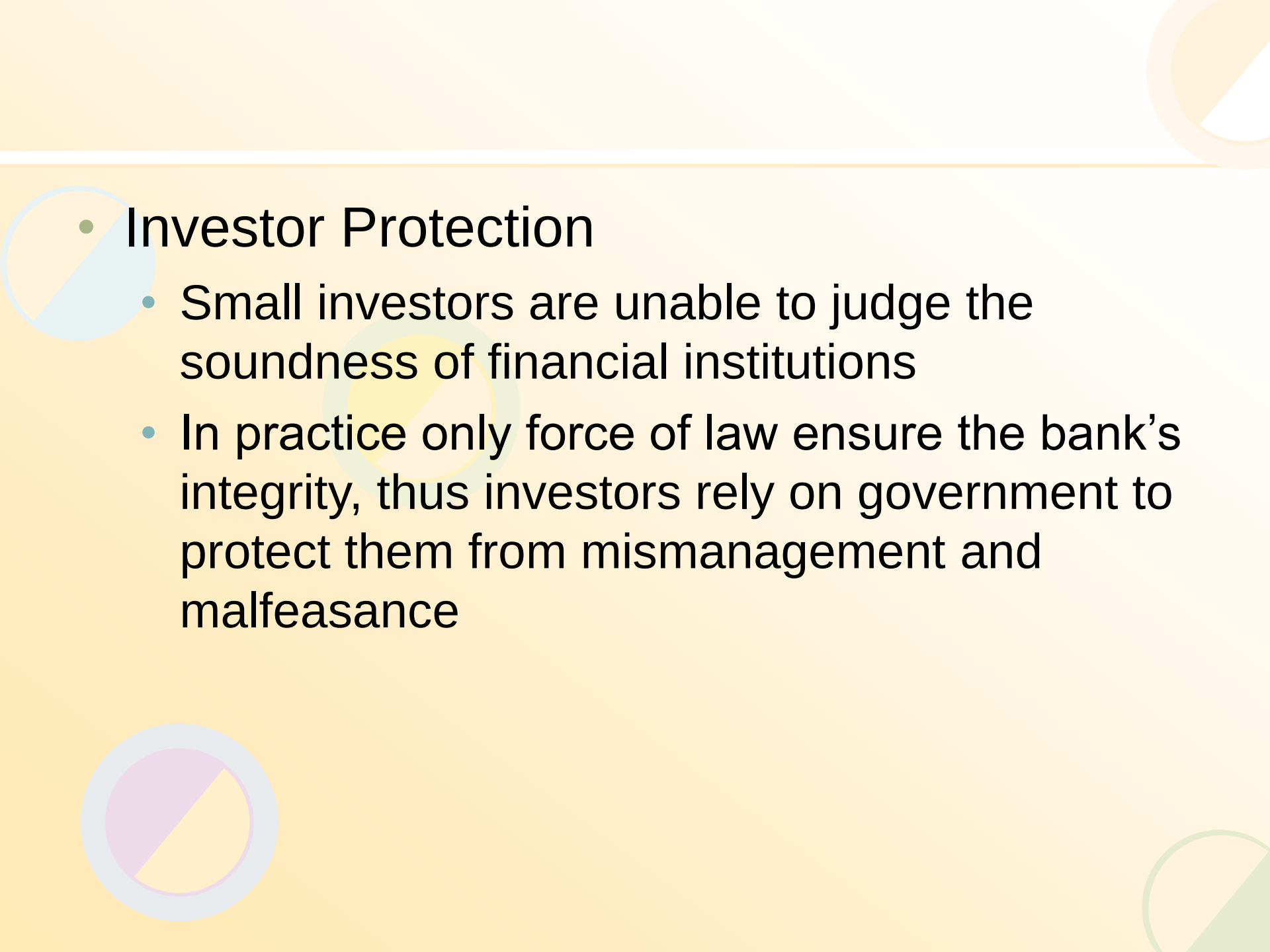


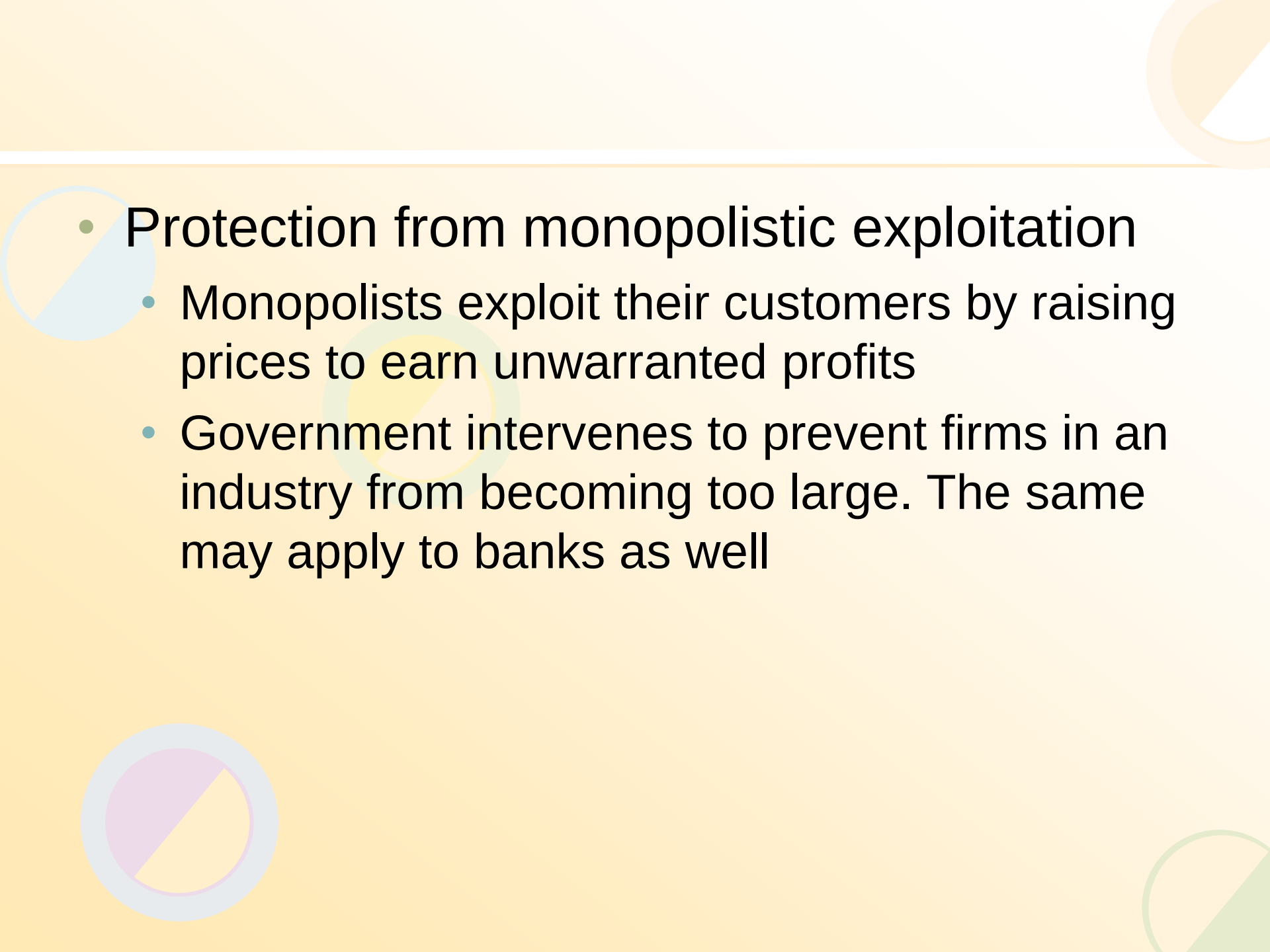
Review of the Previous Lecture

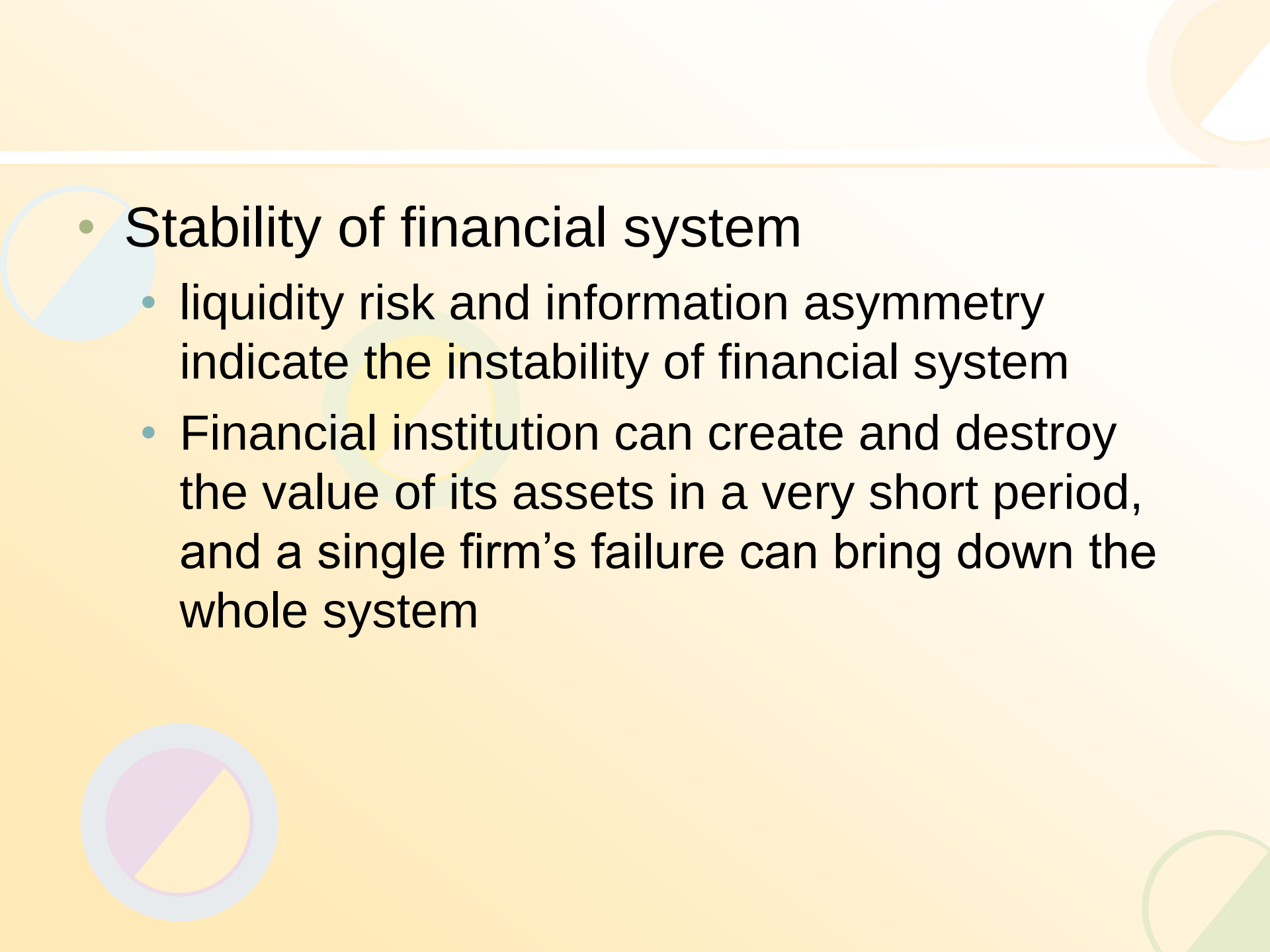
- Non-depository Institutions
 - Insurance Companies
 - Securities Firms
 - Finance Companies
 - Government Sponsored Enterprises
- Banking Crisis
 - Sources of Runs, Panics and Crisis

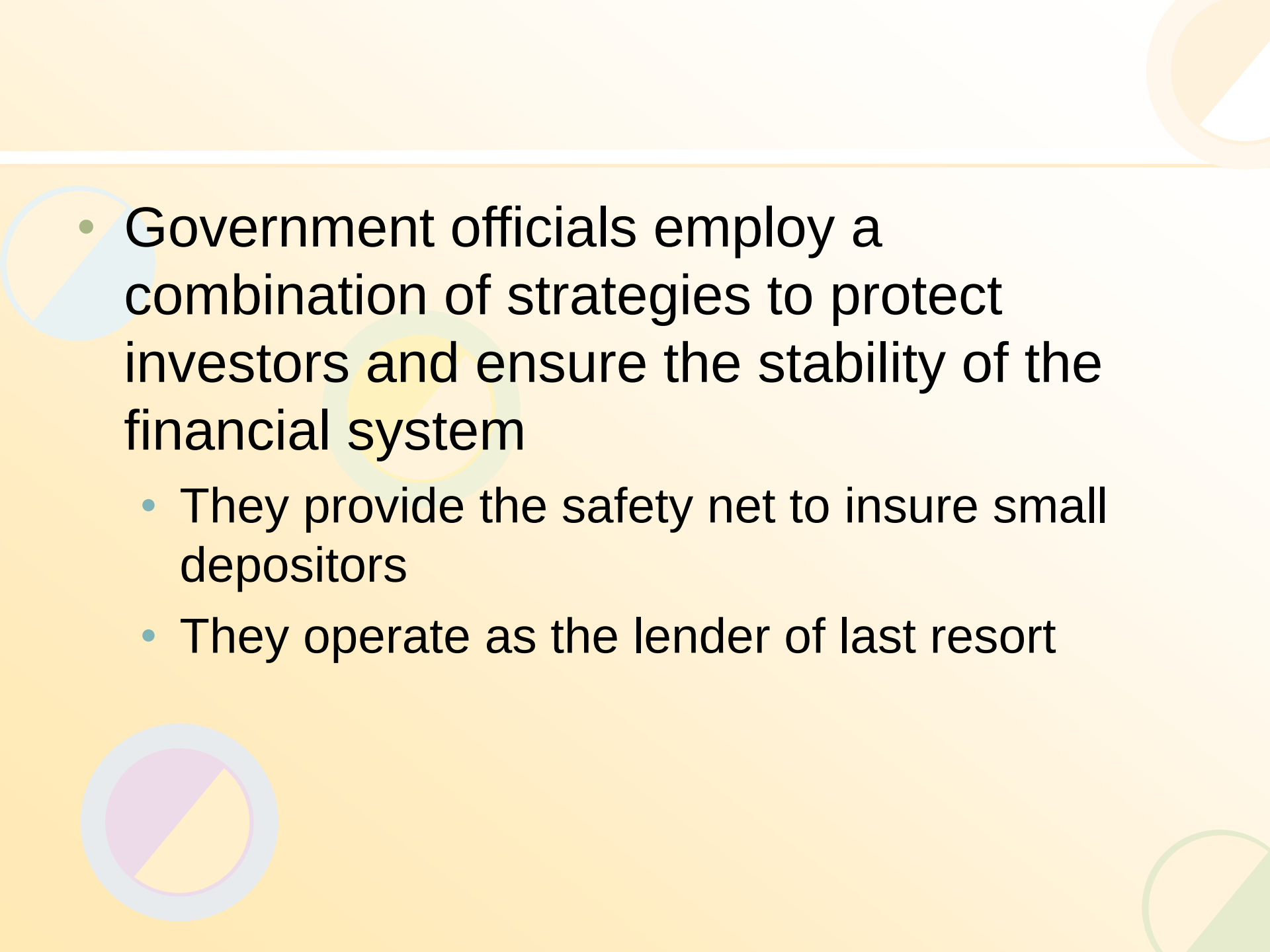
The Government Safety Net

- There are three reasons for the government to get involved in the financial system
 - to protect investors.
 - to protect bank customers from monopolistic exploitation.
 - to ensure the stability of the financial system

- 
- Investor Protection
 - Small investors are unable to judge the soundness of financial institutions
 - In practice only force of law ensure the bank's integrity, thus investors rely on government to protect them from mismanagement and malfeasance

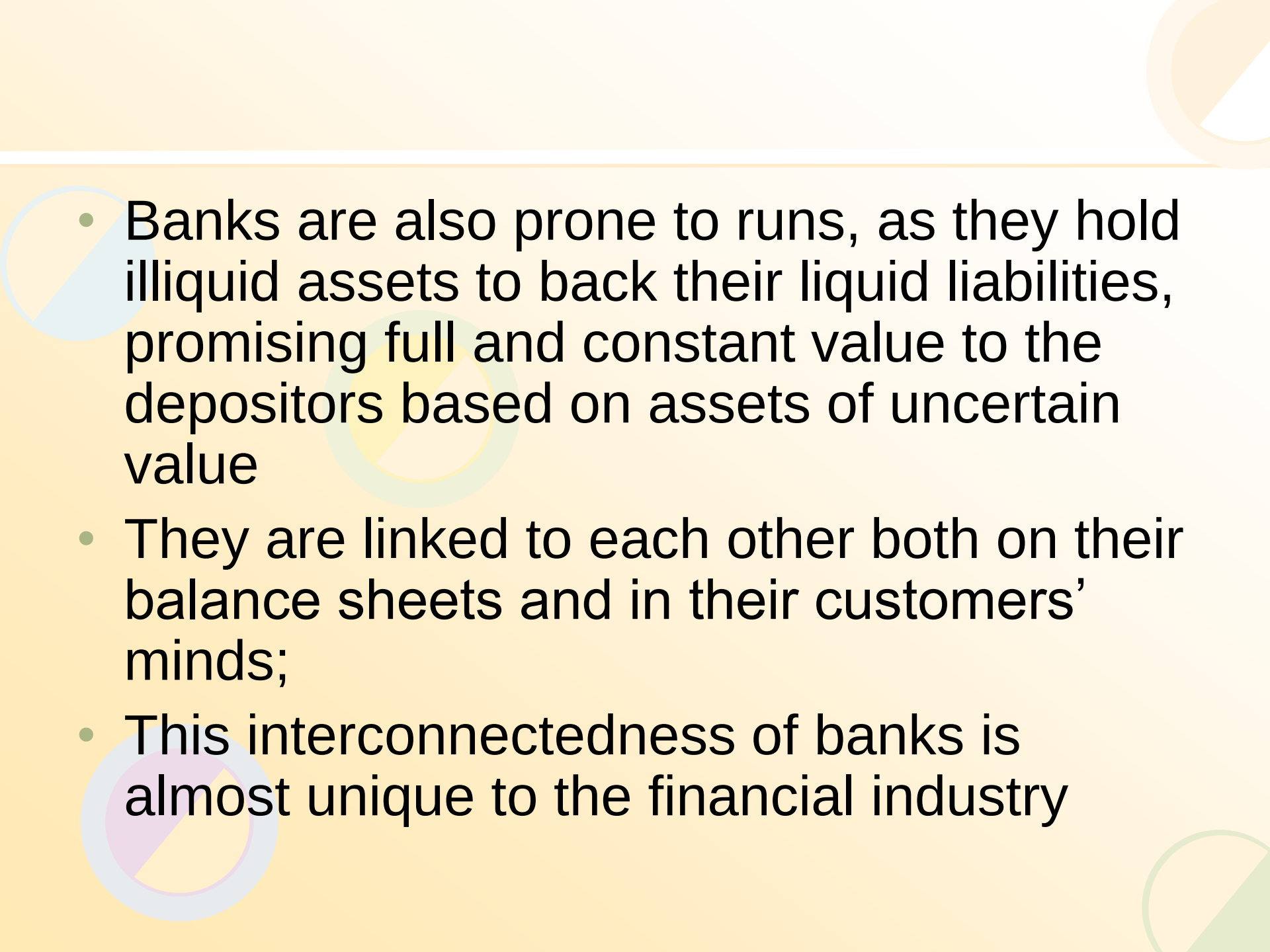
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- Protection from monopolistic exploitation
 - Monopolists exploit their customers by raising prices to earn unwarranted profits
 - Government intervenes to prevent firms in an industry from becoming too large. The same may apply to banks as well

- 
- Stability of financial system
 - liquidity risk and information asymmetry indicate the instability of financial system
 - Financial institution can create and destroy the value of its assets in a very short period, and a single firm's failure can bring down the whole system

- 
- Government officials employ a combination of strategies to protect investors and ensure the stability of the financial system
 - They provide the safety net to insure small depositors
 - They operate as the lender of last resort

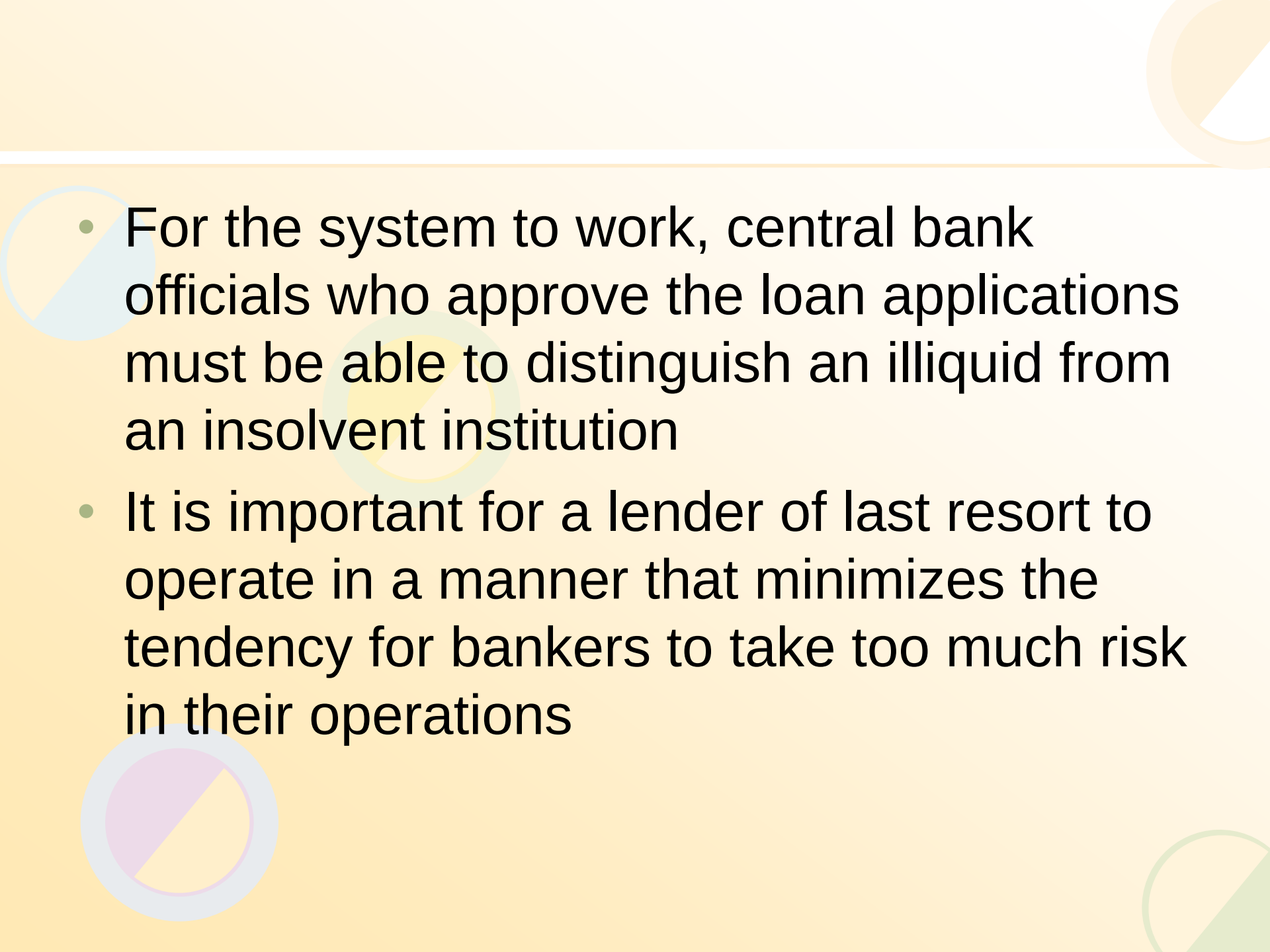
The Unique Role of Depository Institutions

- Depository institutions receive a disproportionate amount of attention from government regulators because
 - they play a central role in the economy
 - they face a unique set of problems
- We all rely heavily on banks for access to the payments system

- 
- Banks are also prone to runs, as they hold illiquid assets to back their liquid liabilities, promising full and constant value to the depositors based on assets of uncertain value
 - They are linked to each other both on their balance sheets and in their customers' minds;
 - This interconnectedness of banks is almost unique to the financial industry

The Government as Lender of Last Resort

- The best way to stop a bank failure from turning into a panic is to make sure solvent institutions can meet their depositors' withdrawal demands
- The existence of a lender of last resort significantly reduces, but does not eliminate, contagion

- 
- For the system to work, central bank officials who approve the loan applications must be able to distinguish an illiquid from an insolvent institution
 - It is important for a lender of last resort to operate in a manner that minimizes the tendency for bankers to take too much risk in their operations

Problems Created by the Government Safety Net

- Protected depositors have no incentive to monitor their banks' behavior, and knowing this, banks take on more risk than they would normally
- In protecting depositors the government creates moral hazard

- Some banks are too big to fail, meaning that their failure would cause havoc in the financial system.
- The managers of such institutions know that if they begin to founder the government will have to bail them out
- The too-big-to-fail policy limits the extent of the market discipline that depositors can impose on banks and compounds the moral hazard problem

Regulation and Supervision of the Financial System

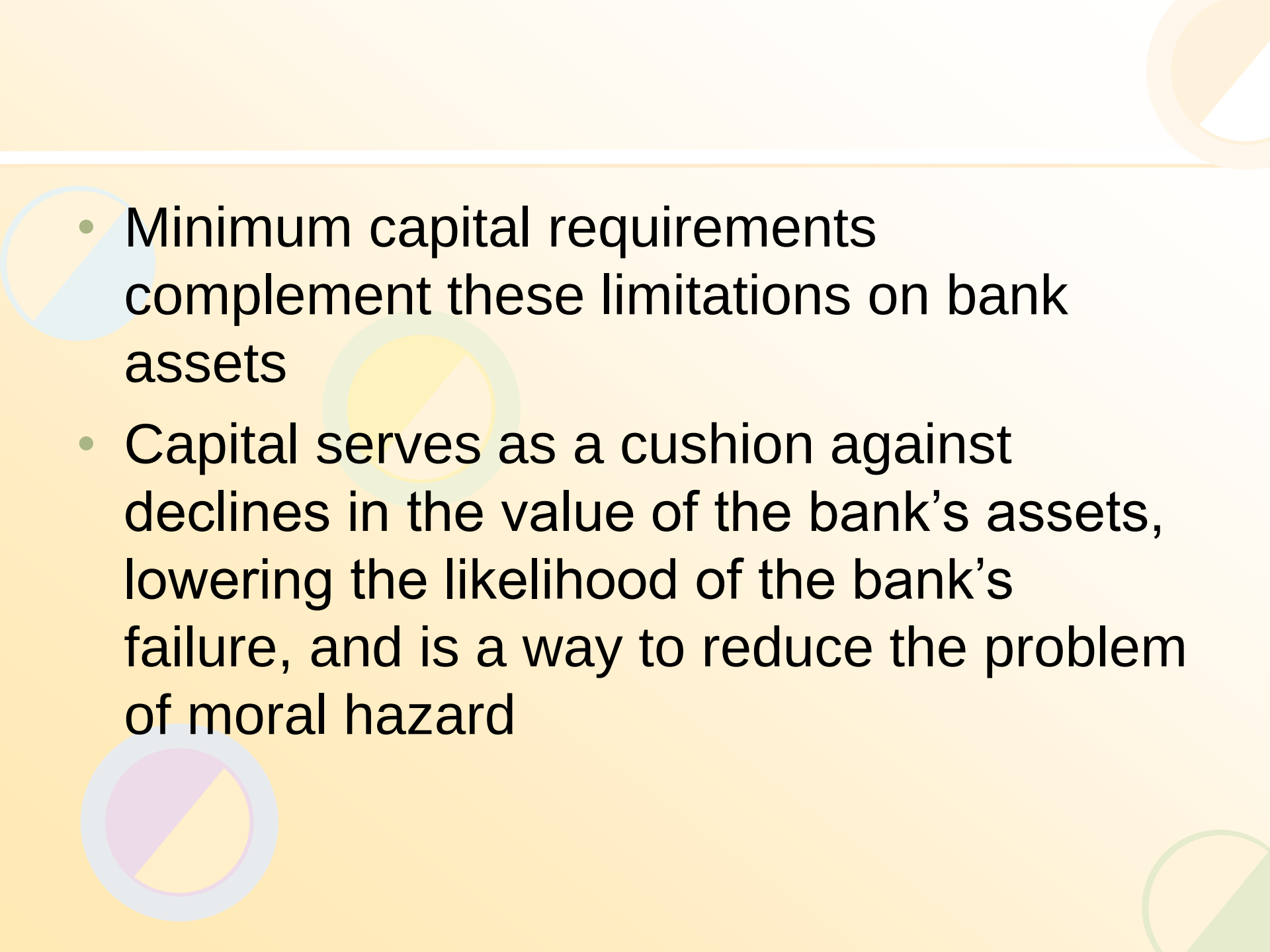
- Government officials employ three strategies to ensure that the risks created by the safety net are contained:
 - regulation establishes rules for bank managers to follow,
 - supervision provides general oversight of financial institutions,
 - examination provides detailed information on the firms' operations

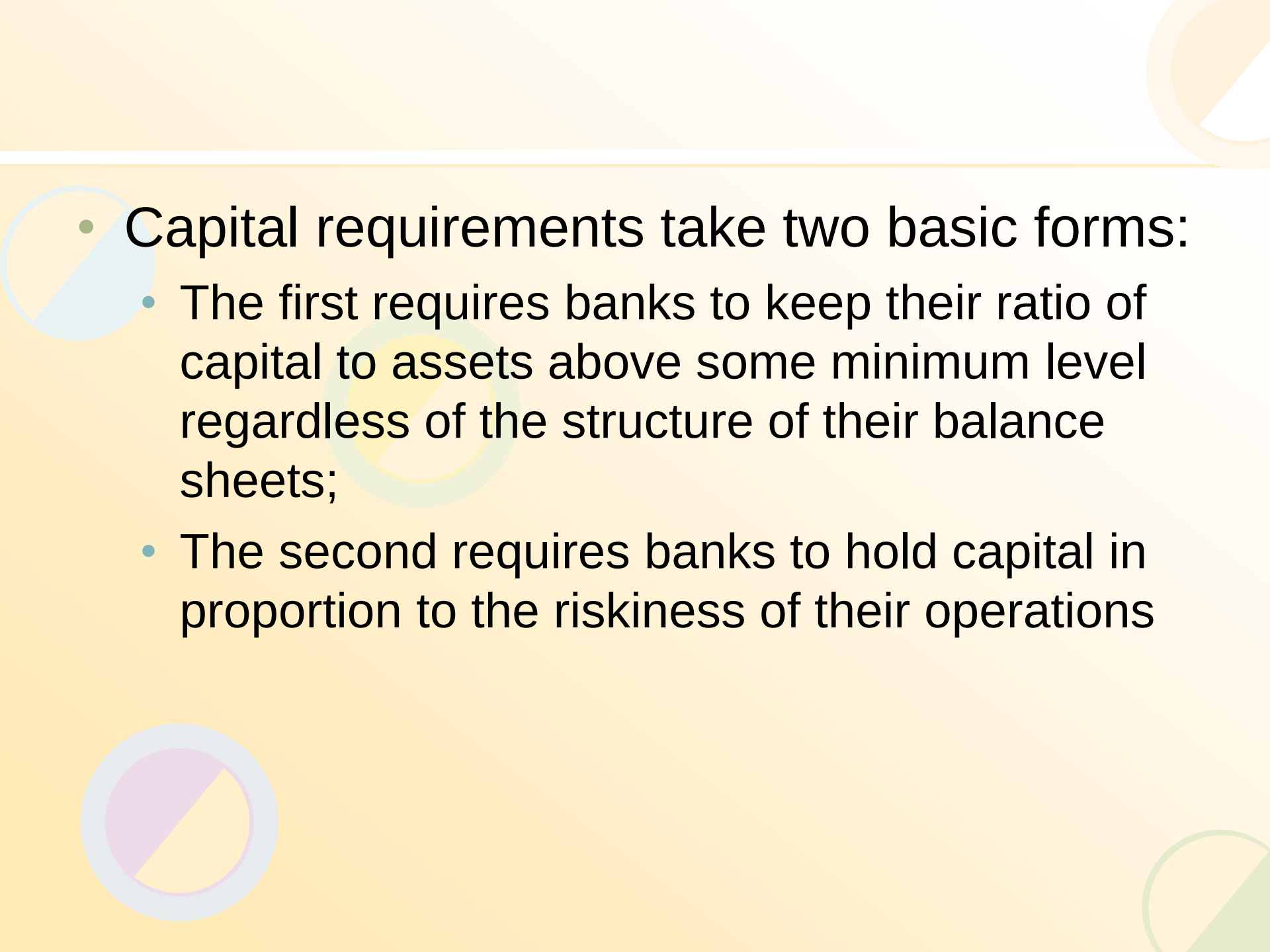
- Regulatory requirements are designed to minimize the cost of failures to the tax-paying public
- One example of regulation is the requirement that banks obtain a charter in order to operate;
 - this provides screening to make sure that the people who own and run banks will not be criminals.
 - Once a bank is operating other regulations control the assets, the amount of capital, and makes information about the bank's balance sheet public

- Government supervisors enforce the regulations;
 - they monitor, inspect, and examine banks to make sure that their business practices conform to regulatory requirements
- State Bank of Pakistan (SBP) is supreme regulatory authority for banking sector in Pakistan
 - www.sbp.org.pk

Asset Holding Restrictions and Minimum Capital Requirements

- The simplest way to prevent bankers from exploiting their safety net is to restrict banks' balance sheets;
 - This can be through restrictions on the kinds of assets banks can hold and requirements that they maintain minimum levels of capital
- The size of the loans a bank can make to particular borrowers is also limited

- 
- Minimum capital requirements complement these limitations on bank assets
 - Capital serves as a cushion against declines in the value of the bank's assets, lowering the likelihood of the bank's failure, and is a way to reduce the problem of moral hazard

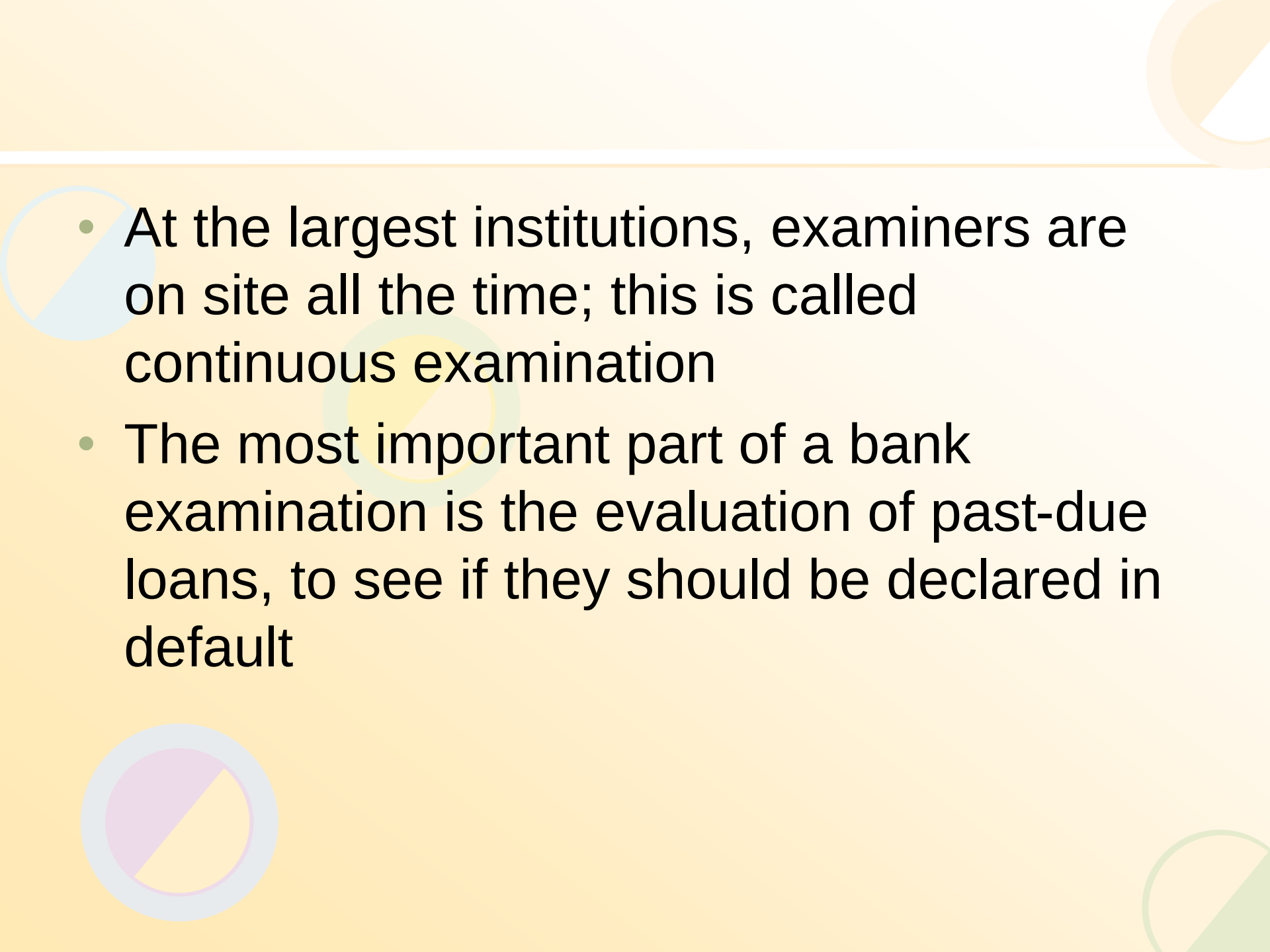
- 
- Capital requirements take two basic forms:
 - The first requires banks to keep their ratio of capital to assets above some minimum level regardless of the structure of their balance sheets;
 - The second requires banks to hold capital in proportion to the riskiness of their operations

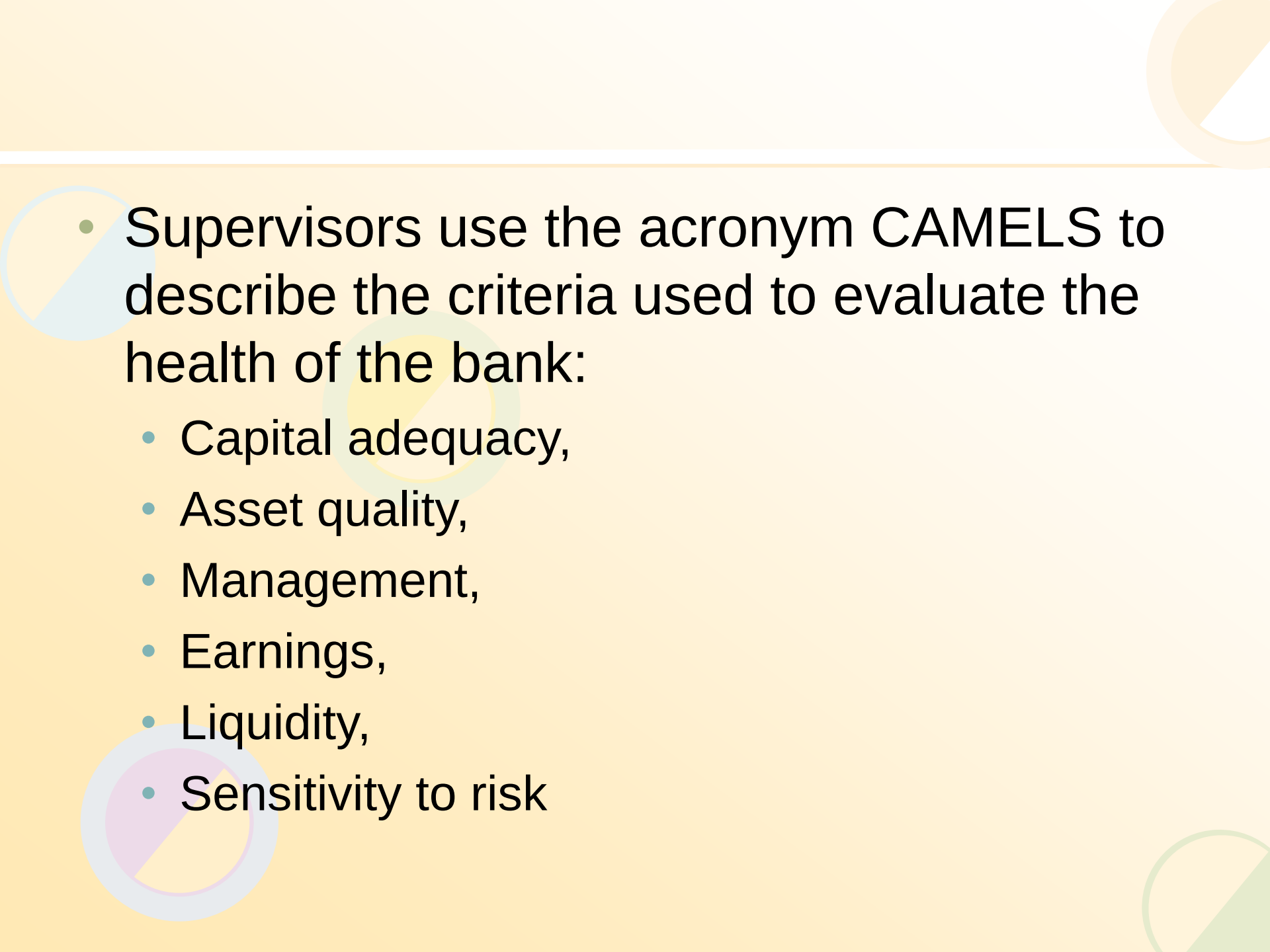
Disclosure Requirements

- Banks must provide information to the financial markets about their balance sheets;

Supervision and Examination

- The government enforces banking rules and regulations through an elaborate oversight process called supervision, which relies on a combination of monitoring and inspection
- Supervision is done remotely, through an examination of the detailed reports banks must submit, as well as through on-site examination

- 
- At the largest institutions, examiners are on site all the time; this is called continuous examination
 - The most important part of a bank examination is the evaluation of past-due loans, to see if they should be declared in default

- 
- Supervisors use the acronym CAMELS to describe the criteria used to evaluate the health of the bank:
 - Capital adequacy,
 - Asset quality,
 - Management,
 - Earnings,
 - Liquidity,
 - Sensitivity to risk

- Current practice is for examiners to act as consultants to banks, advising them on how to get the highest return possible while keeping risk at an acceptable level that ensures the bank will stay in business

Money and Banking

Lecture 30

Review of the Previous Lecture

- Banking Crisis
 - Sources of Runs, Panics and Crisis
- Government Safety Net
 - Government as lender of last resort

Topics under Discussion

- Central Bank:
 - The Government's Bank
 - The Bankers' Bank
 - Objectives

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www.sbp.org.pk

The Government's Bank

- The central bank started out as the government's bank, originally created by rulers to finance wars
- However, the early examples are really the exceptions, as central banking is largely a 20th century phenomenon.
- The central bank occupies a privileged position: it has a monopoly on the issuance of currency

- The central bank creates money and thereby controls the availability of money and credit in a country's economy
- Most central banks go about this by adjusting short-term interest rates, an activity called monetary policy.
- In today's world, central banks use monetary policy to stabilize economic growth and inflation.

- An expansionary or accommodative policy (lower interest rates) raises growth and inflation; tighter or restrictive policy reduces them.
- Governments want to control the printing of money because it is a very profitable business; also, losing control of the amount of currency means losing control of inflation.

The Bankers' Bank

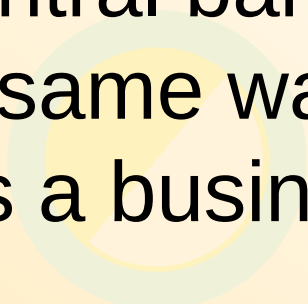
- The most important day-to-day jobs of the central bank are to:
 - provide loans during times of financial stress (the lender of last resort).
 - manage the payments system (settles interbank payments).
 - oversee commercial banks and the financial system (handles the sensitive information about institutions without conflicts of interest).

- By ensuring that sound banks and financial intermediaries can continue to operate, the central bank makes the whole financial system more stable.
- Central banks are the biggest and most powerful players in a country's financial and economic system and are supposed to use this power to stabilize the economy, making us all better off.

- 
- 
- However, central banks that are under extreme political pressure, or that are simply incompetent, can wreak havoc on the economic and financial systems.
 - A central bank does not control :
 - securities markets.
 - the government's budget.
- 



The common arrangement today is for the central bank to serve the government in the same way that a commercial bank serves a business or an individual.



The Functions of a Modern Central Bank

The Government's Bank

A: Manages the finances of the Govt.

B: Controls the availability of money and credit through interest rates

The Bankers' Bank

A: Guarantees that sound banks can do business by lending to them even during crisis

B: Operates a payment system for interbank payments

C: Oversees financial institutions to ensure confidence in their soundness

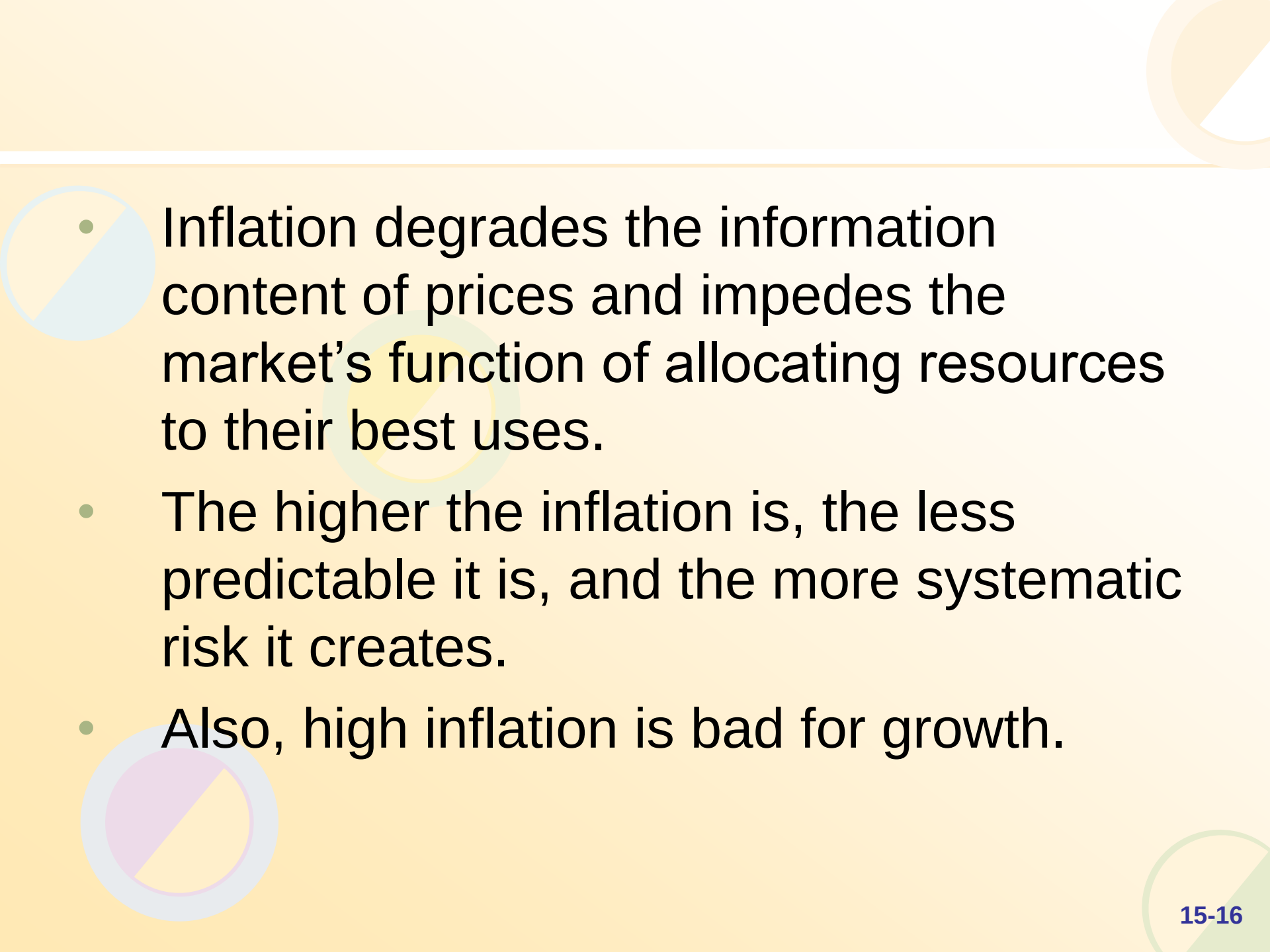
Stability: The Primary Objective of All Central Banks

- When economic and financial systems are left on their own they are prone to episodes of extreme volatility; central bankers work to reduce that volatility
- Central bankers pursue five specific objectives:
 - low and stable inflation
 - high and stable real growth, together with high employment
 - stable financial markets
 - stable interest rates
 - a stable exchange rate

- Instability in any of those would pose an economy-wide economic risk that diversification could not mitigate.
- Thus the job of the central bank is to improve general economic welfare by managing and reducing systematic risk.
- It is probably impossible to achieve all five of these objectives simultaneously, and so tradeoffs must be made

Low, Stable Inflation

- Many central banks take as their primary job the maintenance of price stability; they strive to eliminate inflation.
- The rationale for keeping the economy inflation-free is that money's usefulness as a unit of account and as a store of value is enhanced when its purchasing power is maintained.

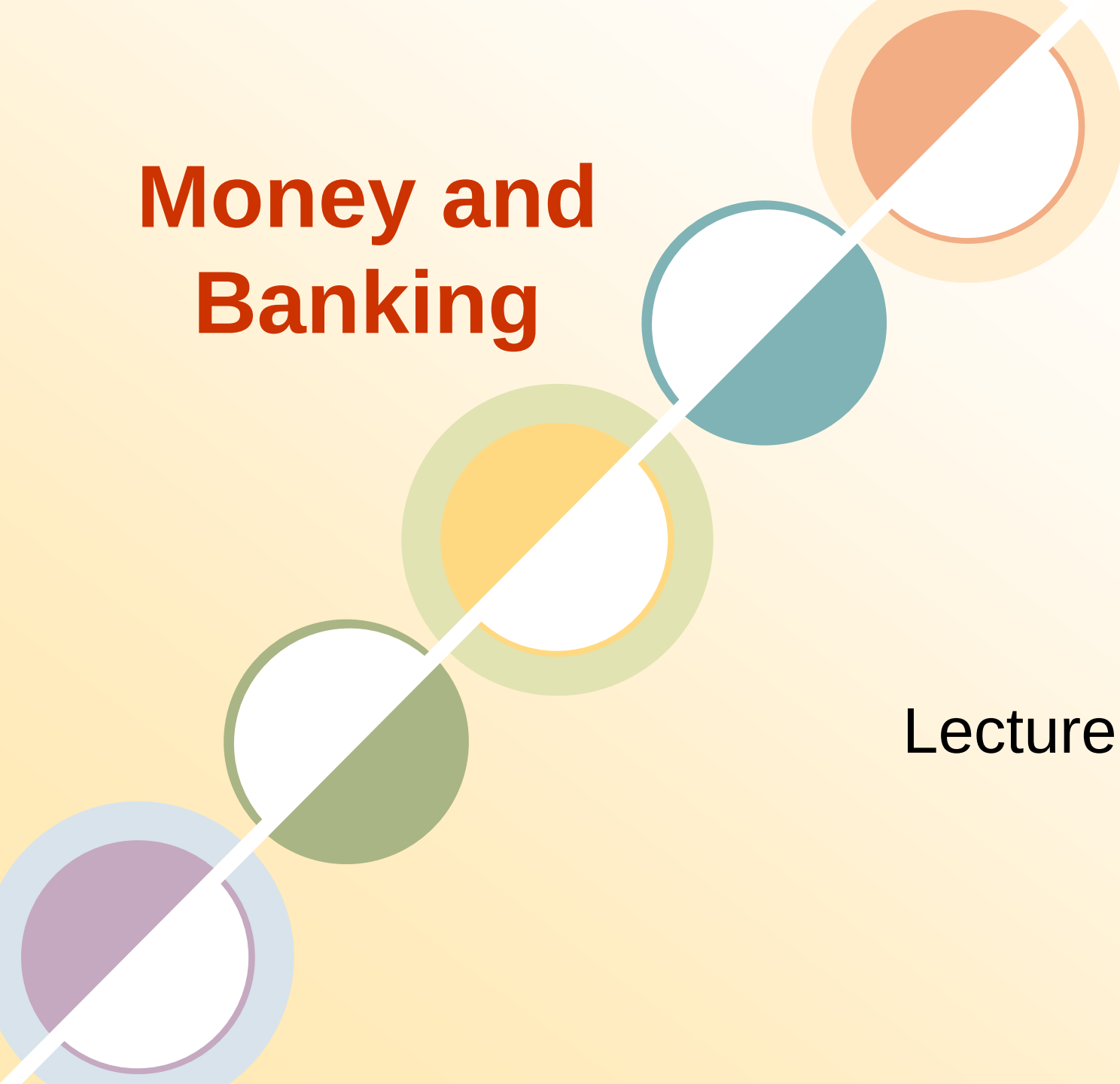
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- Inflation degrades the information content of prices and impedes the market's function of allocating resources to their best uses.
 - The higher the inflation is, the less predictable it is, and the more systematic risk it creates.
 - Also, high inflation is bad for growth.

- While there is agreement that low inflation should be the primary objective of monetary policy, there is no agreement on how low inflation should be.
- Zero inflation is too low, because it brings the risk of deflation (a drop in prices) which in turn results in increased defaults on loans and a threat to the health of banks.

- Furthermore, if inflation were zero, an employer wishing to cut labor costs would need to cut nominal wages, which is difficult to do.
- A small amount of inflation may actually make labor markets work better, at least from the employer's point of view.

Money and Banking

Lecture 31

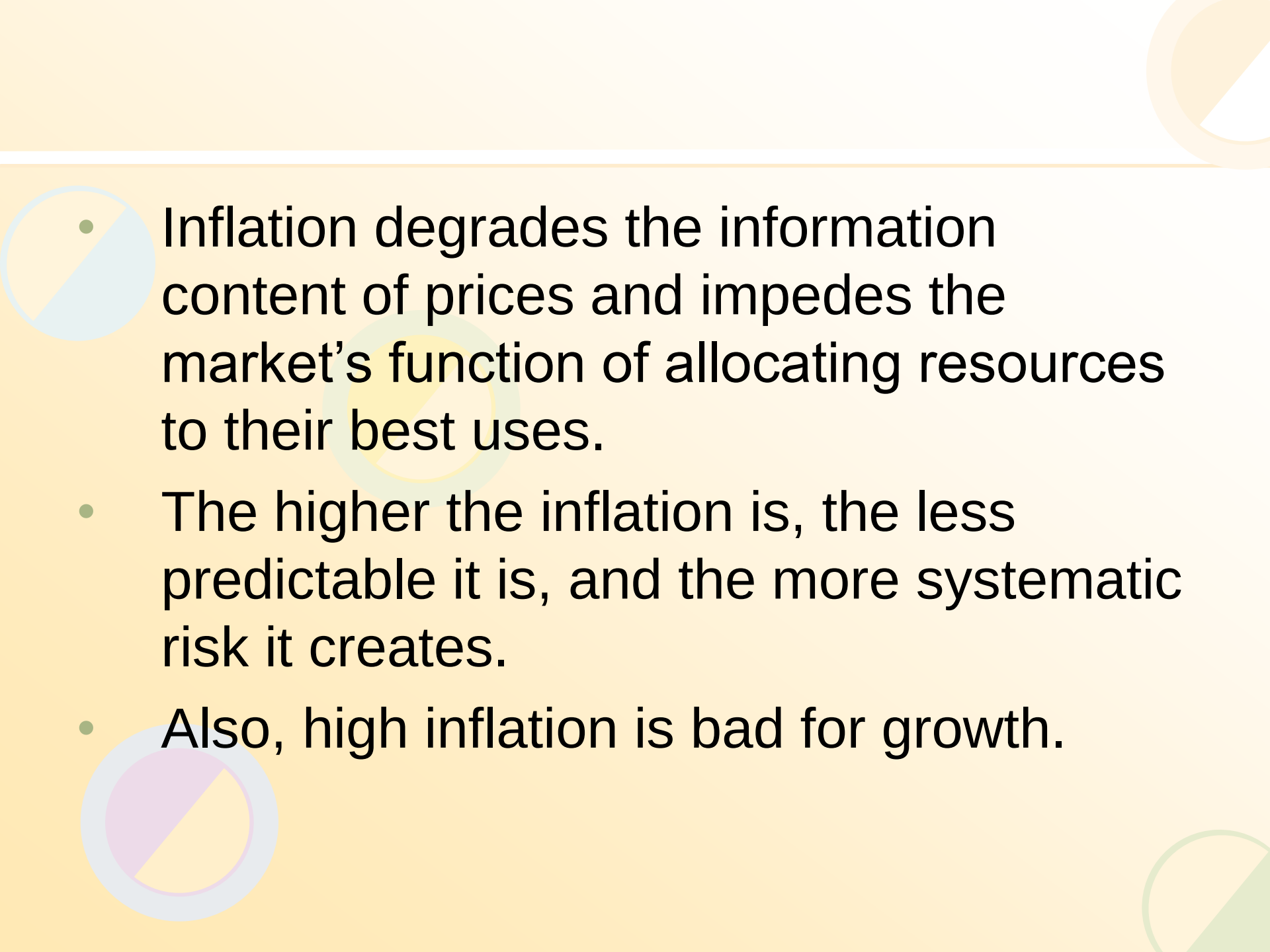


Review of the previous Lecture

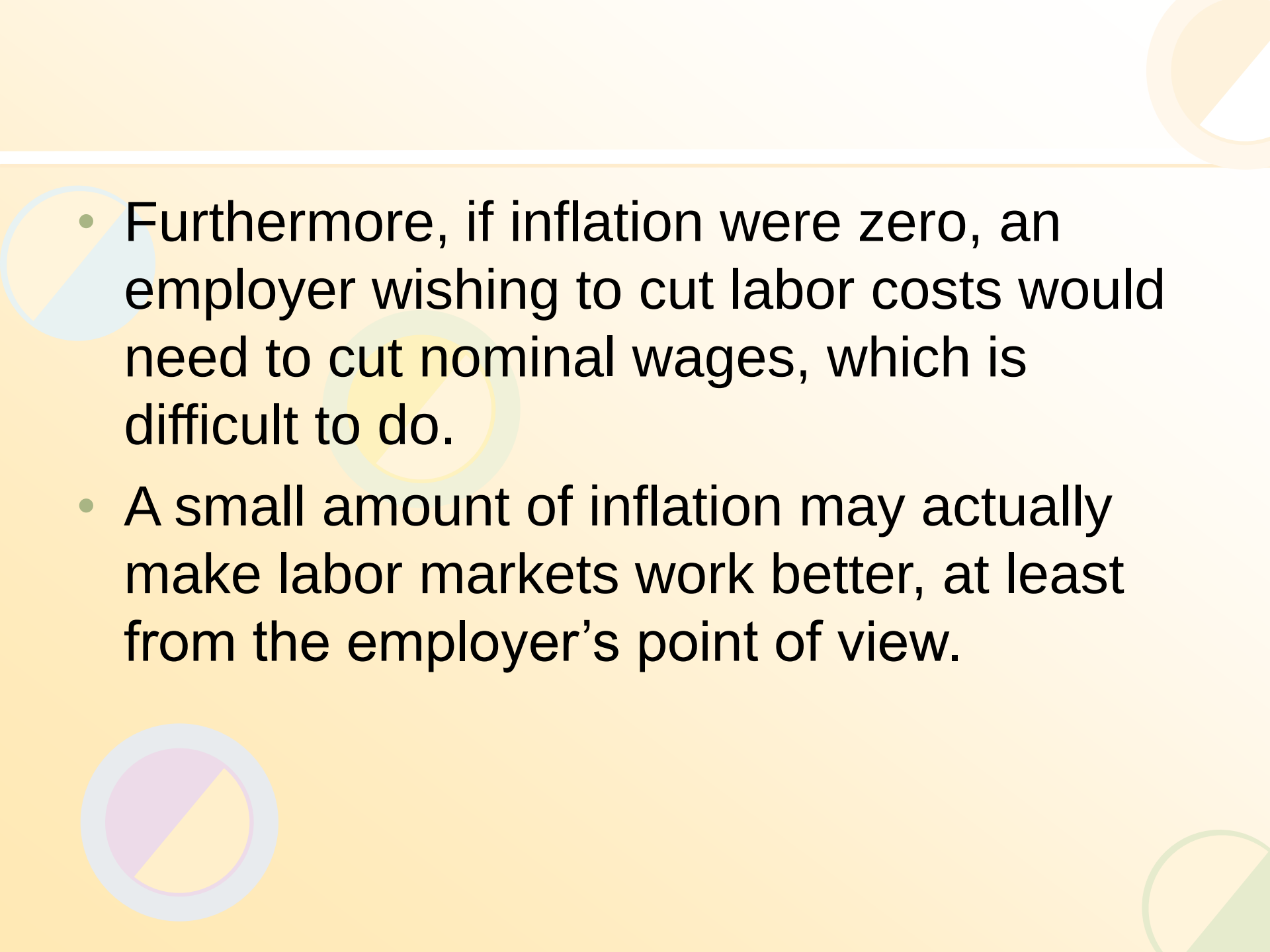
- Central Bank:
 - The Government's Bank
 - The Bankers' Bank
 - Objectives

Low, Stable Inflation

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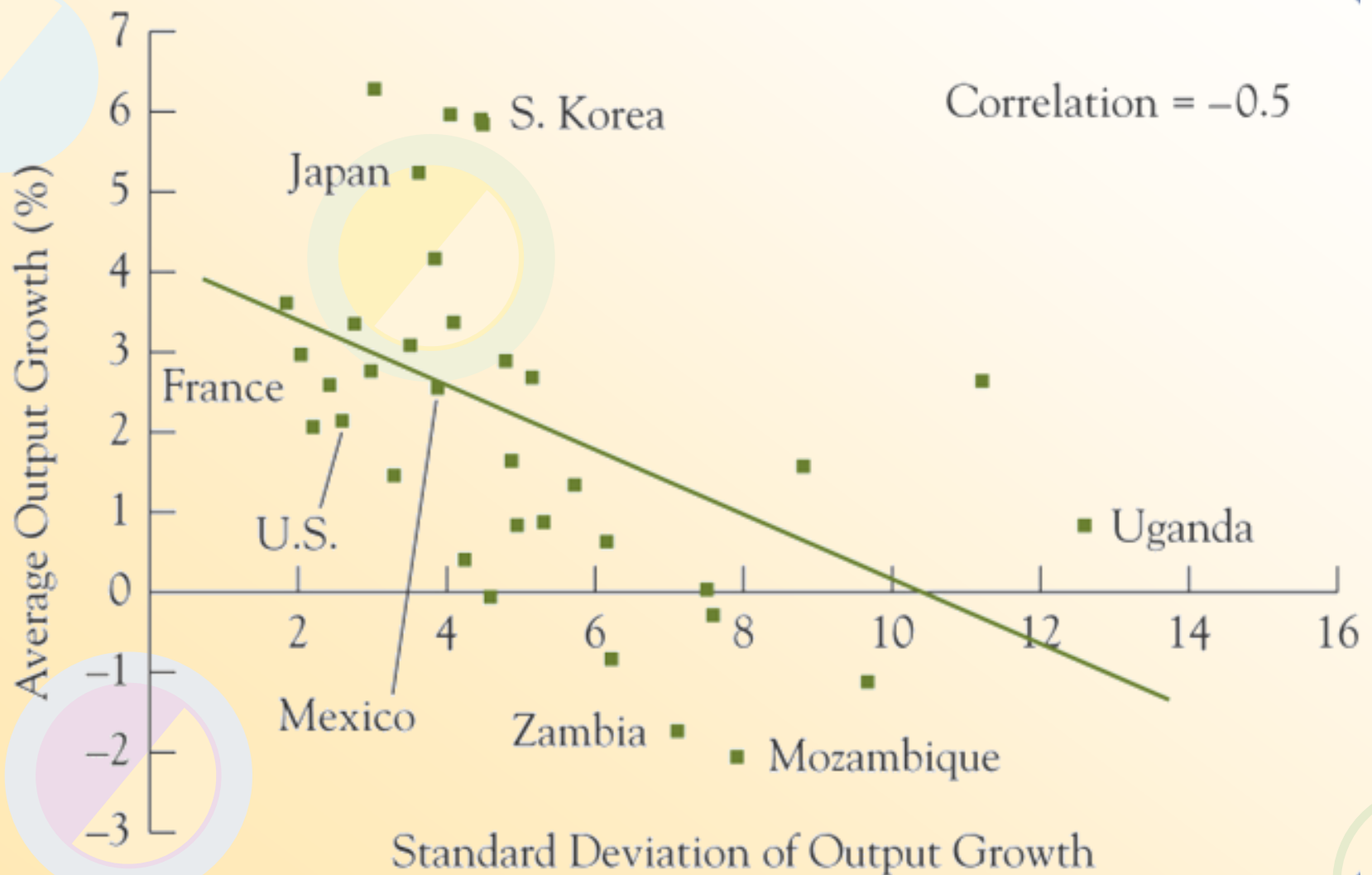
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- 
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High, Stable Real Growth

- Central bankers work to dampen the fluctuations of the business cycle; booms are popular but recessions are not.
- Central bankers work to moderate these cycles and stabilize growth and employment by adjusting interest rates.
- Monetary policymakers can moderate recessions by lowering interest rates and can moderate booms by raising them (to keep growth at a sustainable level).

- Along with growth and employment, stability is also important, because fluctuations in general business conditions are the primary source of systematic risk.



Financial System Stability

- Financial system stability is an integral part of every modern central banker's job.
- The possibility of a severe disruption in the financial markets is a type of systematic risk that central banks must control.

Interest Rate and Exchange Rate Stability

- Interest rate stability and exchange rate stability are a means for achieving the ultimate goal of stabilizing the economy; they are not ends unto themselves.
- Interest rate volatility is a problem because:
 - it makes output unstable as borrowing and expenditure fluctuate with changing rates.
 - it means higher risk and a higher risk premium and makes financial decisions more difficult.

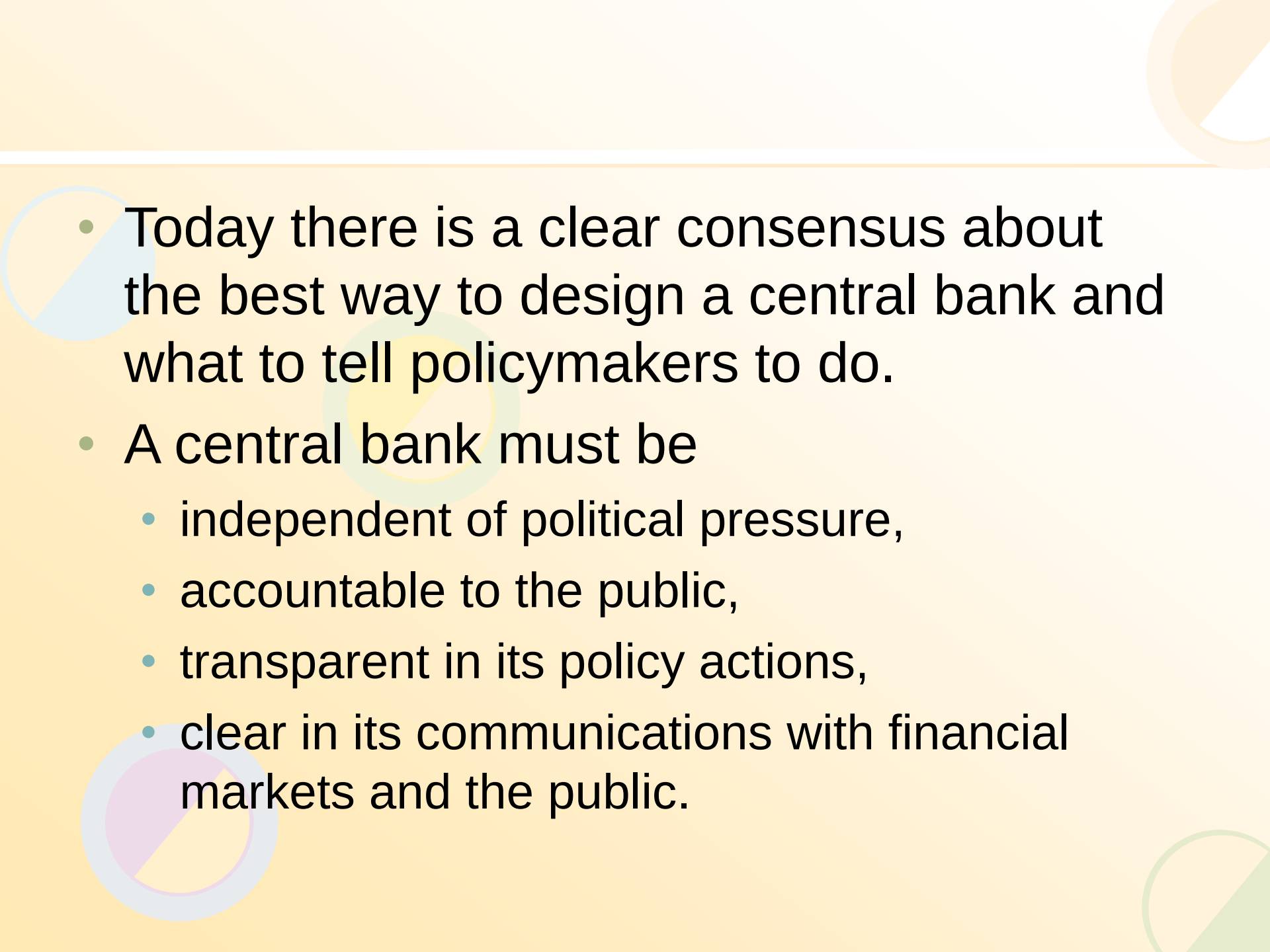
- Even though the exchange rate affects the prices of imports and exports, stabilizing exchange rates is the last item on the list of central bank objectives.
- Different countries have different priorities when it comes to the exchange rate;
 - stable exchange rates are more important in developing countries because imports and exports are central to their economies.

The objectives of a Modern Central Bank

Low Stable Inflation	Inflation creates confusion and makes planning difficult. When inflation is high, growth is low
High Stable growth	Stable predictable growth is higher than unstable, unpredictable growth
Financial System Stability	A stable financial system is necessity for an economy to operate efficiently
Stable Interest Rates	Interest rate volatility creates risk for both lenders and borrowers
Stable Exchange Rates	Variable exchange rates make the revenues from foreign sales and the cost of purchasing imported goods hard to predict

Meeting the Challenge: Creating a Successful Central Bank

- The boom in the past decade with its associated decrease in volatility may have happened because technology sparked a boom just as central banks became better at their jobs.
- Policymakers realized that sustainable growth had gone up, so interest rates could be kept low without worrying about inflation, and central banks were redesigned.

- 
- Today there is a clear consensus about the best way to design a central bank and what to tell policymakers to do.
 - A central bank must be
 - independent of political pressure,
 - accountable to the public,
 - transparent in its policy actions,
 - clear in its communications with financial markets and the public.

Summary

- Central Bank
 - Objectives
 - Principles

Money and Banking

Lecture 32

Review of the Previous Lecture

- Central Bank
 - Roles
 - Government's Bank
 - Bankers' Bank
 - Objectives
 - Low, Stable Inflation
 - High, Stable, Growth
 - Stable Financial System
 - Interest rate stability
 - Exchange rate stability

Meeting the Challenge: Creating a Successful Central Bank

- The boom in the past decade with its associated decrease in volatility may have happened because technology sparked a boom just as central banks became better at their jobs.
- Policymakers realized that sustainable growth had gone up, so interest rates could be kept low without worrying about inflation, and central banks were redesigned.

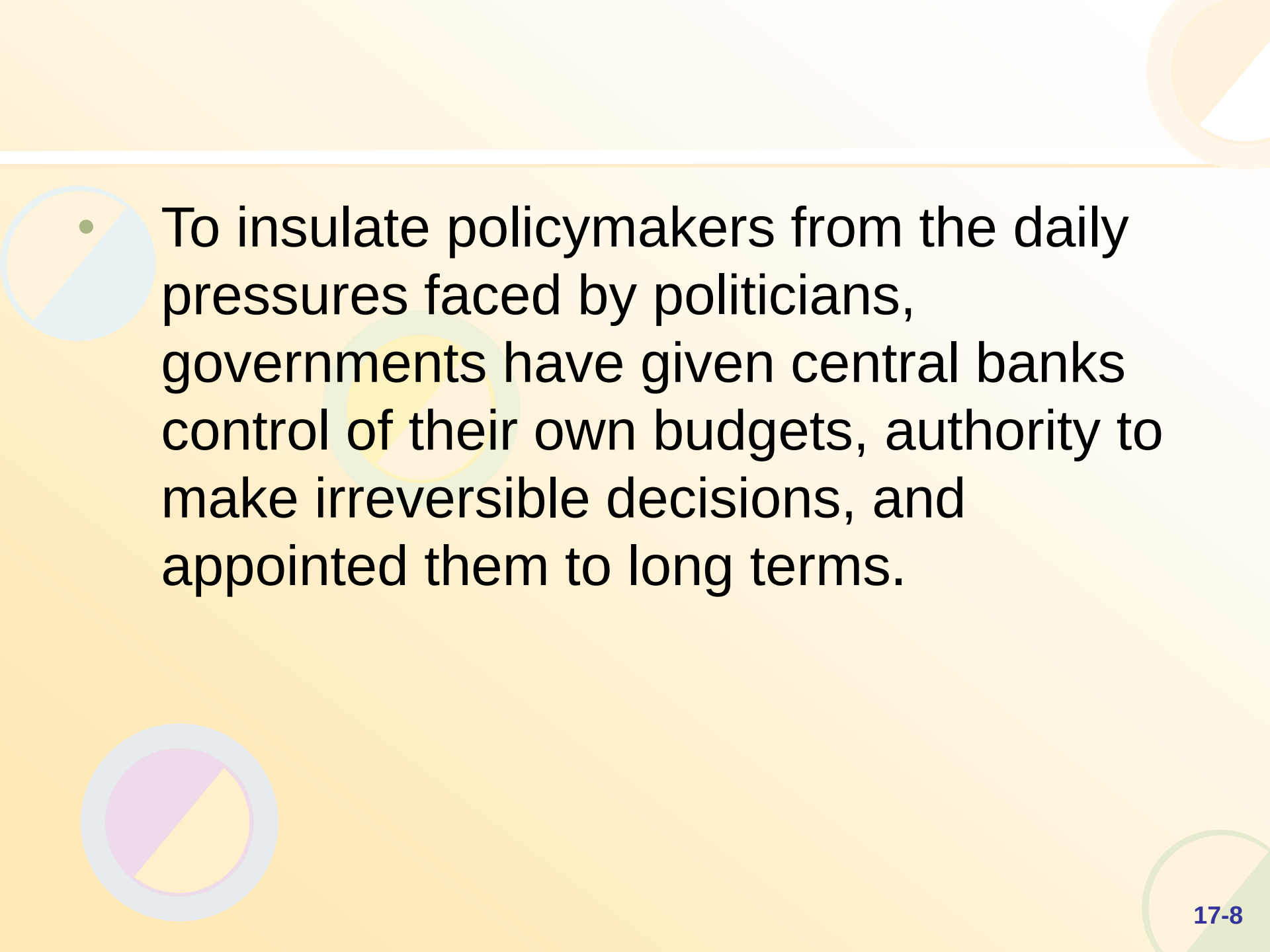
- Today there is a clear consensus about the best way to design a central bank and what to tell policymakers to do.
- A central bank must be
 - independent of political pressure,
 - accountable to the public,
 - transparent in its policy actions,
 - clear in its communications with financial markets and the public.

- In addition, there is general agreement
 - that policy decisions are better made by committee than by individuals,
 - that everyone is well served when policymakers operate within an explicit framework that clearly states their goals and the tradeoffs among them.

The Need for Independence

- The idea that central banks should be independent of political pressure is a new one, because central banks originated as the governments' banks.
- Independence has two components:
 - Monetary policymakers must be free to control their own budgets
 - The bank's policies must not be reversible by people outside the central bank.

- Successful monetary policy requires a long time horizon, which is inconsistent with the need of politicians to focus on short-term goals.
- Given a choice, most politicians will choose monetary policies that are too accommodative, keeping interest rates low and money growth rates high.
- While this raises output and employment in the near term it may result in inflation over the longer term.

- 
- To insulate policymakers from the daily pressures faced by politicians, governments have given central banks control of their own budgets, authority to make irreversible decisions, and appointed them to long terms.

Decision-Making by Committee

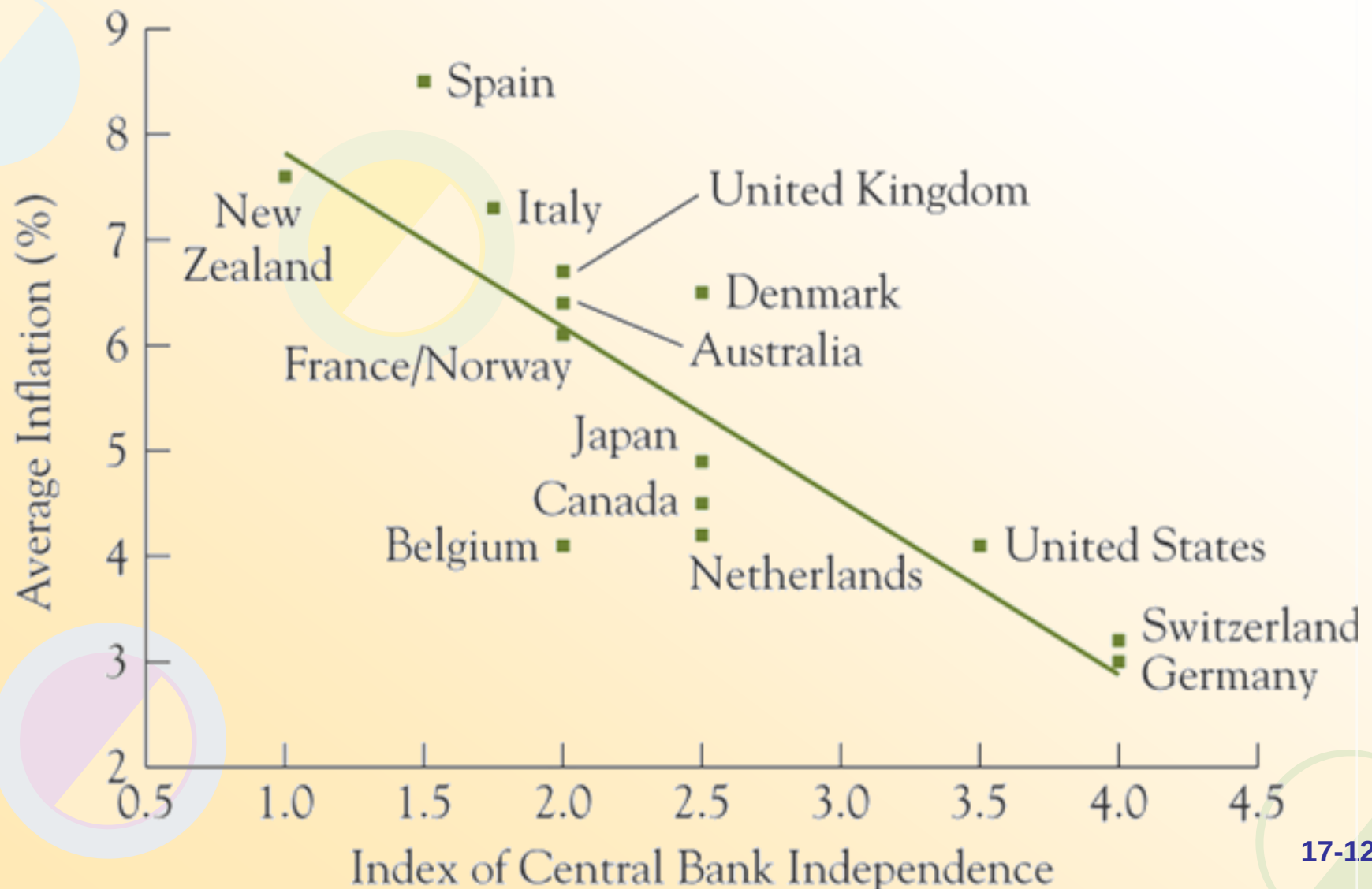
- In the course of normal operations, it is better to rely on a committee than on an individual.
- Pooling the knowledge, experience, and opinions of a group of people reduces the risk that policy will be dictated by an individual's quirks, not to mention that in a democracy, vesting so much power in one individual poses a legitimacy problem.

The Need for Accountability and Transparency

- Central bank independence is inconsistent with representative democracy.
- To solve this problem, politicians have established a set of goals and require the policymakers to report their progress in pursuing these goals.
- Explicit goals foster accountability and disclosure requirements create transparency.

- The institutional means for assuring accountability and transparency differ from one country to the next;
 - in some cases the government sets an explicit numerical target for inflation, while in others the central bank defines the target.
- Similar differences exist in the timing and content of information made public by central banks.

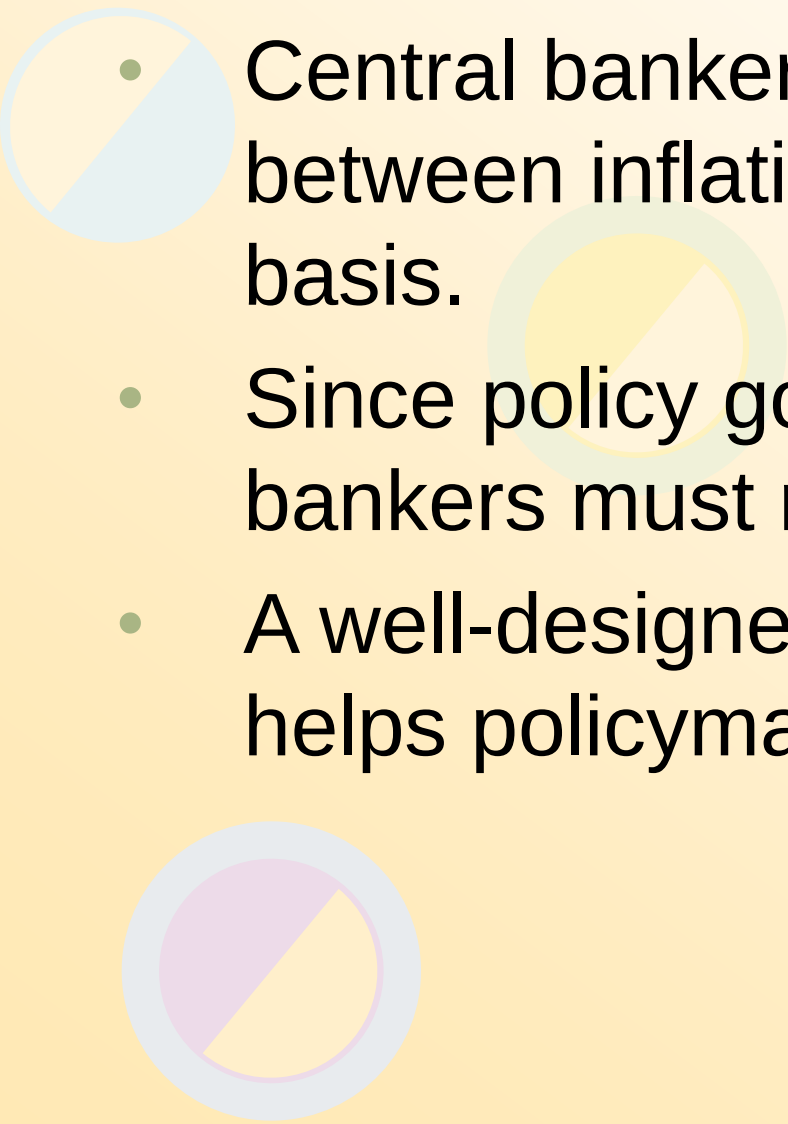
Figure 15.2 Inflation and Central Bank Independence
1973-1988



- Today it is understood that secrecy damages both the policymakers and the economies they are trying to manage, and that policymakers need to be as clear as possible about what they are trying to achieve and how they are going to achieve it.

The Policy Framework, Policy Trade-offs, and Credibility

- The monetary policy framework is made up of the objectives of central banks and the requirements that central banks be independent, accountable, and good communicators.
- The monetary policy framework exists to resolve the ambiguities that arise in the course of the central bank's work and also clarifies the likely responses when goals are in conflict with one another.

- 
- 
- Central bankers face the tradeoff between inflation and growth on a daily basis.
 - Since policy goals often conflict, central bankers must make their priorities clear.
 - A well-designed policy framework also helps policymakers establish credibility.

The Principles of Central Bank Design

Independence

To keep inflation low, monetary decisions must be made free of political influence

Decision
making by
committee

Pooling the knowledge of a number of people yields better decisions than decision making by an individual

Accountability
and
transparency

Policy makers must be held accountable to the public they serve and clearly communicate their objectives, decisions and methods

Policy
framework

Politicians must clearly state their policy goals and the tradeoffs among them

Fitting Everything Together: Central Banks and Fiscal Policy

- The central bank does not control the government's budget; fiscal policy (the decisions about taxes and spending) is the responsibility of elected officials
- While fiscal and monetary policymakers share the same ultimate goal of improving the well-being of the population, conflicts can arise between the two.

- Funding needs create a natural conflict between monetary and fiscal policymakers.
- Fiscal policymakers also tend to ignore the long-term inflationary effects of their actions.
- Politicians often turn to borrowing (instead of taxes) as a way to finance some portion of their spending, but a country can issue only so much debt.

- Inflation is a real temptation to shortsighted fiscal policymakers because it is a way to get money in their hands and it's a way for governments to default on a portion of the debt they owe.
- Responsible fiscal policy is essential to the success of monetary policy.

The Central Bank's Balance Sheet

- The central bank engages in numerous financial transactions, all of which cause changes in its balance sheet.
- Central banks publish their balance sheets regularly. Publication is a crucial part of transparency

The Central Bank's Balance Sheet

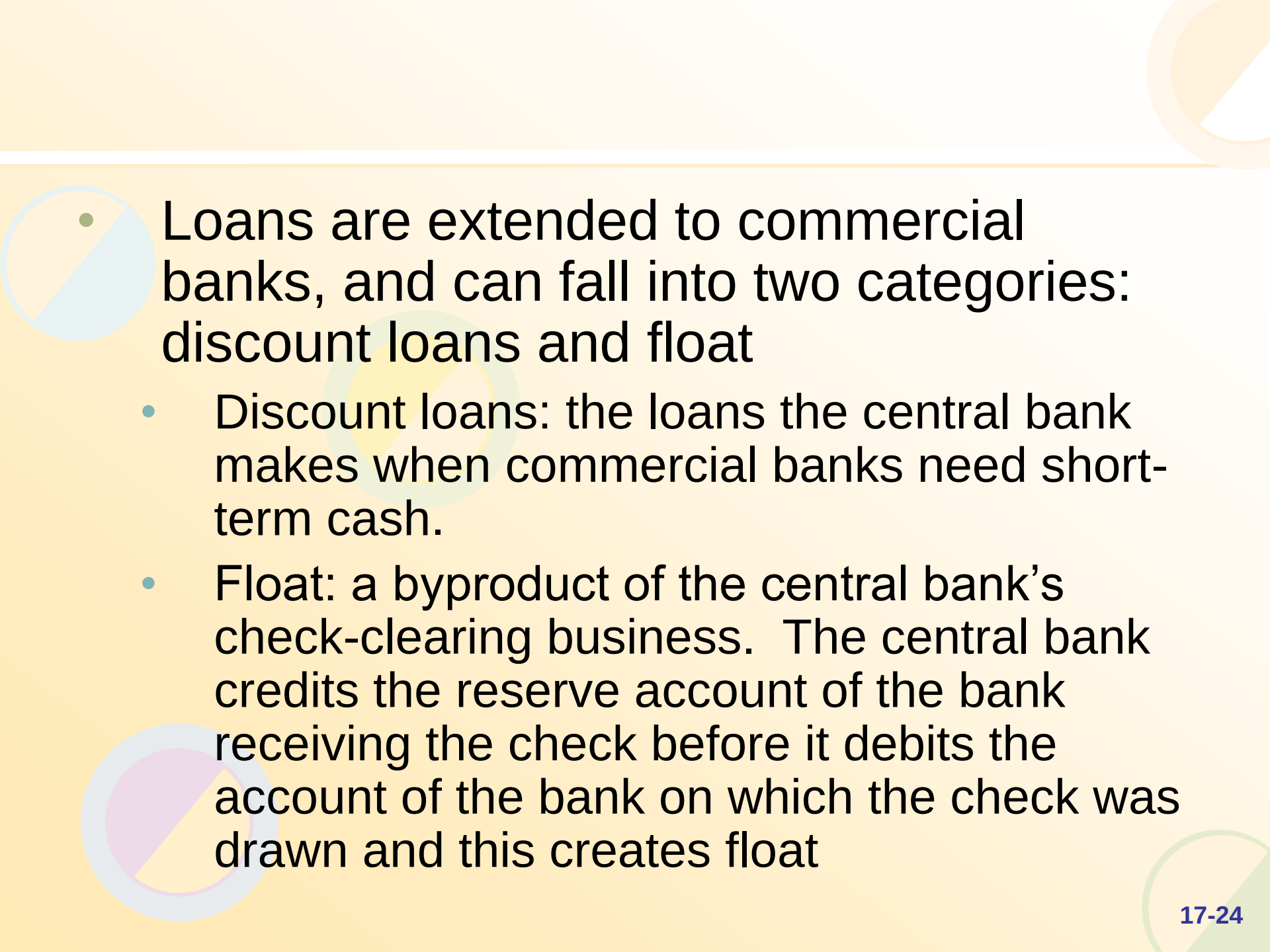
Figure 17.1 The Central Bank's Balance Sheet

Assets		Liabilities
Government's Bank	Securities Foreign exchange reserves	Currency Government's account
Bankers' Bank	Loans	Accounts of the commercial banks (reserves)

Assets

- The central bank's balance sheet shows three basic assets:
 - securities,
 - foreign exchange reserves,
 - loans.

- Securities:
 - The primary assets of most central banks;
 - Independent central banks determine the quantity of securities that they purchase
- Foreign Exchange Reserves:
 - The central bank's and government's balances of foreign currency are held as bonds issued by foreign governments.
 - These reserves are used in foreign exchange market interventions.



Loans are extended to commercial banks, and can fall into two categories: discount loans and float

- Discount loans: the loans the central bank makes when commercial banks need short-term cash.
- Float: a byproduct of the central bank's check-clearing business. The central bank credits the reserve account of the bank receiving the check before it debits the account of the bank on which the check was drawn and this creates float

- Through its holdings of Treasury securities the central bank controls the discount rate and the availability of money and credit.
- Gold reserves, while still an asset of many central banks, are virtually irrelevant these days.

Liabilities

- There are three major liabilities:
 - currency,
 - the government's deposit account,
 - the deposit accounts of the commercial banks.
- The first two items represent the central bank in its role as the government's bank, and the third shows it as the bankers' bank.

- Currency:
 - nearly all central banks have a monopoly on the issuance of currency, and currency accounts for over 90 percent of the central bank's liabilities.
- Government's account:
 - the central bank provides the government with an account into which it deposits funds (primarily tax revenues) and from which it writes checks and makes payments.

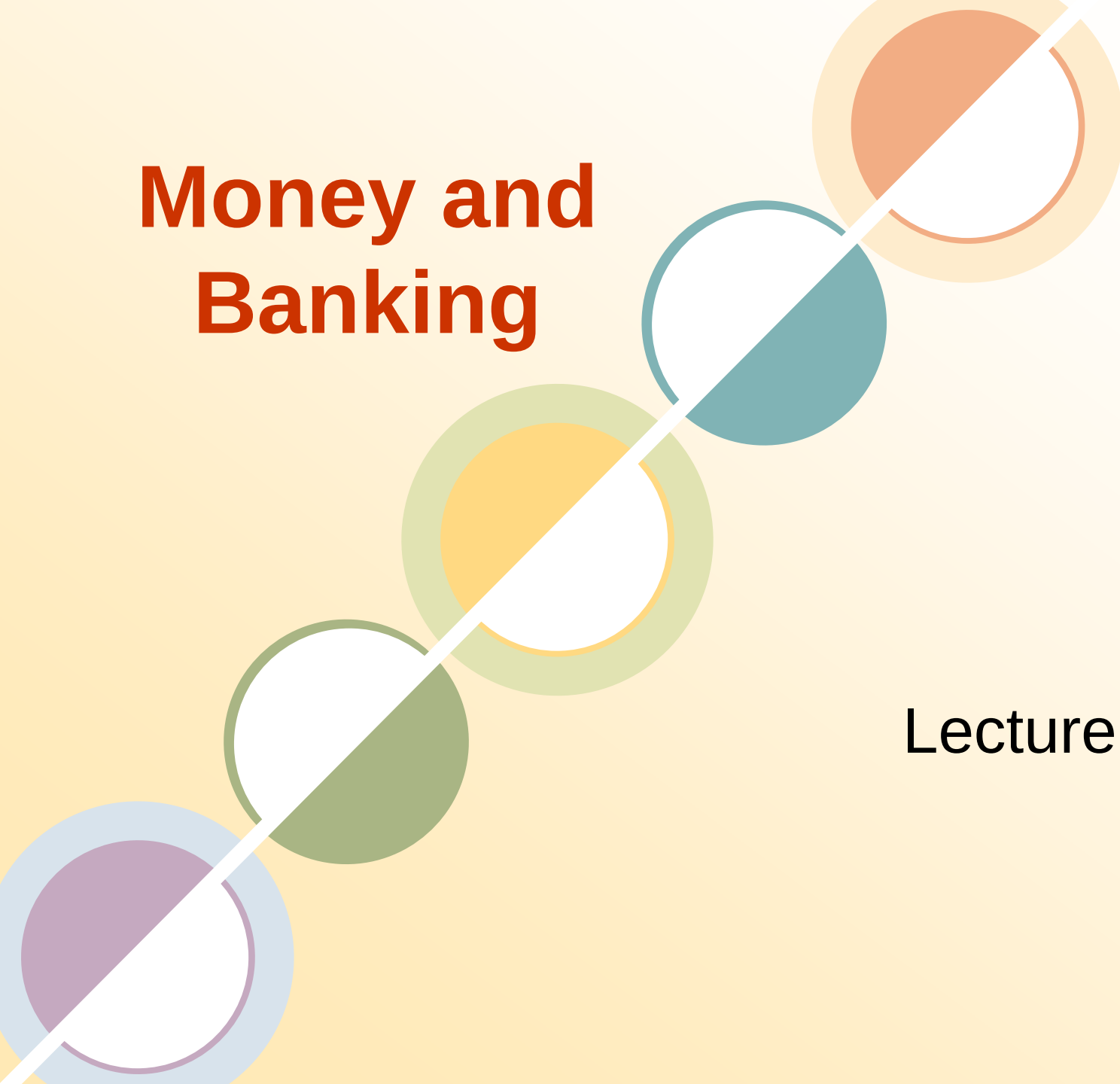
- Reserves:
 - Commercial bank reserves consist of cash in the bank's own vault and deposits at the central bank, which function like the commercial bank's checking account.
- Central banks run their monetary policy operations through changes in banking system reserves.

Chapter 17

End of Chapter

Money and Banking

Lecture 33



Review of the Previous Lecture

- Central Bank
 - Roles
 - Objectives
 - Inflation
 - Growth
 - Financial System
 - Interest rate and exchange rate
 - A successful central bank
 - Independence
 - Accountability
 - Transparency
 - communication

Review of the Previous Lecture

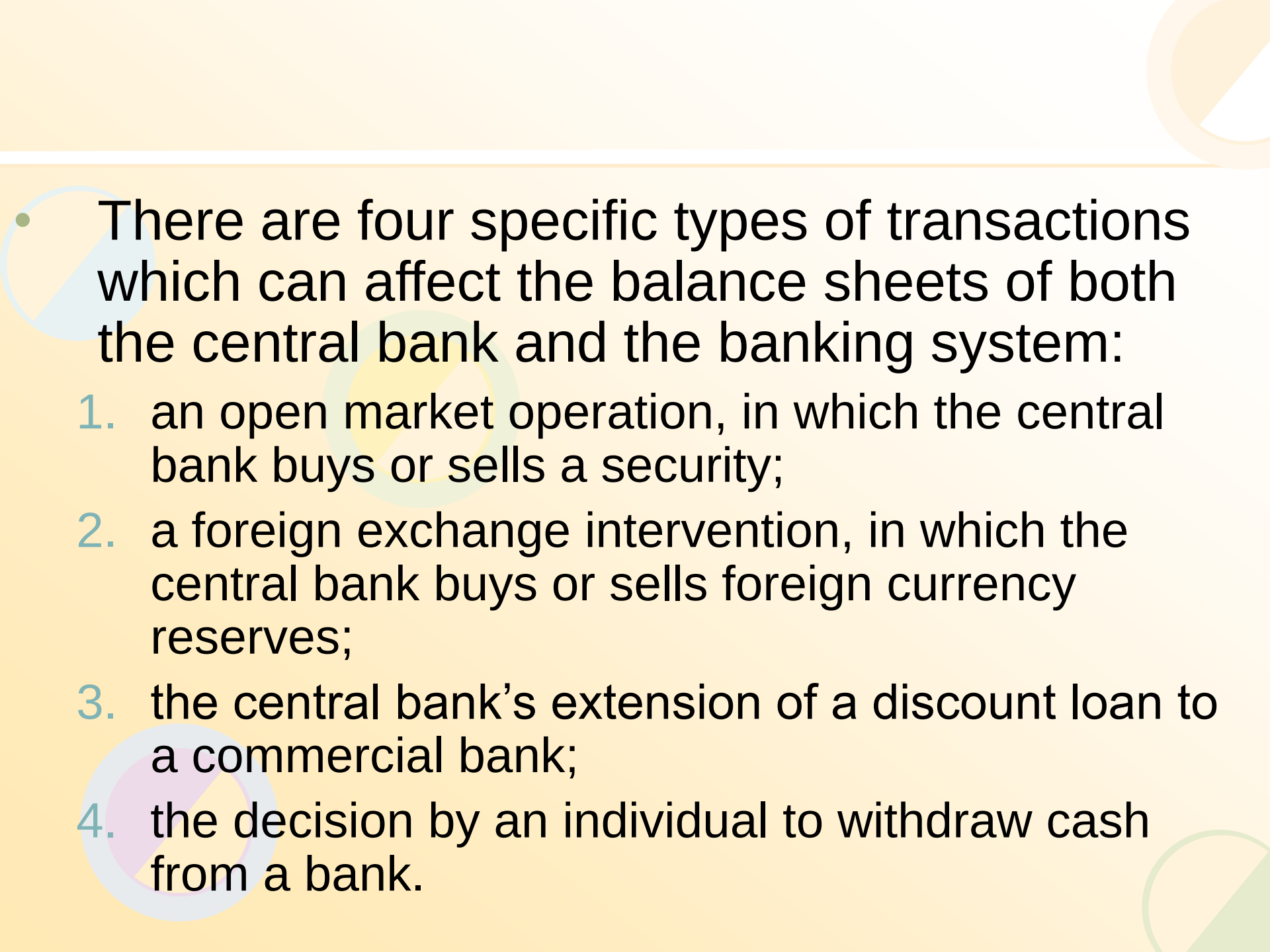
- Central Bank's Balance Sheet
 - Assets
 - securities,
 - foreign exchange reserves,
 - loans
 - Liabilities
 - currency,
 - the government's deposit account,
 - the deposit accounts of the commercial banks

The Monetary Base

- Currency in the hands of the public and the reserves of the banking system are the two components of the monetary base, also called high-powered money.
- Bank Reserves = Vault Cash plus Deposits at the central bank
- The central bank can control the size of the monetary base and therefore the quantity of money

Changing the Size and Composition of the Balance Sheet

- The central bank controls the size of its balance sheet. Policymakers can enlarge or reduce their assets and liabilities at will
- The central bank can buy things, like a bond, and create liabilities to pay for them. It can increase the size of its balance sheet as much as it wants.



- There are four specific types of transactions which can affect the balance sheets of both the central bank and the banking system:

1. an open market operation, in which the central bank buys or sells a security;
2. a foreign exchange intervention, in which the central bank buys or sells foreign currency reserves;
3. the central bank's extension of a discount loan to a commercial bank;
4. the decision by an individual to withdraw cash from a bank.

- Open market operations, foreign exchange interventions, and discount loans, all affect the size of the central bank's balance sheet
- They change the size of the monetary base;
 - cash withdrawals by the public create shifts among the different components of the monetary base, changing the composition of the central bank's balance sheet but leaving its size unaffected

- One simple rule will help in understanding the impact of each of these four transactions on the central bank's balance sheet:
 - When the value of an asset on the balance sheet increases, either the value of another asset decreases (so that the net change is zero) or the value of a liability rises by the same amount (and similarly for an increase in liabilities)

Open Market Operations

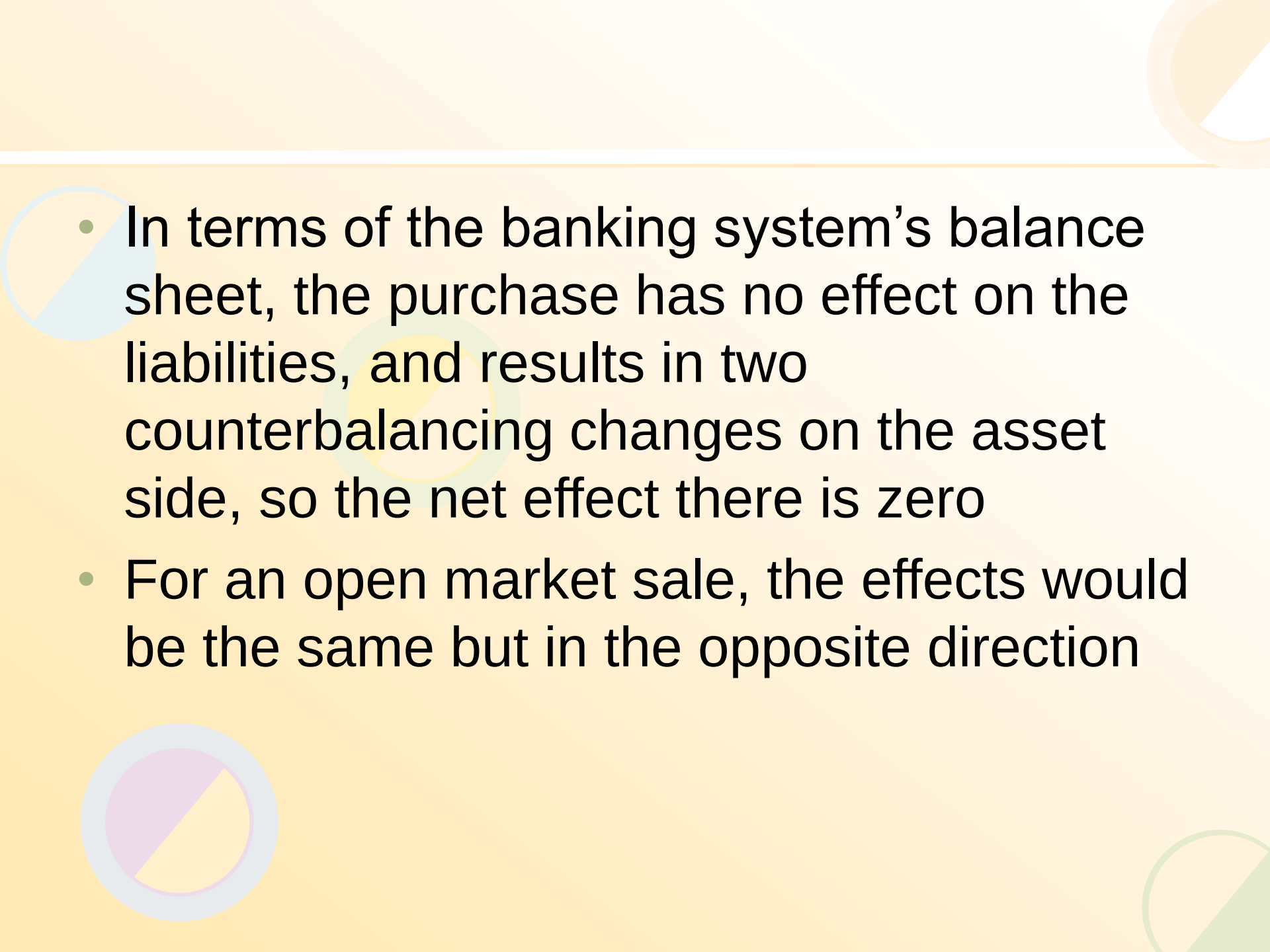
- OMO is when the central bank buys or sells securities in financial markets
- These purchases and sales have a straightforward impact on the central bank's balance sheet:
 - its assets and liabilities increase by the amount of a purchase, and the monetary base increases by the same amount

Change in the Central Bank's Balance Sheet following Purchase of a Treasury Bond

Assets		Liabilities	
Securities (U.S. Treasury bond)	+\$1 billion	Reserves	+\$1 billion

Change in the Banking System's Balance Sheet following the Central Bank's Purchase of a Treasury Bond

Assets		Liabilities
Reserves	+\$1 billion	
Securities (Treasury bond)	-\$1 billion	

- 
- In terms of the banking system's balance sheet, the purchase has no effect on the liabilities, and results in two counterbalancing changes on the asset side, so the net effect there is zero
 - For an open market sale, the effects would be the same but in the opposite direction

Foreign Exchange Intervention

- The impact of a foreign exchange purchase is almost identical to that of an open market purchase:
 - the central bank's assets and liabilities increase by the same amount, as does the monetary base.
- If the central bank buys from a commercial bank, the impact again is like the open market purchase, except the assets involved are different.

Foreign Exchange Intervention

Change in the Central Bank's Balance Sheet following Purchase of Euro- denominated German Government Bonds

Assets		Liabilities	
Foreign exchange reserves (German government bonds in euros)	+\$1 billion	Reserves	+\$1 billion

Foreign Exchange Intervention

Change in the Banking System's Balance Sheet following the Central Bank's Purchase of Euro-denominated German Government Bonds

Assets		Liabilities
Reserves	+\$1 billion	
Securities (German government bonds)	-\$1 billion	

Discount Loans

- The central bank does not force commercial banks to borrow money; the banks ask for loans and must provide collateral, usually a Treasury bond.
- When the central bank makes a loan it creates an asset and a matching increase in its reserve liabilities.

Discount Loans

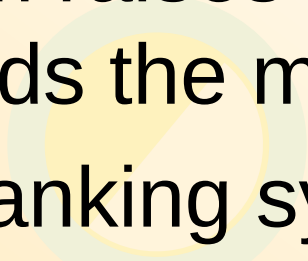
Change in the Central Bank's Balance Sheet following a Discount Loan

Assets		Liabilities	
Discount loans	+\$100 million	Reserves	+\$100 million

Discount Loans

Change in the Banking System's Balance Sheet following a Discount Loan

Assets		Liabilities	
Reserves	+\$100 million	Discount loans	+\$100 million

- 
- 
- 
- 
- The extension of credit to the banking system raises the level of reserves and expands the monetary base.
 - The banking system balance sheet shows an increase in assets (reserves) and an increase in liabilities (the loan).

Cash Withdrawal

- Cash withdrawals affect only the composition, not the size, of the monetary base.
- When people withdraw cash they force a shift from reserves to currency on the central bank's balance sheet.

Change in the Nonbank Public's Balance Sheet following a Cash Withdrawal

Assets		Liabilities
Currency	+\$100	
Checkable deposits	-\$100	

- The withdrawal reduces the banking system's reserves, which is a decrease in its assets, and if the funds come from a checking account, there is a matching decrease in liabilities.

Change in the Banking System's Balance Sheet following a Cash Withdrawal

Assets		Liabilities	
Reserves	-\$100	Checkable deposits	-\$100

- On the central bank's balance sheet both currency and reserves are liabilities, so there is just a change between the two with a net effect of zero.

Change in the Central Bank's Balance Sheet following a Cash Withdrawal

Assets		Liabilities	
		Currency	+\$100
		Reserves	-\$100

Changes in Size and Composition of Central Bank's Balance Sheet and Monetary Base

Transaction	Initiated by	Typical action	Impact
Open market operation	Central bank	Purchase of Treasury bond	Increases reserves, the size of central bank's balance sheet and Monetary base
Foreign Exchange Intervention	Central bank	Purchase of foreign govt. bonds	Increases reserves, the size of central bank's balance sheet and Monetary base
Discount Loans	Commercial bank	Extension of loan to commercial bank	Increases reserves, the size of central bank's balance sheet and Monetary base
Cash withdrawals	Nonbank public	Withdrawal of cash from ATM	Decreases reserves and increases currency, leaving size of central bank's balance sheet and Monetary base unchanged

Money and Banking

Lecture 34

Review of the Previous Lecture

- Changing the Size and Composition of the Balance Sheet
 - Open Market Operation
 - Foreign Exchange Intervention
 - Discount Loans
 - Cash Withdrawals

Deposit Creation in a Single Bank

- If the central bank buys a security from a bank, the bank has excess reserves, which it will seek to lend
- The loan replaces the securities as an asset on the bank's balance sheet.

Change in *First Bank's* Balance Sheet following Central Bank's Purchase of a Treasury Bond

Assets		Liabilities
Reserves	+\$100,000	
Securities	-\$100,000	

- Assuming First bank has granted a loan of \$100,000 to Office Builders Incorporated (OBI)

Change in *First Bank's* Balance Sheet following
Central Bank's Purchase of a Treasury Bond and
Extension of a Loan

Assets		Liabilities
Reserves	+\$100,000	OBI checking account +\$100,000
Securities	-\$100,000	
Loans	+\$100,000	

- OBI paid off its employees and suppliers through checks worth \$100,000

Change in *First Bank's* Balance Sheet following
Central Bank's Purchase of a Treasury Bond,
Extension of a Loan, and Withdrawal by the
Borrower

Assets		Liabilities
Reserves	\$0	Checkable deposits \$0
Securities	-\$100,000	
Loans	+\$100,000	

Deposit Expansion in a System of Banks

- The loan that the First bank made was spent and as the checks cleared, reserves were transferred to other banks
- The banks that receive the reserves will seek to lend their excess reserves, and the process continues until all of the funds have ended up in required reserves

Types of Reserves

- Actual Reserves (R)
- Required Reserves ($RR = r_D D$)
- Excess Reserves (ER)

- Assume

- Bank hold no excess reserves.
- The reserve requirement ratio is 10%
- Currency holding does not change when deposits and loans change.
- When a borrower writes a check, none of the recipients of the funds deposit them back in the bank that initially made the loan.

- Lets say, OBI uses the \$100,000 loan to pay its supplier American Steel Co (ASC), which it deposits in its bank the Second bank.

Change in *Second Bank's* Balance Sheet following American Steel's Deposit

Assets		Liabilities
Reserves	+\$100,000	American Steel's checking account +\$100,000

Change in *Second Bank's* Balance Sheet following a Deposit and Extension of a Loan

Assuming a 10 percent reserve requirement, banks hold no excess reserves, and there are no changes in currency holdings.

Assets		Liabilities
Reserves	+\$10,000	American Steel's checking account
Loan	+\$90,000	+\$100,000

Change in *Third Bank's* Balance Sheet following a Deposit and Extension of a Loan

Assuming a 10 percent reserve requirement, banks hold no excess reserves, and there are no changes in currency holdings.

Assets		Liabilities	
Reserves	+\$9,000	Checking account	+\$90,000
Loan	+\$81,000		

Multiple Deposit Creation

Assuming a 10 percent reserve requirement, banks hold no excess reserves, and there are no changes in currency holdings.



Table 17.2 Multiple Deposit Expansion following a \$100,000 Open Market Purchase Assuming a 10% Reserve Requirement

Bank	Increase in Deposits	Increase in Loans	Increase in Reserves
First Bank	\$ 0	\$100,000	\$ 0
Second Bank	\$100,000	\$90,000	\$10,000
Third Bank	\$90,000	\$81,000	\$9,000
Fourth Bank	\$81,000	\$72,900	\$8,100
Fifth Bank	\$72,900	\$65,610	\$7,290
Sixth Bank	\$65,610	\$59,049	\$6,561
.	.	.	.
.	.	.	.
.	.	.	.
The Banking System	\$1,000,000	\$1,000,000	\$100,000

Deposit Expansion Multiplier

- Assuming
 - no excess reserves are held
 - there are no changes in the amount of currency held by the public,
- the change in deposits will be the inverse of the required deposit reserve ratio (r_D) times the change in required reserves, or
$$\Delta D = (1/r_D) \Delta RR$$

- Alternatively

- $RR = r_D D$ or $\Delta RR = r_D \Delta D$

- So for every dollar increase in reserves, deposits increase by

$$\frac{1}{r_D}$$

- The term $(1/r_D)$ represents the simple deposit expansion multiplier.
 - A decrease in reserves will generate a deposit contraction in a multiple amount too

- $R_D = 10\%$ (0.10), and $\Delta RR = \$100,000$

- $\Delta D = \frac{1}{.1} \$100,000$

- $\Delta D = \$1,000,000$

Deposit Expansion with Excess Reserves and Cash Withdrawals

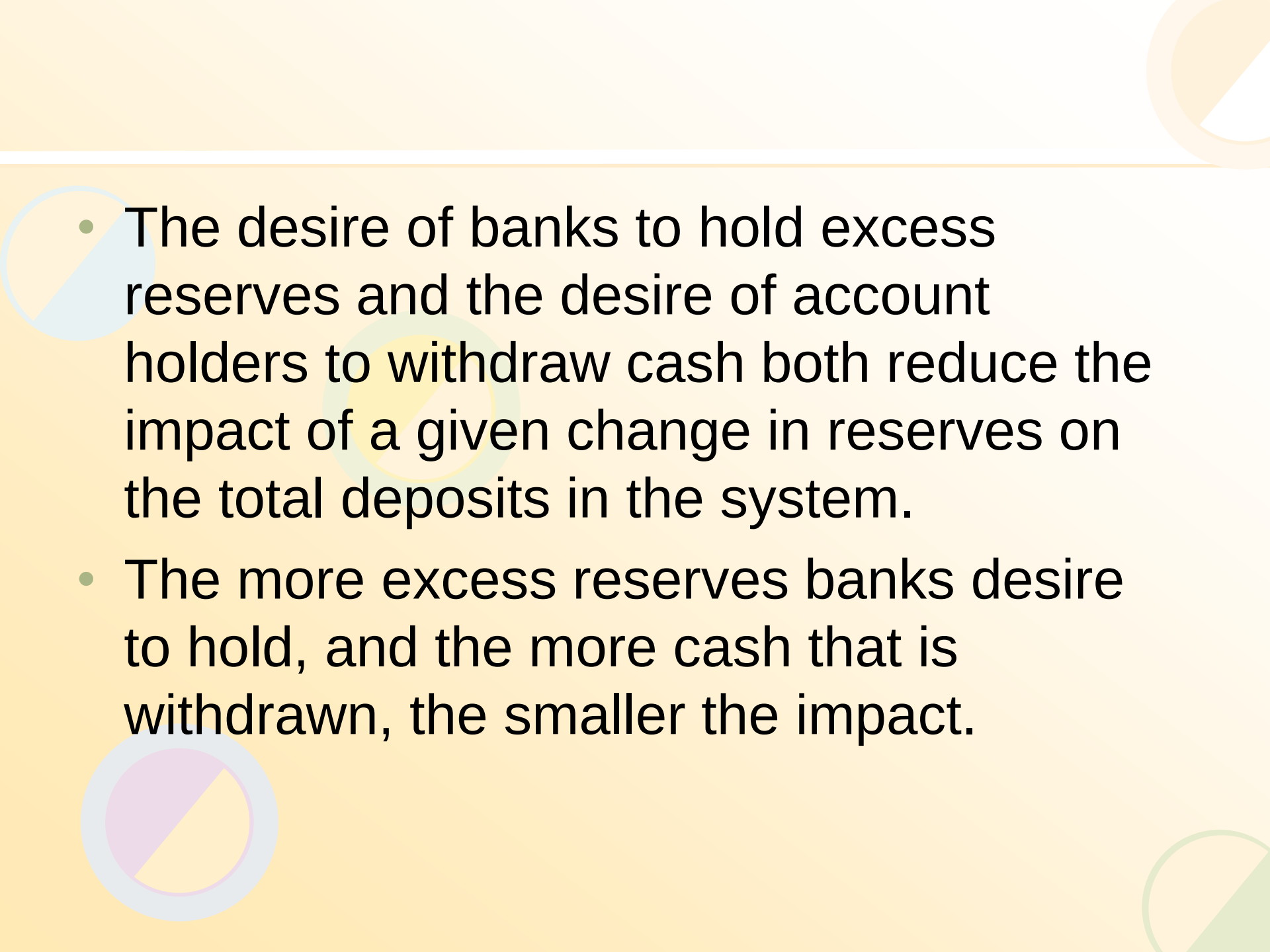
- The simple deposit expansion multiplier was derived assuming no excess reserves are held and that there is no change in currency holdings by the public.
- These assumptions are now relaxed as
 - 5% withdraw of cash.
 - Excess reserves of 5% of deposits

- Continuing with our previous example, if American Steel Co (ASC) removes 5% of its new funds in cash, which leaves \$95,000 in the checking account and \$95,000 in the Second bank's reserve account.
- Bank wishes to hold excessive reserves of 5% of deposits, it would keep reserves of 15% of \$95,000 or \$14,250 and making a loan of \$80,750

Change in *Second Bank's* Balance Sheet following a Deposit and Extension of a Loan

Assuming excess reserves and cash holdings. Note: American Steel also has \$5,000 in cash.

Assets		Liabilities
Required reserves	+\$9,500	American Steel's checking account +\$95,000
Excess reserves	+\$4,750	
Loan	+\$80,750	

- 
- The desire of banks to hold excess reserves and the desire of account holders to withdraw cash both reduce the impact of a given change in reserves on the total deposits in the system.
 - The more excess reserves banks desire to hold, and the more cash that is withdrawn, the smaller the impact.

Money Multiplier

- The money multiplier shows how the quantity of money (checking account plus currency) is related to the monetary base (reserves in the banking system plus currency held by the nonbank public)
- Taking m for money multiplier and MB for monetary base, the Quantity of Money, M is

$$M = m \times MB$$

(This is why the MB is called High Powered Money)

- Consider the following relationships

Money = Currency + Checkable deposits

$$\mathbf{M = C + D}$$

Monetary Base = Currency + Reserves

$$\mathbf{MB = C + R}$$

Reserves = Req. Res. + Exc. Res

$$\mathbf{R = RR + ER}$$

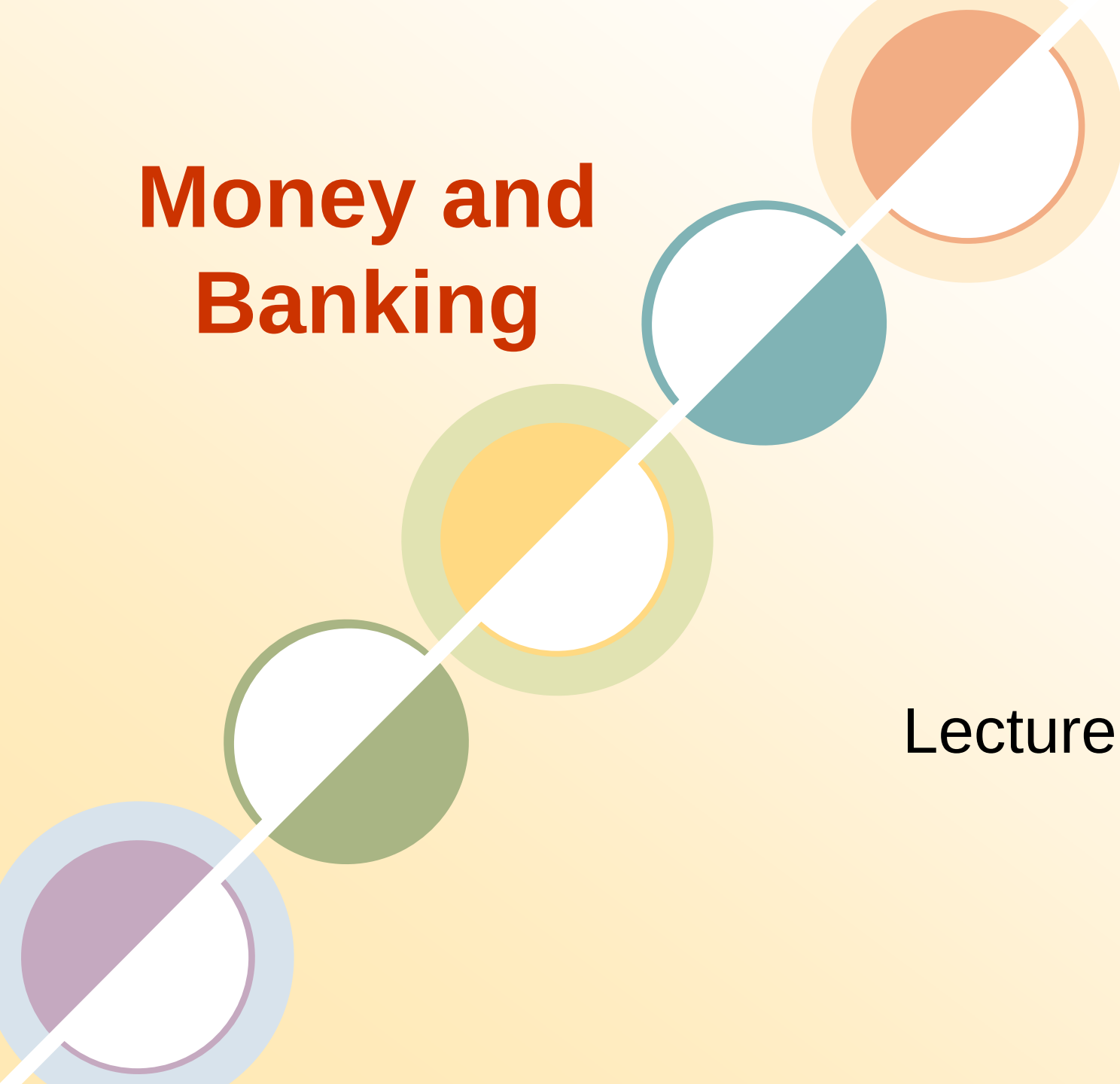
- The amount of excess reserves a bank holds depends on the costs and benefits of holding them,
 - the cost is the interest foregone
 - the benefit is the safety from having the reserves in case there is an increase in withdrawals
- The higher the interest rate, the lower banks' excess reserves will be; the greater the concern over possible deposit withdrawals, the higher the excess reserves will be

Introducing Excess Reserve Ratio $\{ER/D\}$

- $R = RR + ER$
 $= r_D D + \{ER/D\}D$
 $= (r_D + \{ER/D\})D$

Money and Banking

Lecture 35



Review of the Previous Lecture

- Deposit Creation in a Single Bank
- Deposit Creation in a System of Banks
- Deposit Expansion Multiplier
- Deposit Expansion with Excess Reserves and Cash Withdrawals
- Money Multiplier

Money Multiplier

Remember, we discussed that

- Assuming
 - no excess reserves are held
 - there are no changes in the amount of currency held by the public,
- the change in deposits will be the inverse of the required deposit reserve ratio (r_D) times the change in required reserves, or $\Delta D = (1/r_D) \Delta RR$

- Alternatively

- $RR = r_D D$ or $\Delta RR = r_D \Delta D$

- So for every dollar increase in reserves, deposits increase by

$$\frac{1}{r_D}$$

- The term $(1/r_D)$ represents the simple deposit expansion multiplier.
 - A decrease in reserves will generate a deposit contraction in a multiple amount too

Money Multiplier

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Monetary Base = Currency + Reserves

$$\mathbf{MB = C + R}$$

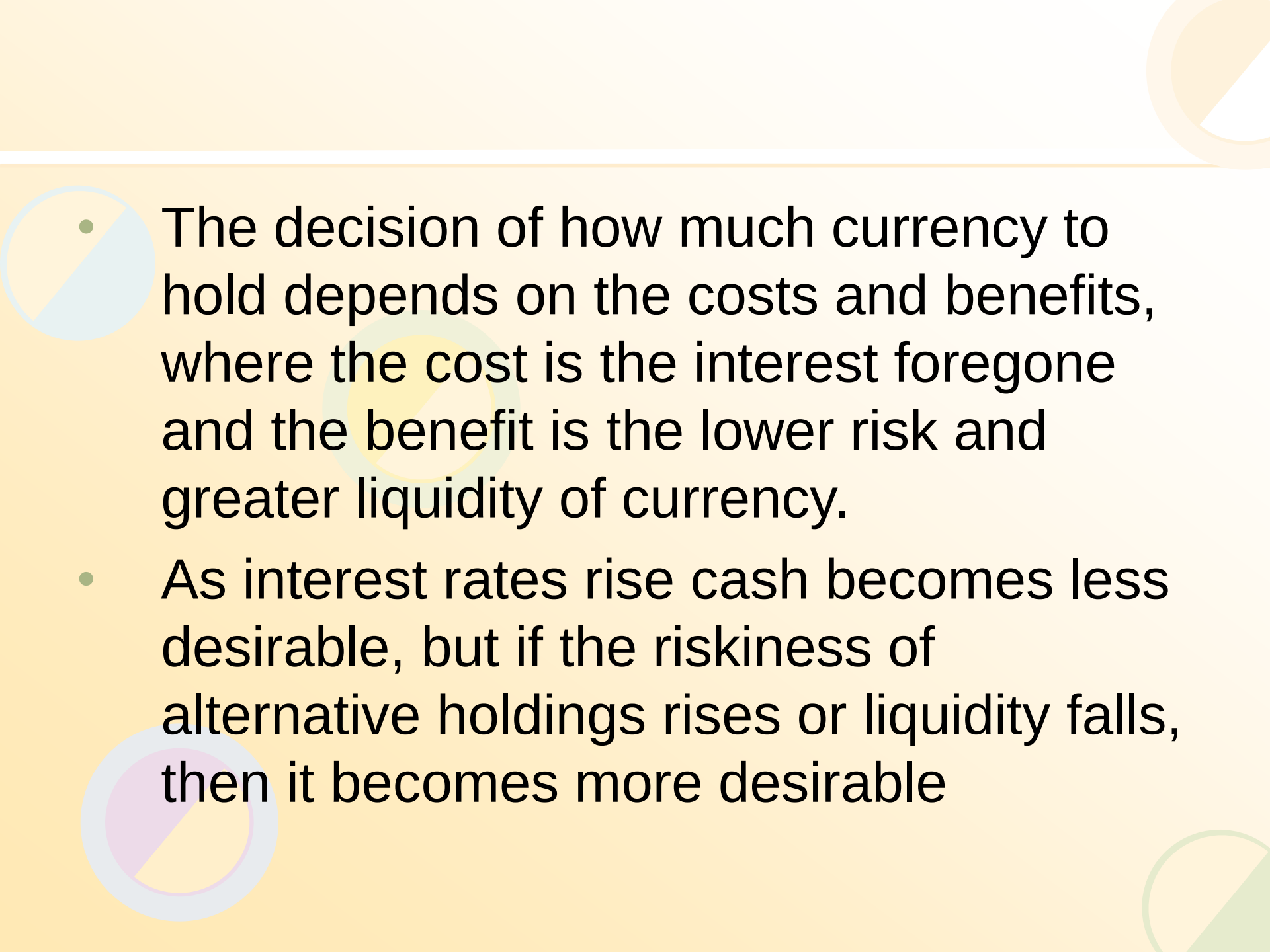
Reserves = Req. Res. + Exc. Res

$$\mathbf{R = RR + ER}$$

- The amount of excess reserves a bank holds depends on the costs and benefits of holding them,
 - the cost is the interest foregone
 - the benefit is the safety from having the reserves in case there is an increase in withdrawals
- The higher the interest rate, the lower banks' excess reserves will be; the greater the concern over possible deposit withdrawals, the higher the excess reserves will be

Introducing Excess Reserve Ratio $\{ER/D\}$

- $R = RR + ER$
 $= r_D D + \{ER/D\}D$
 $= (r_D + \{ER/D\})D$

- 
- The decision of how much currency to hold depends on the costs and benefits, where the cost is the interest foregone and the benefit is the lower risk and greater liquidity of currency.
 - As interest rates rise cash becomes less desirable, but if the riskiness of alternative holdings rises or liquidity falls, then it becomes more desirable

Now taking Currency Ratio as $\{C/D\}$

- $MB = C + R$
 $= \{C/D\}D + (r_D + \{ER/D\})D$
 $= (\{C/D\} + r_D + \{ER/D\})D$
- This shows that Monetary base has three uses
 - Required reserves
 - Excessive reserves
 - Cash in the hands of nonbank public

Deposit Expansion with Excess Reserves and Cash Withdraws.

$$D = \frac{1}{\{C/D\} + r_D + \{ER/D\}} \times MB$$

$$M = \frac{\{C/D\} + 1}{\{C/D\} + r_D + \{ER/D\}} \times MB$$

The Quantity of Money (M) Depends on:

- The Monetary base (MB), Controlled by the central bank.
- Reserve Requirements
- Bank's desired to hold excess reserves.
- The public's demand for currency.

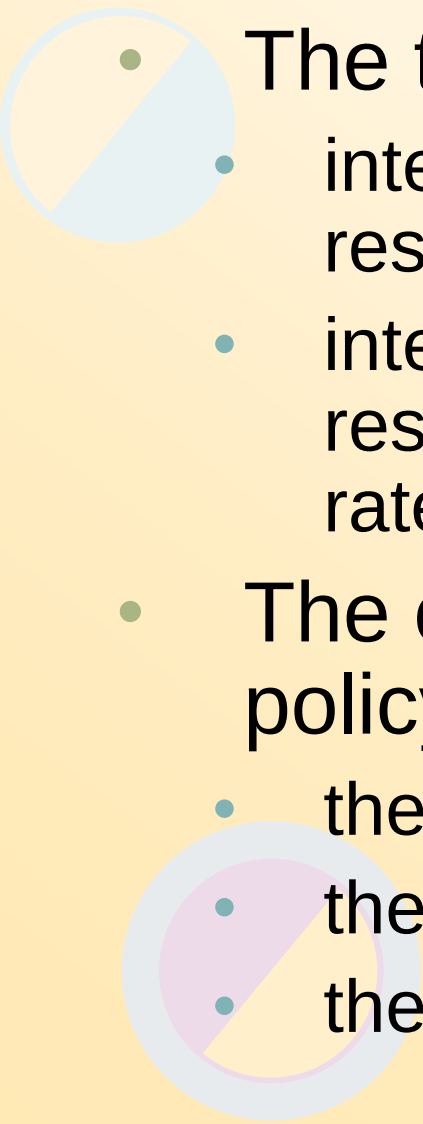
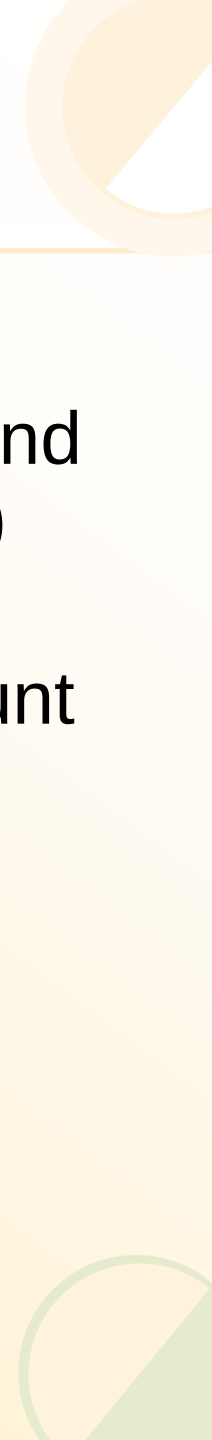
- The quantity of money changes directly with the base, and for a given amount of the base, an increase in either the reserve requirement or the holdings of excess reserves will decrease the quantity of money.
- But currency holdings affect both the numerator and the denominator of the multiplier, so the effect is not immediately obvious. Logic tells us that an increase in currency decreases reserves and so decreases the money supply

Factors Affecting the Quantity of Money

Factor	Who Controls It	Change	Impact on M
Monetary base	Central bank	Increase	Increase
Required reserve-to-deposit ratio	Bank regulators	Increase	Decrease
Excess reserve-to-deposit ratio	Commercial banks	Increase	Decrease
Currency-to-deposit ratio	Nonbank public	Increase	Decrease

The Central Bank's Monetary Policy Toolbox

- Central bank controls the quantity of reserves that commercial banks hold
- Besides the quantity of reserves, the central bank can control either the size of the monetary base or the price of its components

- 
- The two prices it concentrates on are
 - interest rate at which banks borrow and lend reserves overnight (the federal funds rate)
 - interest rate at which banks can borrow reserves from the central bank (the discount rate).
 - The central bank has three monetary policy tools, or instruments:
 - the target federal funds rate,
 - the discount rate, and
 - the reserve requirement.
- 
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The Target Federal Funds Rate and Open Market Operations

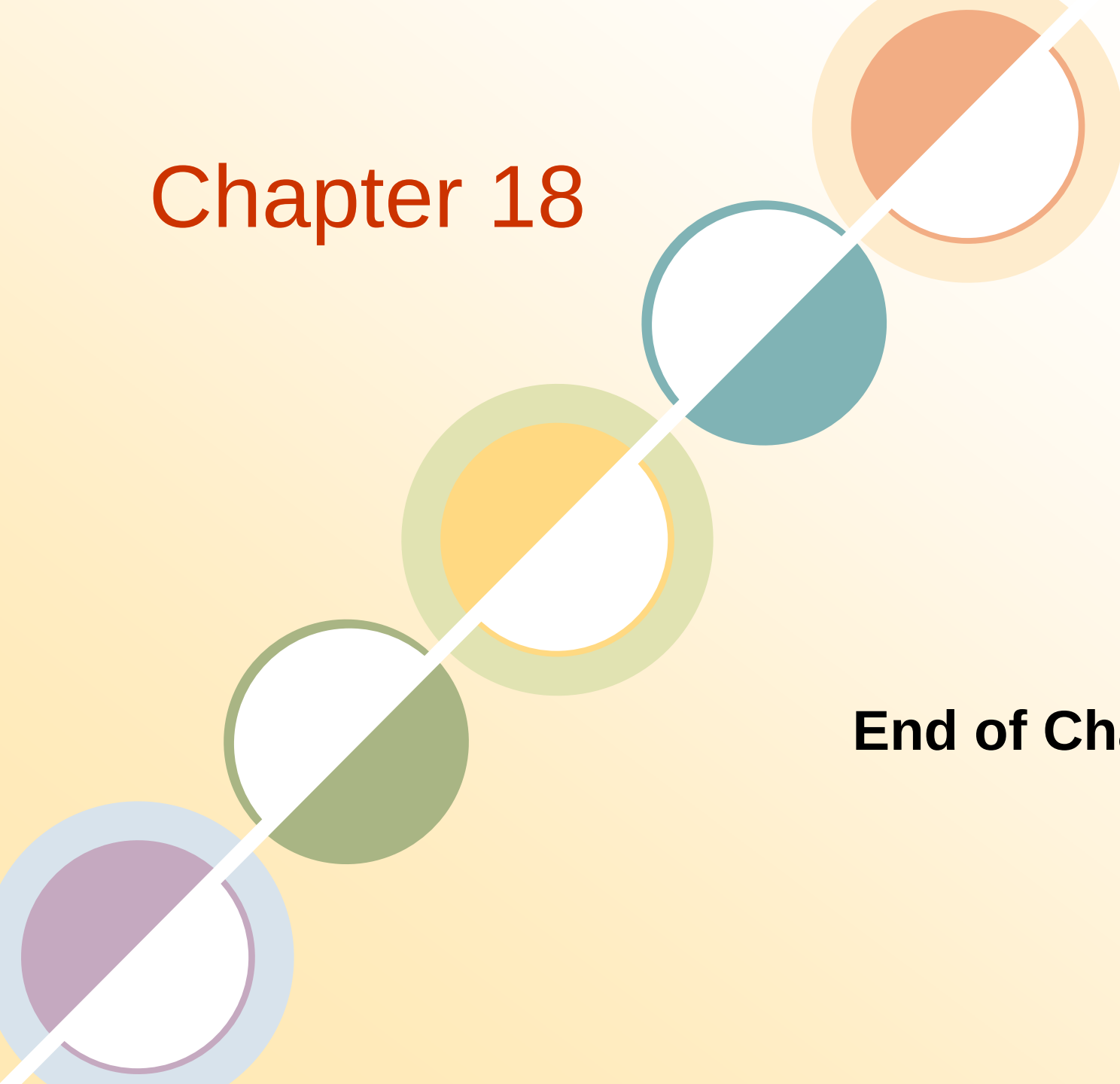
- The target federal funds rate is the central bank's primary policy instrument.
- The federal funds rate is determined in the market, rather than being controlled by the central bank.
- The name "federal funds" comes from the fact that the funds banks trade their deposit balances at the federal reserves or central bank.

- Central bank holds the capacity to force the market federal funds rate to equal the target rate all the time by participating directly in the market for overnight reserves, both as a borrower and as a lender.
- As a lender, the central bank would need to make unsecured loans to commercial banks, and as a borrower, the central bank would in effect be paying interest on excess reserves

- The central bank chooses to control the federal funds rate by manipulating the quantity of reserves through open market operations: the central bank buys or sells securities to add or drain reserves as required.

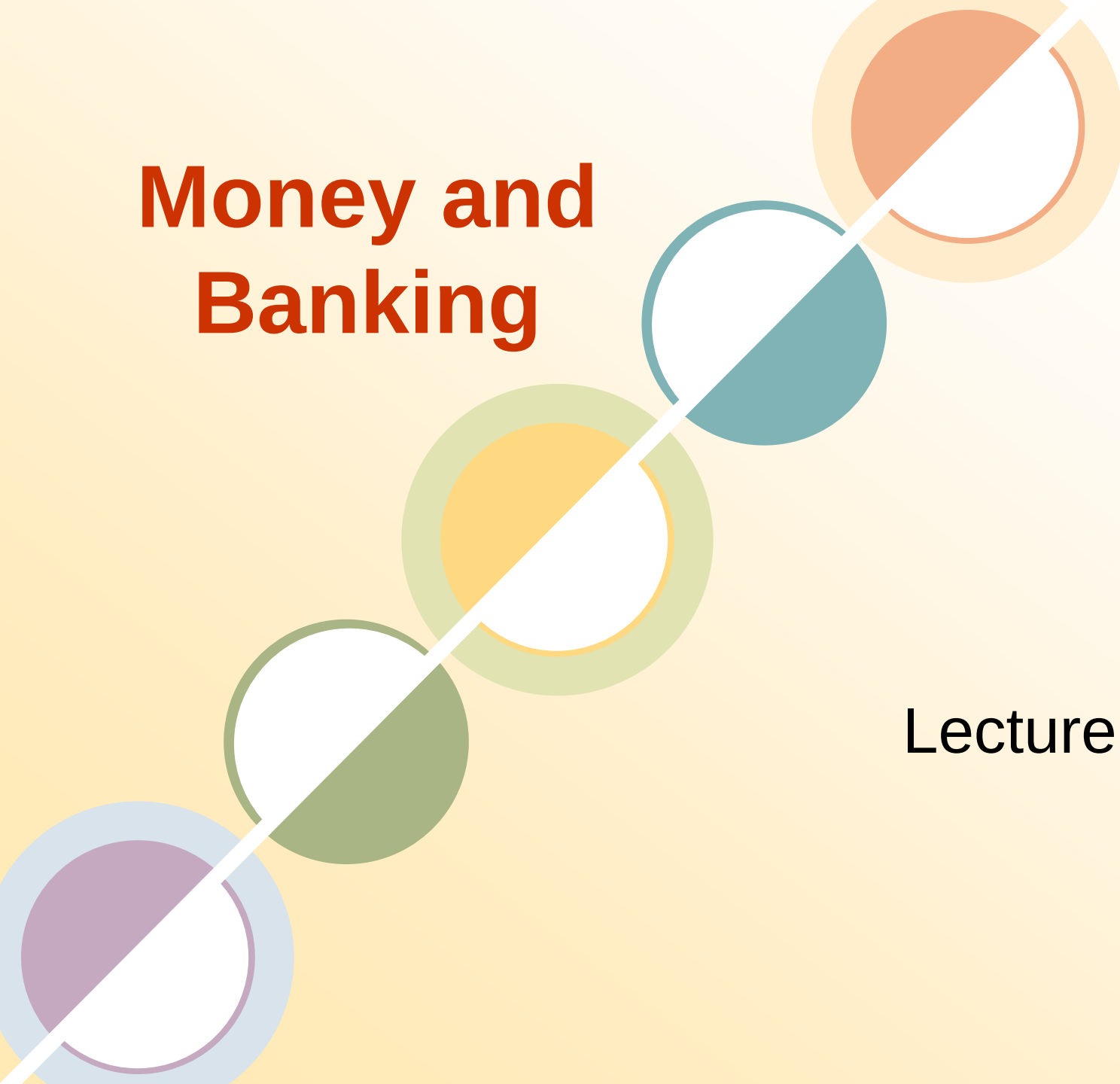
Chapter 18

End of Chapter



Money and Banking

Lecture 36



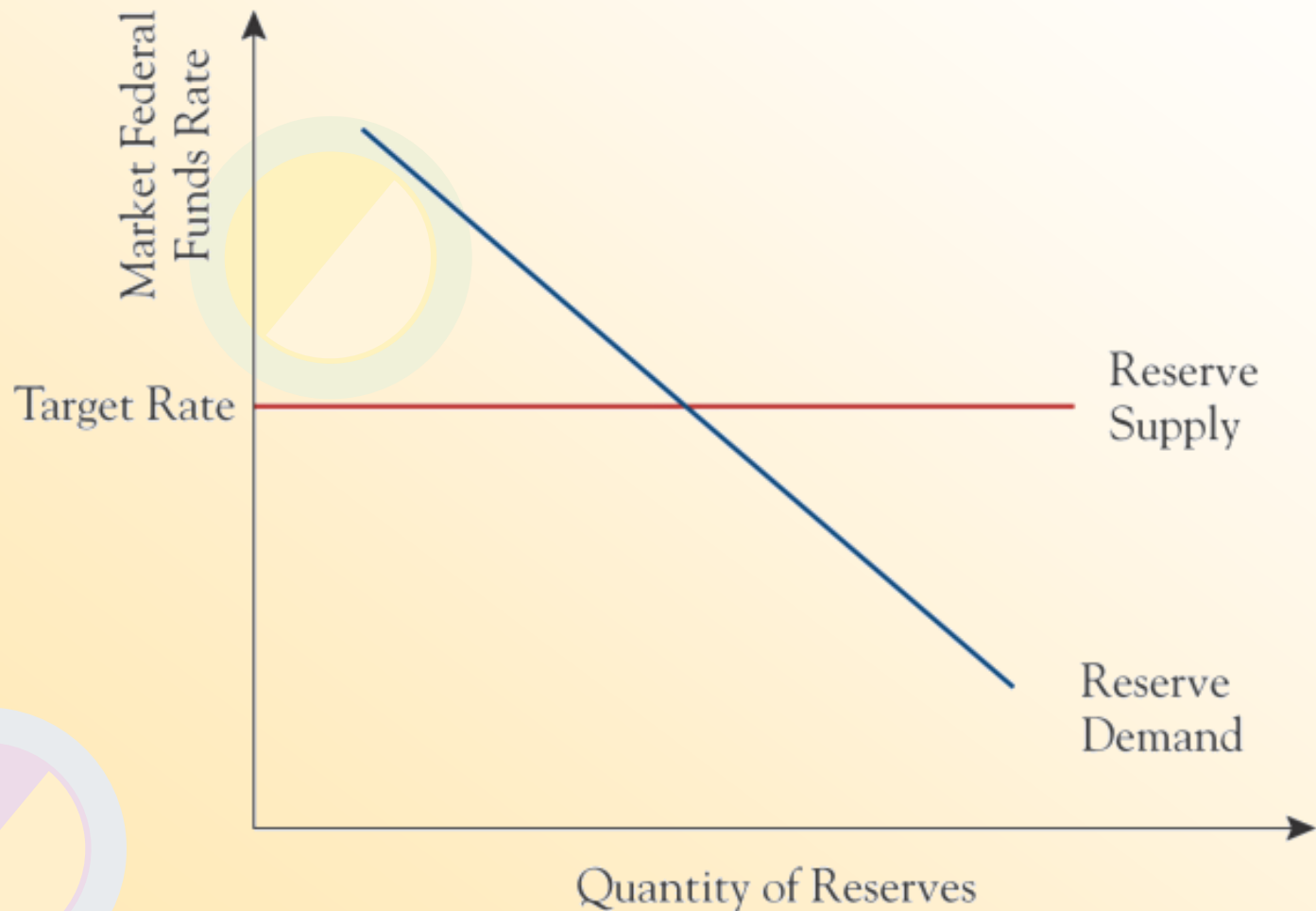
Review of the Previous Lecture

- Deposit Multiplier and Money Multiplier
- Central Bank's Monetary Policy Toolbox
 - Target Federal Funds Rate and Open Market Operation

Target Federal Funds Rate and Open Market Operation

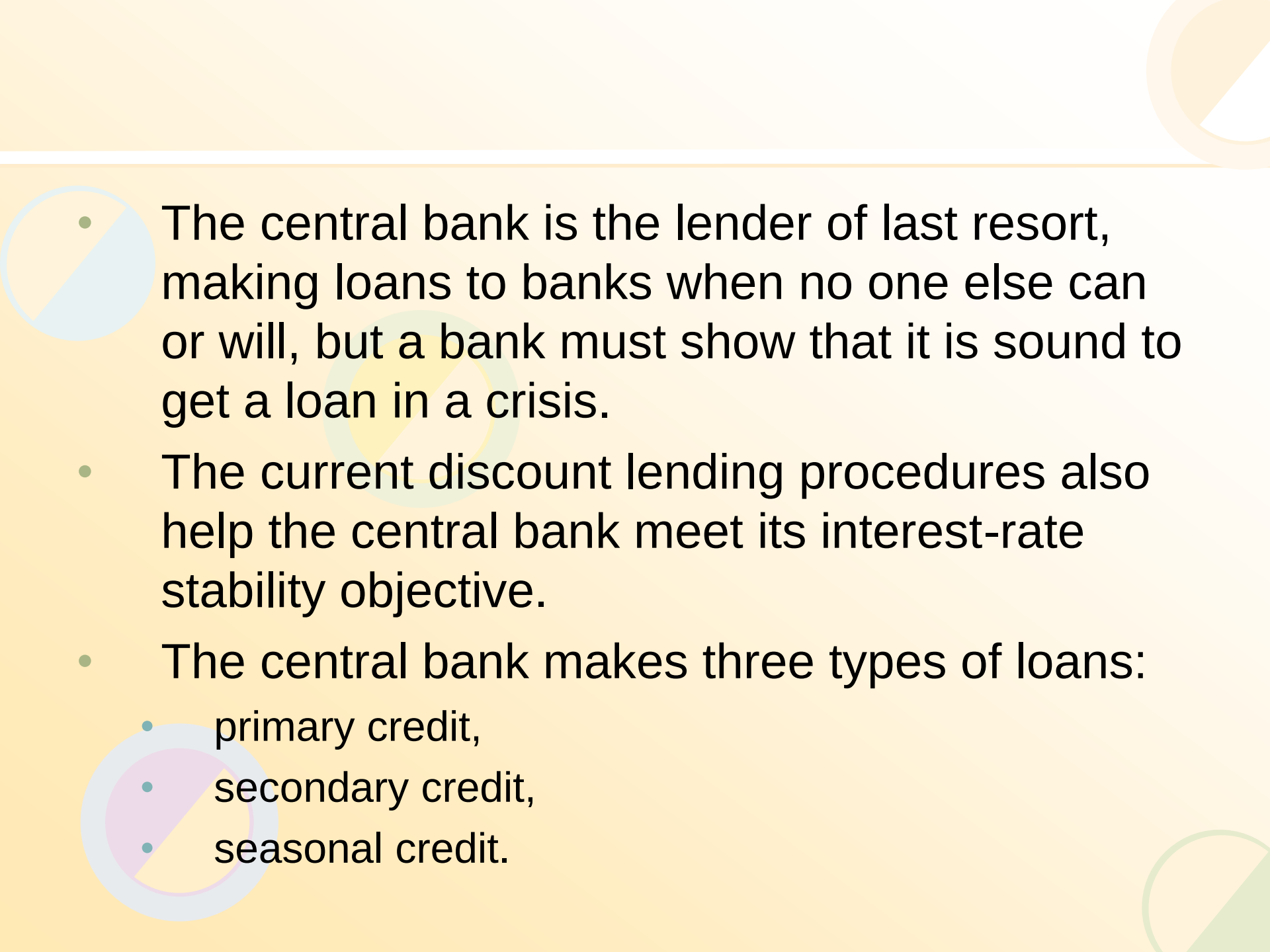
- The central bank chooses to control the federal funds rate by manipulating the quantity of reserves through open market operations: the central bank buys or sells securities to add or drain reserves as required.

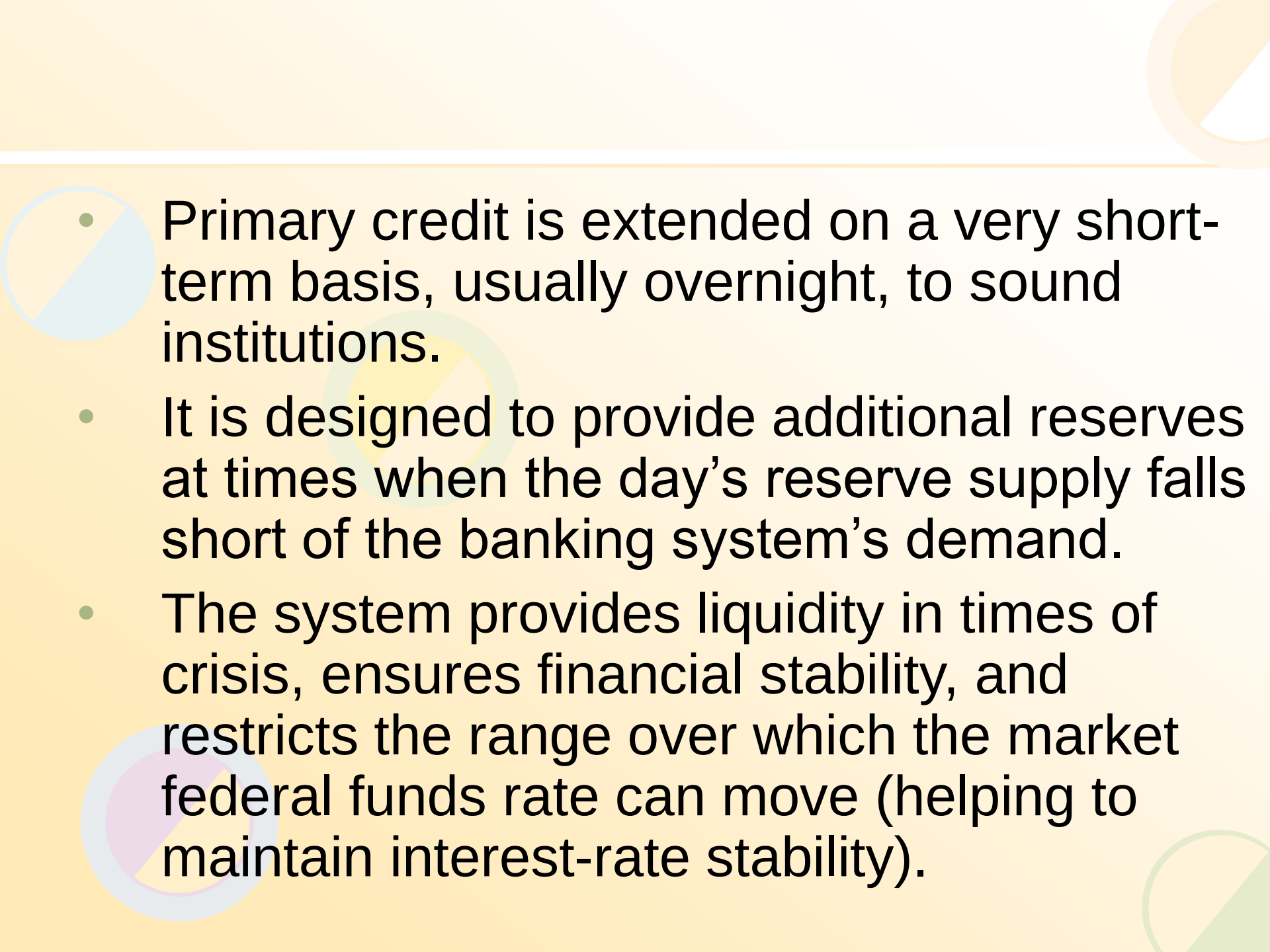
The Market for Bank Reserves



Discount Lending, the Lender of Last Resort and Crisis Management

- Lending to commercial banks is not an important part of the central bank's day-to-day monetary policy.
- However, such lending is the central bank's primary tool for ensuring short-term financial stability, for eliminating bank panics, and preventing the sudden collapse of institutions that are experiencing financial difficulties.

- 
- The central bank is the lender of last resort, making loans to banks when no one else can or will, but a bank must show that it is sound to get a loan in a crisis.
 - The current discount lending procedures also help the central bank meet its interest-rate stability objective.
 - The central bank makes three types of loans:
 - primary credit,
 - secondary credit,
 - seasonal credit.

- 
- Primary credit is extended on a very short-term basis, usually overnight, to sound institutions.
 - It is designed to provide additional reserves at times when the day's reserve supply falls short of the banking system's demand.
 - The system provides liquidity in times of crisis, ensures financial stability, and restricts the range over which the market federal funds rate can move (helping to maintain interest-rate stability).

- Secondary credit is available to institutions that are not sufficiently sound to qualify for primary credit.
- Banks may seek secondary credit due to a temporary shortfall in reserves or because they have longer-term problems that they need to work out.
- Seasonal credit is used primarily by small agricultural banks to help in managing the cyclical nature of farmers' loans and deposits

Reserve Requirements

- By adjusting the reserve requirement, the central bank can influence economic activity because changes in the requirement affect deposit expansion.
- Unfortunately, the reserve requirement turns out not to be very useful because small changes in the reserve requirement have large (really too large) impacts on the level of deposits.

- Today, the reserve requirement exists primarily to stabilize the demand for reserves and help the central bank to maintain the market federal funds rate close to target; it is not used as a direct tool of monetary policy

The central bank's Monetary Policy Toolbox

The Tools of Monetary Policy

	What Is It?	How Is It Controlled?	What Is Its Impact?
Target Federal Funds Rate	Interest rate charged on overnight loans between banks.	Supply of reserves adjusted through open market operations to meet expected demand at the target rate.	Changes interest rates throughout the economy.
Discount Rate	Interest rate charged by the Central bank on loans to commercial banks.	Set as a premium over the target federal funds rate.	Provides short-term liquidity to banks in times of crisis and aids in controlling the federal funds rate.
Reserve Requirement	Fraction of deposits that banks must keep either on deposit at the Central bank or as cash in their vaults.	Set by the Central bank within a legally imposed range.	Stabilizes the demand for reserves.

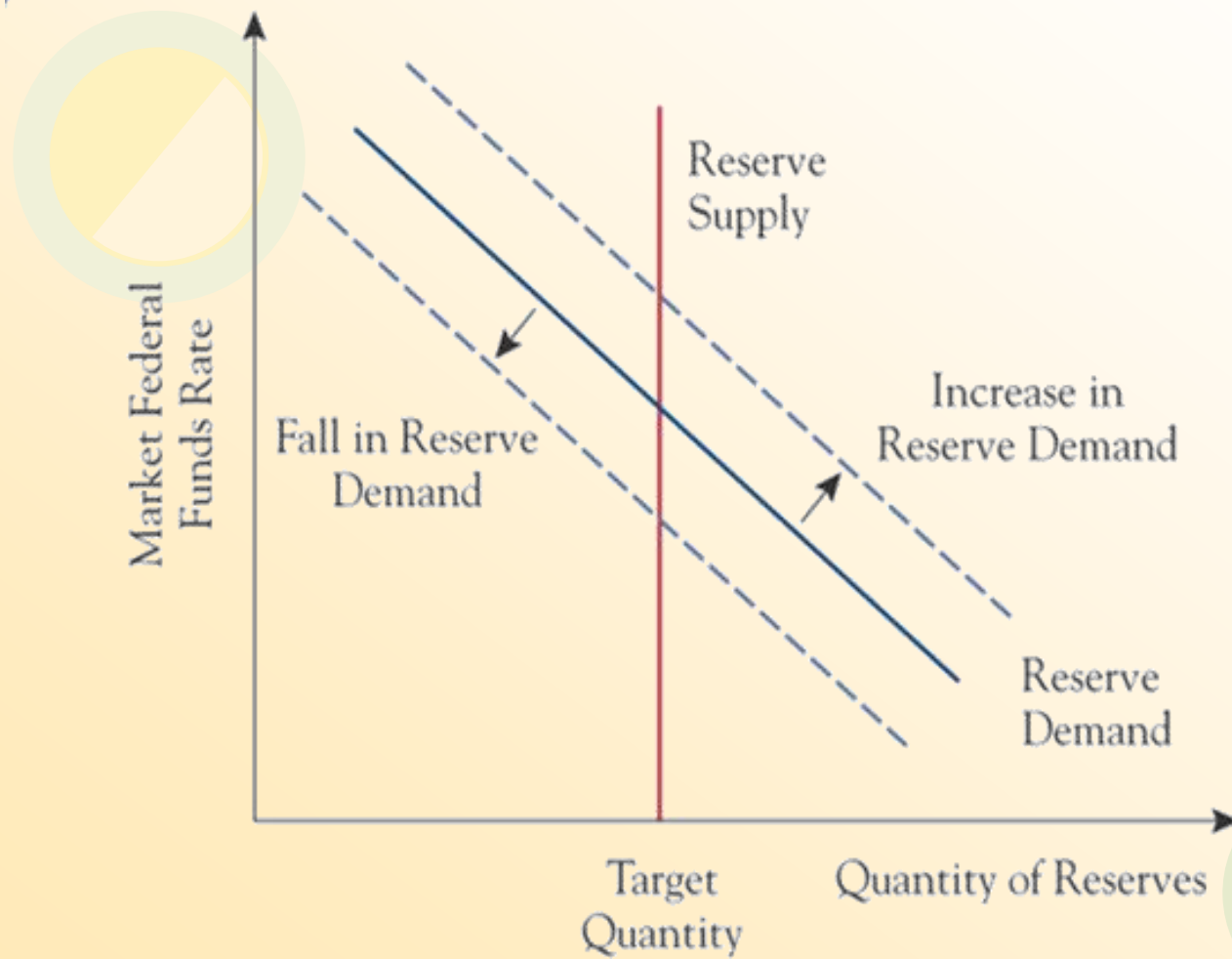
Linking Tools to Objectives

- Desirable Features of a Policy Instrument
 - Easily observable by everyone
 - Controllable and quickly changed
 - Tightly linked to the policymakers' objectives
- These requirements leave policymakers with few choices, and over the years central banks have switched between controlling the quantity and controlling the prices.

The Market for Bank Reserves when the Central bank Targets the Quantity of Reserves

When the Central bank targets the quantity of reserves, a shift in reserve demand causes the market federal funds rate to move. An increase in reserve demand forces the interest rate up, while a fall in reserve demand forces the interest rate down.

Reserve targets make interest rates volatile.



Targets and Instruments

- Operating instruments refer to actual tools of policy, instruments that the central bank controls directly.
- Intermediate target refers to instruments that are not directly under the control of the central bank but that lie between their policymaking tools and their objectives.

Targets and Instruments

- Over the last two centuries, central bankers largely abandoned intermediate targets, having realized that they didn't make much sense.
- Instead, policymakers focus on how their actions directly affect their target objectives

Instruments, Targets, and Objectives

Using policy instruments to target objectives directly



Money and Banking

Lecture 37

Review of the Previous Lecture

- Central Bank's Monetary Policy Toolbox
 - Open Market Operations
 - Discount Lending
 - Reserve Requirements
- Linking tools to Objectives

Why We Care About Monetary Aggregates

- Every country with high inflation has high money growth; thus to avoid sustained episodes of high inflation, a central bank must be concerned with money growth.
- It is impossible to have high, sustained inflation without monetary accommodation.

Monetary Aggregates

Figure 20.1 Inflation and Money Growth, All Countries, 1981-2003

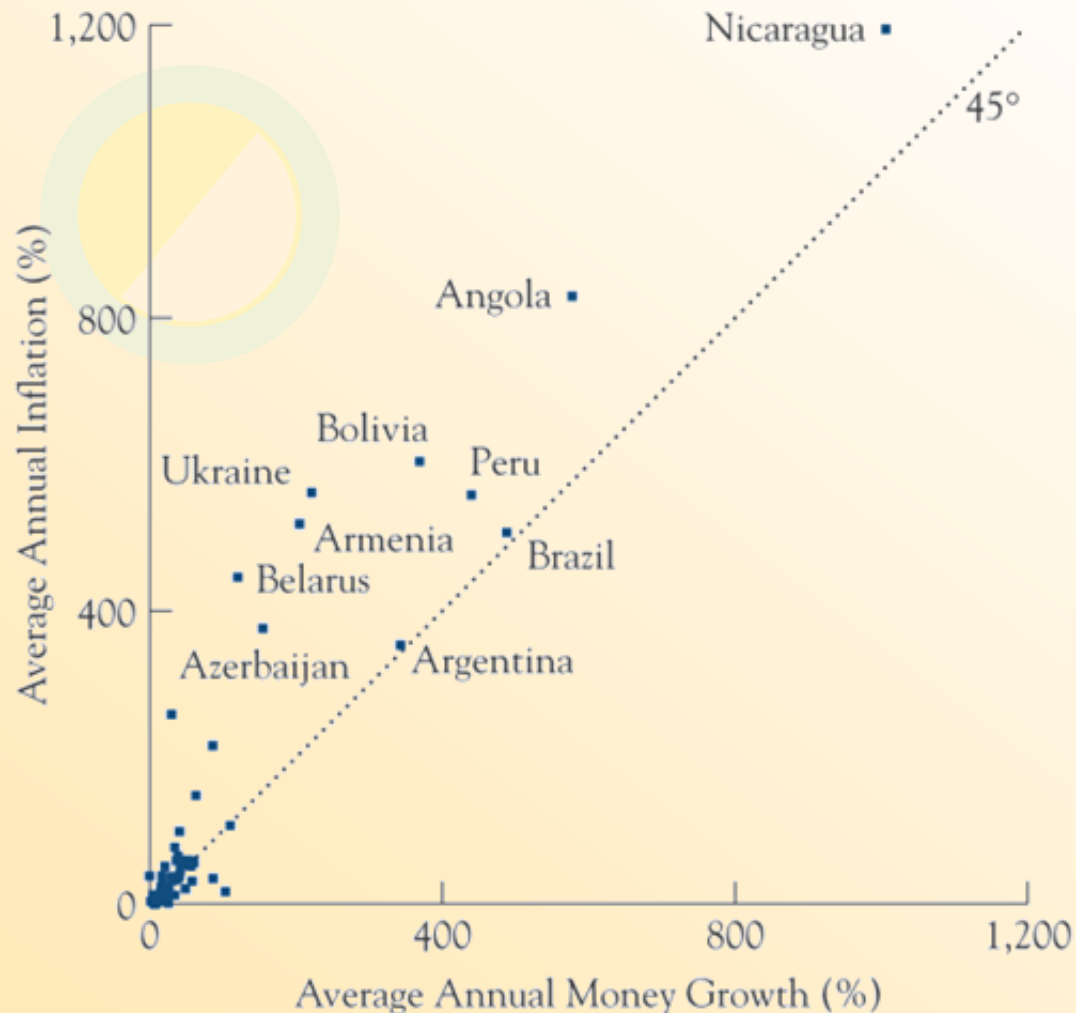


Figure 20.2 Inflation and Money Growth:
Moderate-Inflation Countries, 1981-2003

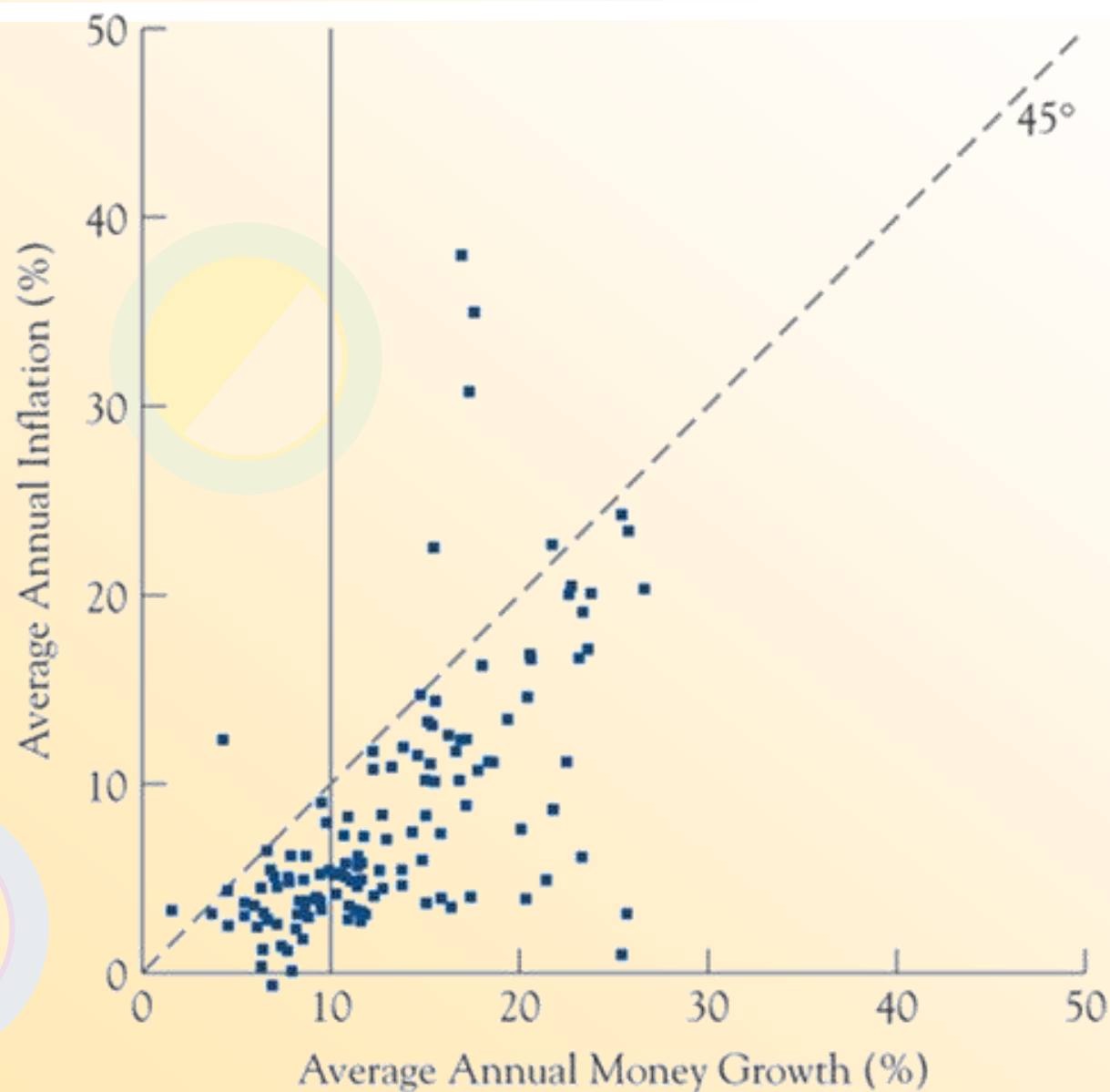
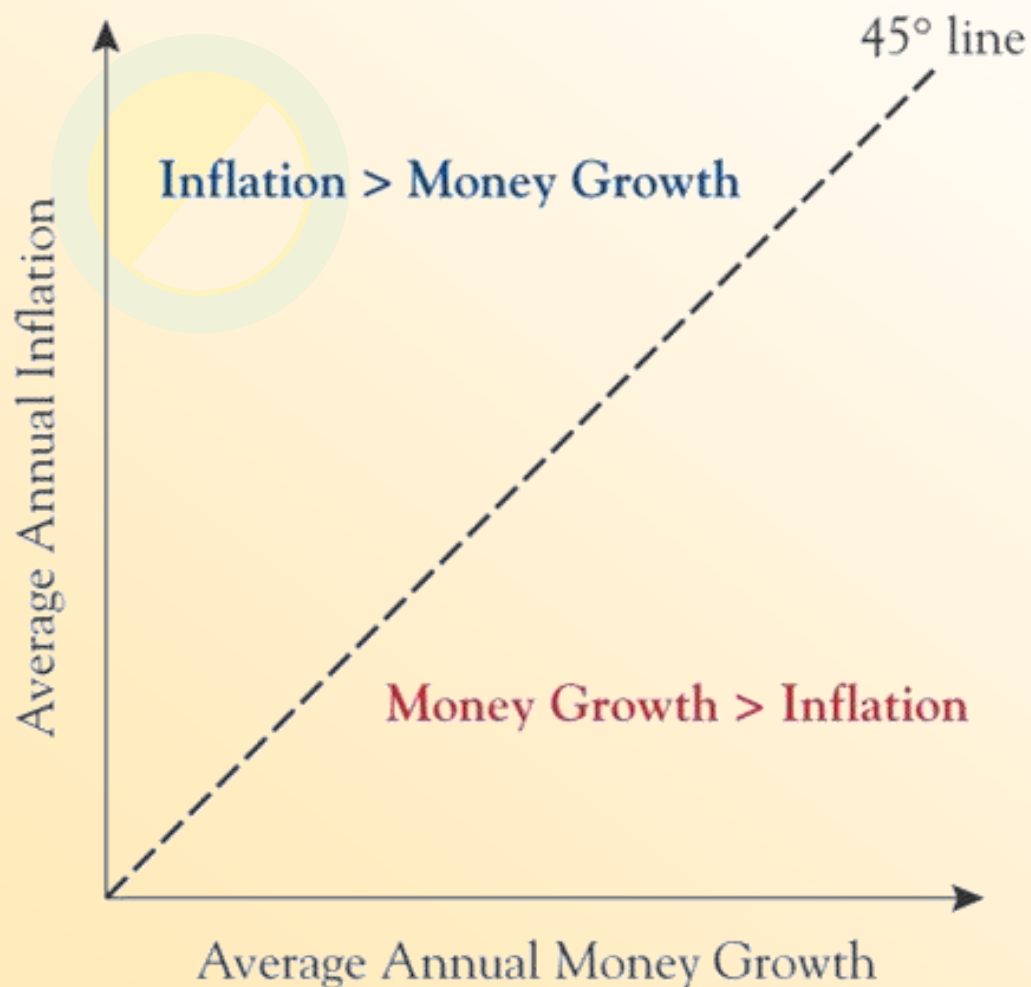


Figure 20.3 Money Growth and Inflation



- When the currency that people are holding loses value much rapidly, they will work to spend what they have as quickly as possible
- This will have the same effect on inflation as an increase in money growth

Monetary Aggregates

- It is impossible to have high, sustained inflation without monetary accommodation.
- Something beyond just differences in money growth accounts for the differences in inflation across countries.

Velocity and the Equation of Exchange

- To understand the relationship between inflation and money growth we need to focus on money as a means of payment.
- Consider an example of four students
 - Ali has Rs. 100 in cash
 - Bilal has a Rs. 100 calculator
 - Chohan has 2 tickets worth Rs. 50 each for a cricket match
 - Dilawer has a set of 25 drawing pencils worth Rs. 4 each

- Ali needs a calculator which he buys from Bilal.
- Bilal wishes to see the match so he buys the tickets from Chohan
- Chohan uses the proceeds to purchase the drawing pencils from Dilawer

- Total Value of the transactions is
 - $(\text{Rs. } 100 \times 1 \text{ calculator}) + (\text{Rs. } 50 \times 2 \text{ tickets}) + (\text{Rs. } 4 \times 25 \text{ pencils}) = \text{Rs. } 300$
- Generally
 - $\text{No. of Rupees} \times \text{No. of time each Re is used} = \text{Rs. Value of Transactions}$

- The number of times each rupee is used (per unit of time) in making payments is called the velocity of money; the more frequently each rupee is used, the higher the velocity of money

- Applying to economy wide transactions:
 - Quantity of Money x Velocity of Money = Nominal GDP
- Using data on the quantity of money and nominal GDP we can compute the velocity of money; each monetary aggregate has its own velocity

- If we represent
 - Money with M
 - Velocity with V
 - Price level with P
 - Real GDP with Y
- Nominal GDP = $P \times Y$
- Substituting, we get

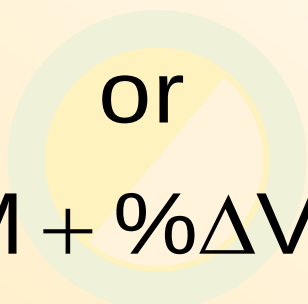
$$M \times V = P \times Y$$

- The equation of exchange, $MV=PY$ provides the link between money and prices if we rewrite it in terms of percentage changes

The Quantity Theory and the Velocity of Money


$$MV = PY$$

or


$$\% \Delta M + \% \Delta V = \% \Delta P + \% \Delta Y$$

Money Growth + Velocity Growth = Inflation + Output Growth

The Quantity Theory of Money

- In the early 20th century, Irving Fisher wrote down the equation of exchange and derived the implication that

$$\text{money growth} + \text{velocity growth} = \text{inflation} + \text{real growth}$$

- Assuming
 - no important changes occur in payment methods or the cost of holding money,
 - real output is determined solely by economic resources and production technology,
- then changes in the aggregate price level are caused solely by changes in the quantity of money.

- In other words
 - assume that $\% \Delta V = 0$ and $\% \Delta Y = 0$.
 - doubling the quantity of money doubles the price level.
- Inflation is a monetary phenomenon (Milton Friedman).

- In our example of four students, number of rupees needed equaled total rupee value of the transaction divided by no. of times each rupee was used

$$\text{Money demand} = \frac{\text{Total value of transaction}}{\text{velocity of Money}}$$

- For the economy as a whole,
Money demand = $\frac{\text{Nominal GDP}}{\text{Velocity}}$

$$M^d = 1/V \times PY$$

- Money Supply (M^s) is determined by central bank and the behavior of the banking system
- Equilibrium means $M^d = M^s = M$
- Rearranging the Money demand function gives $MV = PY$

- The quantity theory of money tells us why high inflation and high money growth go together, and explains why countries can have money growth that is higher than inflation (because they are experiencing real growth).

Figure 20.3 Money Growth and Inflation

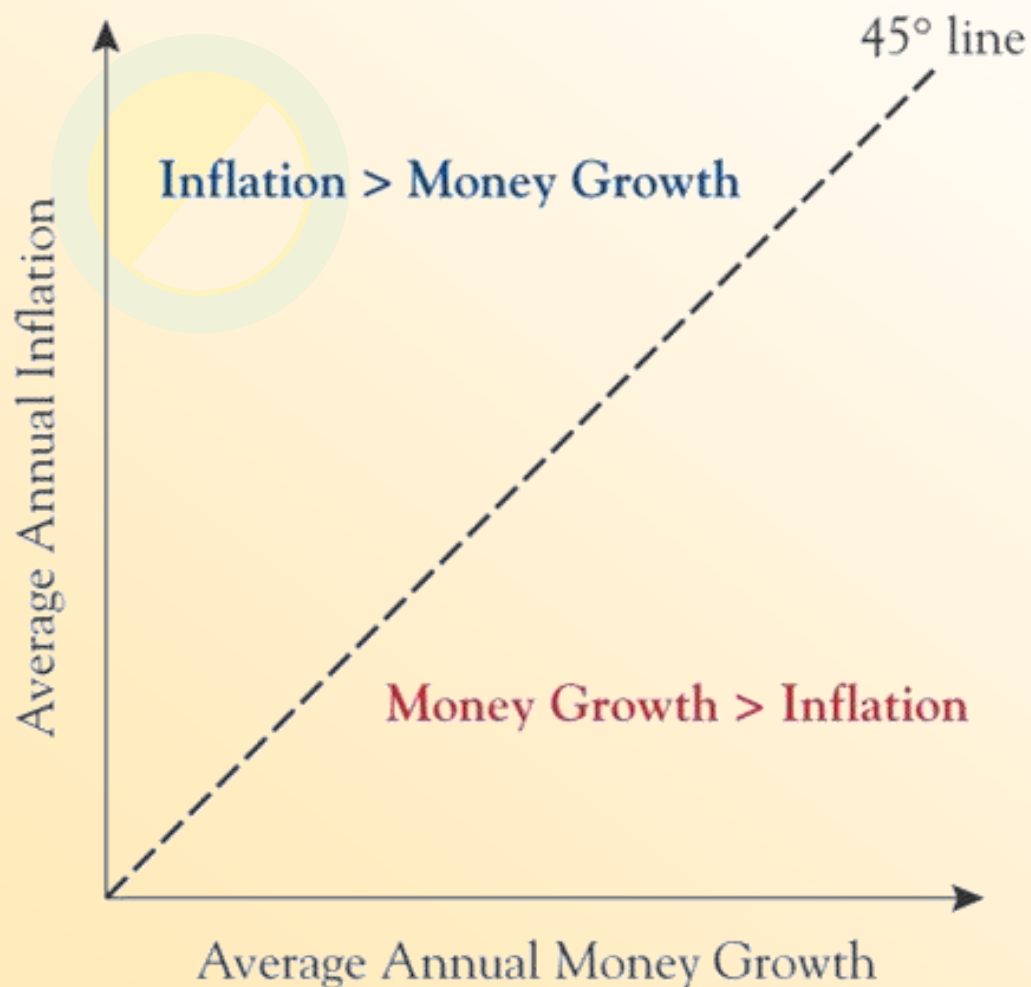
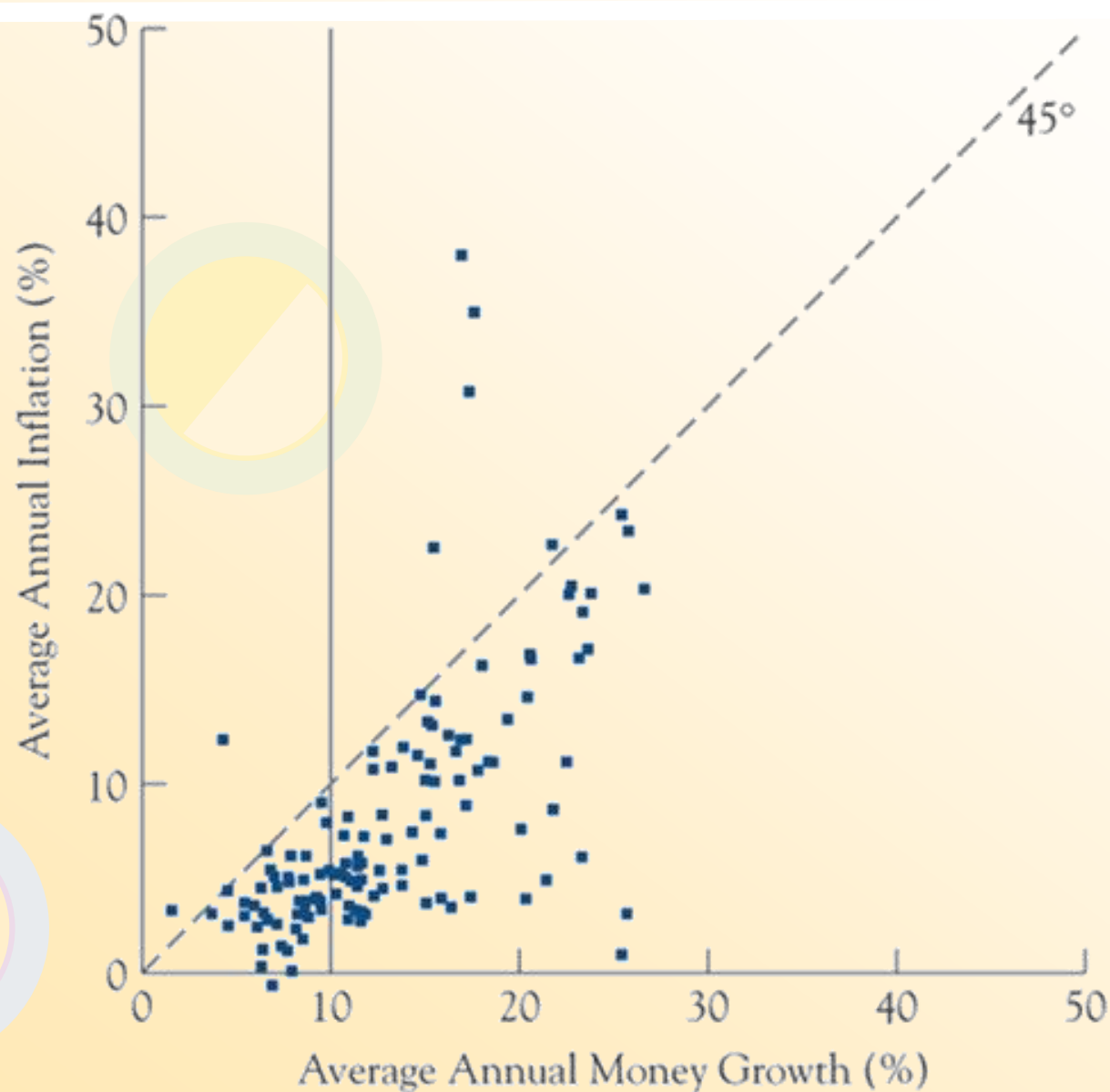
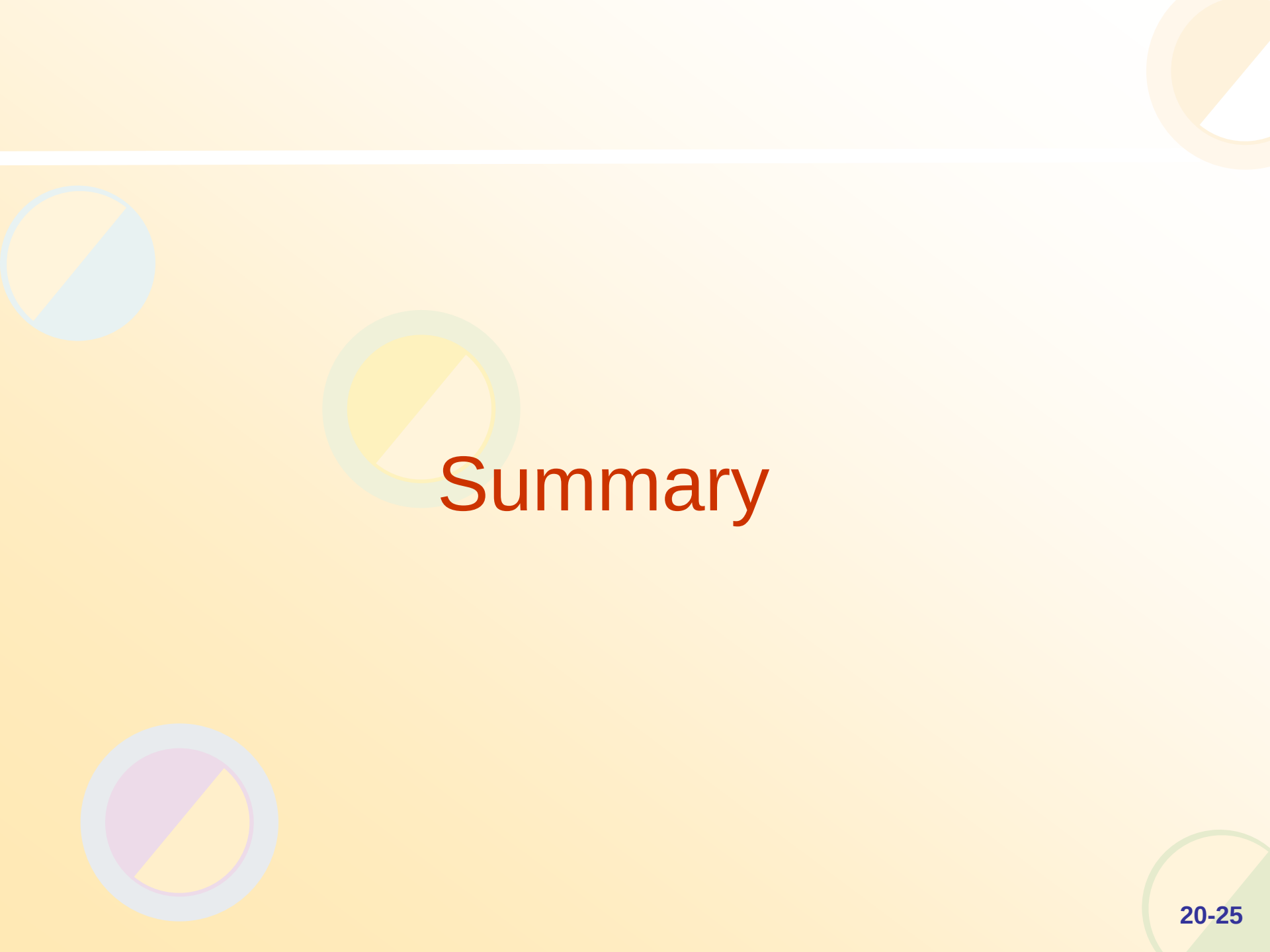


Figure 20.2 Inflation and Money Growth:
Moderate-Inflation Countries, 1981-2003



The Facts about Velocity

- Fisher's logic led Milton Friedman to conclude that central banks should simply set money growth at a constant rate.
- Policymakers should strive to ensure that the monetary aggregates grow at a rate equal to the rate of real growth plus the desired level of inflation.



Summary

Money and Banking

Lecture 38

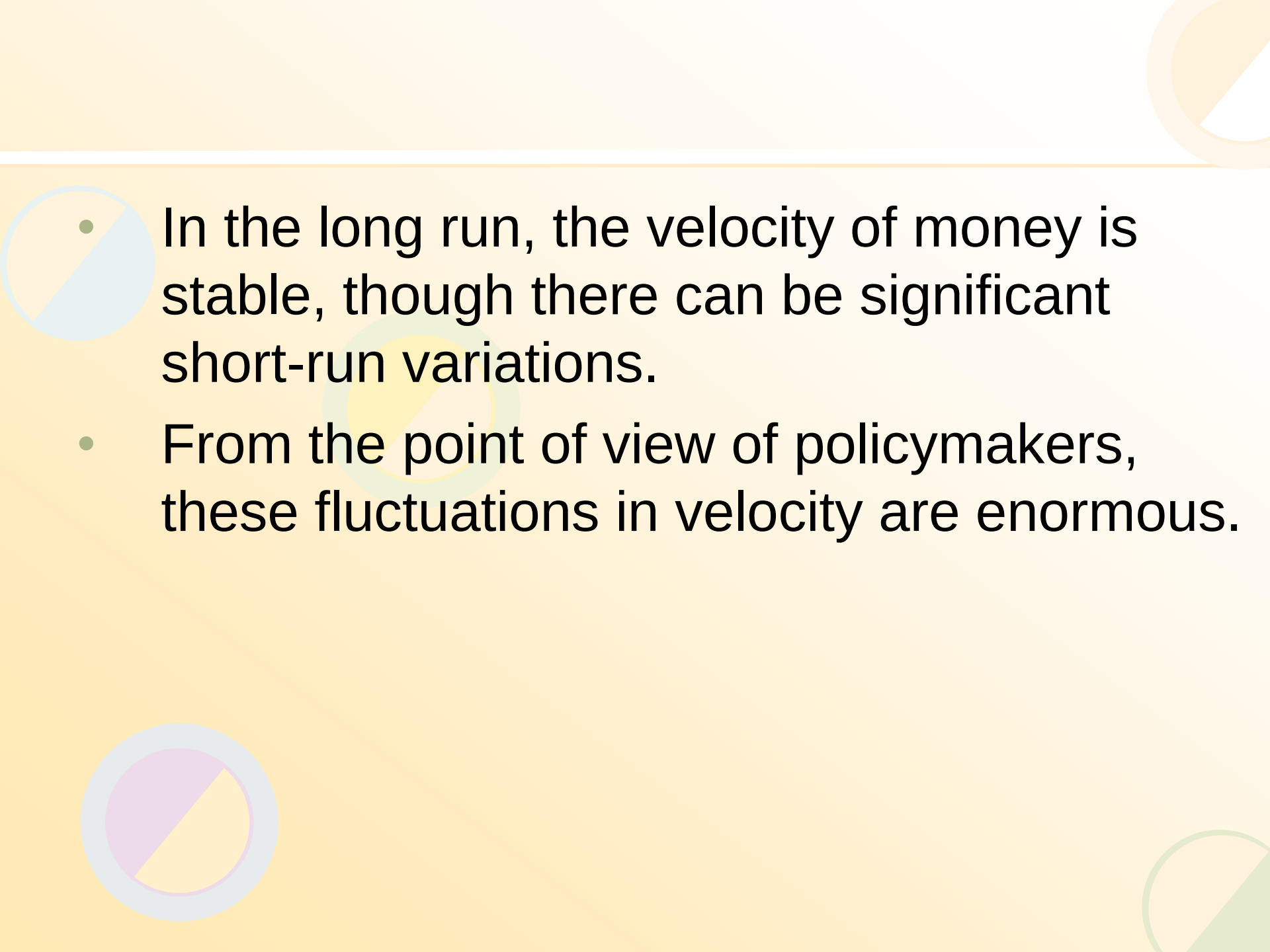
Review of the Previous Lecture

- Monetary Aggregates
- Velocity and the Equation of Exchange
- The Quantity Theory of Money

The Facts about Velocity

- Fisher's logic led Milton Friedman to conclude that central banks should simply set money growth at a constant rate.
- Policymakers should strive to ensure that the monetary aggregates grow at a rate equal to the rate of real growth plus the desired level of inflation.

- Knowing that the multiplier is a variable, Friedman suggested changes in regulations that would
 - limit banks' discretion in creating money
 - tighten the relationship between the monetary aggregates and the monetary base.
- However, even with Friedman's recommendations, the central bank would stabilize inflation by keeping money growth constant only if velocity were constant.

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- In the long run, the velocity of money is stable, though there can be significant short-run variations.
 - From the point of view of policymakers, these fluctuations in velocity are enormous.

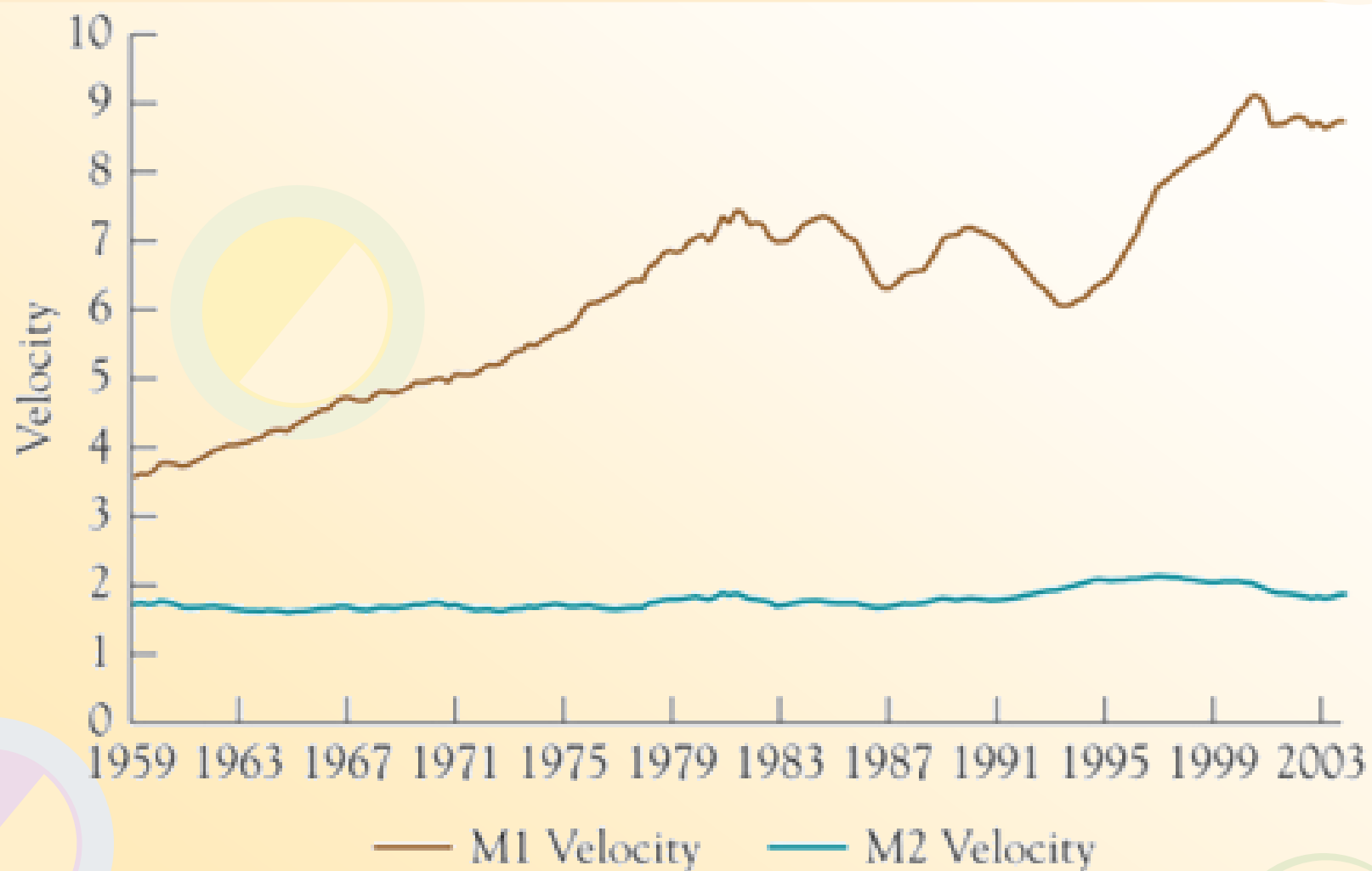
- Assuming that central bank can accurately control the growth rate of M2 as well as accurately forecast real growth.

$$\text{Money Growth} + \text{Velocity Growth} = \text{Inflation} + \text{Real Growth}$$

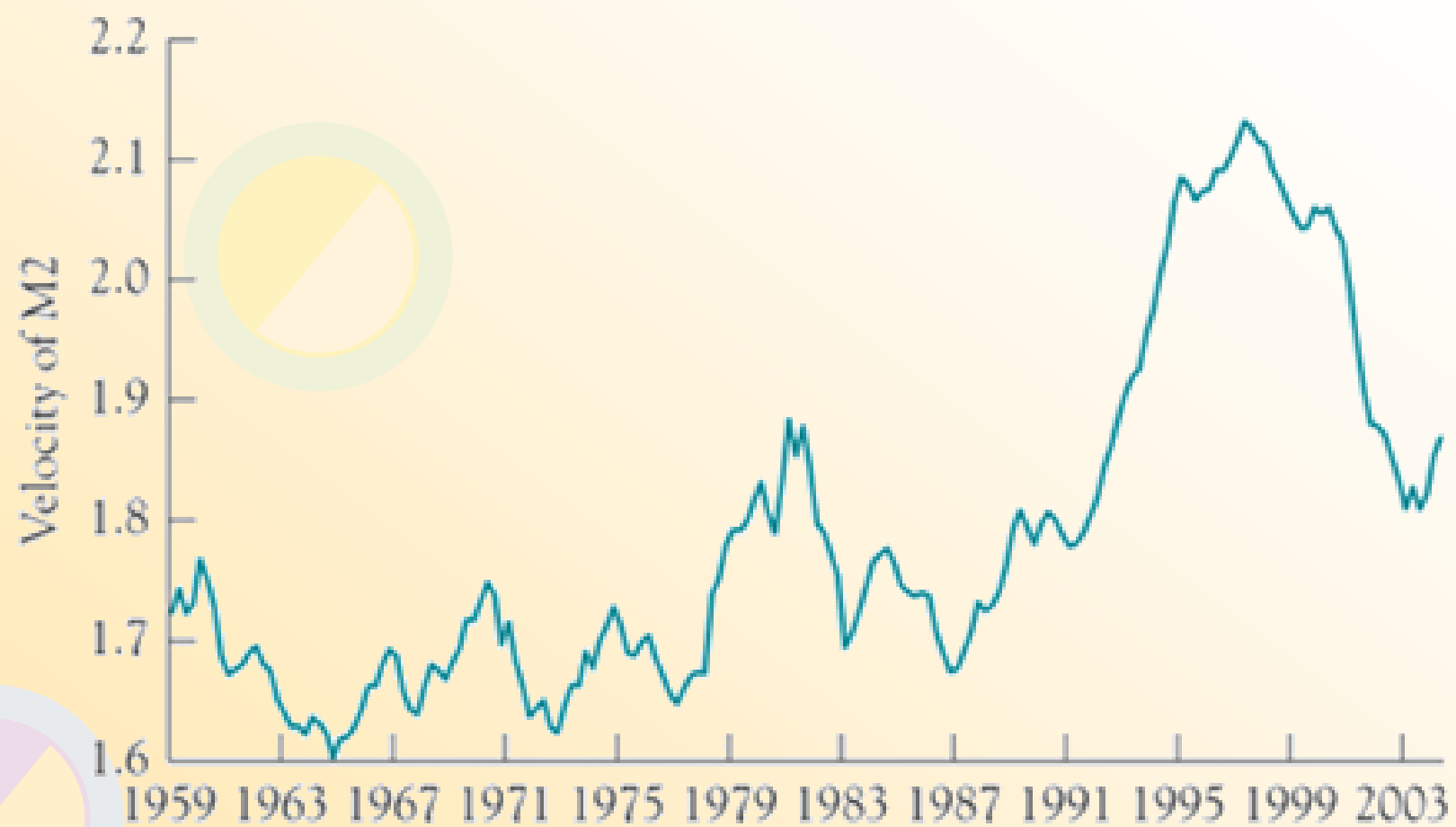
- With an inflation objective of 2% and a real growth forecast of 3.5%, equation of exchange tells us that policy makers should set money growth 5.5% minus the growth rate of velocity.
- If velocity increases by 3% then money growth needs to be 2.5%
- If it falls by 3% then money growth needs to be 8.5%

- When inflation is low, short run velocity growth can be several times the policy makers' inflation objectives.
- So to use money growth targets to stabilize inflation, policy makers must understand how velocity changes
- Fluctuations in velocity are tied to changes in people's desire to hold money and so in order to understand and predict changes in velocity, policymakers must understand the demand for money.

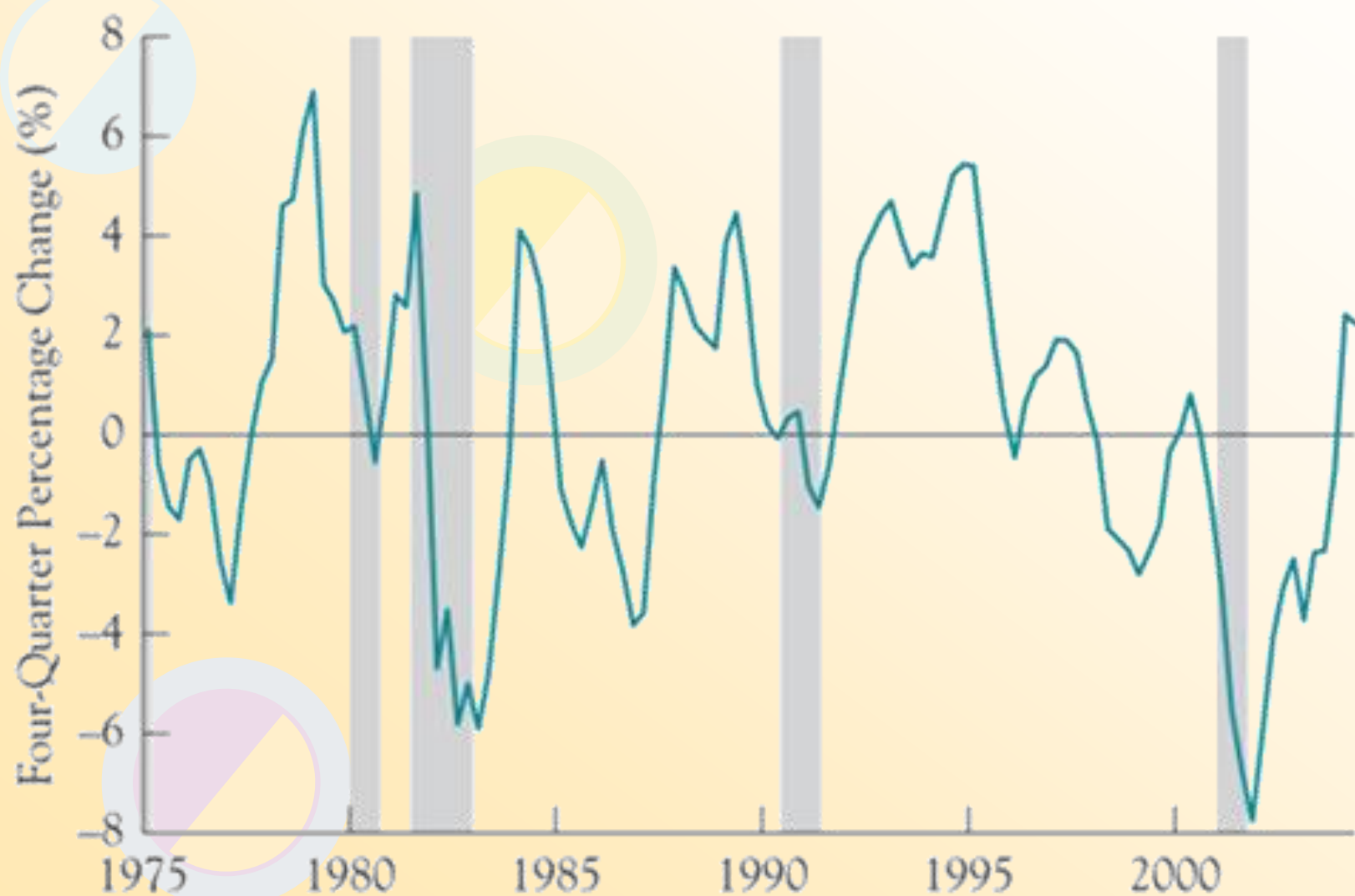
Velocity of M1 and M2 (on the same vertical scale)



5. Velocity of M2 (on its own vertical scale)



6 Percentage Change in the Short-Run Velocity of M2, 1960-2004



The Transactions Demand for Money

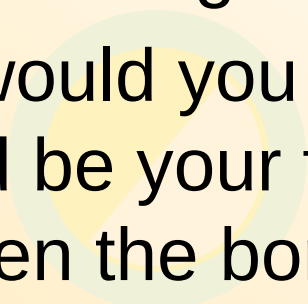
- The quantity of money people hold for transactions purposes depends on
 - their nominal income,
 - the cost of holding money,
 - the availability of substitutes.
- Nominal money demand rises with nominal income, as more income means more spending, which requires more money

- Holding money allows people to make payments, but has cost of interest foregone.

- There may also be costs in switching between interest-bearing assets and money.

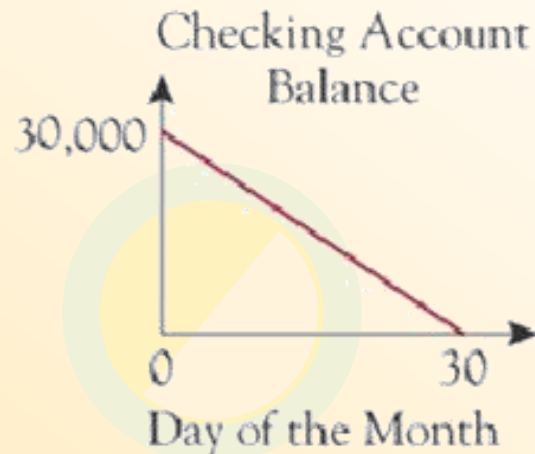
- Example

- If your monthly earning is Rs.30,000 (deposited in bank each month) and assuming you spend Rs.1,000 each day, after 15 days your checking account balance will decline to Rs.15,000 and to zero on 30th day
 - Your bank offers you a choice of leaving the entire 30,000 in the account or shifting funds back and forth between checking and a bond fund.

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- The bond fund pays interest but adds a service charge of Rs.20 for each withdrawal.
 - How would you manage your funds and what should be your frequency of shifting the funds between the bond fund and checking account?
 - Consider the following alternatives.

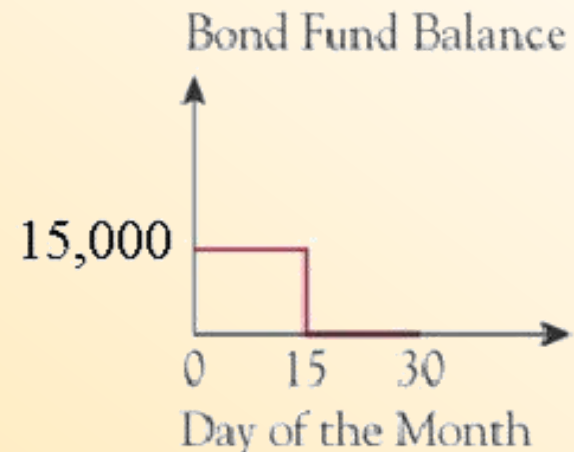
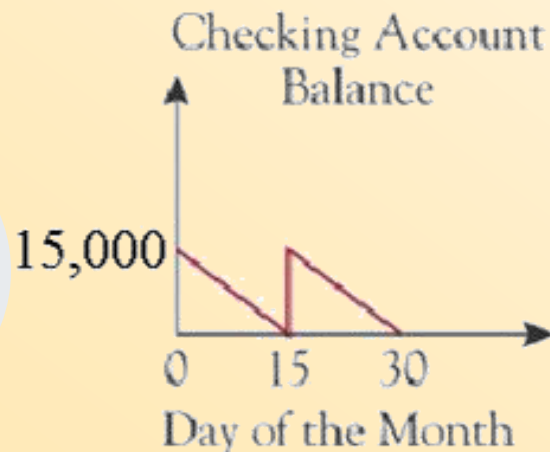
Two Alternatives for Managing Your Cash Balance

Strategy 1: Leave entire balance in checking account



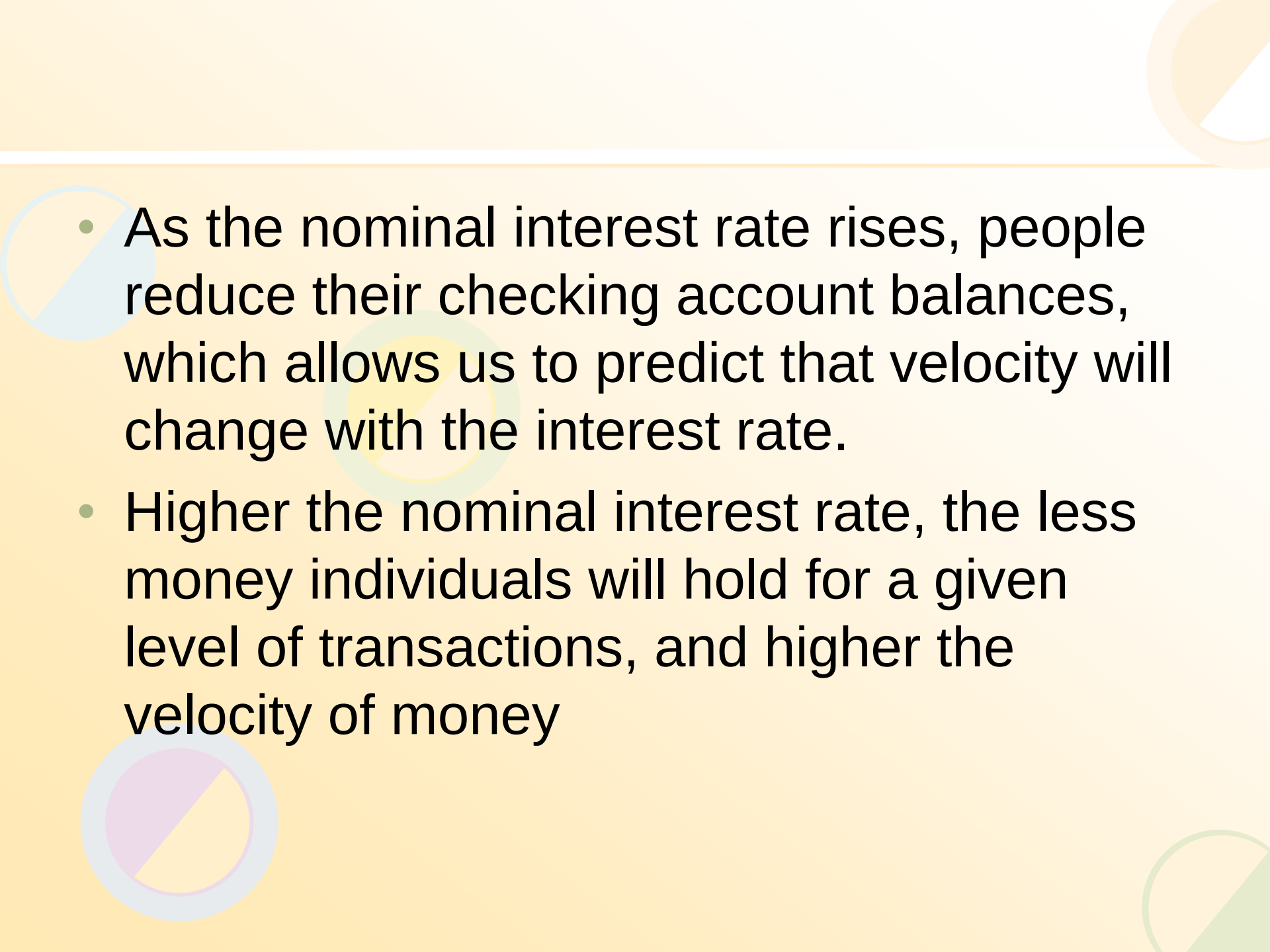
Bond fund balance is zero throughout the month.

Strategy 2: Transfer half to bond fund, then transfer back at mid-month



- Your choice depends upon the interest rate you receive on the bond fund
 - If interest income is at least as much as the service charge then you will split your pay check at the beginning of the month.
 - Otherwise you will not want to invest in bond fund.
 - If you shift half your funds once , at the middle of the month, you'll have Rs.15,000 in bond fund during the first half of the month and Rs.0 during the second half, so your average balance will be Rs.7,500.

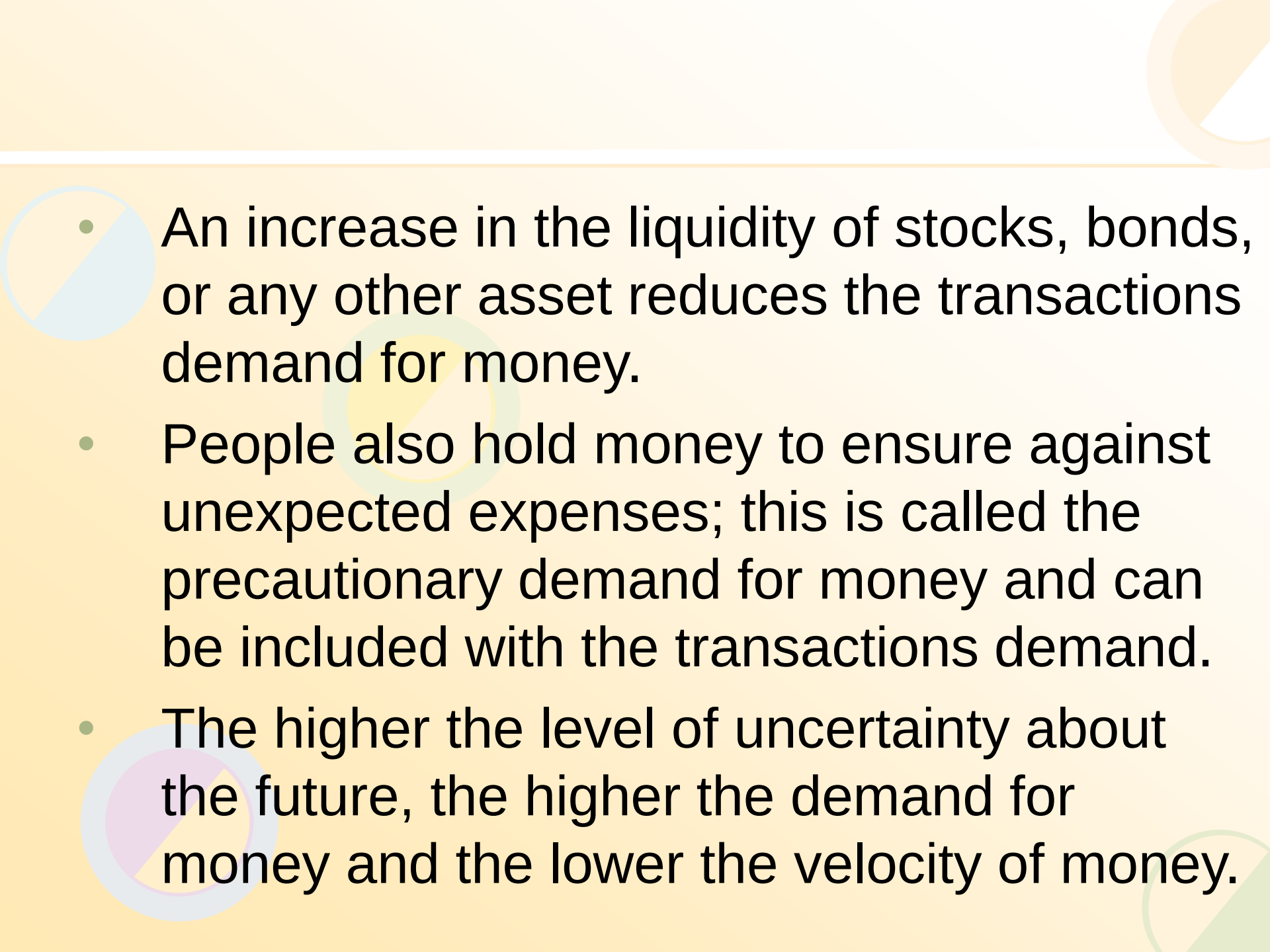
- Making shift will cost you Rs.20 so if the interest on Rs.7,500 is greater than 20, you should make the shift.
 - At monthly interest rate of 0.27%, Rs.7,500 will produce an income of Rs.20
($20 / 7500 = 0.0027$)
 - So if bond fund offers a higher rate you should make the shift.

- 
- As the nominal interest rate rises, people reduce their checking account balances, which allows us to predict that velocity will change with the interest rate.
 - Higher the nominal interest rate, the less money individuals will hold for a given level of transactions, and higher the velocity of money

- The transactions demand for money is also affected by technology, as financial innovation allows people to limit the amount of money they hold.
- The lower the cost of shifting money between accounts, the lower the money holdings and the higher the velocity.

- Suppose your bank offers free automatic transfer account. You sign up for it but continue using your old check and debit account
- Your take home pay is the same Rs.30,000. each time you make a purchase, your bank automatically shifts the amount of purchase from your bond fund to your checking account where it remains for one day before being paid to your creditor.

- Spending your Rs.30,000 in 30 days, your average money holding will be Rs.1000 far below Rs.1,500 you would hold if you simply left the 30,000 in your checking account and spent it at a rate of Rs.1,000 per day
- So lower the cost of shifting funds from your bond fund to your checking account, the lower your money holdings at a given level of income and the higher the velocity of your money

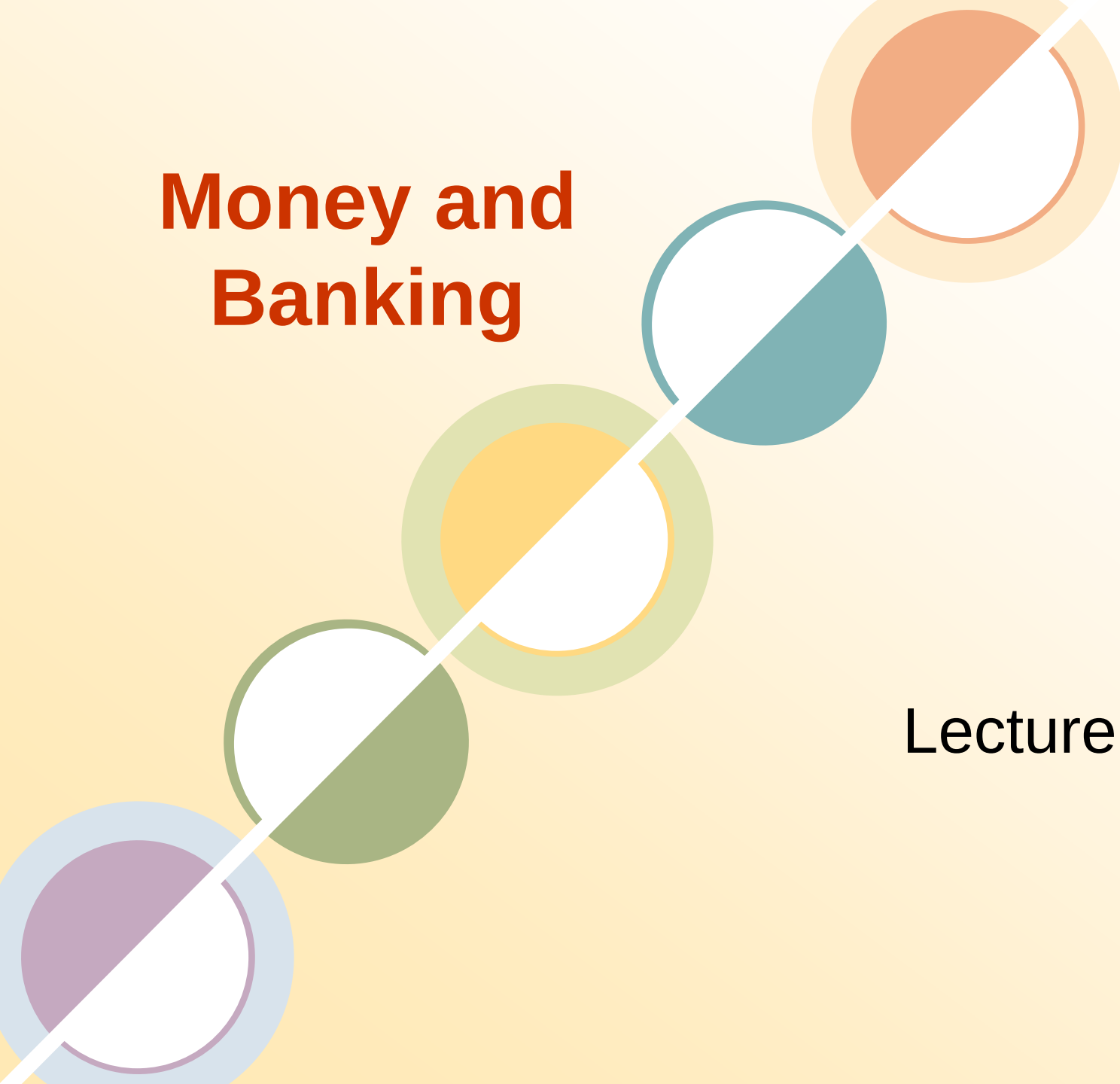
- 
- An increase in the liquidity of stocks, bonds, or any other asset reduces the transactions demand for money.
 - People also hold money to ensure against unexpected expenses; this is called the precautionary demand for money and can be included with the transactions demand.
 - The higher the level of uncertainty about the future, the higher the demand for money and the lower the velocity of money.

The Portfolio Demand for Money

- Money is just one of many financial instruments that we can hold in our investment portfolios.
- Expectations that interest rates will change in the future are related to the expected return on a bond and also affect the demand for money.

Money and Banking

Lecture 39



Review of the Previous Lecture

- Monetary Aggregates
- Equation of Exchange
 - $MV = PY$
- Quantity Theory of Money
 - Facts about Velocity of Money
- Demand for Money
 - Transactions Demand for money

The Portfolio Demand for Money

- Money is just one of many financial instruments that we can hold in our investment portfolios.
- Expectations that interest rates will change in the future are related to the expected return on a bond and also affect the demand for money.

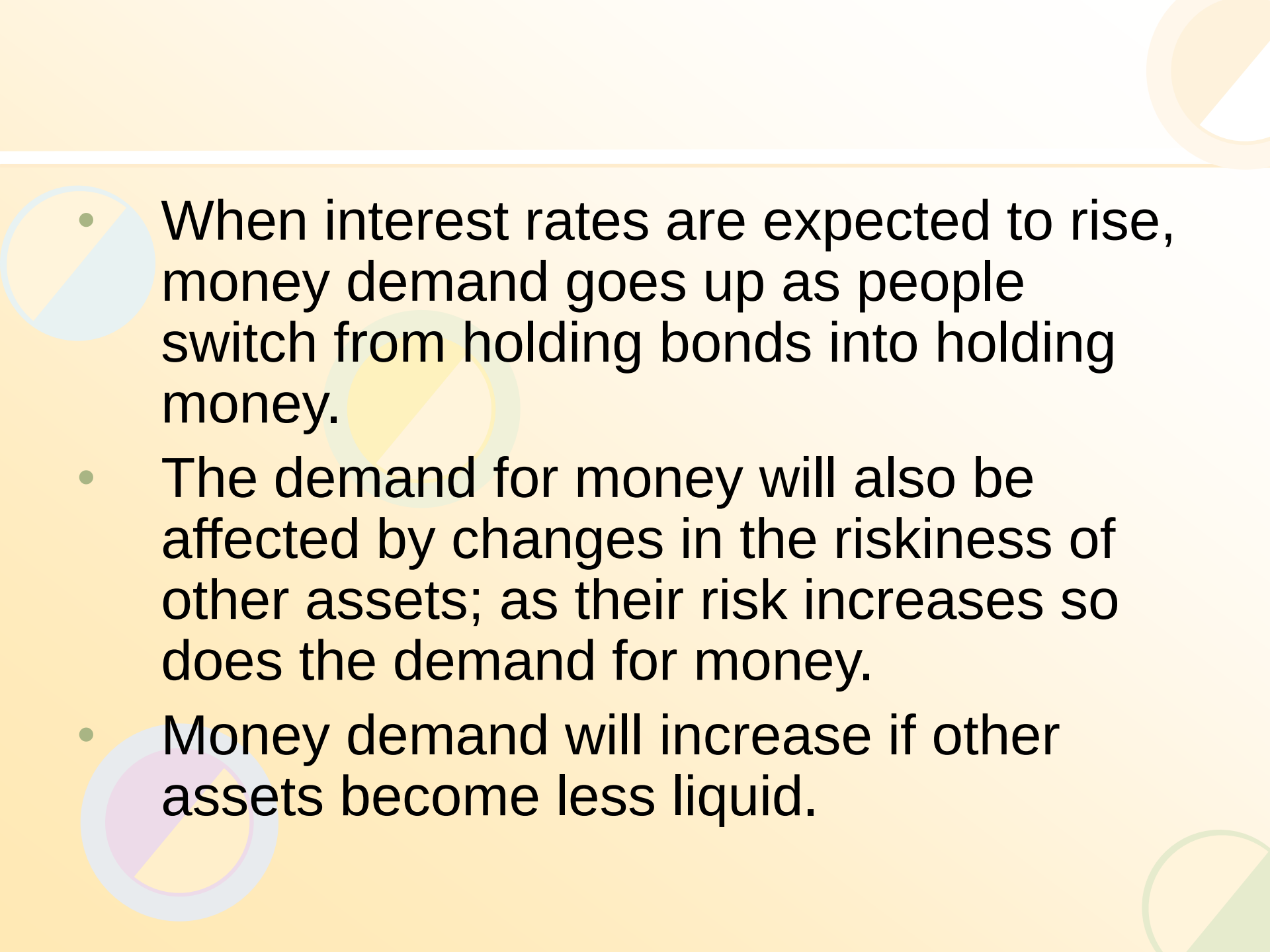
- 
- When interest rates are expected to rise, money demand goes up as people switch from holding bonds into holding money.
 - The demand for money will also be affected by changes in the riskiness of other assets; as their risk increases so does the demand for money.
 - Money demand will increase if other assets become less liquid.

Table 20.1 Determinants of Money Demand: Factors That Cause Individuals to Hold More Money

Transactions Demand for Money

Nominal income	The higher nominal income, the higher the demand for money.
Interest rates	The lower interest rates, the higher the demand for money.
Availability of alternative means of payment	The less available alternatives means of payment, the higher the demand for money.

Portfolio Demand for Money

Wealth	As wealth rises, the demand for money goes up.
Return relative to alternatives	As the return on alternatives falls, the demand for money goes up.
Expected future interest rates	As expected future interest rates rise, the demand for money goes up.
Risk relative to alternatives	As the riskiness of alternatives rises, the demand for money goes up.
Liquidity relative to alternatives	As the liquidity of alternatives falls, the demand for money goes up.

Targeting Money Growth in a Low-Inflation Environment

- In the long run, inflation is tied to money growth.
- In a high-inflation environment moderate variations in the growth of velocity are a mere annoyance.
- the only solution to inflation in a high inflation environment is to reduce money growth.

Targeting Money Growth in a Low-Inflation Environment

In a low-inflation environment, the ability to use money growth as a policy guide depends on the stability of the velocity of money.

Targeting Money Growth in a Low-Inflation Environment

Two criteria for the use of money growth as a direct monetary policy target:

- A stable link between the monetary base and the quantity of money
- A predictable relationship between the quantity of money and inflation

Targeting Money Growth in a Low-Inflation Environment

These allow policymakers to

- predict the impact of changes in the central bank's balance sheet on the quantity of money
- translate changes in money growth into changes in inflation.

Output and Inflation in the Long Run

• Potential Output

- Potential output is what the economy is capable of producing when its resources are used at normal rates.
- Potential output is not a fixed level, because the amount of labor and capital in an economy can grow, and improved technology can increase the efficiency of the production process
- Unexpected events can push current output away from potential output, creating an output gap
- In the long run, current output equals potential output.

Output and Inflation in the Long Run

- Long-Run Inflation

- In the long run, since current output equals potential output, real growth must equal growth in potential output.
- Ignoring changes in velocity, in the long run, inflation equals money growth minus growth in potential output.
- Though central banks focus on controlling short term nominal interest rates, they keep an eye on money growth
 - When they try to adjust level of reserves in banking system to maintain interest rate, it affects money growth. Which in turn determines inflation

Money Growth, Inflation, and Aggregate Demand

- Aggregate demand tells us how spending (demand) by households, firms, the government, and foreigners changes as inflation goes up and down.
- The level of aggregate demand is tied to monetary policy through the equation of exchange ($MV=PY$) because the amount of money in the economy limits the ability to make payments .

Money Growth, Inflation, and Aggregate Demand

- Rearranging the equation of exchange

$$Y^{ad} = \frac{MV}{P}$$

- where Y^{ad} = aggregate demand,
- M = the quantity of money,
- V = the velocity of money, and
- P = the price level.
- From this expression it is clear that an increase in the price level reduces the purchasing power of money, which means less purchases are made, pushing down aggregate demand

Money Growth, Inflation, and Aggregate Demand

Inflation



**Money Growth
Unchanged
and less than
inflation**



$$\frac{M}{P}$$



**Velocity
Unchanged**

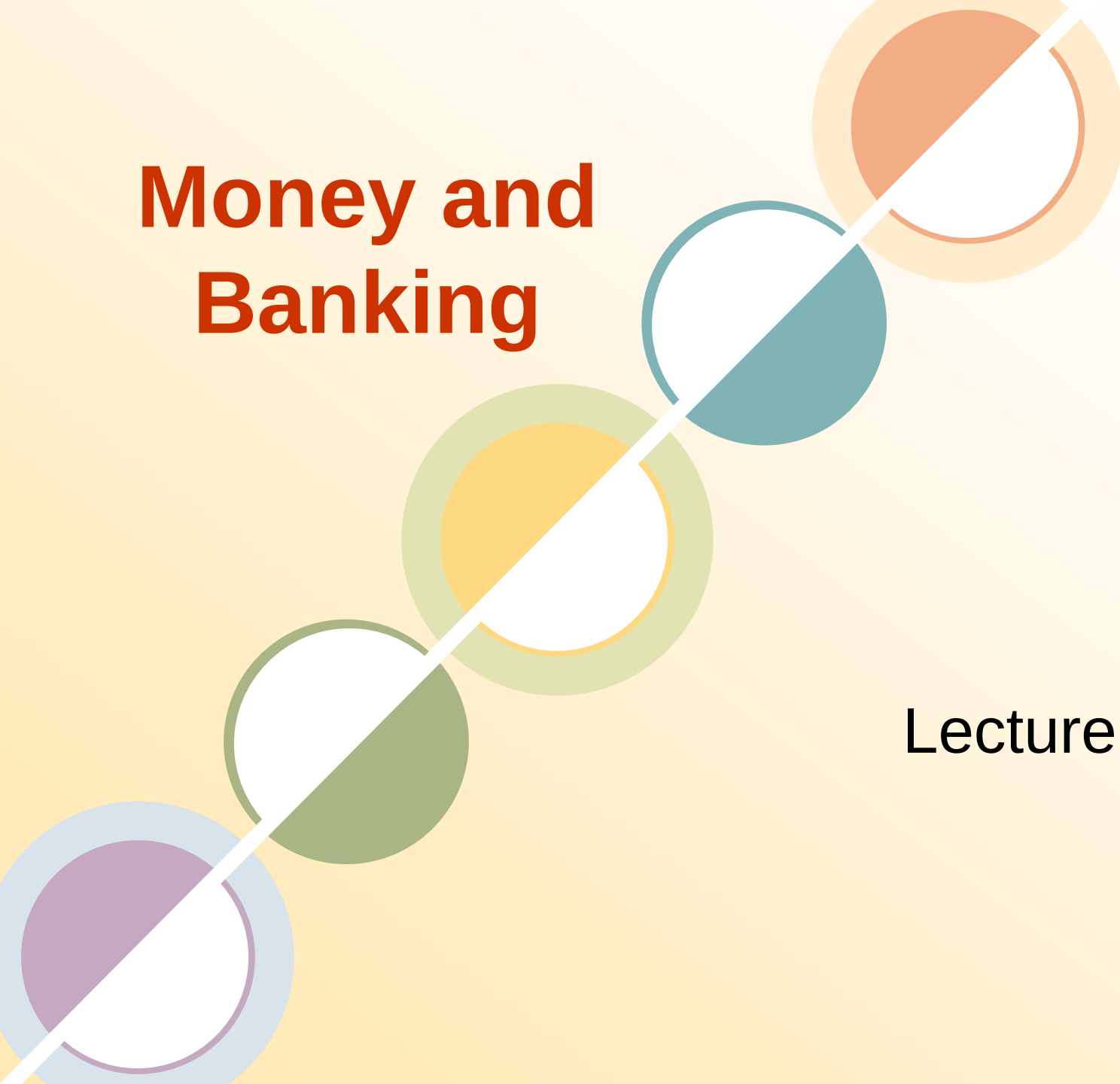
**Aggregate
Demand**

Y^{ad}



Money and Banking

Lecture 40



Review of the Previous Lecture

- Monetary Aggregates
- Equation of Exchange
- Quantity Theory of Money
- Demand for Money
- Targeting Money growth in Low Inflation Environment
- Output and inflation in the Long run

Money Growth, Inflation, and Aggregate Demand

- Rearranging the equation of exchange

$$Y^{ad} = \frac{MV}{P}$$

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Money Growth, Inflation, and Aggregate Demand

Inflation



**Money Growth
Unchanged
and less than
inflation**



$$\frac{M}{P}$$



**Velocity
Unchanged**

**Aggregate
Demand**

Y^{ad}



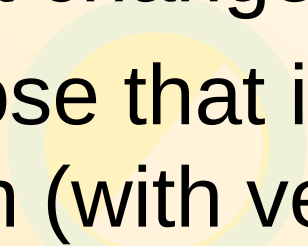
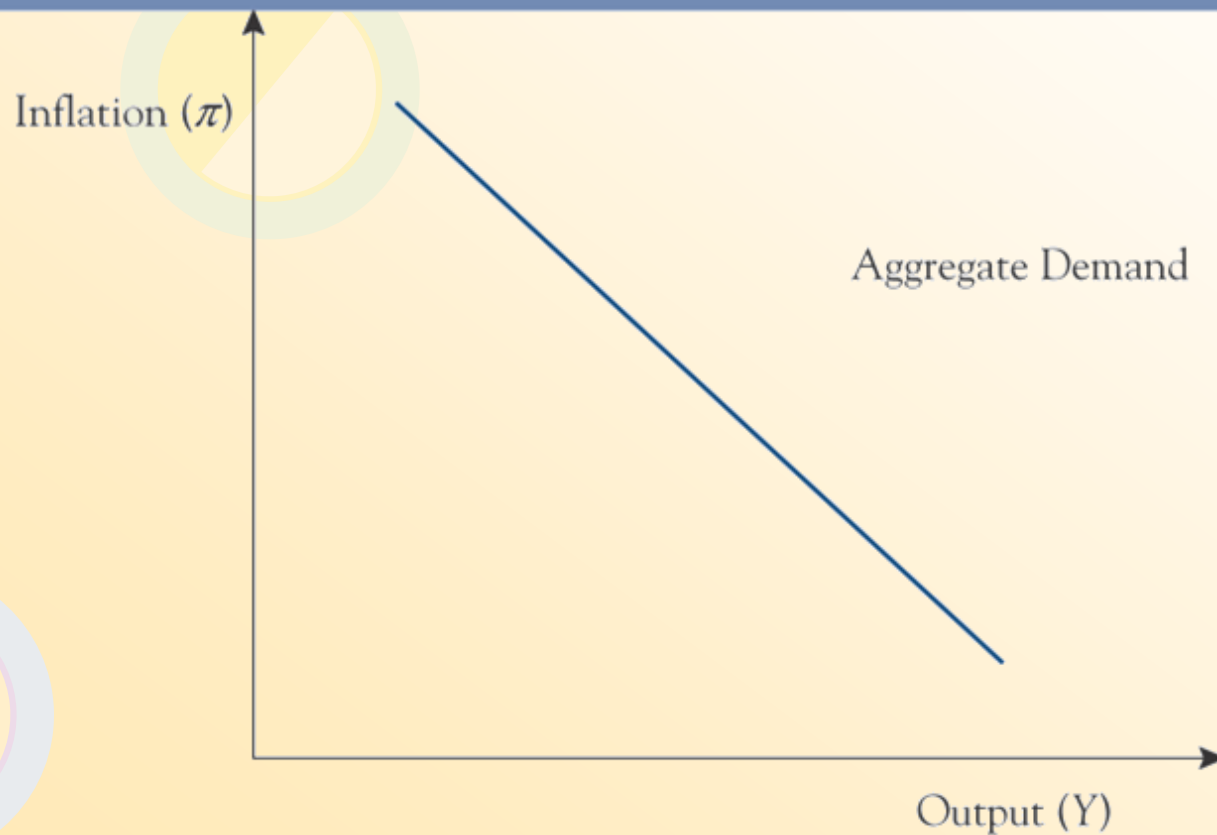
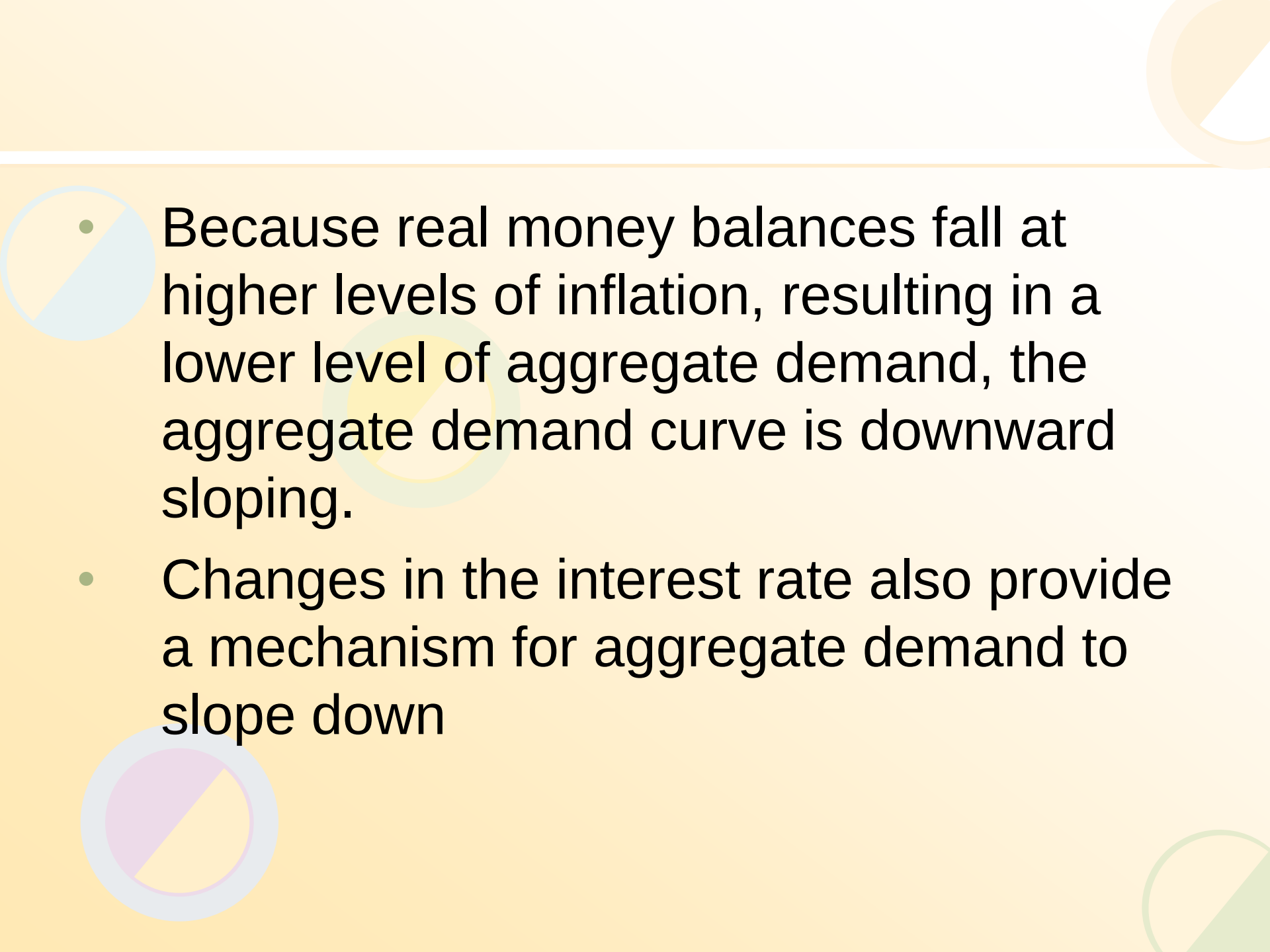
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- To shift the focus to inflation, we need to look at changes in the price level.
 - Suppose that inflation exceeds money growth (with velocity held constant). Real money balances will fall and so will aggregate demand

Figure 21.1 The Aggregate Demand Curve

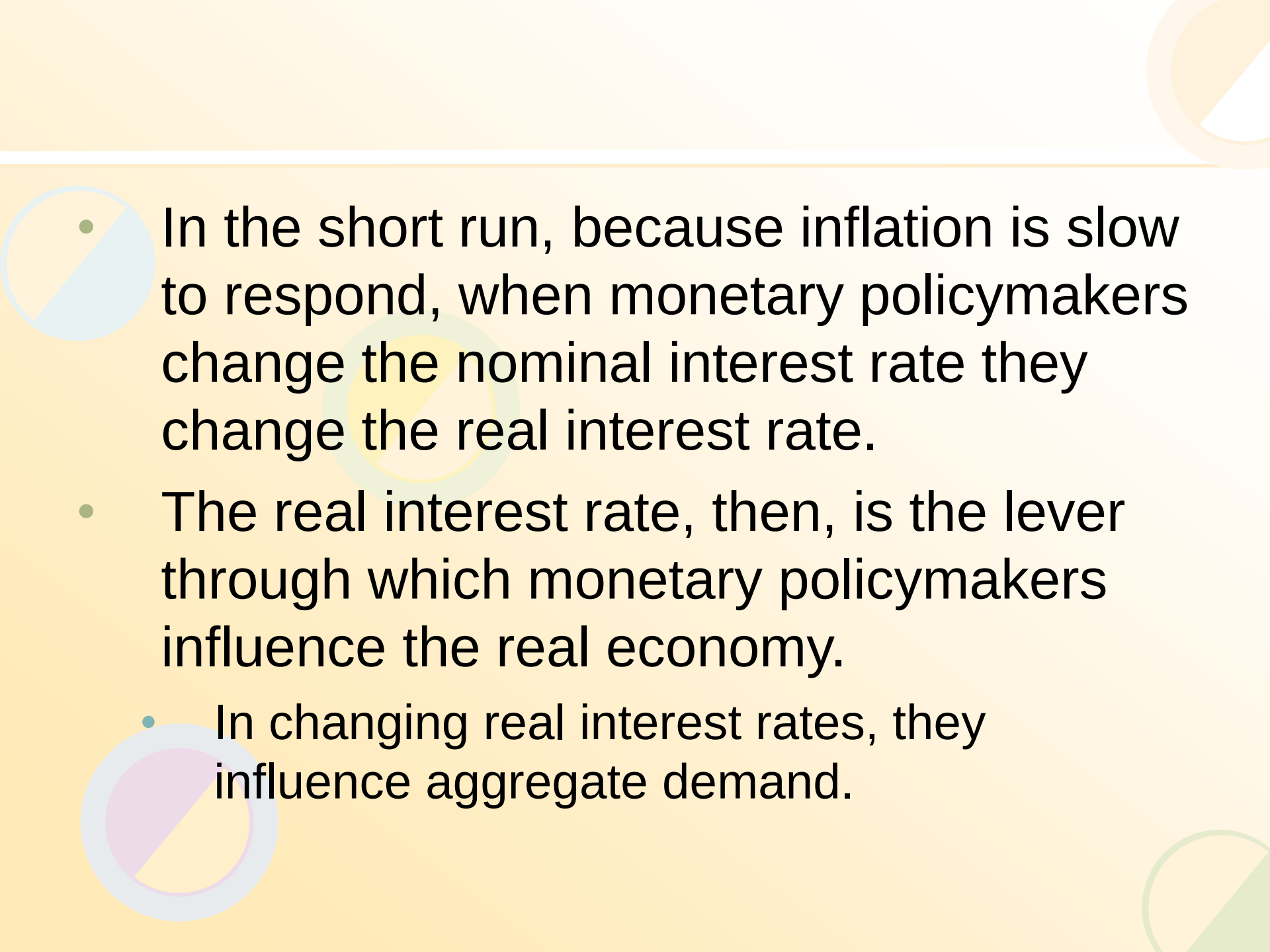
At higher levels of inflation, real money balances fall, resulting in a lower level of aggregate demand.



- 
- Because real money balances fall at higher levels of inflation, resulting in a lower level of aggregate demand, the aggregate demand curve is downward sloping.
 - Changes in the interest rate also provide a mechanism for aggregate demand to slope down

Monetary Policy and the Real Interest Rate

- Central bankers control short-term nominal interest rates by controlling the market for reserves.
- But the economic decisions of households and firms depend on the real interest rate;
 - To alter the course of the economy, central banks must influence the real interest rate as well

- 
- In the short run, because inflation is slow to respond, when monetary policymakers change the nominal interest rate they change the real interest rate.
 - The real interest rate, then, is the lever through which monetary policymakers influence the real economy.
 - In changing real interest rates, they influence aggregate demand.

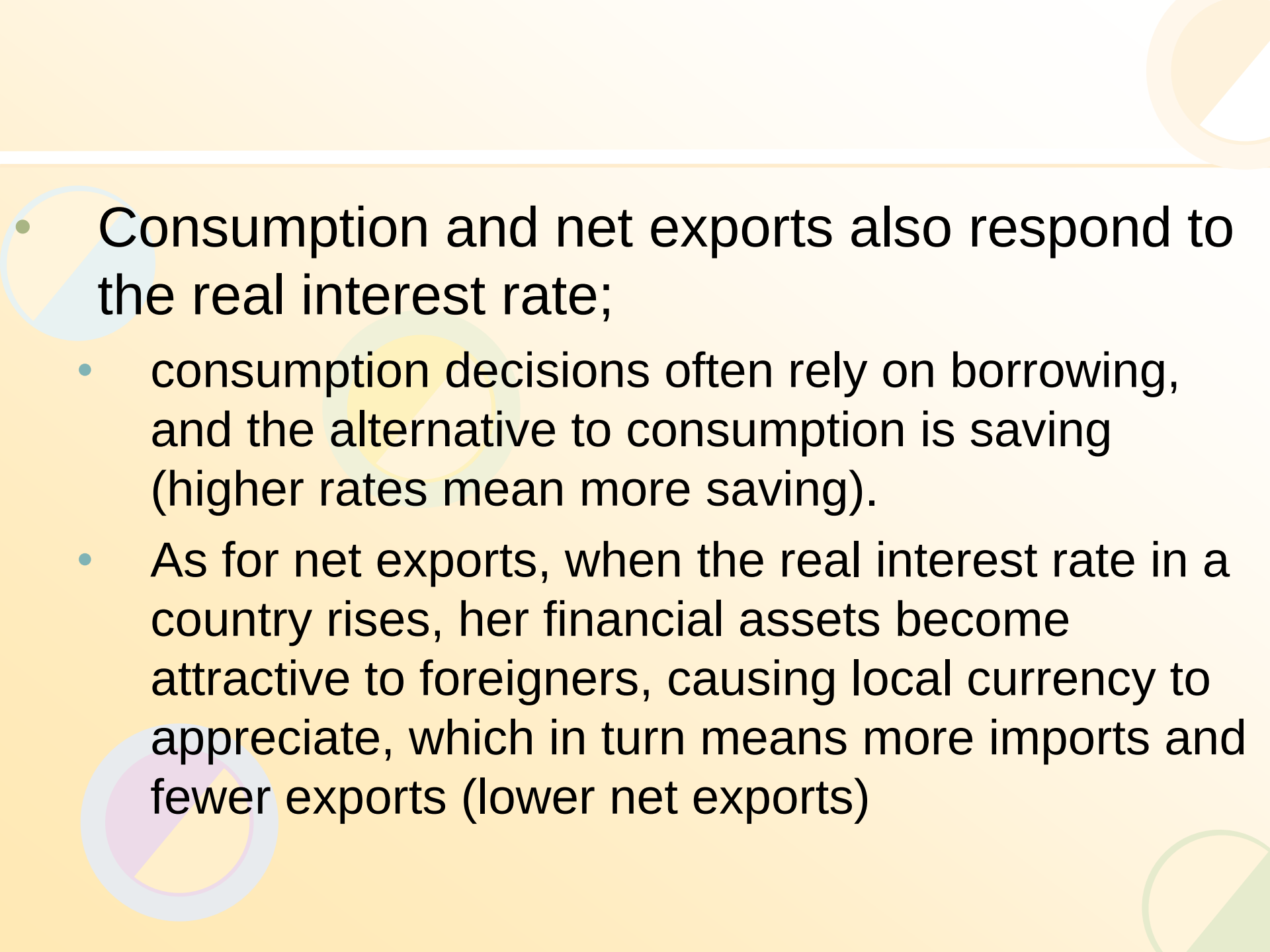
Aggregate Demand and the Real Interest Rate

- Aggregate demand is divided into four components:
 - consumption,
 - investment,
 - government purchases,
 - net exports

Aggregate Demand = Consumption + Investment + Govt.'s Purchases + Net Exports

$$Y^{ad} = C + I + G + NX$$

- It is helpful to think of aggregate demand as having two parts, one that is sensitive to real interest rate changes and one that is not
- Investment is the most important of the components of aggregate demand that are sensitive to changes in the real interest rate.
 - An investment can be profitable only if its internal rate of return exceeds the cost of borrowing



- Consumption and net exports also respond to the real interest rate;

- consumption decisions often rely on borrowing, and the alternative to consumption is saving (higher rates mean more saving).
- As for net exports, when the real interest rate in a country rises, her financial assets become attractive to foreigners, causing local currency to appreciate, which in turn means more imports and fewer exports (lower net exports)

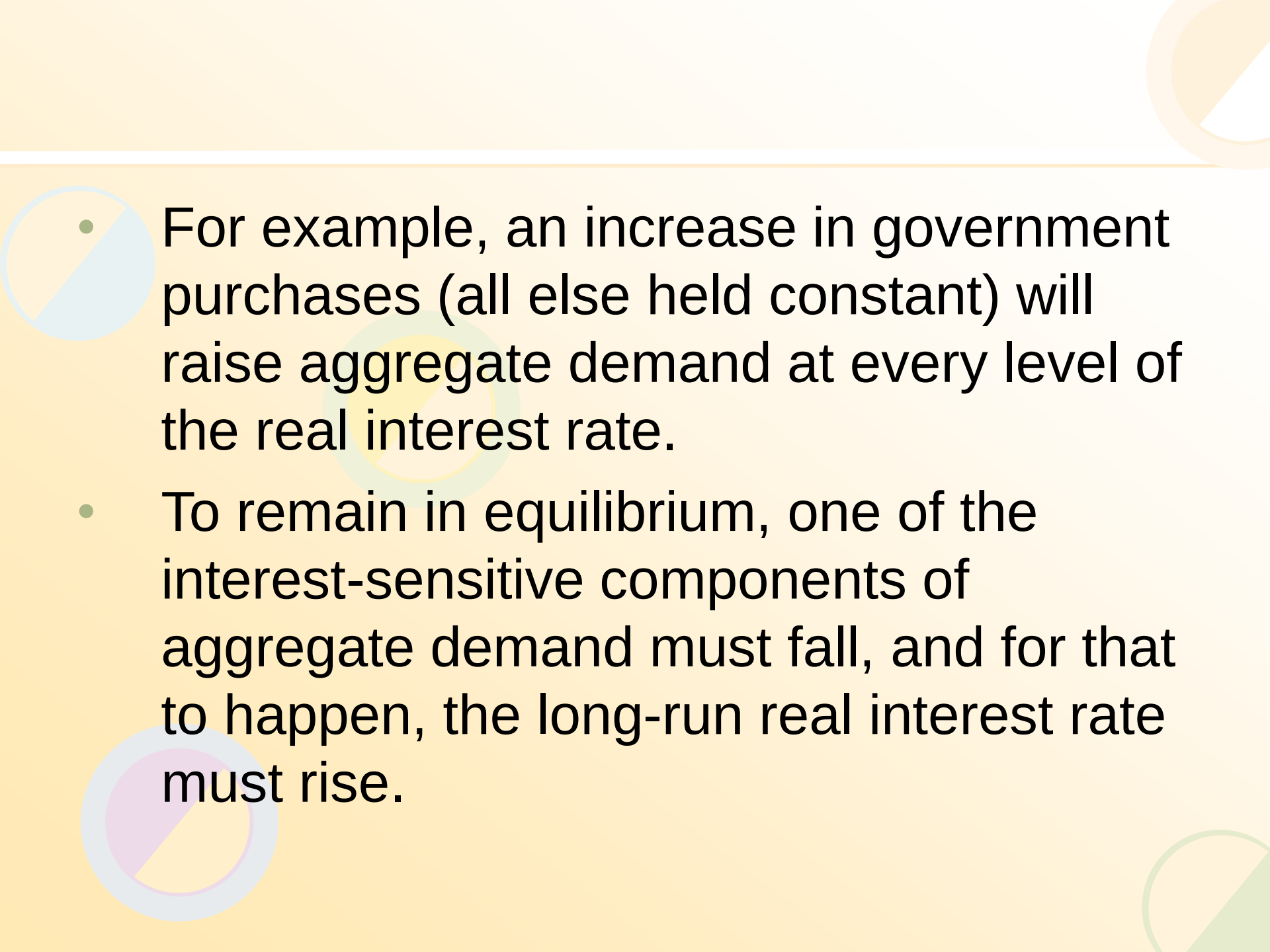
- While changes in real interest rate may have an impact on the government's budget by raising the cost of borrowing, the effect is likely to be small and ignorable.
- Thus, considering consumption, investment, and net exports, an increase in the real interest rate reduces aggregate demand (the effect on the 4th component, government spending, is small enough to be ignored).

Table 21.1 Impact of Rise in the Real Interest Rate on Components of Aggregate Demand

Component of Aggregate Demand	Effect of a Rise in the Real Interest Rate	Impact on Component of Aggregate Demand
Consumption (<i>C</i>)	Reward to saving rises	Consumption <i>falls</i>
Investment (<i>I</i>)	Cost of financing rises	Investment <i>falls</i>
Net Exports (<i>NX</i>)	Demand for domestic assets rises, causing a currency appreciation, raising the price of exports and reducing the cost of imports	Exports fall; imports Rise; net exports <i>fall</i>
Aggregate Demand (Y^{ad})	<i>C</i> , <i>I</i> , and <i>NX</i> all fall	Aggregate demand falls

The Long-Run Real Interest Rate

- There must be some level of the real interest rate at which aggregate demand equals potential output; this is the long-run real interest rate.
- The long-run real interest rate equates aggregate demand with potential output.
- The rate will change if a component of aggregate demand that is not sensitive to the real interest rate goes up (or down) or if potential output changes.

- 
- For example, an increase in government purchases (all else held constant) will raise aggregate demand at every level of the real interest rate.
 - To remain in equilibrium, one of the interest-sensitive components of aggregate demand must fall, and for that to happen, the long-run real interest rate must rise.

- The same would be true for increases in other components of aggregate demand that are not interest sensitive.
- A change in potential output has an inverse effect on the long-run real interest rate;
 - when potential output rises, aggregate demand must rise with it, which requires a decrease in the real interest rate

Inflation, the Real Interest Rate, and the Monetary Policy Reaction Curve

- Policymakers set their short-run nominal interest rate targets in response to economic conditions in general and inflation in particular.
- When current inflation is high or current output is running above potential output, central bankers will raise nominal interest rates; when current inflation is low or current output is well below potential, they will lower interest rates

The Monetary Policy Reaction Curve

- While they state their policies in terms of nominal rates they do so knowing that changes in the nominal interest rate will eventually translate into changes in the real interest rate, and it is those changes that influence the economic decisions of firms and households
- Experts agree that any (coherent) monetary policy can be written as an inflation target plus a response to supply shocks

Money and Banking

Lecture 41

Review of the Previous Lecture

- Output and Inflation in the Long Run
- Money growth, Inflation and Aggregate Demand
 - Monetary Policy and Real Interest Rate
 - Aggregate Demand and Real Interest Rate
- Long Run Real Interest Rate

Deriving the Monetary Policy Reaction Curve

- To ensure that deviations of inflation from the target are only temporary, monetary policymakers respond to change in inflation by changing the real interest rate in the same direction.
- The monetary policy reaction curve is set so that when current inflation equals target inflation, the real interest rate equals the long-run real interest rate.

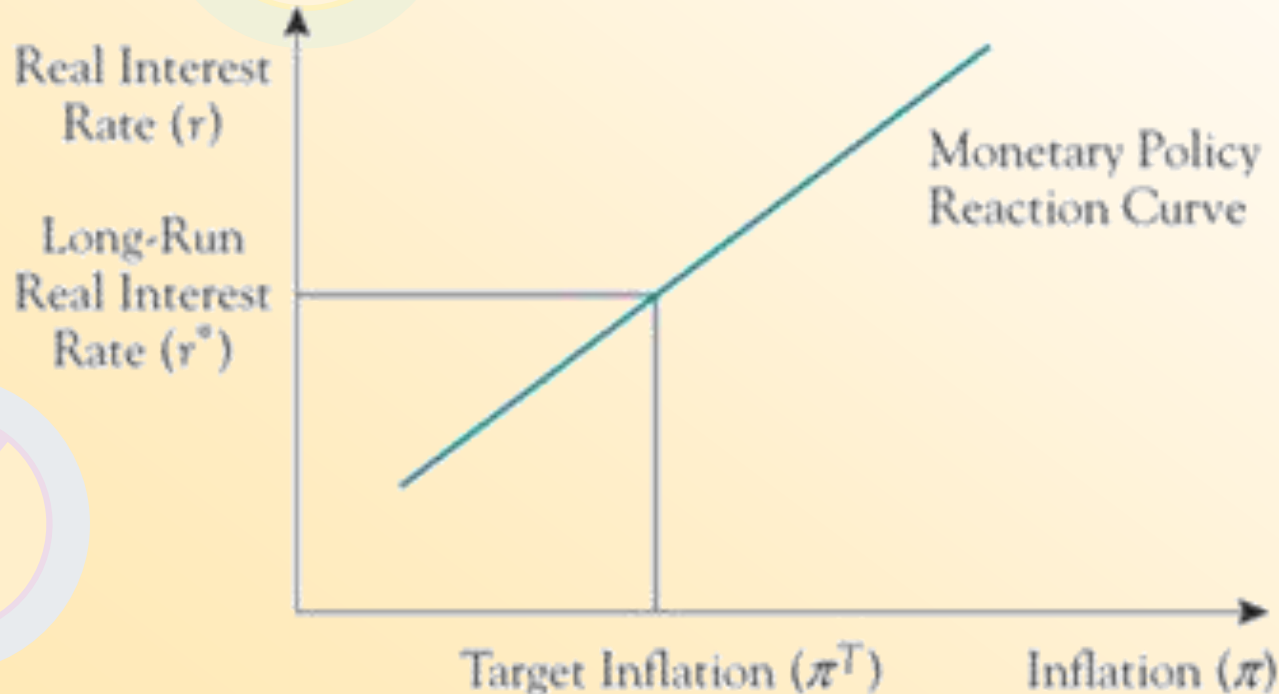
- The slope of the curve depends on policymakers' objectives;
 - when central bankers decide how aggressively to pursue their inflation target, and how willing they are to tolerate temporary changes in inflation, they determine the slope of the curve

The Monetary Policy Reaction Curve

Figure 21.4 Monetary Policy

A. The Monetary Policy Reaction Curve

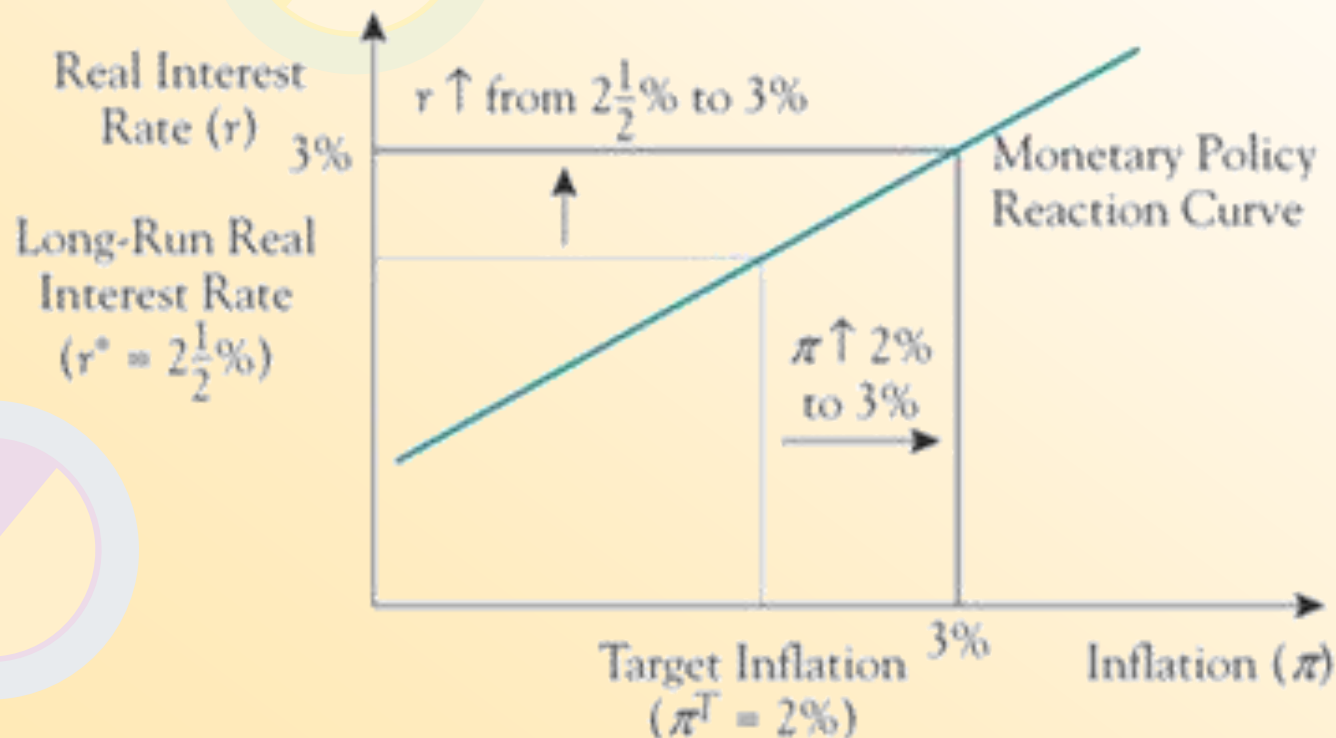
Monetary policymakers react to changes in current inflation by changing the real interest rate. Increases in current inflation lead them to raise the real interest rate, while decreases lead them to lower it. The monetary policy reaction curve is located so that the central bank's target inflation is consistent with the long-run real interest rate, which equates aggregate demand with potential output.



The Monetary Policy Reaction Curve

B. Movements Along the Monetary Policy Reaction Curve

The long-run real interest rate in an economy is roughly 2.5 percent, and the Central bank's implicit inflation target is approximately 2 percent. The monetary policy reaction curve implies that a one-percentage-point increase in inflation calls for a half-percentage-point increase in the real interest rate—a movement along the monetary policy reaction curve. That means that an increase in inflation from 2 to 3 percent calls for an increase in the real interest rate from 2.5 to 3 percent.



Shifting the Monetary Policy Reaction Curve

- Policymakers who are aggressive in keeping current inflation near target will have a steep curve, meaning that a small change in inflation will be met with a large change in the real interest rate
- A relatively flat curve means that central bankers are less concerned than they might be with keeping current inflation near target over the short term.

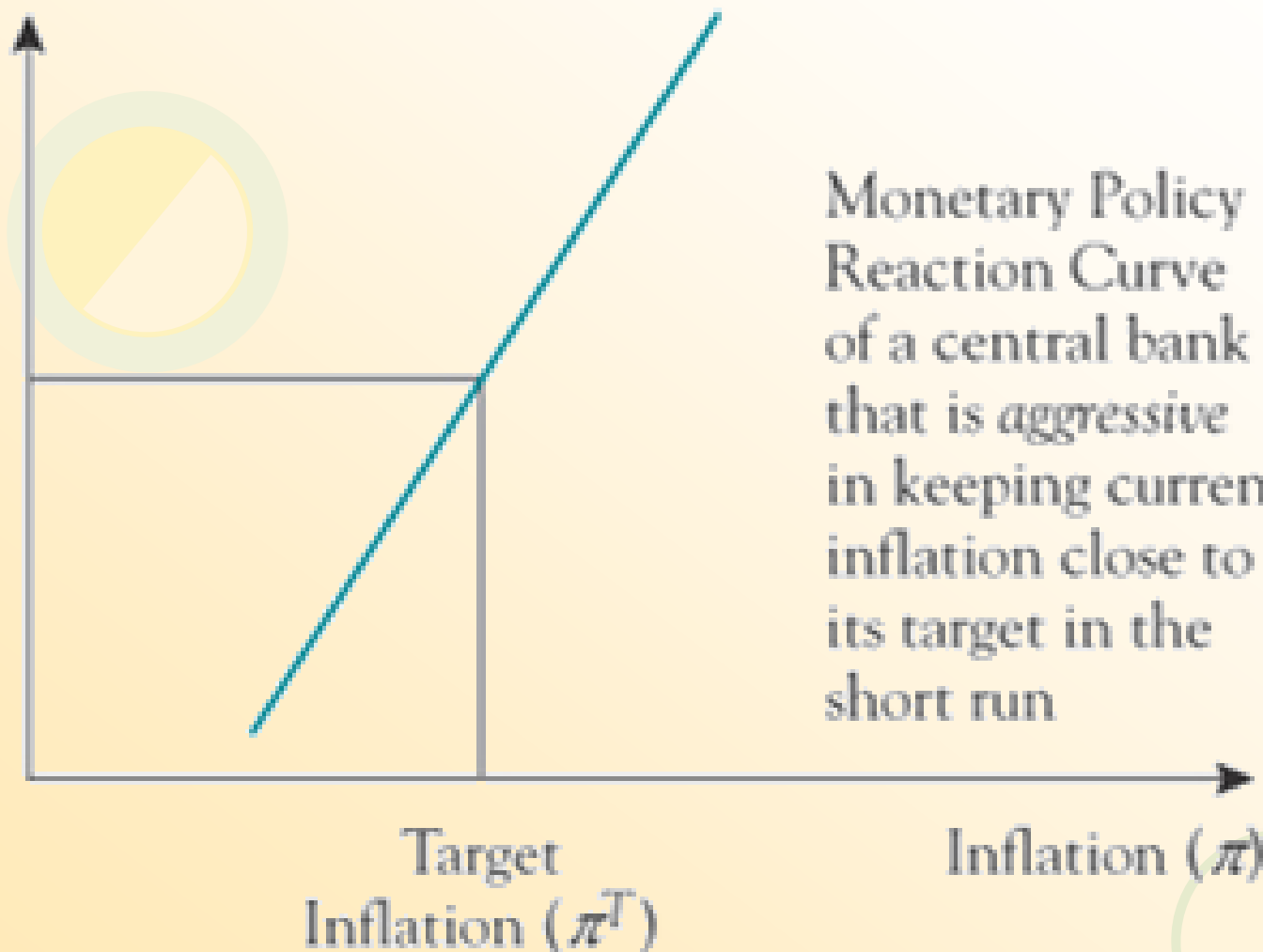
- The monetary policy reaction curve is set so that when current inflation equals target inflation, the real interest rate equals the long-run real interest rate.
 - $r = r^*$ when $\pi = \pi^T$.

- When policymakers adjust the real interest rate they are either moving along a fixed monetary policy reaction curve or shifting the curve.
- A movement along the curve is a reaction to a change in current inflation; a shift in the curve represents a change in the level of the real interest rate at every level of inflation

A.

Real Interest
Rate (r)

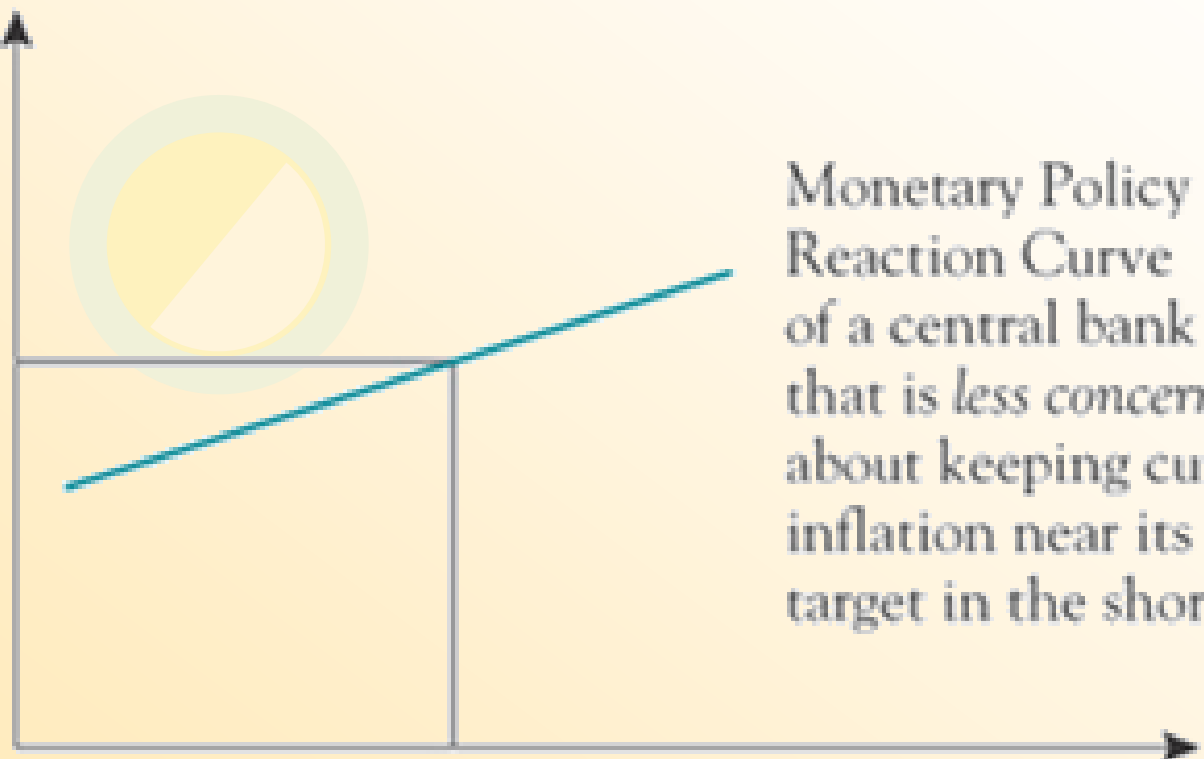
Long-Run
Real Interest
Rate (r^*)



B.

Real Interest
Rate (r)

Long-Run
Real Interest
Rate (r^*)



Monetary Policy
Reaction Curve
of a central bank
that is *less concerned*
about keeping current
inflation near its
target in the short run

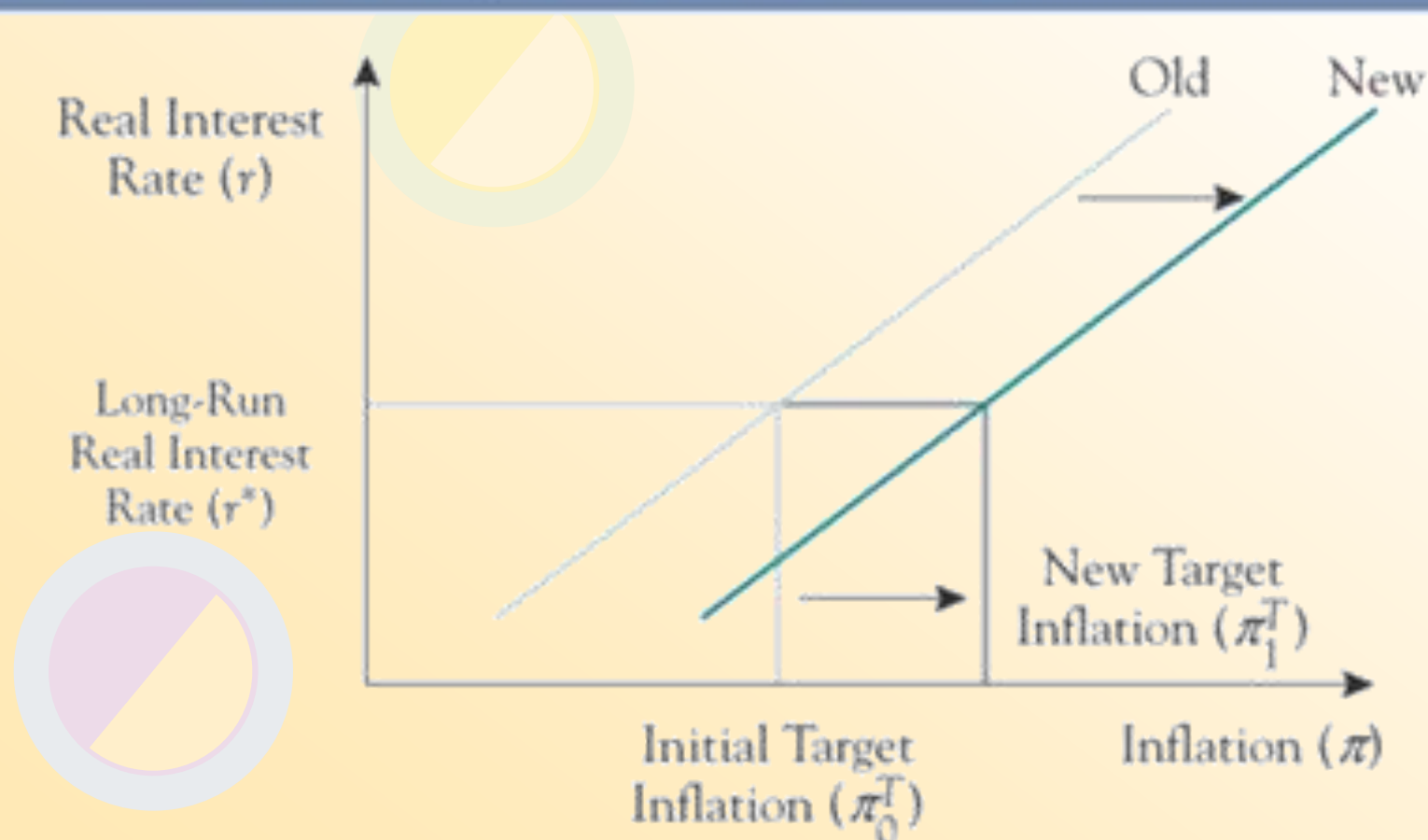
Target
Inflation (π^T)

Inflation (π)

- If either target inflation or the long-run real interest rate change, then the entire curve will shift
- With a higher inflation target, the central bank will set a lower current real interest rate at every level of current inflation, shifting the monetary policy reaction curve to the right (a reduction would have the opposite effect).

A. An Increase in the Central Bank's Inflation Target

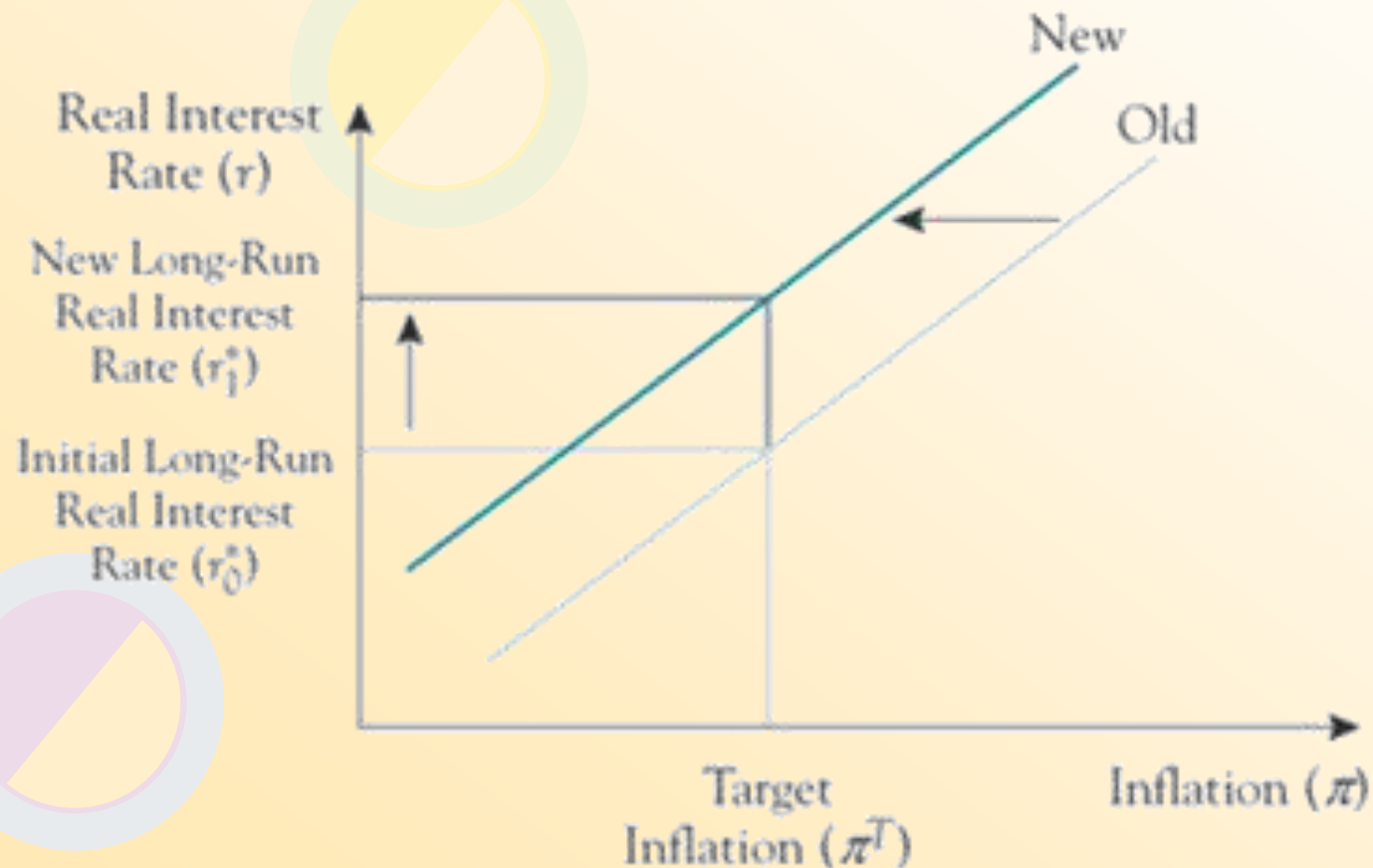
An increase in the central bank's inflation target from an initial level of π_0^T to the new level of π_1^T shifts the monetary policy reaction curve to the right, lowering the real interest rate at every level of current inflation.



- The long-run real interest rate is determined by the structure of the economy;
 - If it were to rise as a result of an increase in government purchases (or some other component of aggregate demand that is not sensitive to the real interest rate) then the monetary policy reaction curve would shift left

B. An Increase in the Long-Run Real Interest Rate

An increase in the economy's long-run real interest rate from r_0^* to r_1^* shifts the monetary policy reaction curve to the left, raising the real interest rate at every level of current inflation.



- Any shift in the monetary policy reaction curve can be characterized as either a change in target inflation or a shift in the long-run real interest rate

The Monetary Policy Reaction Curve

Table 21.2 The Monetary Policy Reaction Curve

What is it?	The relationship between current inflation and the real interest rate set by monetary policymakers.
What determines its location?	Drawn so that, when current inflation equals target inflation, policymakers will set the real interest rate equal to the long-run real interest rate.
What determines its slope?	Policymakers' attitude toward inflation. The more aggressive policymakers are in keeping current inflation close to target level, and the less tolerant they are of temporary changes in inflation, the steeper the slope.
When does it shift?	In response to changes in either the long-run real interest rate or the central bank's inflation target. An increase in the long-run real interest rate shifts the curve to the left. An increase in the inflation target shifts the curve to the right.



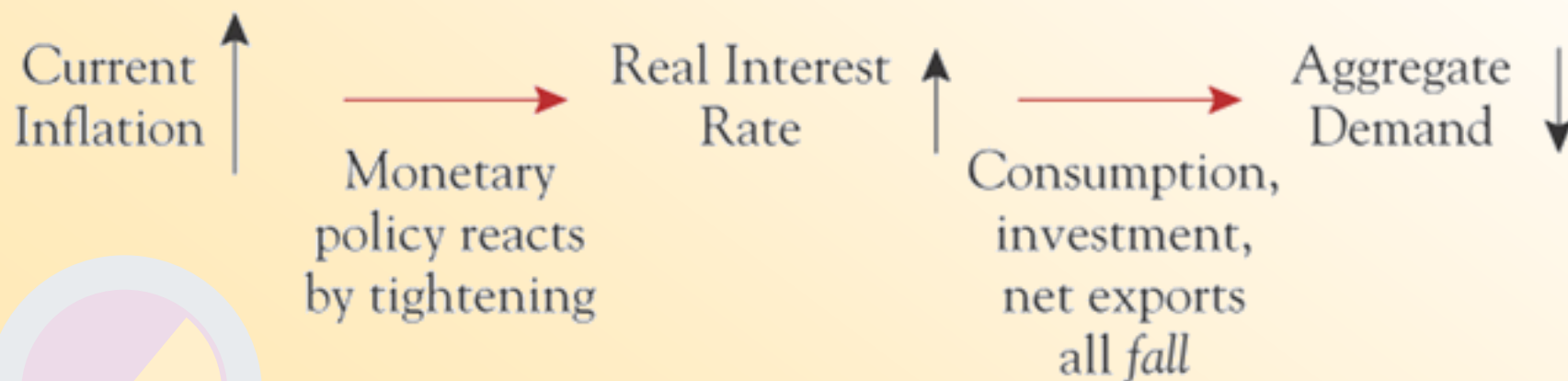
The Aggregate Demand Curve

- When current inflation rises
 - Monetary policymakers raise the real interest rate, moving upward along the monetary policy reaction curve
 - The higher real interest rate reduces consumption, investment, and net exports causing aggregate demand (output) to fall.

The Aggregate Demand Curve

Figure 21.7 The Link between Current Inflation and Aggregate Demand

When current inflation rises, policymakers react by raising the real interest rate, which reduces consumption, investment, and net exports. The result is a reduction in aggregate demand.

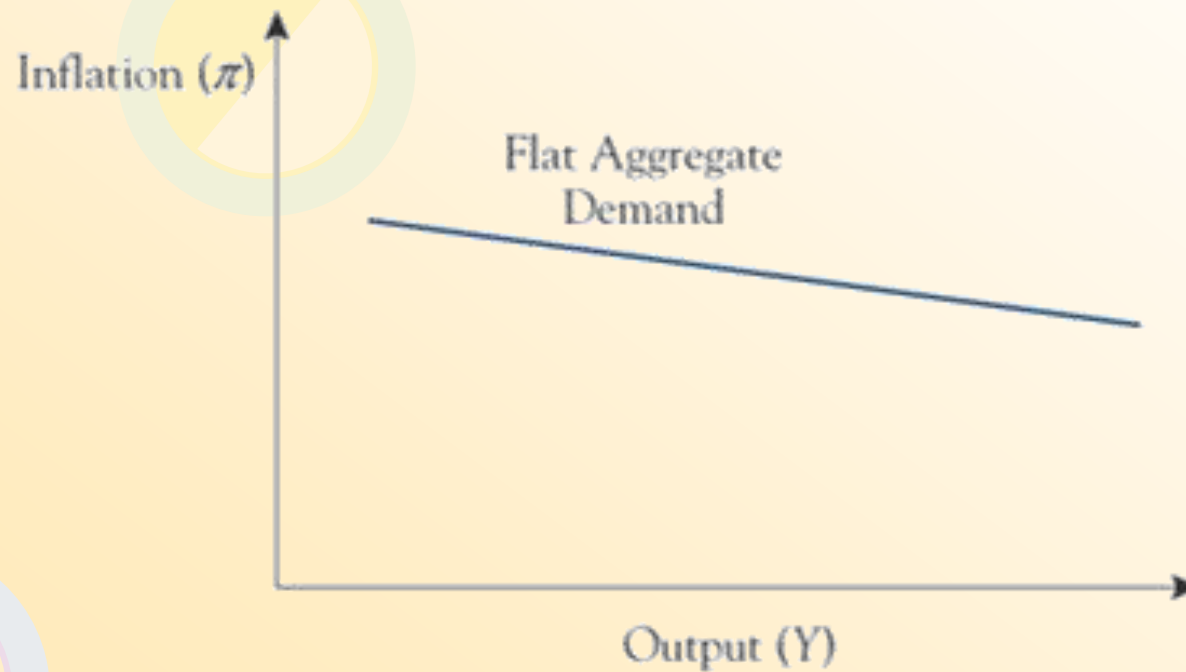


- Changes in current inflation move the economy along a downward-sloping aggregate demand curve
- This is in addition to the effect of higher inflation on real money balances noted earlier

- The slope of the aggregate demand curve tells us how sensitive current output is to a given change in current inflation.
- The aggregate demand curve will be relatively
 - flat if current output is very sensitive to inflation (a change in current inflation causes a large movement in current output)
 - steep if current output is not very sensitive to inflation

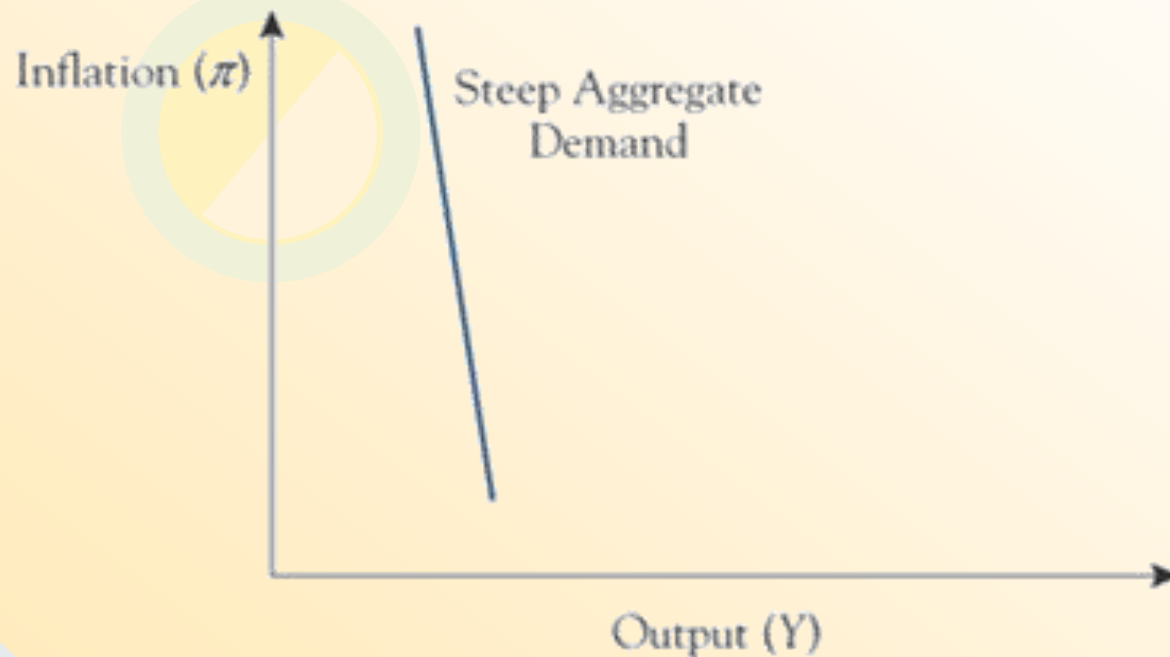
The Aggregate Demand Curve

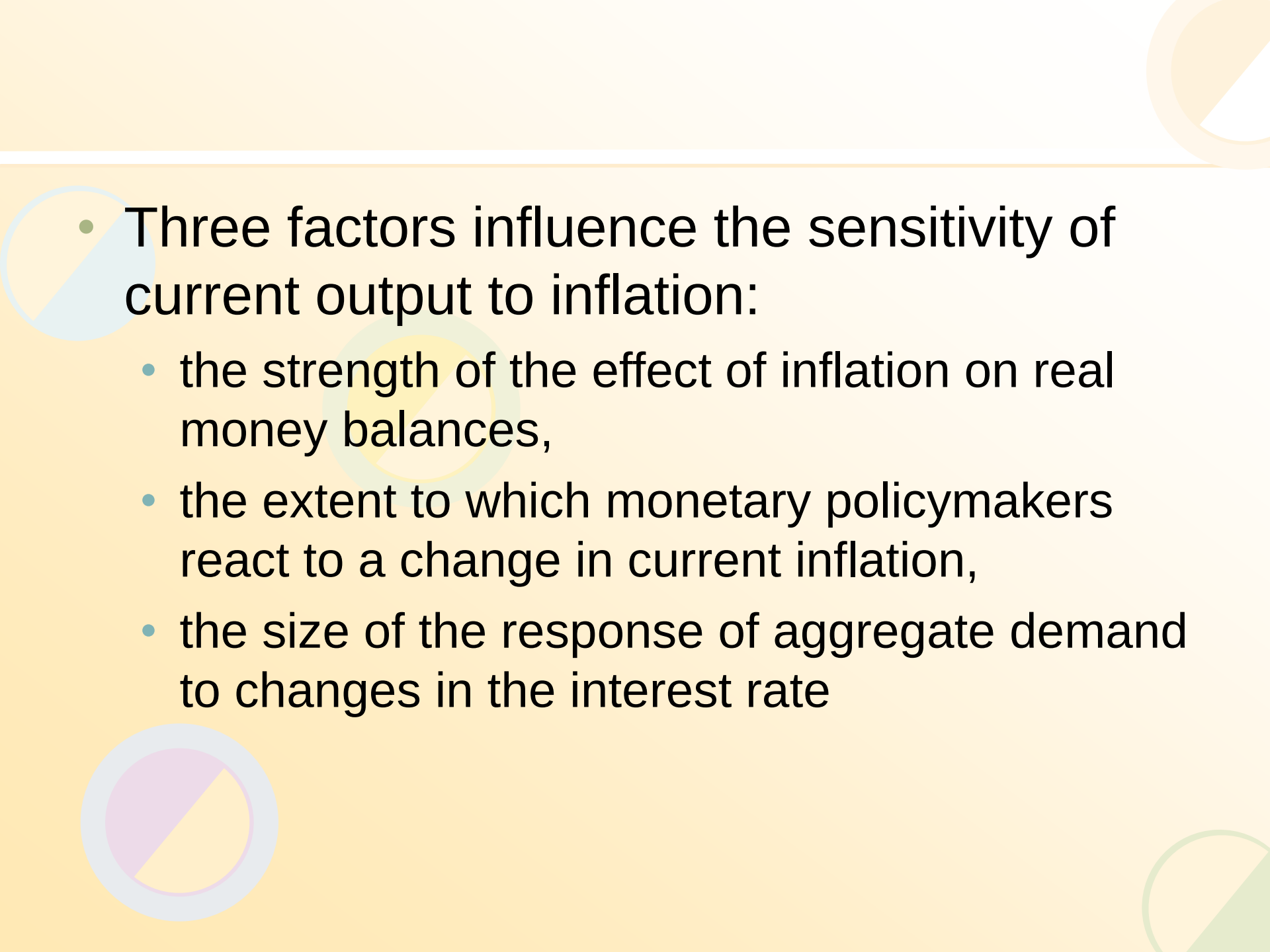
A. When the *monetary policy reaction curve is steep*, the central bank is aggressive in keeping current inflation near its target level, the *aggregate demand curve is flat*.



The Aggregate Demand Curve

B. When the *monetary policy reaction curve is flat*, the central bank is less concerned about keeping current inflation near its target level, the *aggregate demand curve is steep*.



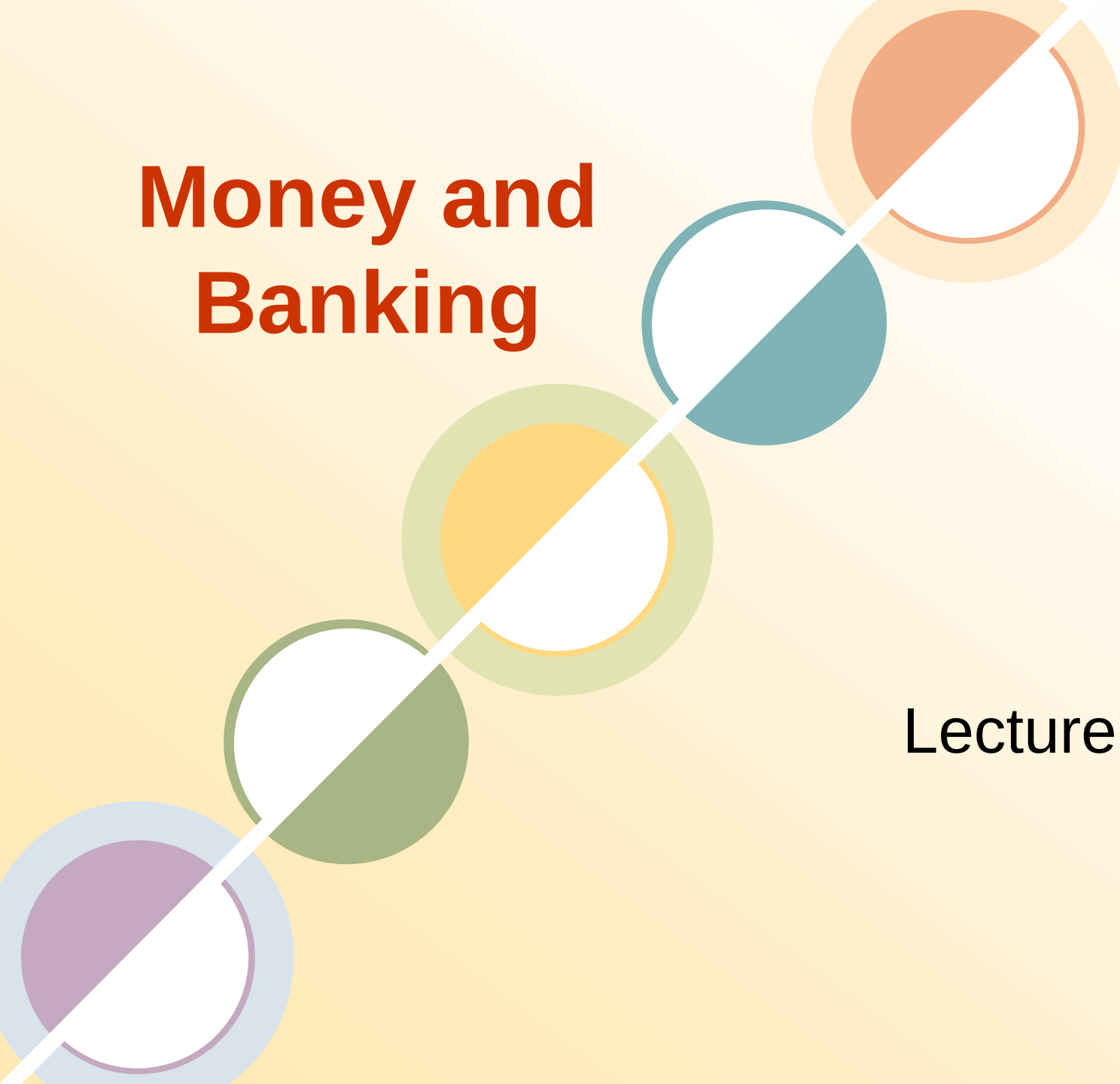
- 
- Three factors influence the sensitivity of current output to inflation:
 - the strength of the effect of inflation on real money balances,
 - the extent to which monetary policymakers react to a change in current inflation,
 - the size of the response of aggregate demand to changes in the interest rate

- The second factor relates to the slope of the monetary policy reaction curve
 - If policymakers react aggressively to a movement of current inflation away from its target level with a large change in the real interest rate, the monetary policy reaction curve will be steep and the aggregate demand curve is flat
 - If policymakers respond more cautiously, the monetary policy reaction curve is flat and the aggregate demand curve is steep

- The slope of the aggregate demand curve depends in part on the preferences of the central bank;
 - how aggressive policymakers are in responding to deviations of inflation from the target level

Money and Banking

Lecture 42



Review of the Previous Lecture

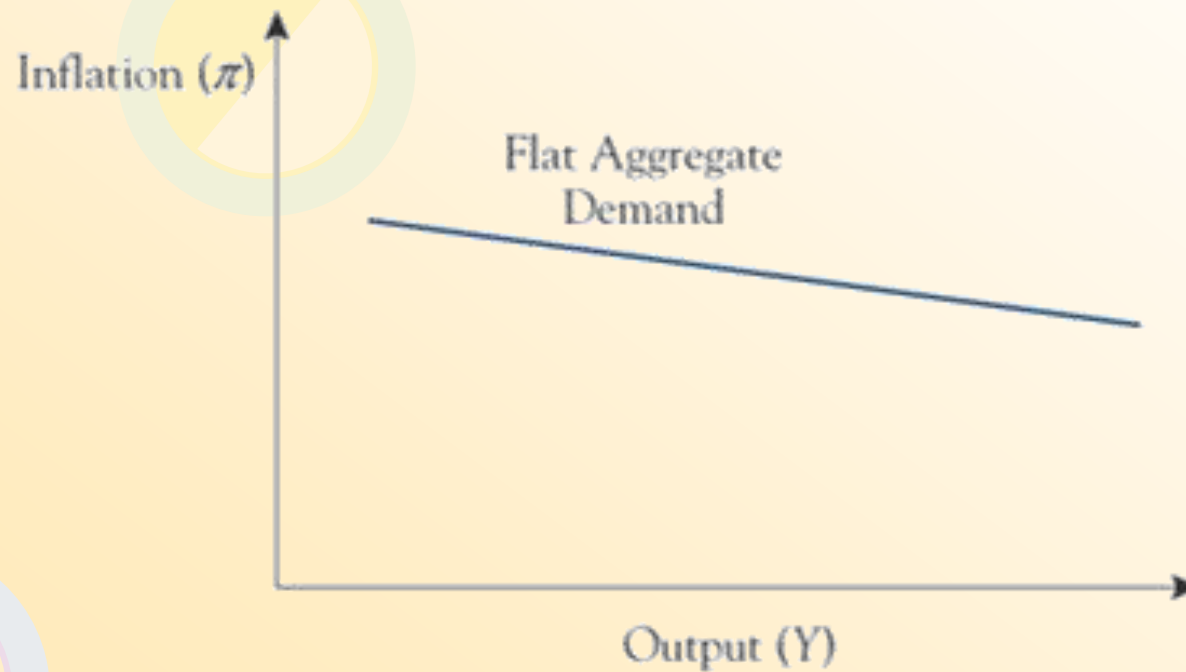
- Money growth, Inflation and Aggregate Demand
- Long Run Real Interest Rate
- Monetary Policy Reaction Curve
- Aggregate Demand Curve

The Aggregate Demand Curve

- The slope of the aggregate demand curve tells us how sensitive current output is to a given change in current inflation.
- The aggregate demand curve will be relatively
 - flat if current output is very sensitive to inflation (a change in current inflation causes a large movement in current output)
 - steep if current output is not very sensitive to inflation

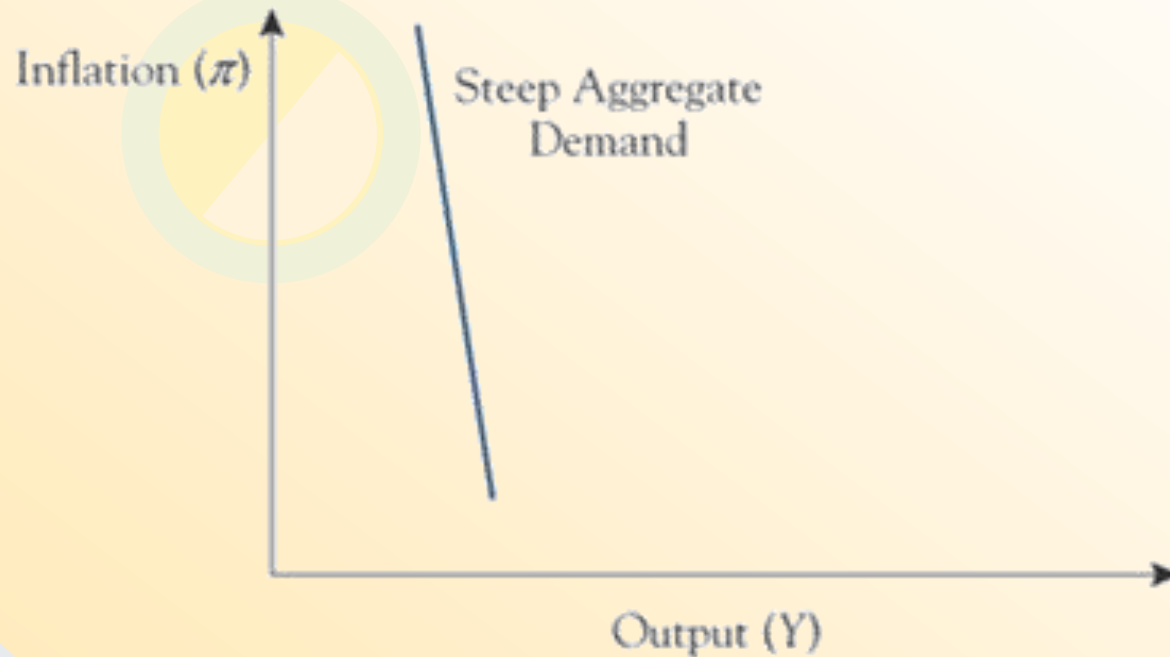
The Aggregate Demand Curve

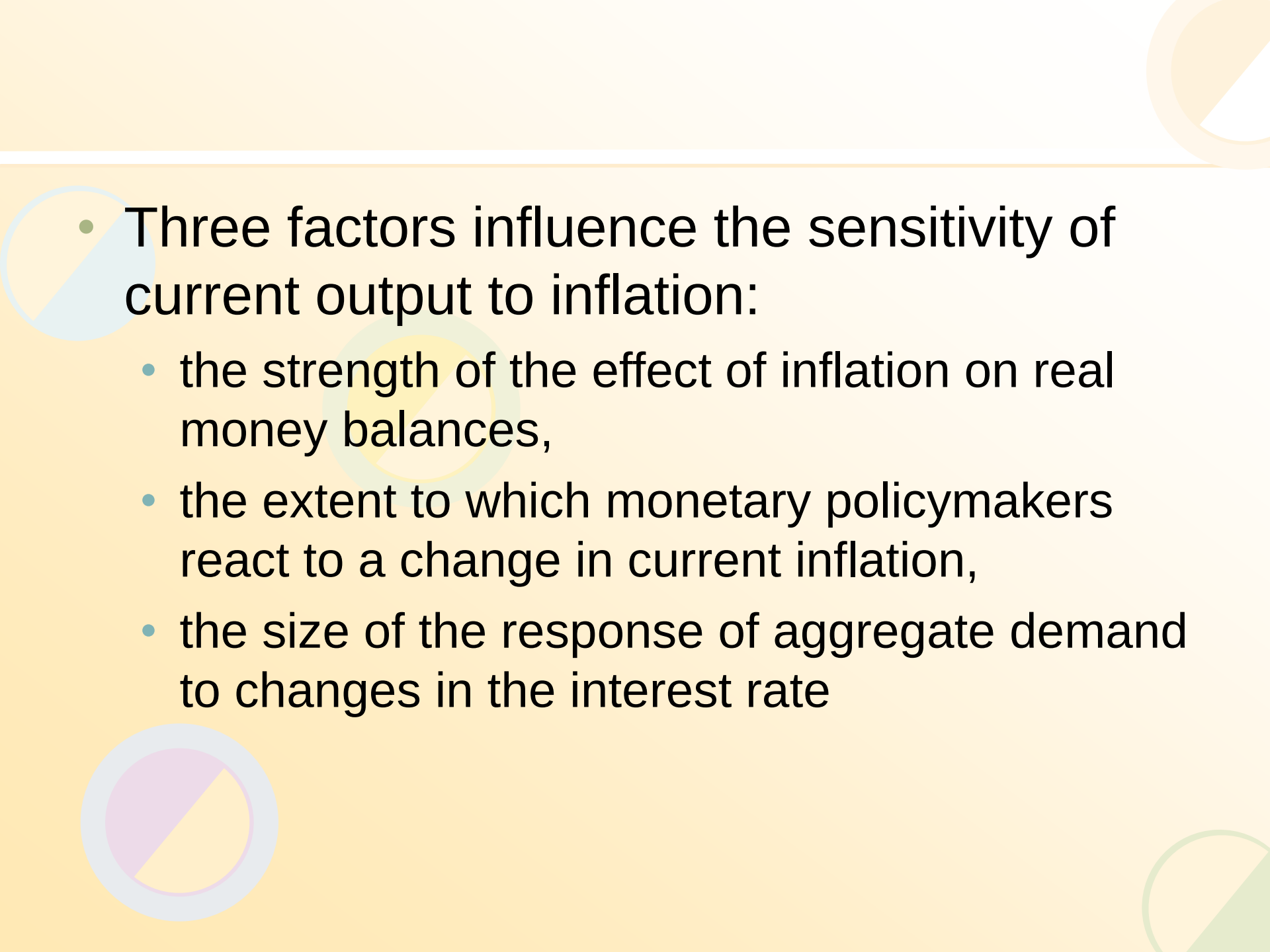
A. When the *monetary policy reaction curve is steep*, the central bank is aggressive in keeping current inflation near its target level, the *aggregate demand curve is flat*.



The Aggregate Demand Curve

B. When the *monetary policy reaction curve is flat*, the central bank is less concerned about keeping current inflation near its target level, the *aggregate demand curve is steep*.

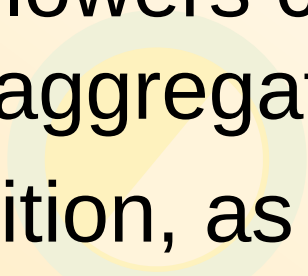


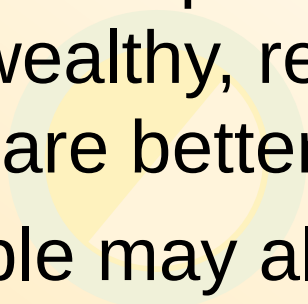
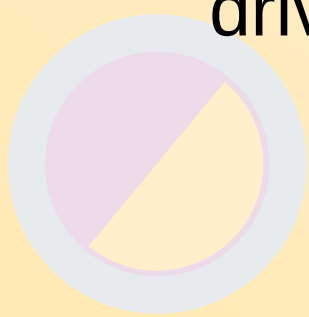
- 
- Three factors influence the sensitivity of current output to inflation:
 - the strength of the effect of inflation on real money balances,
 - the extent to which monetary policymakers react to a change in current inflation,
 - the size of the response of aggregate demand to changes in the interest rate

- The second factor relates to the slope of the monetary policy reaction curve
 - If policymakers react aggressively to a movement of current inflation away from its target level with a large change in the real interest rate, the monetary policy reaction curve will be steep and the aggregate demand curve is flat
 - If policymakers respond more cautiously, the monetary policy reaction curve is flat and the aggregate demand curve is steep

- The slope of the aggregate demand curve depends in part on the preferences of the central bank;
 - how aggressive policymakers are in responding to deviations of inflation from the target level

- Why the Aggregate Demand Curve Slopes Down?
- There are two reasons why the aggregate demand curve slopes down:
 - First, because higher inflation reduces real money balances (thus reducing purchases),
 - Second, because higher inflation induces policymakers to raise the real interest rate, depressing various components of aggregate demand

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- Rising inflation also reduces wealth, which lowers consumption and drives down aggregate demand.
 - In addition, as inflation rises the uncertainty about inflation rises, which makes equities a more risky investment and drops their value, also reducing wealth

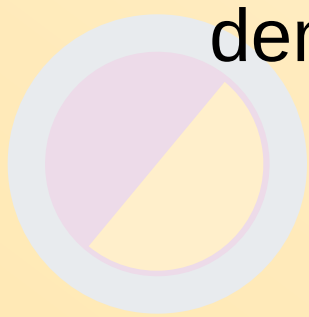
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- Another reason is that inflation can have a greater impact on the poor than it does on the wealthy, redistributing income to those who are better off
 - People may also save more as a result of the increased risk associated with inflation.
 - Also, rising inflation makes foreign goods cheaper in relation to domestic goods, driving imports up and net exports down.

Shifting the Aggregate Demand Curve

- In our derivation of the aggregate demand curve, we held constant both the location of the monetary policy reaction curve and those components of aggregate demand that do not respond to the real interest rate.
- Changes in any of those components, as well as changes in the location of the monetary policy reaction curve, will shift the aggregate demand curve.



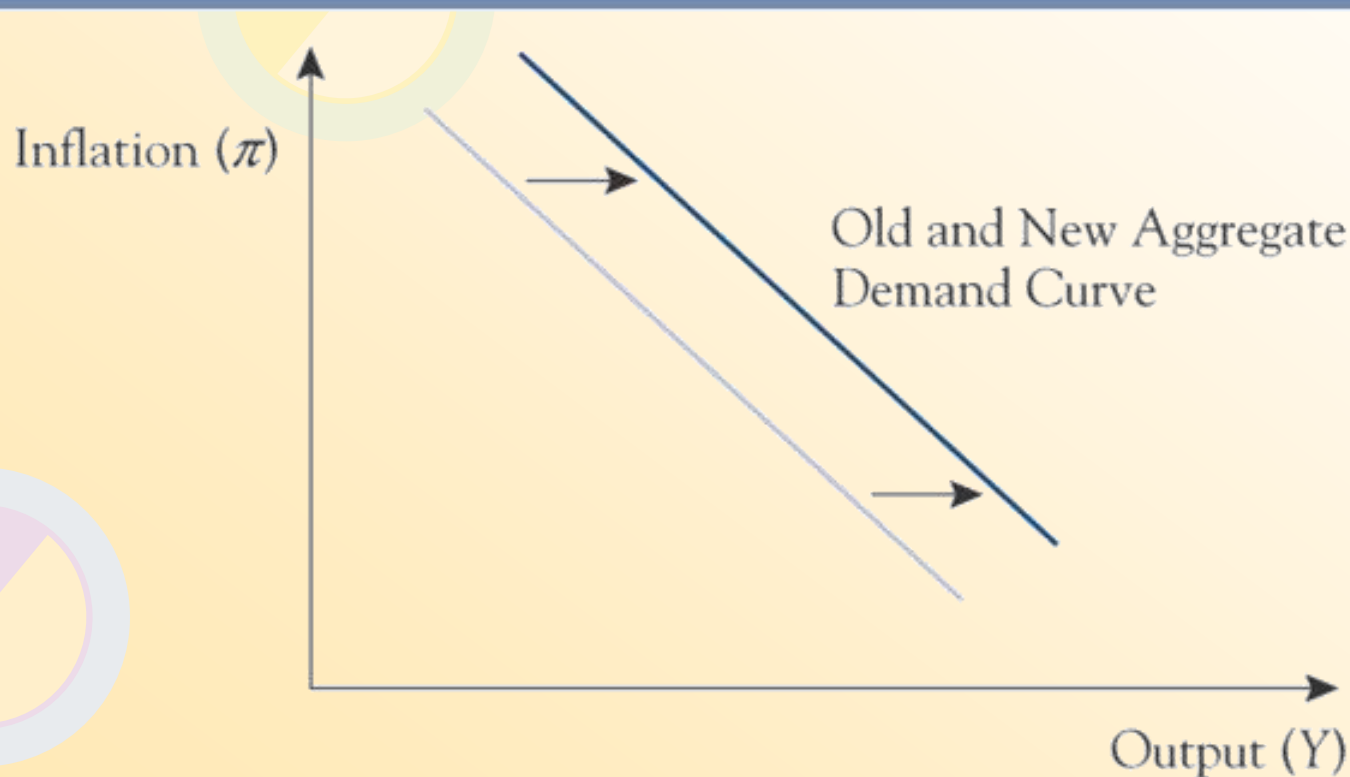
Shifts in the Monetary Policy Reaction Curve

- Whenever the monetary policy reaction curve shifts, the aggregate demand curve will shift as well
 - Changes in the long-run real interest rate, which is a consequence of the structure of the economy, will also shift aggregate demand
- 

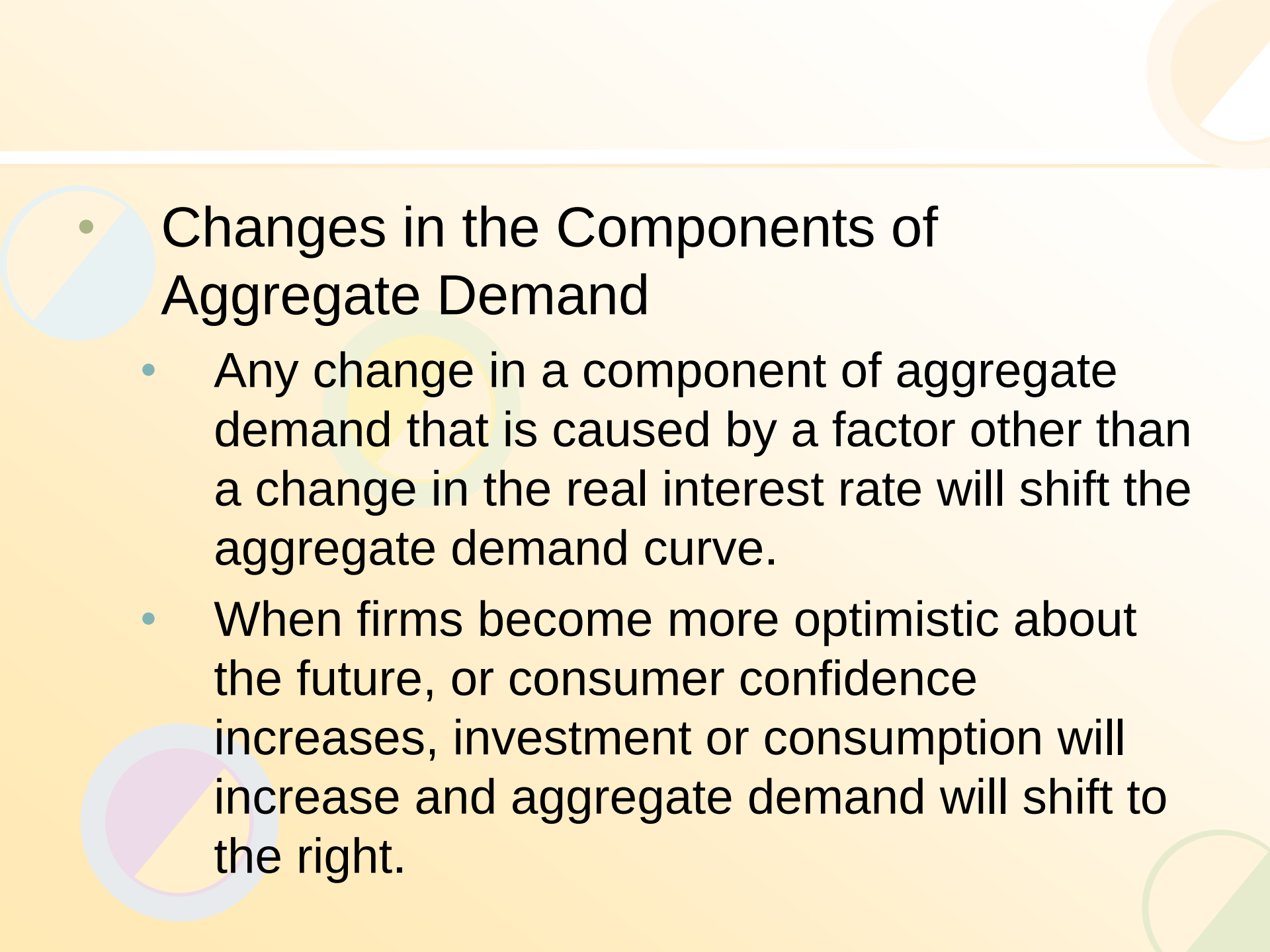
The Aggregate Demand Curve

Figure 21.9 Shifting the Aggregate Demand Curve

Increases in aggregate demand arising from a change in monetary policy, such as a higher inflation target, will shift the aggregate demand curve to the right. (Increases in interest-rate-insensitive components of aggregate demand, such as government purchases, will also shift the aggregate demand curve to the right.)

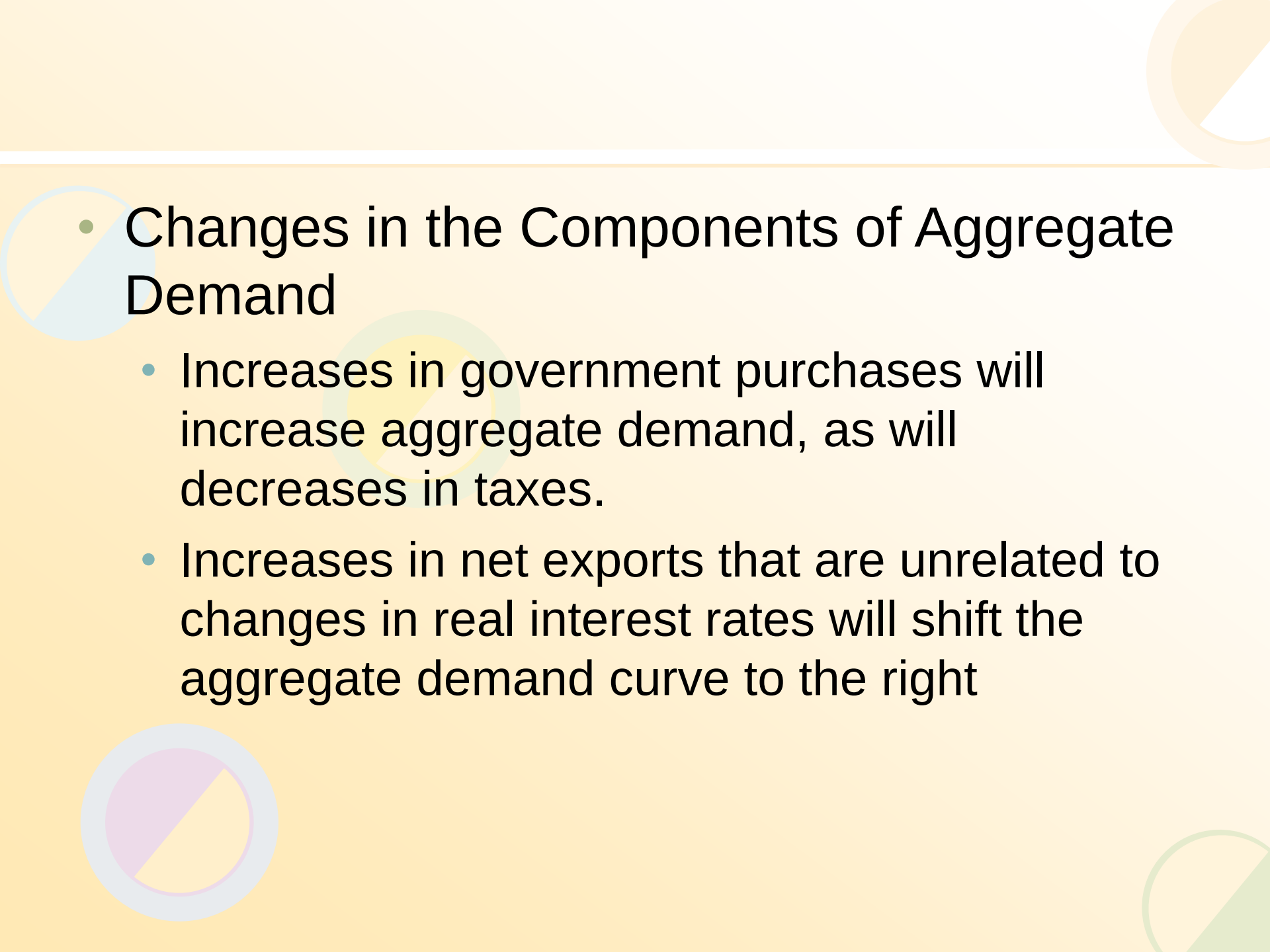


- Either a fall in target inflation or a rise in the long-run real interest rate will shift the monetary policy reaction curve to the left and the aggregate demand curve to the left.



Changes in the Components of Aggregate Demand

- Any change in a component of aggregate demand that is caused by a factor other than a change in the real interest rate will shift the aggregate demand curve.
- When firms become more optimistic about the future, or consumer confidence increases, investment or consumption will increase and aggregate demand will shift to the right.

- 
- Changes in the Components of Aggregate Demand
 - Increases in government purchases will increase aggregate demand, as will decreases in taxes.
 - Increases in net exports that are unrelated to changes in real interest rates will shift the aggregate demand curve to the right

The Aggregate Demand Curve

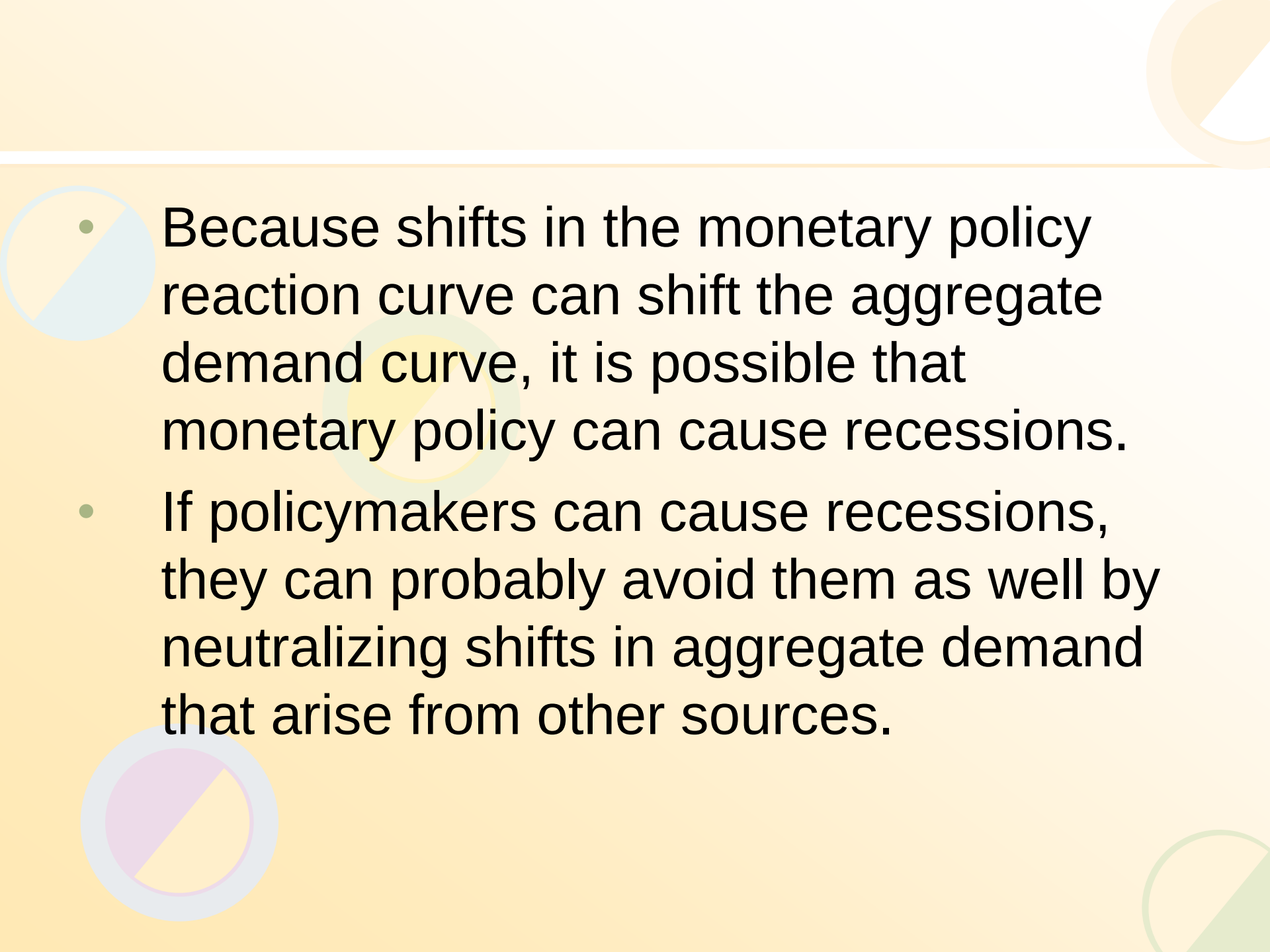
Factors that Shift the Aggregate Demand Curve to the Right

Changes that shift the Monetary Policy Reaction Curve to the right:

- An **increase** in the central bank's inflation target.
- A **decline** in the long-term real interest rate.

Changes that shift the Components of Aggregate Demand to the right:

- An **increase** in consumption that is unrelated to a change in the real interest rate.
- An **increase** in investment that is unrelated to a change in the real interest rate.
- An **increase** in government purchases.
- A **decrease** in taxes.
- An **increase** in net exports that is unrelated to a change in the real interest rate.

- 
- Because shifts in the monetary policy reaction curve can shift the aggregate demand curve, it is possible that monetary policy can cause recessions.
 - If policymakers can cause recessions, they can probably avoid them as well by neutralizing shifts in aggregate demand that arise from other sources.

- The analysis up to this point has assumed that inflation does not change over time; but in reality inflation and output are jointly determined, and monetary policy plays a role in the short-run movements of both.

Summary

- Aggregate Demand Curve
- Shifts in Aggregate Demand Curve

Money and Banking

Lecture 43

Review of the Previous Lecture

- Money growth, Inflation and Aggregate Demand
- Long Run Real Interest Rate
- Monetary Policy Reaction Curve
- Aggregate Demand Curve
 - Shifts in Aggregate Demand

The Aggregate Supply Curve

- The aggregate supply curve tells us where on the aggregate demand curve the economy will end up, explaining the relationship between inflation and real output in the process.
 - The short run aggregate supply curve tells us where the economy will settle at any particular time
 - Long run curve tells us the levels of inflation and output the economy is moving toward.

The Aggregate Supply Curve

- Inflation persistence.
 - Inflation tends to change slowly; when it is low one year it tends to be low the next year, and when it is high it tends to stay high. This is called inflation persistence
 - If inflation remains steady over shorter periods, while real output adjusts, then the short-run aggregate supply curve must be flat at the current level of inflation

Inflation statistics of Pakistan

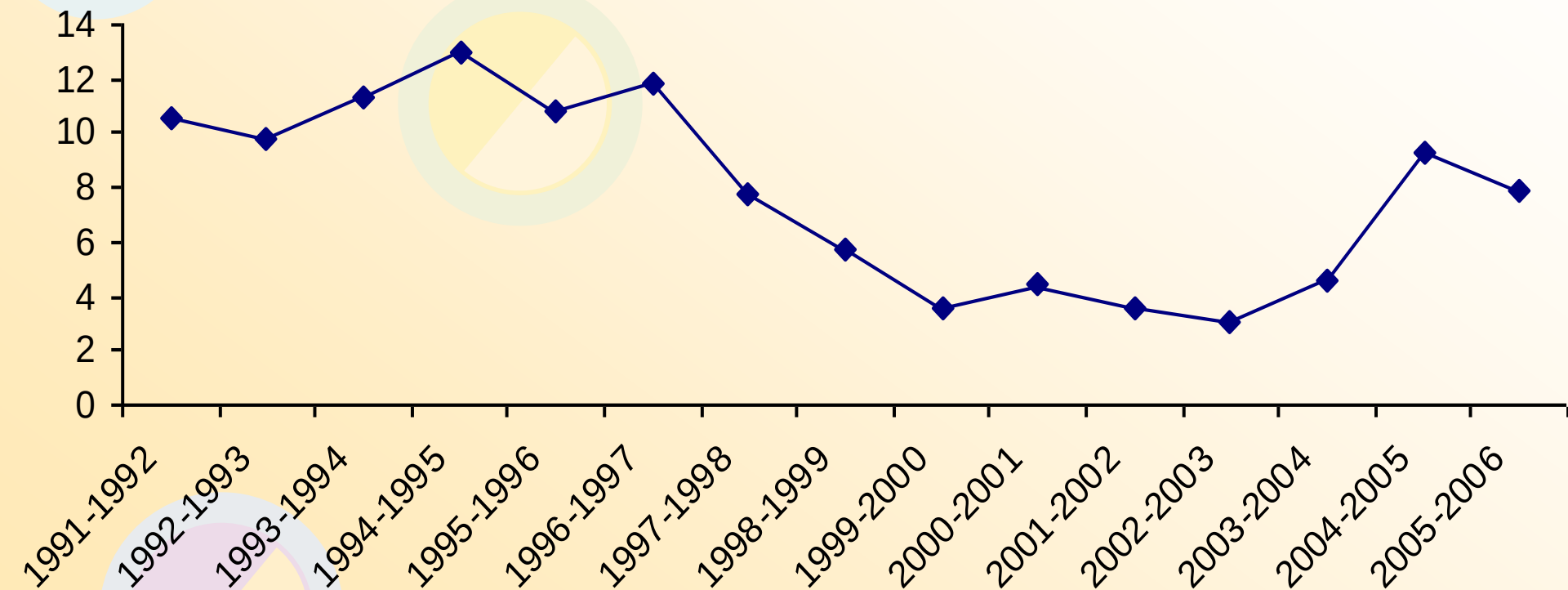
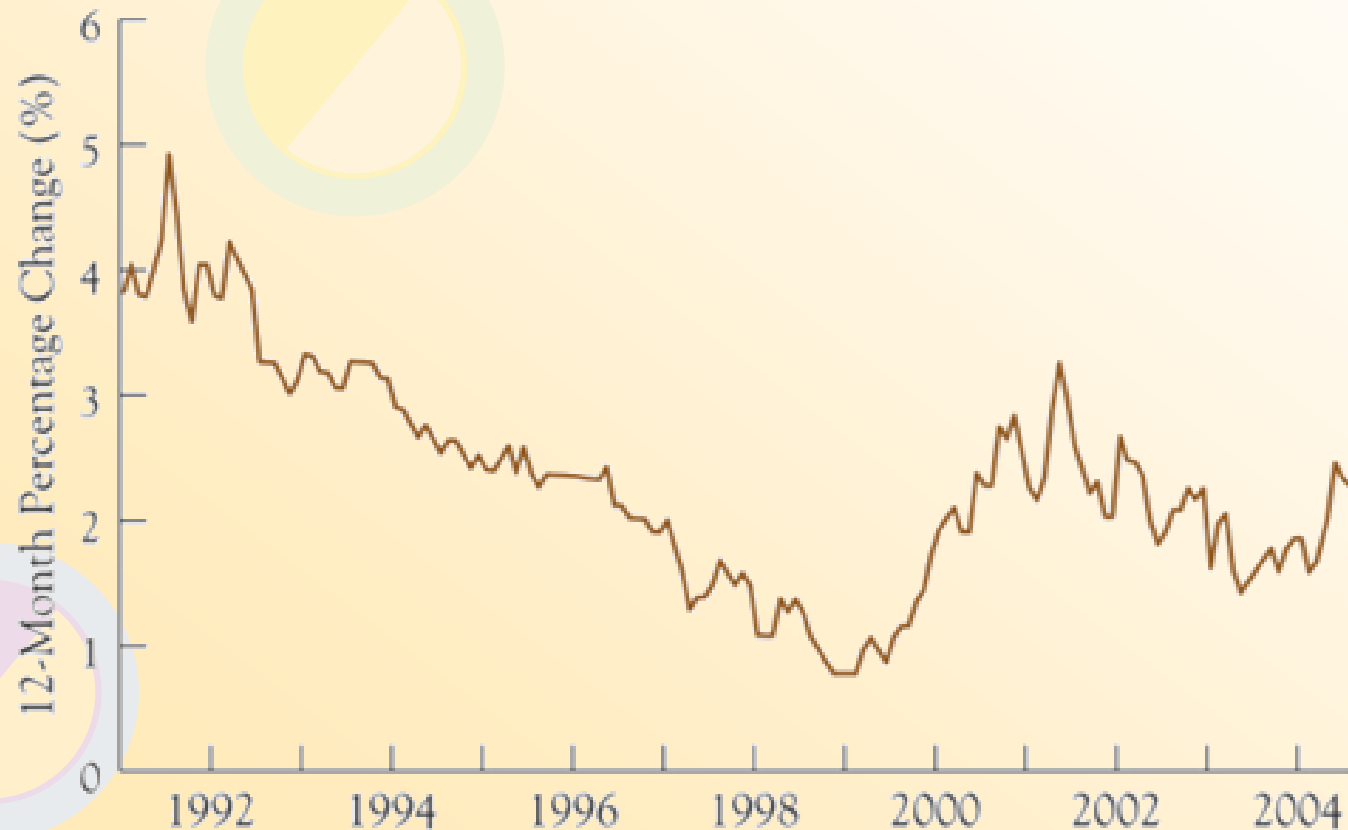


Figure 22.2 Inflation in the Euro Area, 1991-2004

This figure shows the 12-month change in the harmonized index of consumer prices for the 12 countries of the euro area.



- Inflation is persistent for two reasons
 - First, when people expect inflation to continue, they adjust their prices and wages accordingly.
 - When people expect inflation in the near future they raise wages and prices in a way that causes the inflation they expect to occur.
 - Therefore, current inflation is at least partially determined by expected inflation

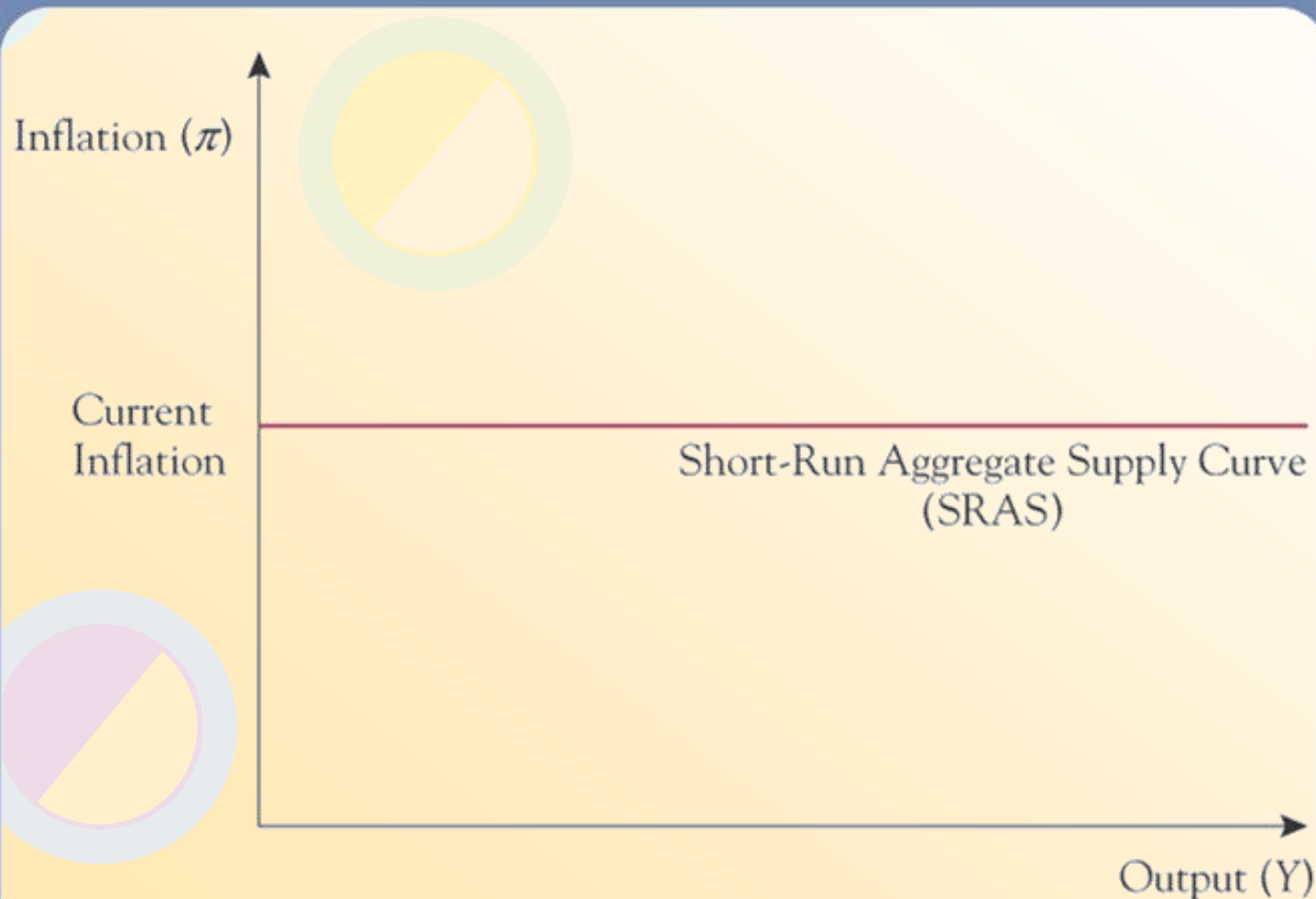
- Inflation is persistent for two reasons
 - Second, not all wage and price decisions are made at the same time.
 - Price and wage adjustments are staggered, and this slows down the adjustment process causing persistence in inflation

- The Short-Run Aggregate Supply Curve:
 - The fact that inflation is persistent means that it is fixed in the short run, so the short-run aggregate supply curve is horizontal at the current level of inflation.

The Aggregate Supply Curve

Figure 22.3 Short-Run Aggregate Supply Curve

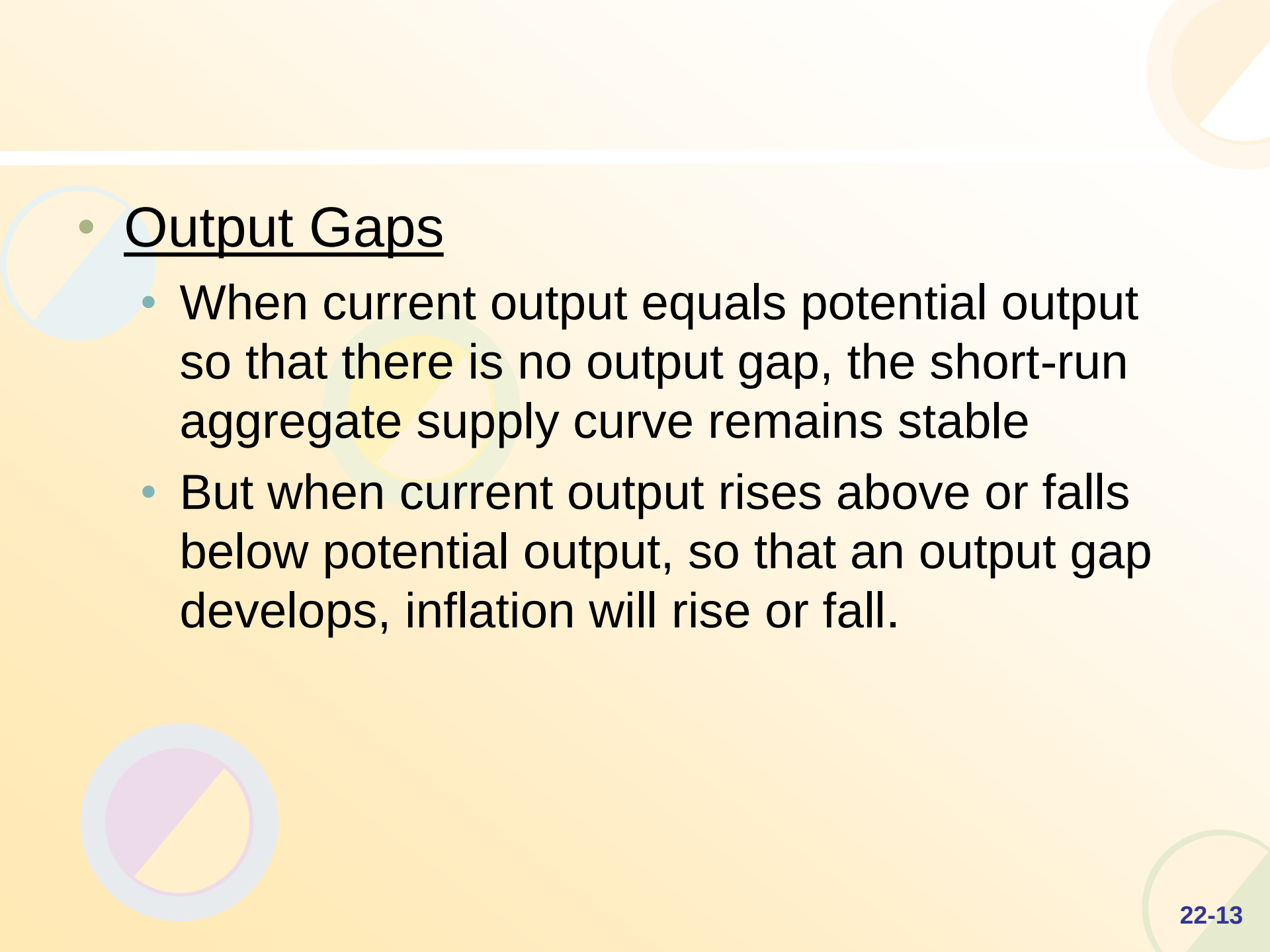
Inflation persistence means the short-run aggregate supply curve is horizontal.



- Firms simply do not adjust the rate of their price increases in the short run; instead, they adjust the quantities they produce and sell.
- Over periods of several years or more, inflation does change, shifting the short-run aggregate supply curve up or down

Shifts in the Short-Run Aggregate Supply Curve

- There are two reasons why the short run aggregate supply curve can shift
 - Deviations of current output from potential output, causing changes in inflation
 - Changes in external factors driving production costs



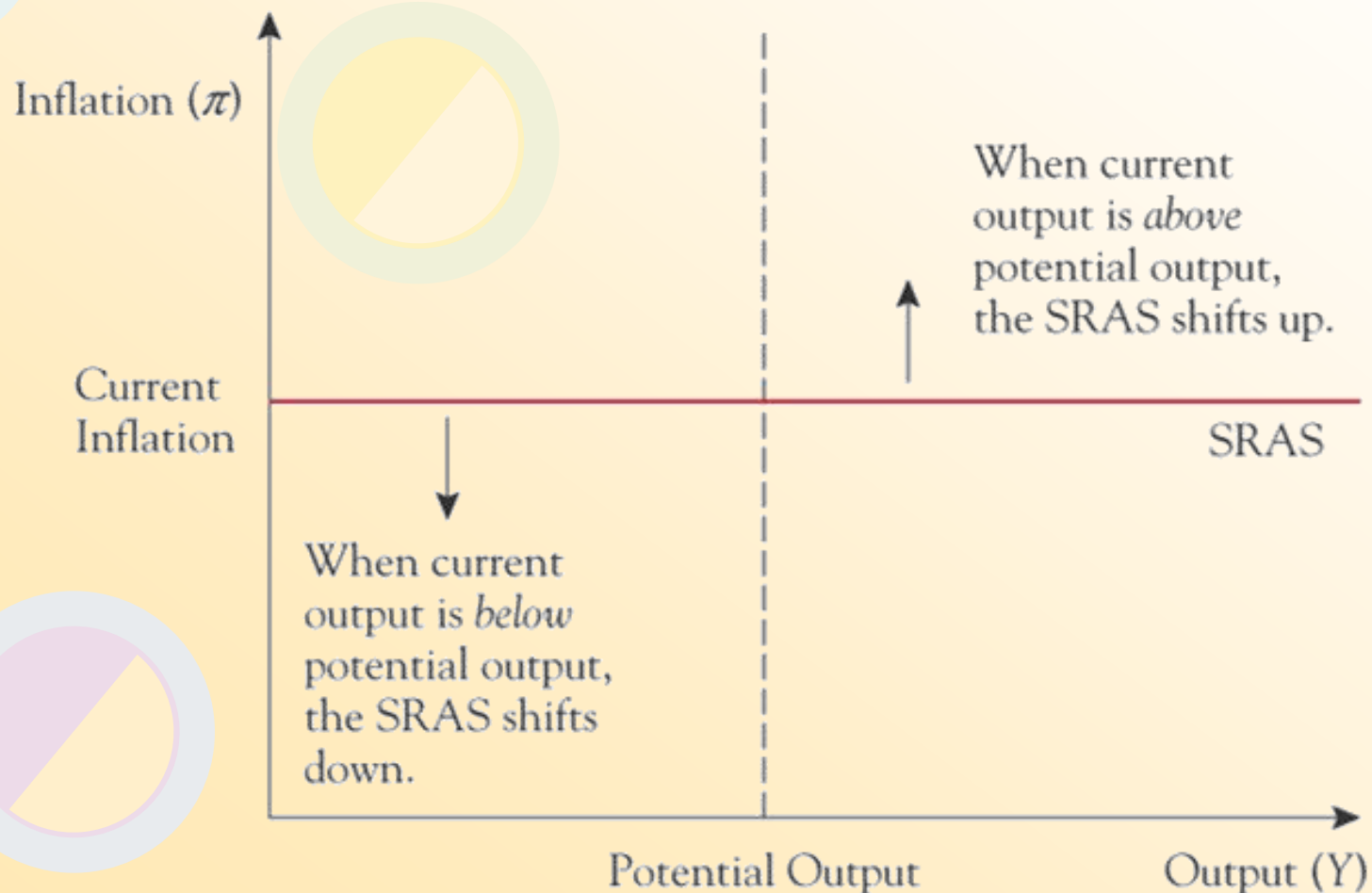
• Output Gaps

- When current output equals potential output so that there is no output gap, the short-run aggregate supply curve remains stable
- But when current output rises above or falls below potential output, so that an output gap develops, inflation will rise or fall.

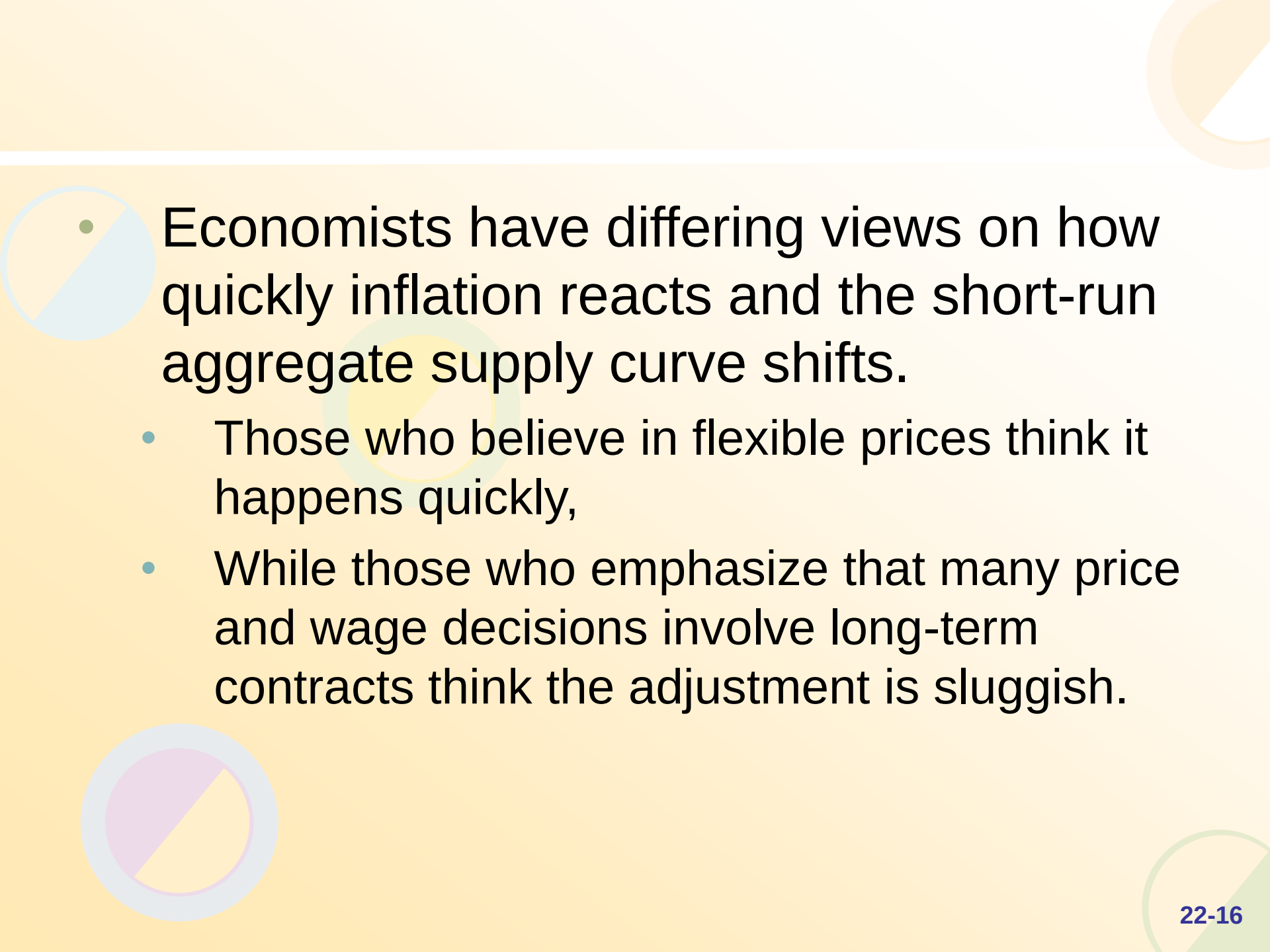
Figure 22.4 Shifts in the Short-Run Aggregate Supply Curve

Response to an Output Gap

When current output rises above potential output, creating an expansionary output gap, the short-run aggregate supply curve shifts upward. When current output falls below potential output, creating a recessionary output gap, the short-run aggregate supply curve shifts downward.



- When current output is below potential output, part of the economy's capacity is idle, and firms tend to raise their prices and wages less than they did when current output equaled potential output.
- When current output exceeds potential output, the opposite happens; firms increase their prices and wages more than they would if they were operating at normal levels
- Thus when current output deviates from potential output, inflation adjusts, and the effect takes time to be felt.

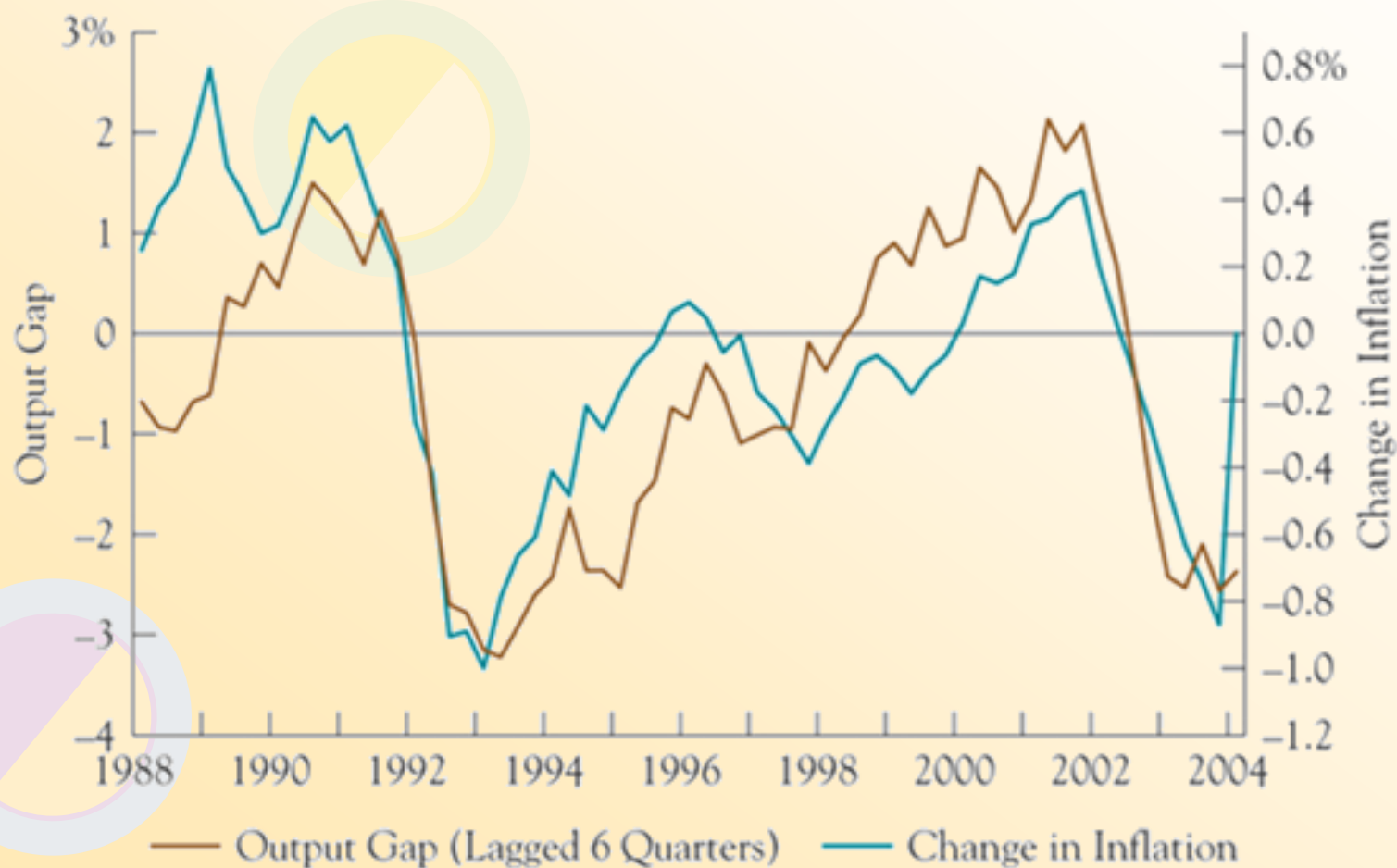


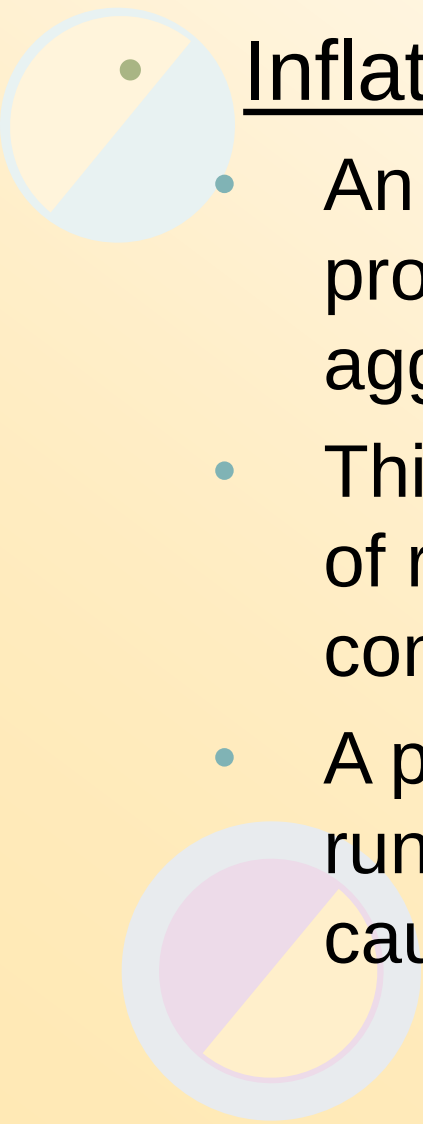
Economists have differing views on how quickly inflation reacts and the short-run aggregate supply curve shifts.

- Those who believe in flexible prices think it happens quickly,
- While those who emphasize that many price and wage decisions involve long-term contracts think the adjustment is sluggish.

Figure 22.5 Inflation and the Output Gap, 1988-2004

In this figure the output gap is measured as the percentage deviation of current real gross domestic product from potential real GDP. The change in inflation is measured as the year-to-year change in the two-year consumer price index, excluding food and energy.





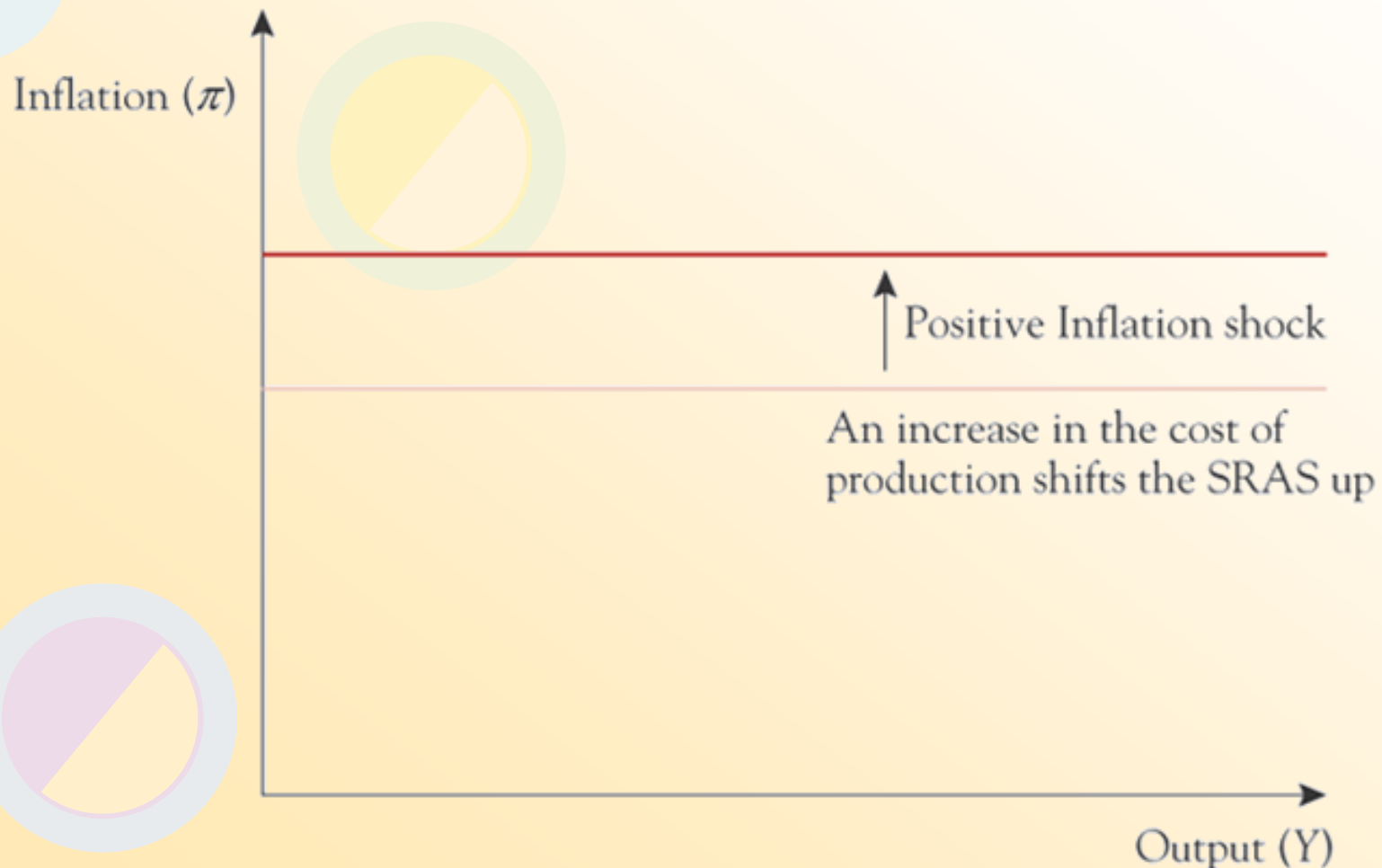
Inflation Shocks:

- An inflation shock is a change in the cost of producing output and causes the short-run aggregate supply curve to shift.
- This can be the result of changes in the cost of raw materials or labor, or (as is most common) a change in the price of energy.
- A positive inflation shock causes the short-run aggregate supply curve to shift upward, causing inflation to rise

Figure 22.6 Shifts in the Short-Run Aggregate Supply Curve

Response to an Inflation Shock

A rise in labor costs, the price of raw materials, or the expected future level of inflation creates an *inflation shock* that shifts the short-run aggregate supply curve upward, causing inflation to rise.

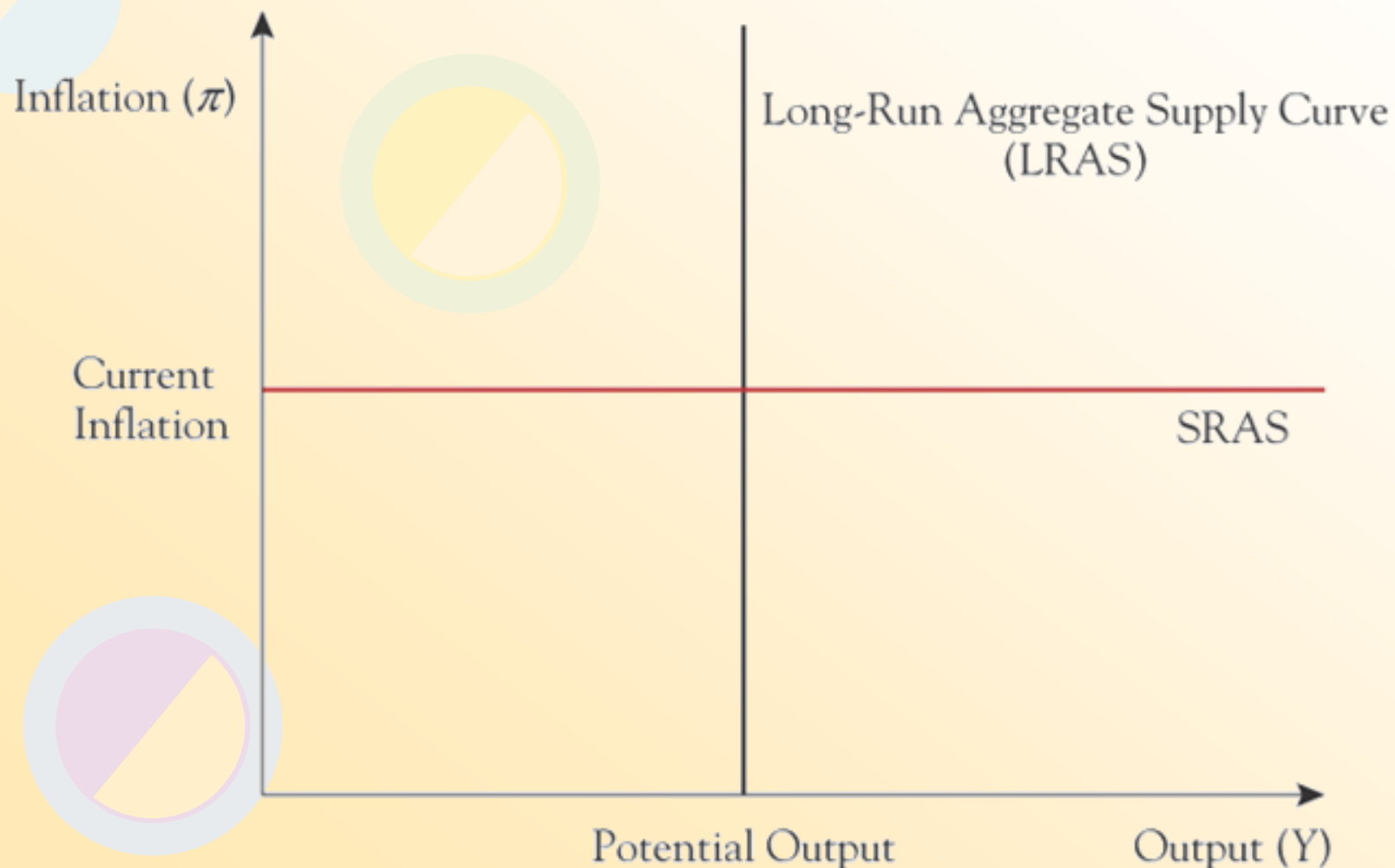


The Long-Run Aggregate Supply Curve

- In the long run the economy moves to the point where current output equals potential output, while inflation is determined by money growth
- The long-run aggregate supply curve is vertical at the point where current output equals potential output.

Figure 22.7 The Long-Run and Short-Run Aggregate Supply Curves

The long-run aggregate supply curve is vertical at the point where current output equals potential output.

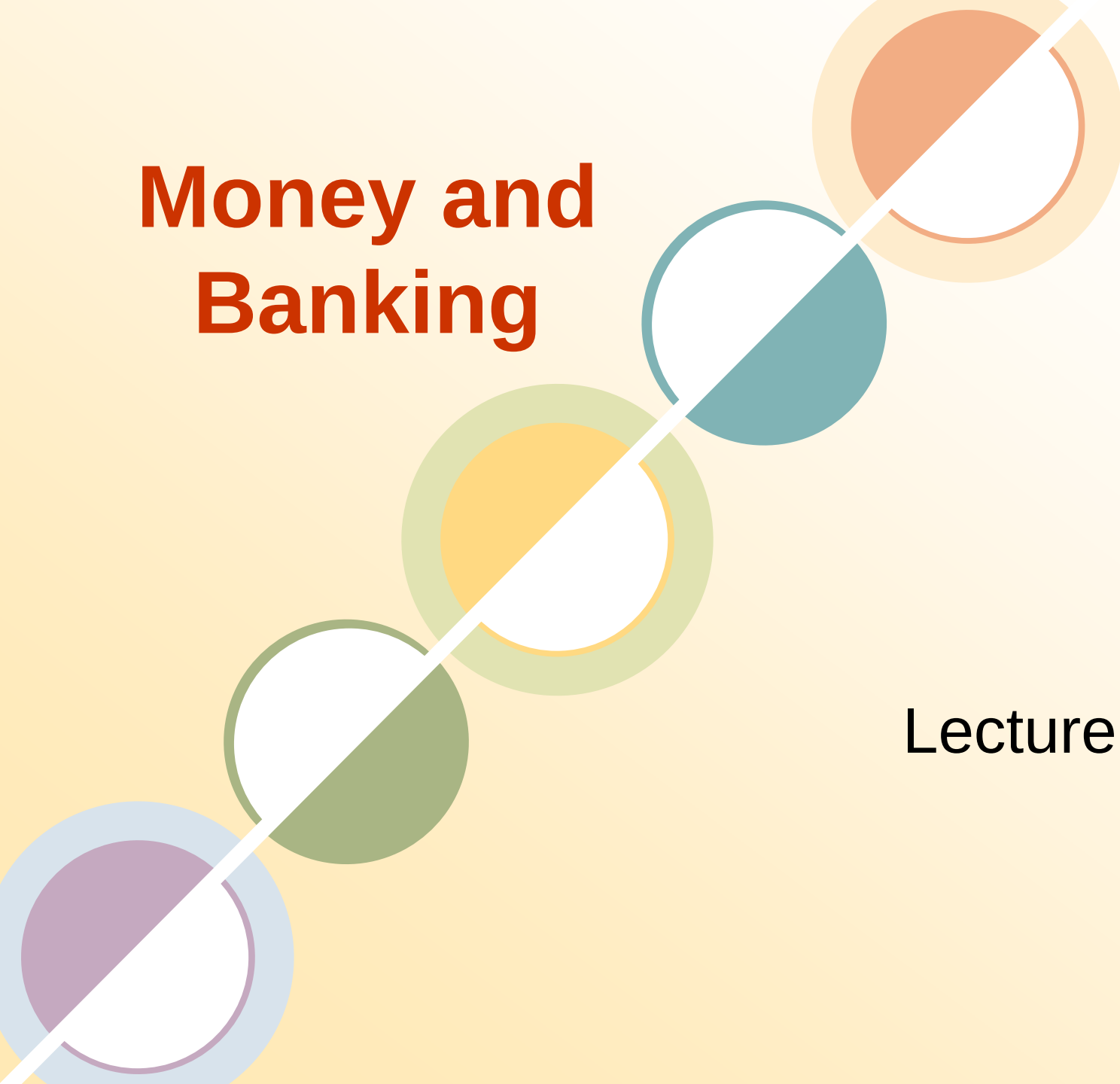


- Changes in expected inflation operate like cost shocks, shifting the short-run aggregate supply curve up and down.
- For the economy to remain in long-run equilibrium, then, in addition to current output equaling potential output, current inflation must equal expected inflation

- At any point along the long-run aggregate supply curve, current output equals potential output and current inflation equals expected inflation
- Potential output is constantly rising as a result of investment and technological improvements (the sources of economic growth), which increase the normal output level.
- Changes in the economy's productive capacity will shift the long-run aggregate supply curve; increases will shift it right and decreases will shift it left.

Money and Banking

Lecture 44



Review of the Previous Lecture

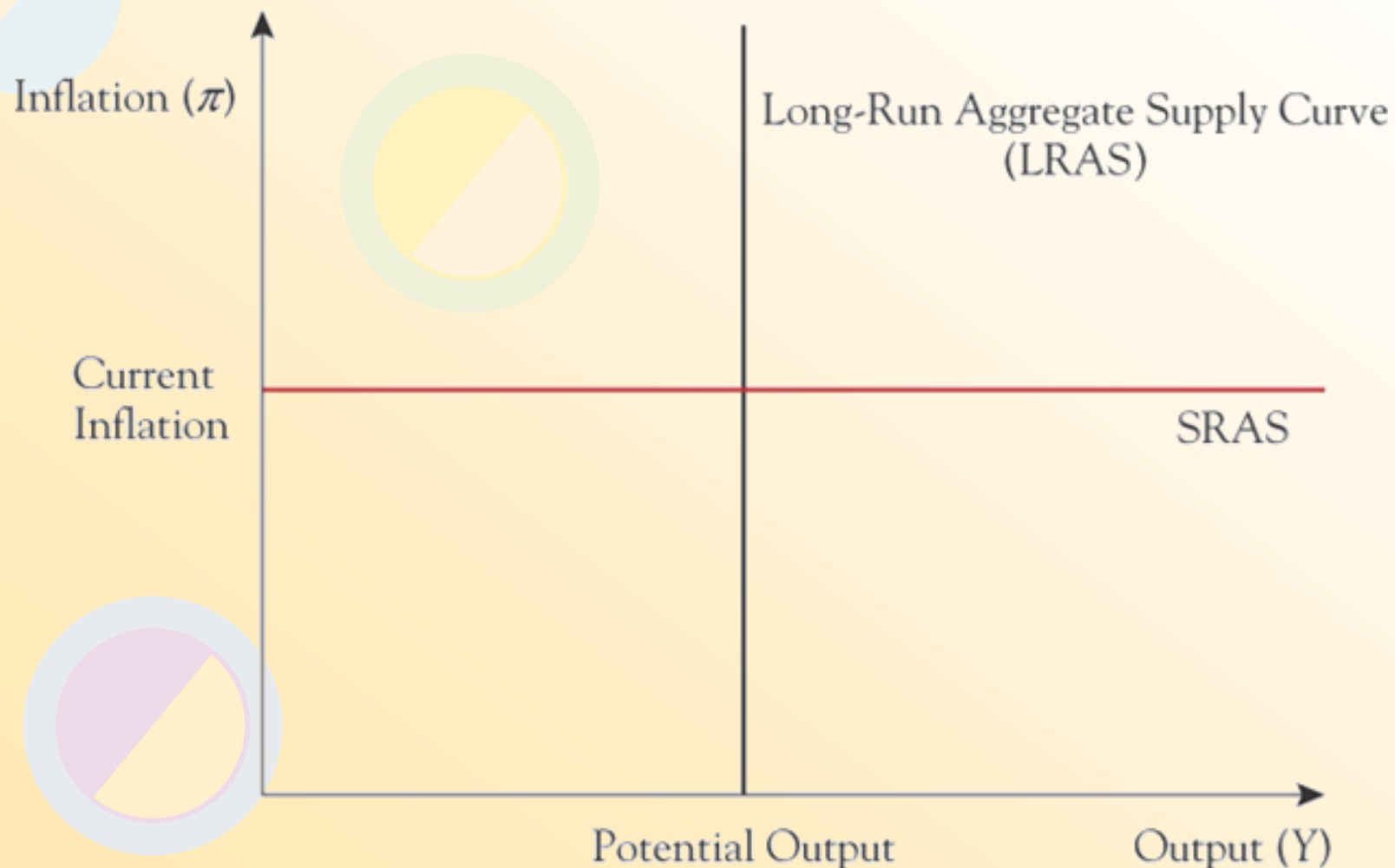
- Aggregate Supply Curve
 - Short-run Aggregate Supply Curve
 - Shifts in Short-run Aggregate Supply Curve

The Long-Run Aggregate Supply Curve

- In the long run the economy moves to the point where current output equals potential output, while inflation is determined by money growth
- The long-run aggregate supply curve is vertical at the point where current output equals potential output.

Figure 22.7 The Long-Run and Short-Run Aggregate Supply Curves

The long-run aggregate supply curve is vertical at the point where current output equals potential output.



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- Changes in expected inflation operate like cost shocks, shifting the short-run aggregate supply curve up and down.
 - For the economy to remain in long-run equilibrium, then, in addition to current output equaling potential output, current inflation must equal expected inflation

- At any point along the long-run aggregate supply curve, current output equals potential output and current inflation equals expected inflation
- Potential output is constantly rising as a result of investment and technological improvements (the sources of economic growth), which increase the normal output level.
- Changes in the economy's productive capacity will shift the long-run aggregate supply curve; increases will shift it right and decreases will shift it left.

Equilibrium and the Determination of Output and Inflation

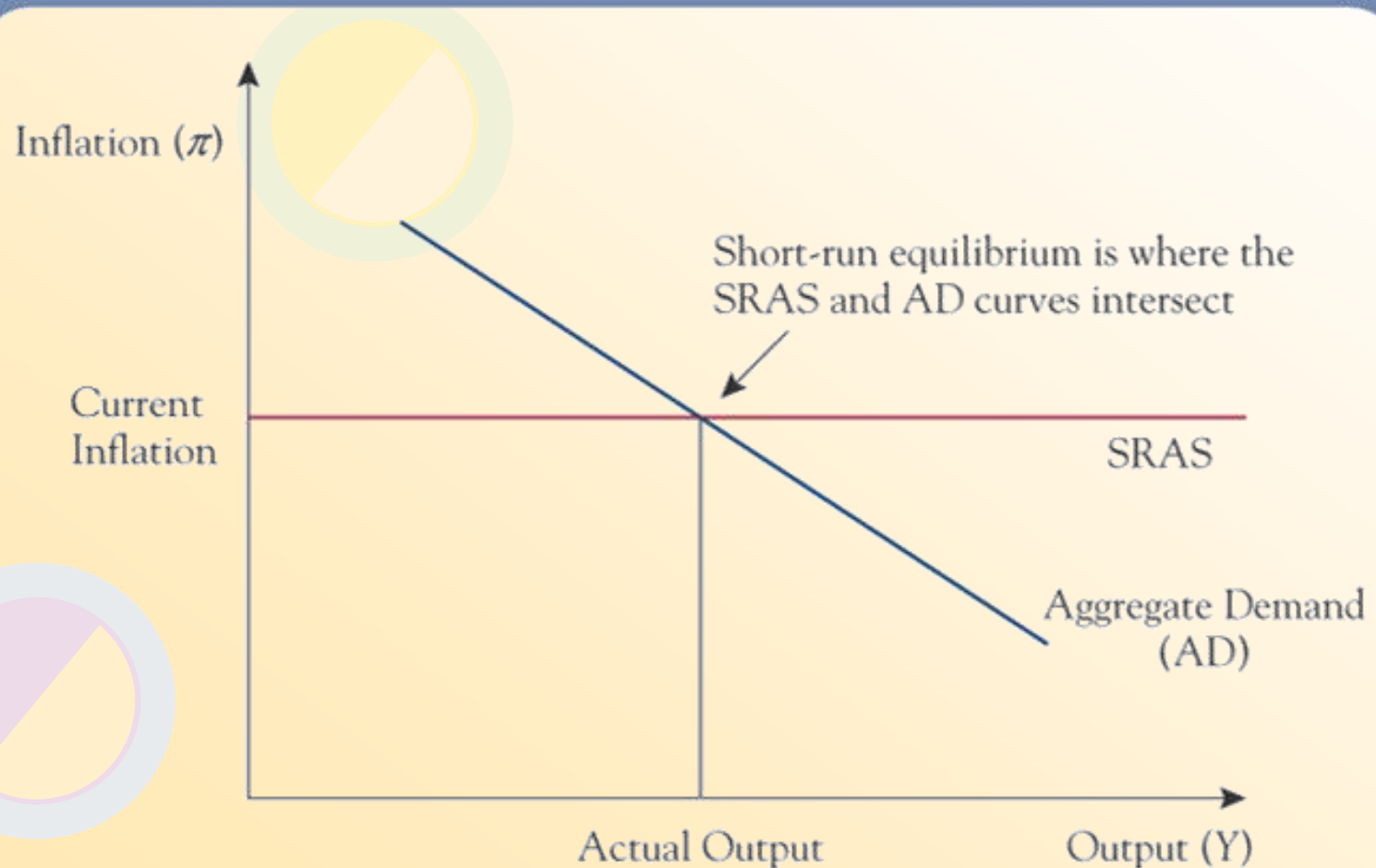
A. Short-Run Equilibrium

- Short-run equilibrium is determined by the intersection of the aggregate demand curve with the short-run aggregate supply curve.

Equilibrium and the Determination of Output and Inflation

Figure 22.9 Short-Run Equilibrium of Output and Inflation

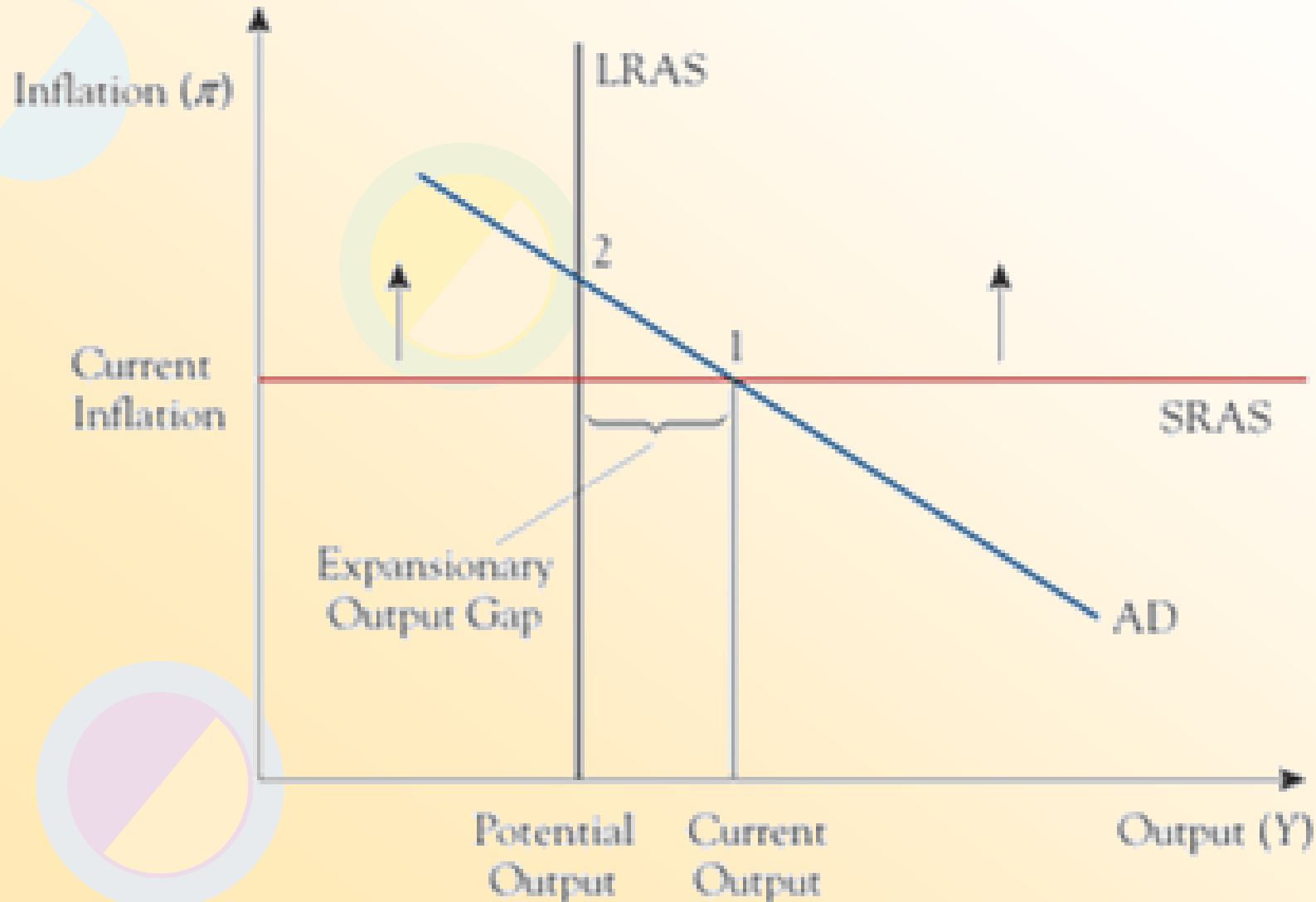
Inflation and actual output are determined by the intersection of the short-run aggregate supply curve with the aggregate demand curve.



B. Adjustment to Long-Run Equilibrium

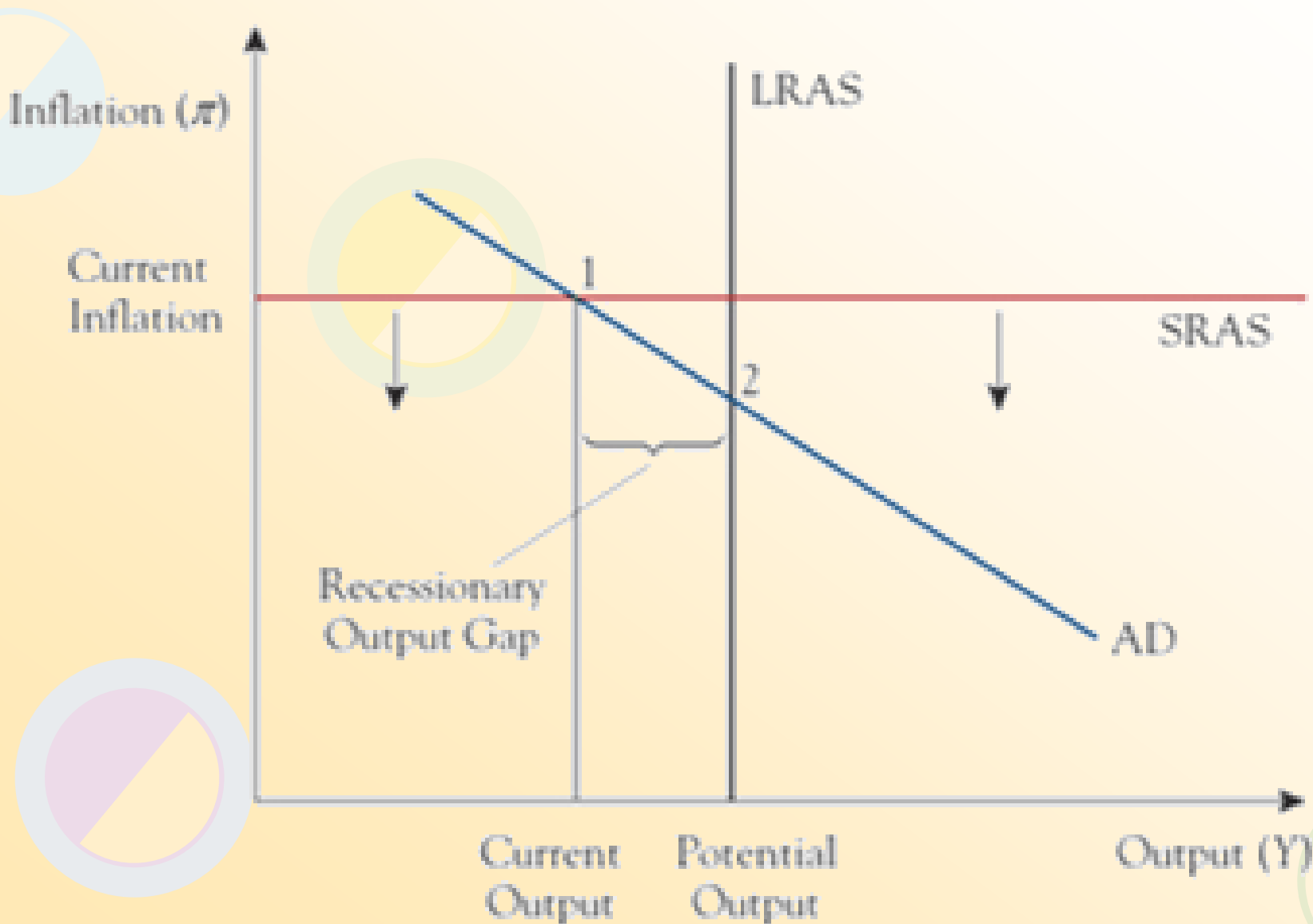
- When current output exceeds potential, the resulting expansionary gap exerts upward pressure on inflation, shifting the short-run aggregate supply curve upward, a process that continues until output returns to potential; at this point inflation stops changing

Equilibrium and the Determination of Output and Inflation



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- If current output is lower than potential output, the resulting recessionary gap places downward pressure on inflation, causing the short-run aggregate supply curve to shift downward, and once again the process continues until current output returns to potential
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Equilibrium and the Determination of Output and Inflation



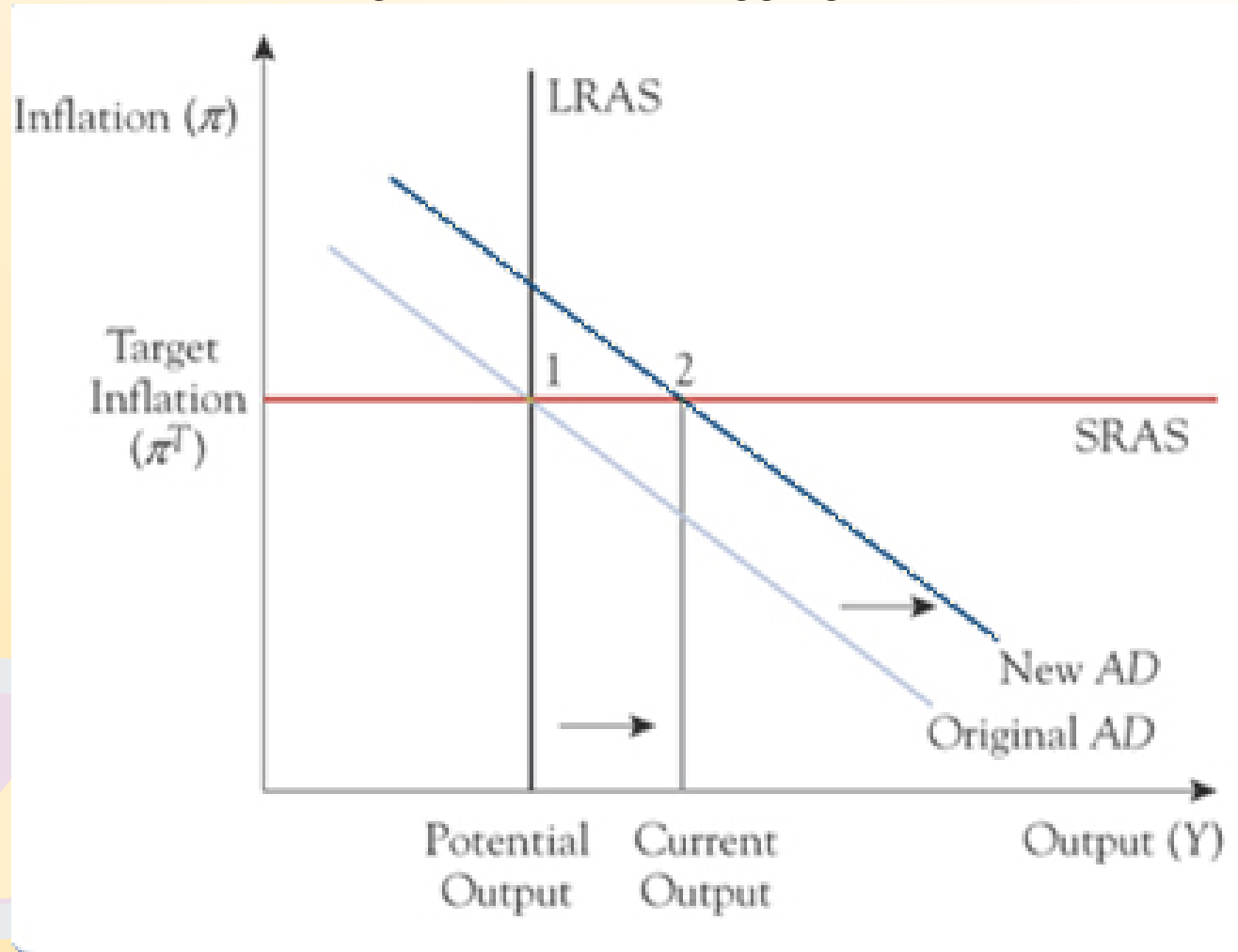
- This shows that the economy does indeed have a self-correcting mechanism and that the manner in which the short-run aggregate supply curve shifts in response to output gaps reinforces our conclusion that the long-run aggregate supply curve is vertical
- In long-run equilibrium, current output equals potential output and current inflation is steady and equal to target inflation, which equals expected inflation

The Impact of Shifts in Aggregate Demand on Output and Inflation

- Suppose aggregate demand shifted right as a result of an increase in government purchases.
 - At first, current output rises but inflation does not change.
 - But the higher level of output creates an expansionary gap and the short-run aggregate supply curve starts to shift upward and inflation rises

The Impact of a Shift in Aggregate Demand on Output and Inflation

Short-Run Equilibrium Inflation and Output
Following an Increase in Aggregate Demand



The Impact of a Shift in Aggregate Demand on Output and Inflation

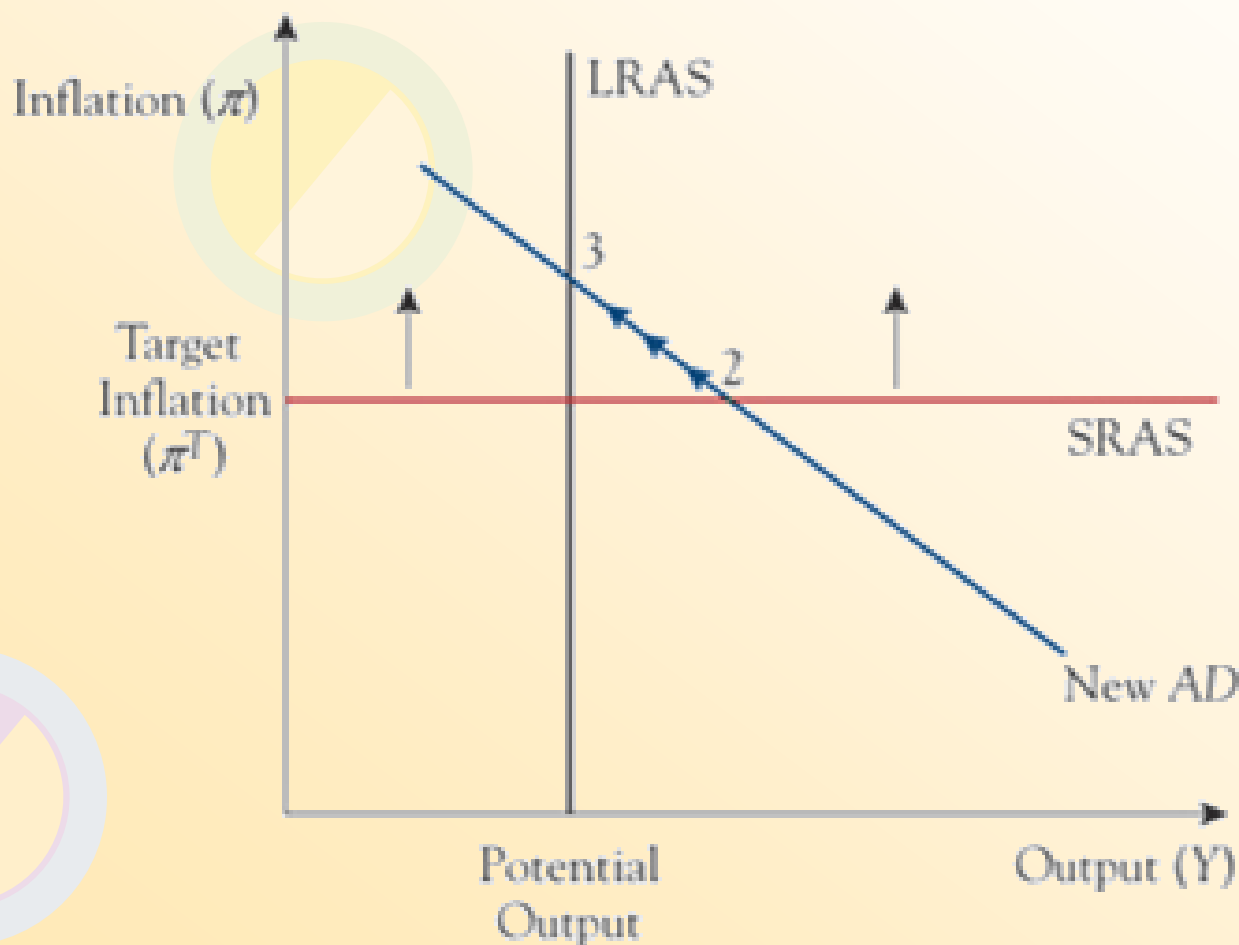
- **1. Start at Long-Run Equilibrium**
 - $Y = \text{Potential Output}$
 - $\pi = \text{Target Inflation}$
- **2. Aggregate Demand Shifts Right**
 - Original AD shifts to New AD
 - $Y > \text{Potential Output}$
 - Inflation Is Unchanged

Short-run equilibrium moves from point 1 to point 2

- Higher inflation moves policymakers along their reaction curve, leading them to raise the real interest rate and moving the economy upward along the new aggregate demand curve. Output then begins to fall back toward its long-run equilibrium level.
- The economy will settle at the point at which the new aggregate demand curve crosses the long-run aggregate supply curve and current output again equals potential output

The Impact of a Shift in Aggregate Demand on Output and Inflation

Adjustment of Short-Run Equilibrium Inflation and Output Following an Increase in Aggregate Demand



The Impact of a Shift in Aggregate Demand on Output and Inflation

- **Adjustment:**
- **1. At the Short-Run Equilibrium point 2:**
 - $Y > \text{Potential Output}$
- **2. SRAS begins to shift up**
 - Output begins to fall
 - Inflation begins to rise as economy moves along New AD
- **3. With no policy response, economy moves to point 3, where**
 - Current inflation $>$ Target inflation

The Impact of a Shift in Aggregate Demand on Output and Inflation

- If central bankers simply sit and watch as the aggregate demand curve shifts to the right, inflation will rise
- So long as monetary policymakers remain committed to their original inflation target, they will need to do something to get the economy back to the point where it began—point “1”

The Impact of a Shift in Aggregate Demand on Output and Inflation

- An increase in government purchases raises the long-term real interest rate.
- Policymakers will compensate by shifting their monetary policy reaction curve to the left, increasing the real interest rate at every level of inflation

The Impact of a Shift in Aggregate Demand on Output and Inflation

- When the monetary policy reaction curve shifts, the aggregate demand curve shifts with it.
- The aggregate demand curve will shift to the left, bringing the economy back to long-run equilibrium.

The Impact of a Shift in Aggregate Demand on Output and Inflation

- An increase in aggregate demand causes a *temporary* increase in both output and inflation.
- A decline in aggregate demand causes a *temporary* decline in both output and inflation

The Impact of a Shift in Aggregate Demand on Output and Inflation

- This discussion implies that whenever we see a *permanent* increase in inflation, it must be the result of monetary policy.
- That is, if inflation goes up or down and remains at its new level, the only explanation is that central bankers must be allowing it to happen.
- They have changed their inflation target, whether or not they acknowledge the change explicitly.

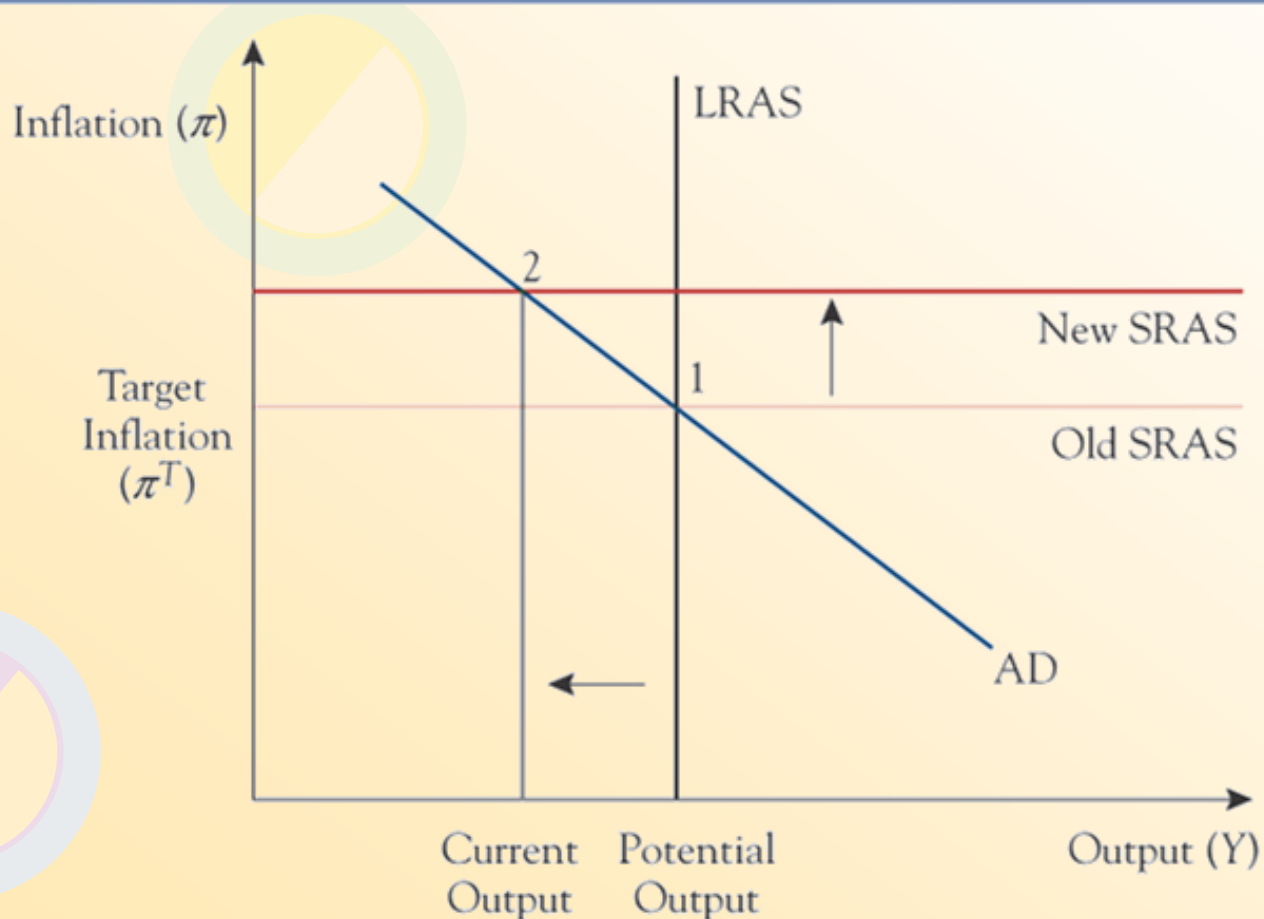
The Impact of Inflation Shocks on Output and Inflation

- An inflation shock shifts the short-run aggregate supply curve (such as an oil price increase)
- A positive shock moves it to a higher level, and the result is higher inflation and lower output, a situation called “stagflation”.

Impact of Inflation Shocks on Output and Inflation

Figure 22.12 The Effects of a Positive Inflation Shock on Short-Run Equilibrium

A positive inflation shock shifts the short-run aggregate supply curve upward, moving short-run equilibrium from point 1 to point 2. Inflation rises and output falls.



- But the decline in output exerts downward pressure on inflation, causing the short-run aggregate supply curve to shift down
- Inflation falls and output rises until the economy returns to the point where current output equals potential output and inflation equals the central bank's target.

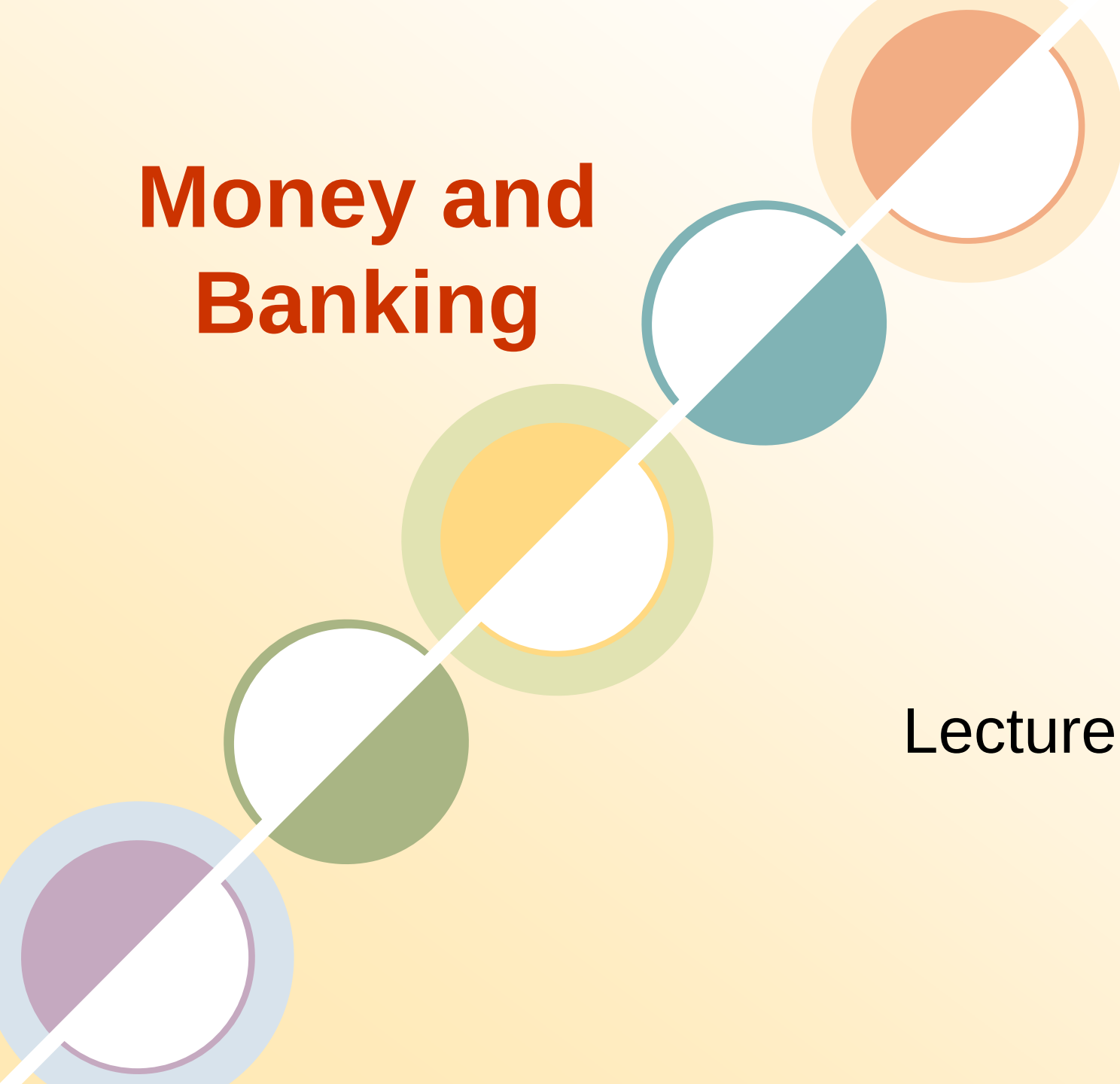
- An inflation shock has no effect on the economy's long-run equilibrium point; only a change in potential output or a change in the central bank's inflation target can accomplish that

Summary

- Long-run Aggregate Supply Curve
- Equilibrium and Determination of Output and Inflation
- Impact of Shift in Aggregate Demand on Output and Inflation
- Impact of Inflation Shocks on Output and Inflation

Money and Banking

Lecture 45



Review of the Previous Lecture

- Long-run Aggregate Supply Curve
- Equilibrium and Determination of Output and Inflation
- Impact of Shift in Aggregate Demand on Output and Inflation
- Impact of Inflation Shocks on Output and Inflation

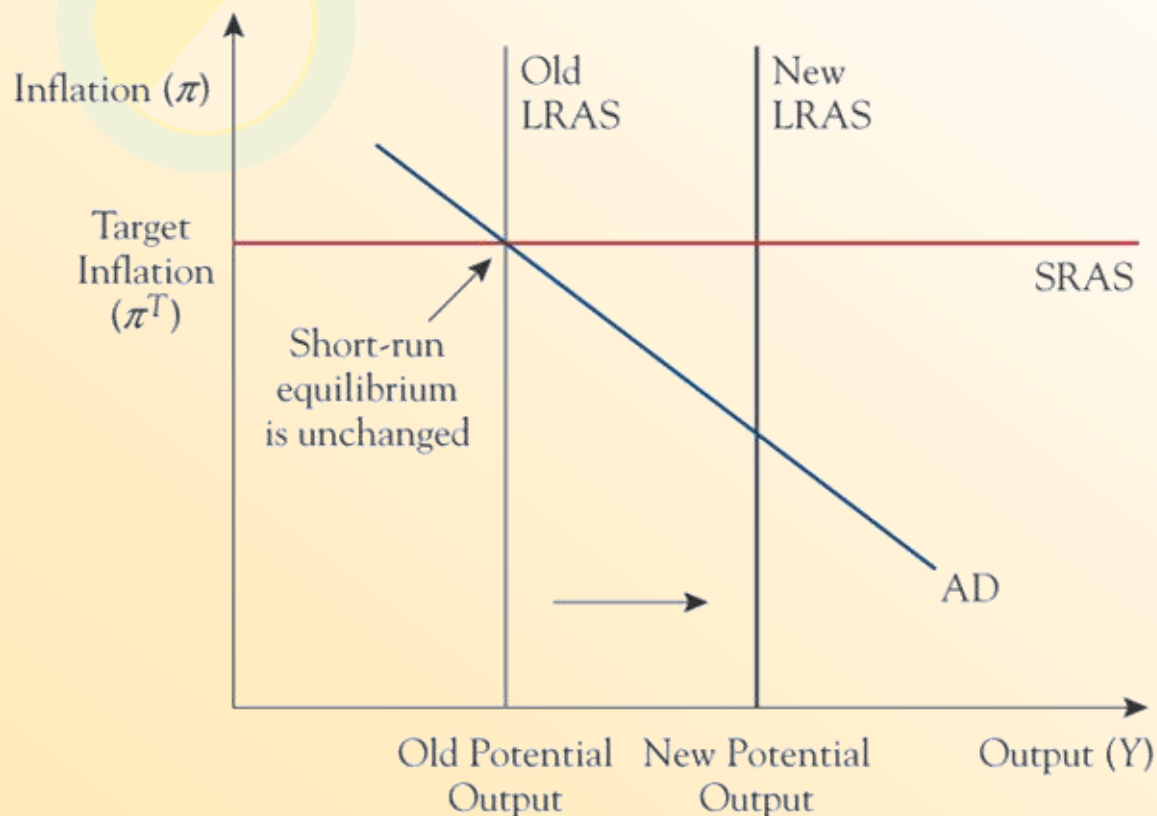
Shifts in Potential Output and Real Business Cycle Theory

- Changes in potential output shift the long-run aggregate supply curve
- At first the shift has no impact on the short-run aggregate supply curve, so inflation and output remain stable
- But with time, the increase in potential output will mean that current output is now below potential output, creating a recessionary output gap, which puts downward pressure on inflation, shifting the short-run aggregate supply curve downward

Shifts in Potential Output and Real Business Cycle Theory

Figure 22.13 The Effects of an Increase in Potential Output on Inflation and Output

An increase in potential output shifts the long-run aggregate supply curve (LRAS) to the right. In the short run, current output remains unchanged. But since current output is now below potential output, the resulting recessionary gap places downward pressure on inflation, and output eventually begins to rise.



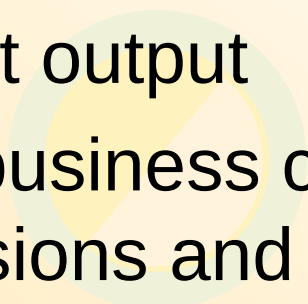
Shifts in Potential Output and Real Business Cycle Theory

- What happens next depends on what policymakers do; they can:
 - Take advantage of the downward pressure on inflation to reduce their inflation target
 - Initiate actions that ensure that inflation does not fall.
- In either case, notice that the higher level of potential output means a lower long-term real interest rate.

Shifts in Potential Output and Real Business Cycle Theory

- Business cycle fluctuations can therefore be explained in terms of shifts in aggregate demand that change its point of intersection with a flat short-run aggregate supply curve
- An alternative explanation for business cycle fluctuations focuses on shifts in potential output, a view called real business cycle theory

- Real business cycle theory
 - Real business cycle theory starts with the assumption that prices and wages are flexible, so that inflation adjusts rapidly (the short-run aggregate supply curve shifts quickly in response to deviations of current output from potential output)
 - This assumption implies that the short-run aggregate supply curve is irrelevant: equilibrium output and inflation are determined by the point on the aggregate demand curve where current output equals potential output

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- Any shift in the aggregate demand curve, regardless of its source, will change inflation but not output
 - Real business cycle theorists explain recessions and booms by looking at fluctuations in potential output, focusing on changes in productivity and their impact on GDP

The Impact of a Shift in Aggregate Demand and Aggregate Supply on Output and Inflation

	Increase in Aggregate Demand
Source	Consumer Confidence up Business Optimism up Govt. Purchases up Taxes Down Exchange Rate Depreciates
Short-Run Effects	Y Increases π Is unchanged
Path of Adjustment	1. Expansionary output gap puts upward pressure on inflation 2. As Inflation begins to rise, output begins to fall
Long-Run Effects	Y = original potential output π = target (may change)

The Impact of a Shift in Aggregate Demand and Aggregate Supply on Output and Inflation

	Positive Inflation Shock
Source	Labor Costs up Raw Material Prices up Expected Inflation up
Short-Run Effects	Y falls π rises
Path of Adjustment	1. Recessionary output gap puts downward pressure on inflation 2. As Inflation begins to fall, output begins to rise
Long-Run Effects	Y = original potential output π = target (may change)

The Impact of a Shift in Aggregate Demand and Aggregate Supply on Output and Inflation

	Increase in Potential Output
Source	Capital in Production up Labor in Production up Productivity up
Short-Run Effects	Y unchanged π Unchanged
Path of Adjustment	1. Recessionary output gap puts downward pressure on inflation 2. As Inflation begins to fall, output begins to rise
Long-Run Effects	Y = new potential output π = target (may change)

Stabilization Policy

- Monetary Policy
 - Policymakers can shift the aggregate demand curve by shifting their monetary policy reaction curve, but they cannot shift the short-run aggregate supply curve
 - They can neutralize movements in aggregate demand, but they cannot eliminate the effects of an inflation shock

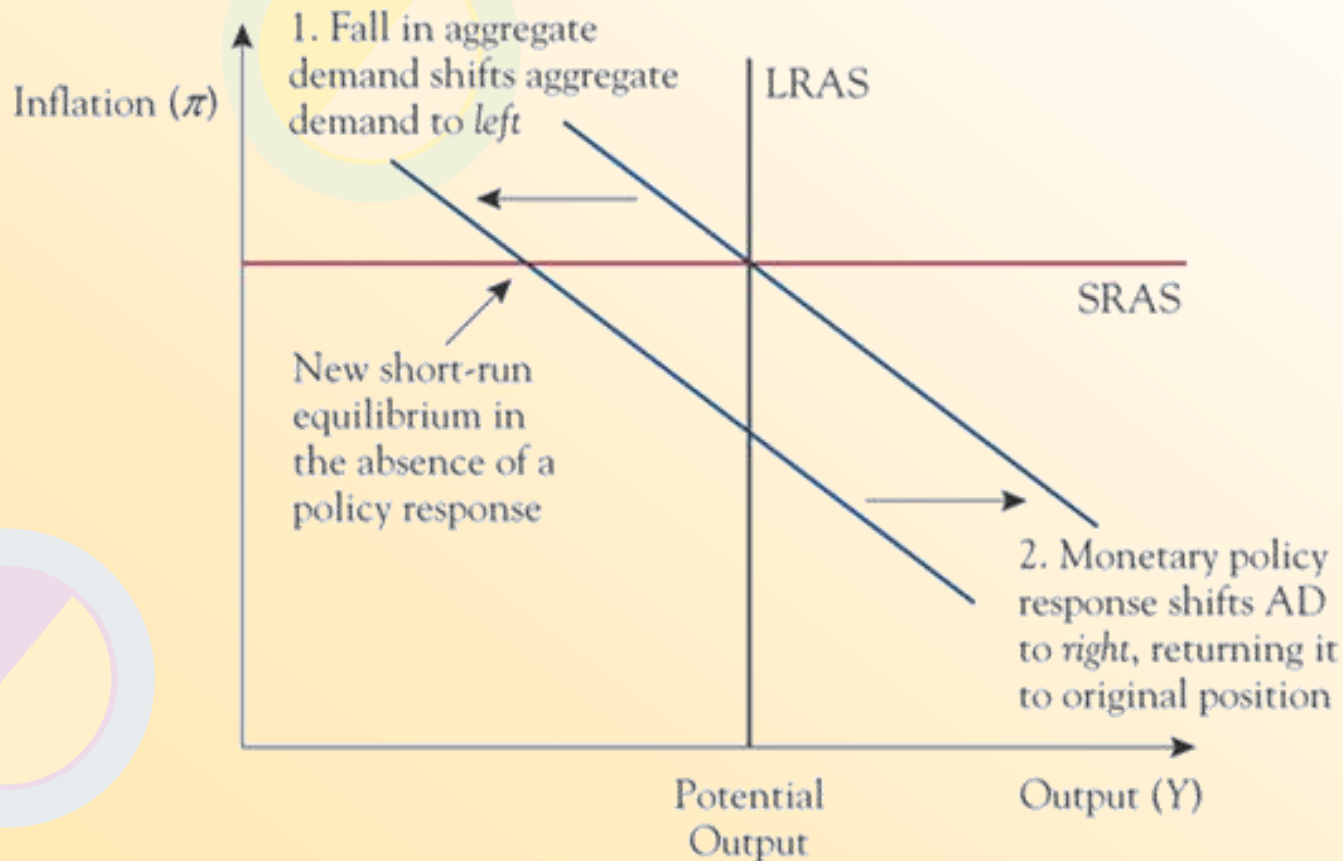
Shifts in Aggregate Demand

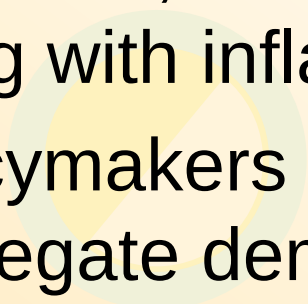
- If households and businesses become more pessimistic, driving down aggregate demand, the economy moves into a recession as the new short-run equilibrium point is at a current output less than potential output.
- Policymakers will conclude that the long-run real interest rate has gone down and will shift their monetary policy reaction curve to the right, reducing the level of the real interest rate at every level of inflation
- This shifts the aggregate demand curve back to its initial position

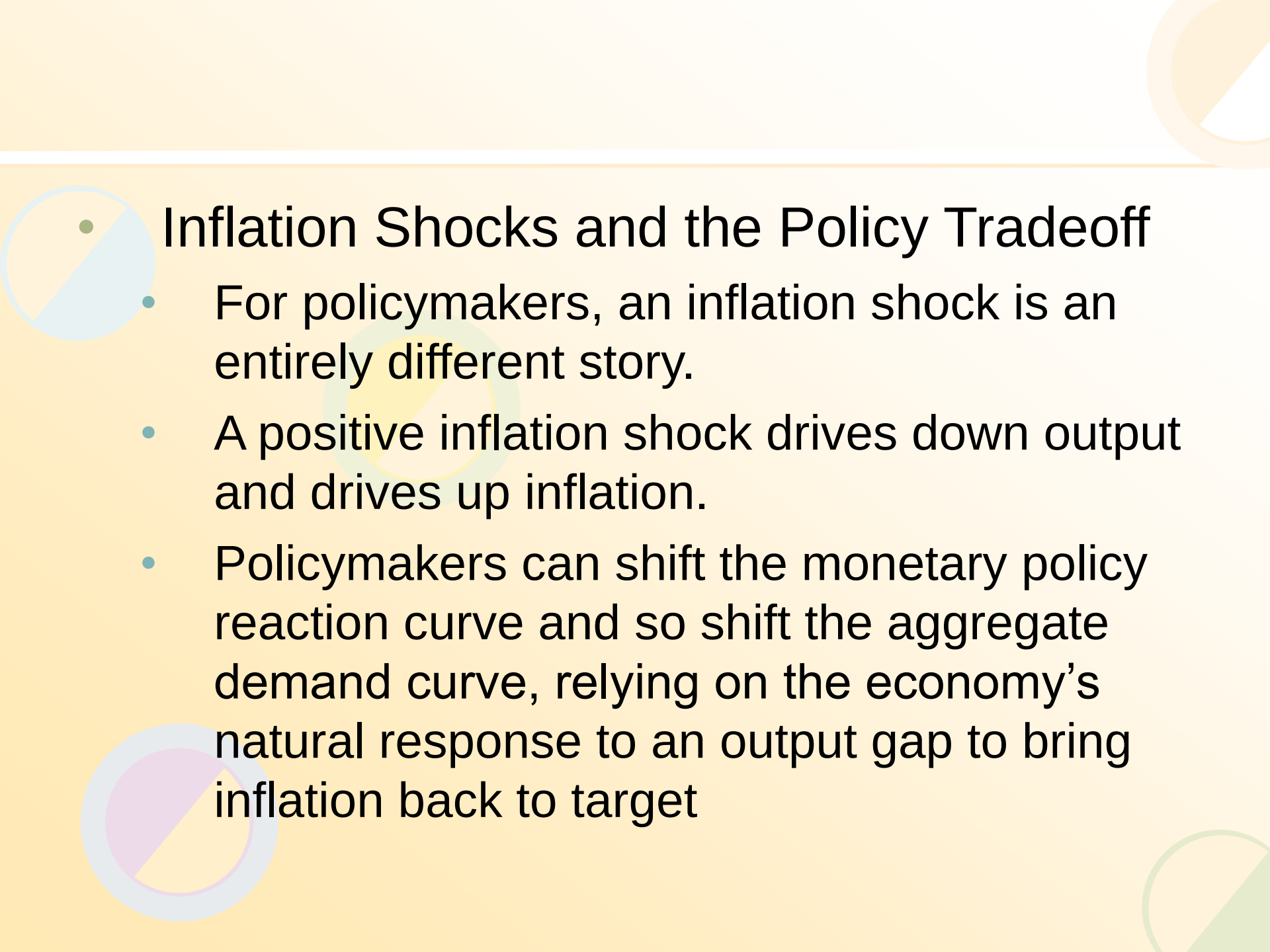
Stabilization Policy

Figure 22.14 Stabilizing a Shift in Aggregate Demand

Following a drop in consumer or business confidence, aggregate demand shifts to the left. To stabilize the economy, the central bank can ease policy, shifting the monetary policy reaction curve to the right. This reduces the real interest rate at every level of inflation and shifts the aggregate demand curve back to where it started. Their action leaves current output and inflation unchanged.



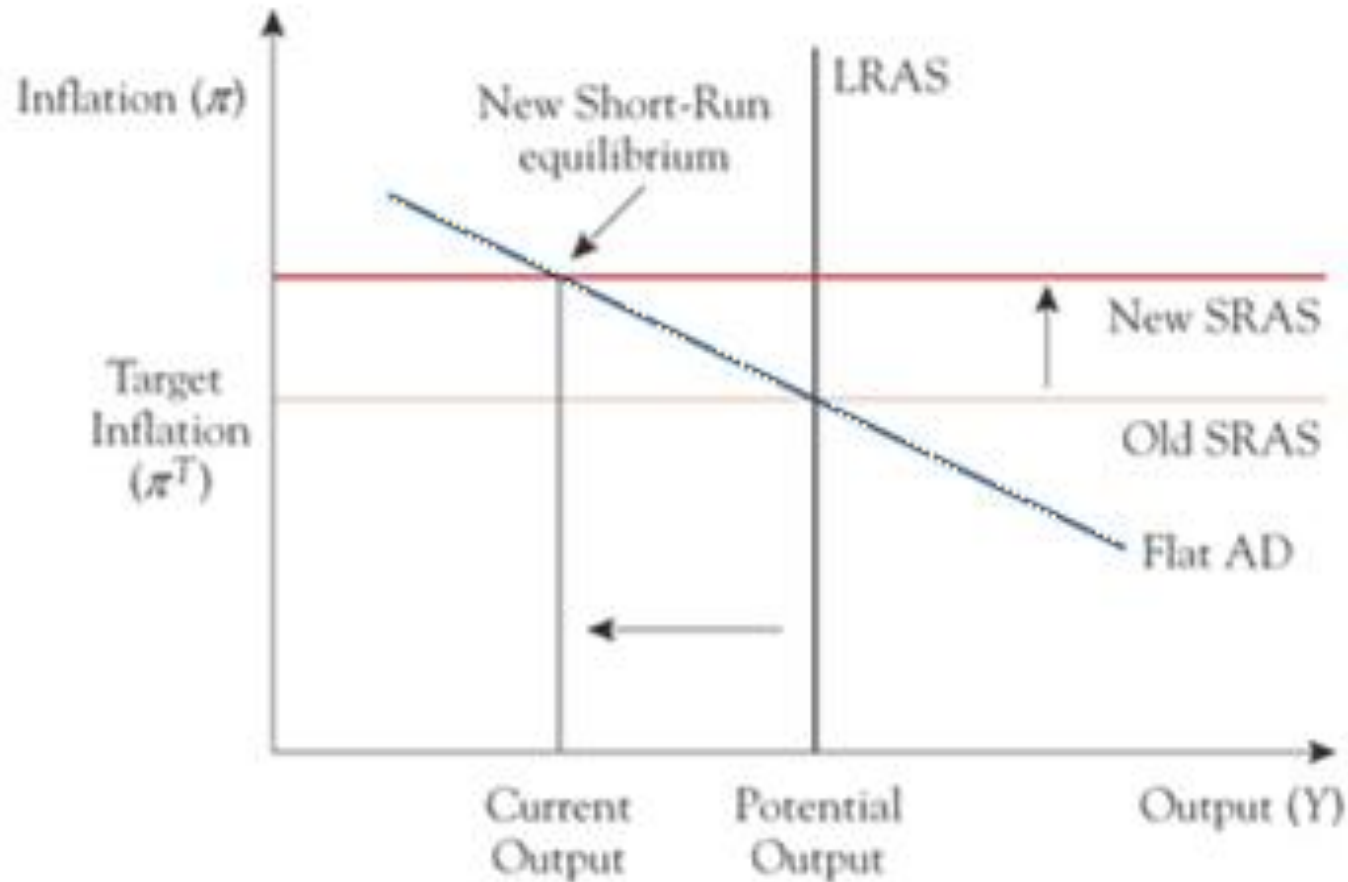
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- In the absence of a policy response, output would fall; instead, output remains steady along with inflation.
 - Policymakers have neutralized the shift in aggregate demand, keeping current output equal to potential output and current inflation equal to target inflation.



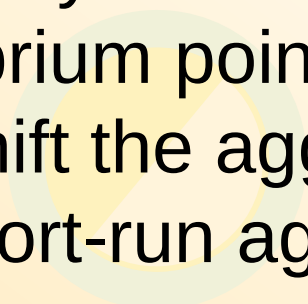
• Inflation Shocks and the Policy Tradeoff

- For policymakers, an inflation shock is an entirely different story.
- A positive inflation shock drives down output and drives up inflation.
- Policymakers can shift the monetary policy reaction curve and so shift the aggregate demand curve, relying on the economy's natural response to an output gap to bring inflation back to target

Stabilization Policy

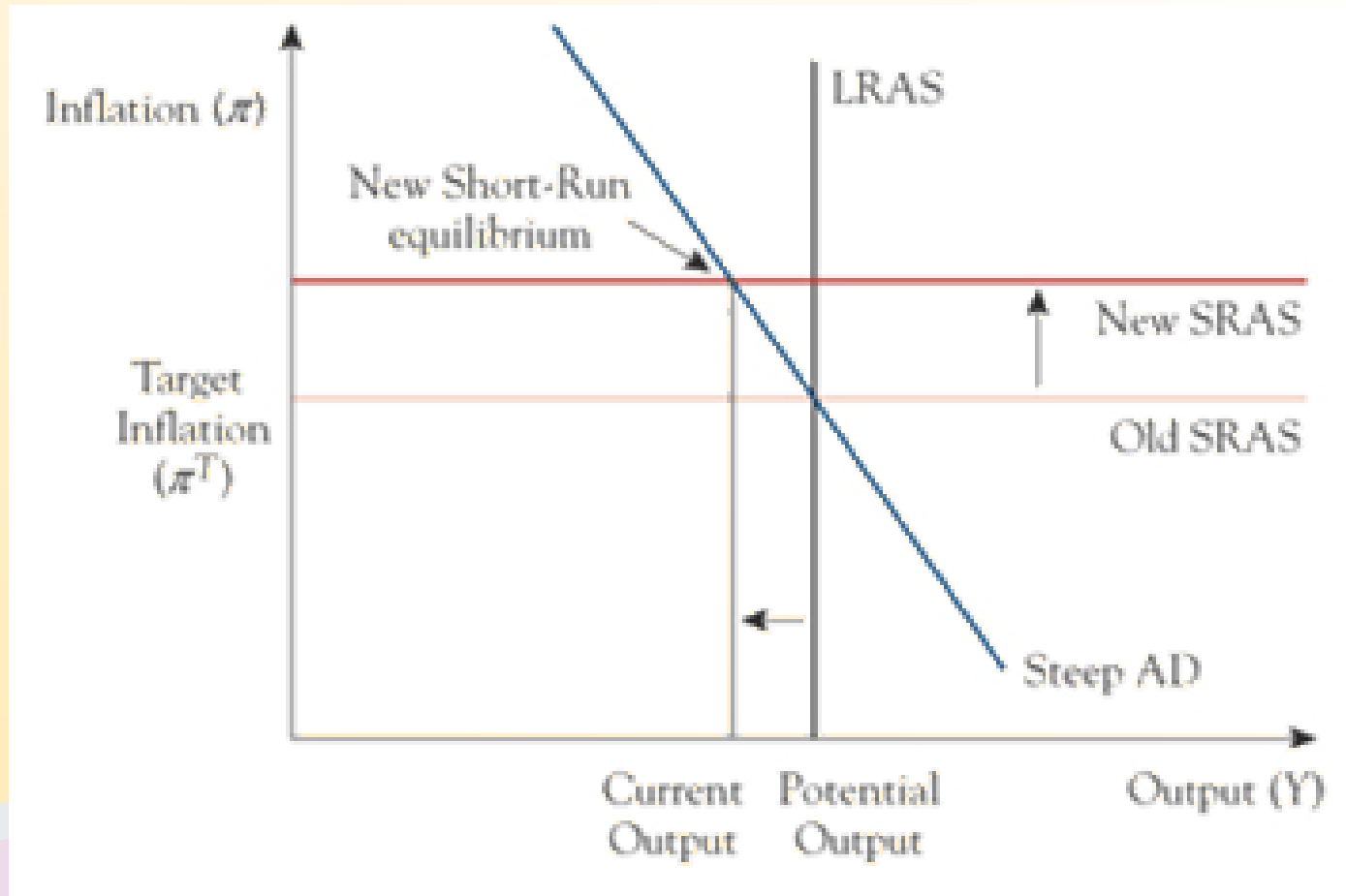


The central bank can Respond Aggressively to keep Current Inflation Near Target

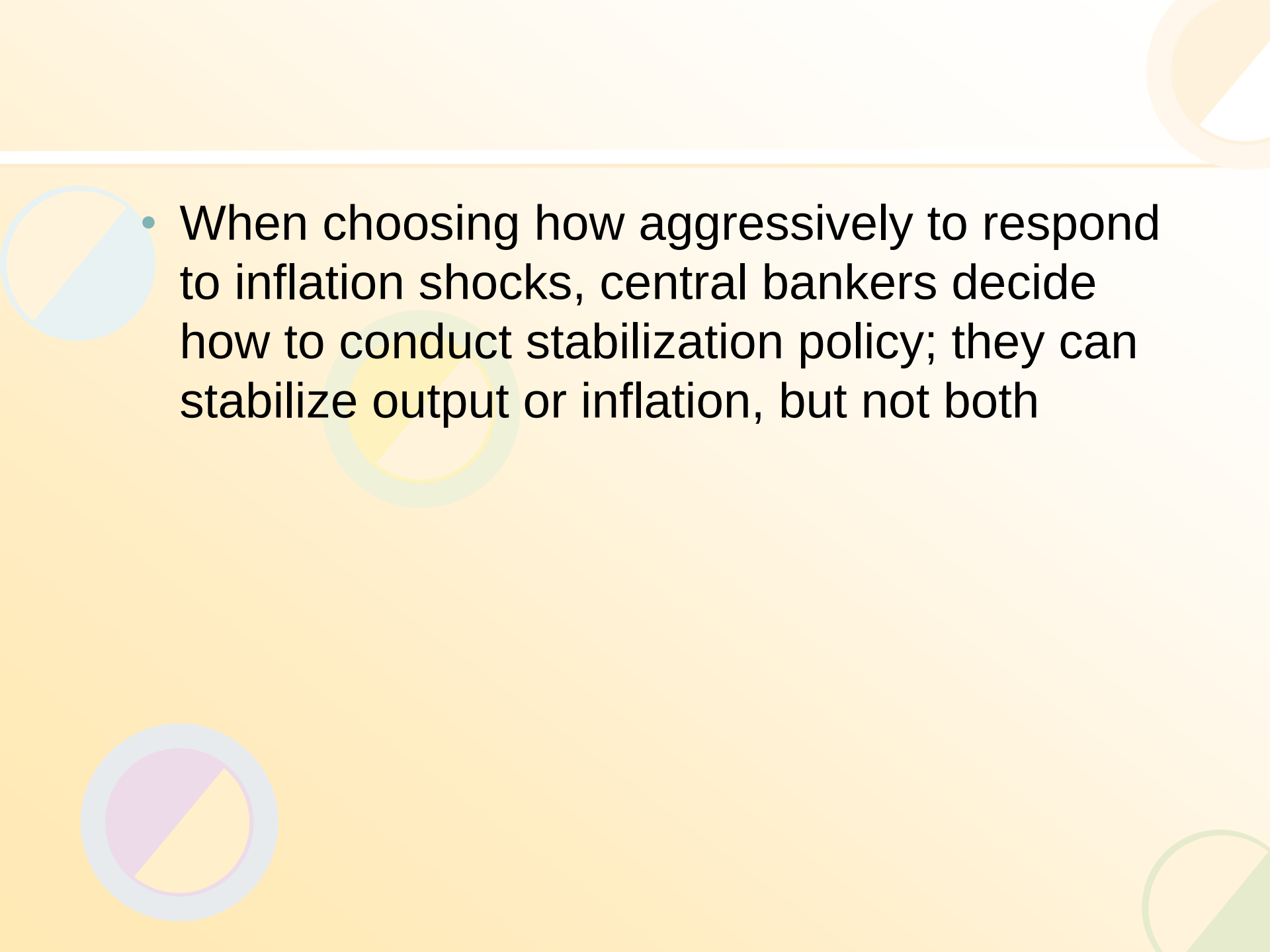
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- But this tool cannot be used to bring the economy back to its original long-run equilibrium point, because monetary policy can shift the aggregate demand curve but not the short-run aggregate supply curve
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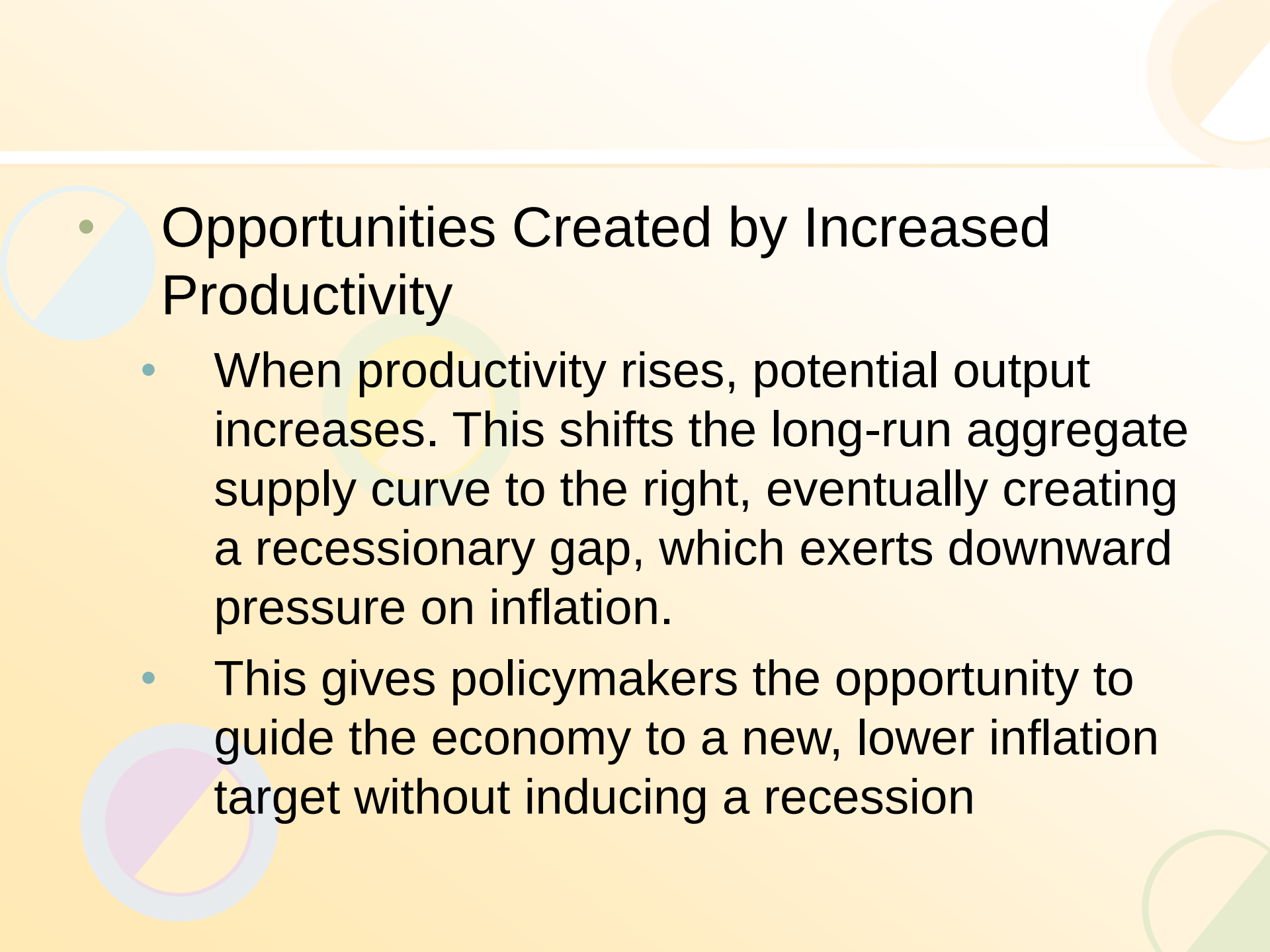
- However, monetary policymakers can choose the slope of their monetary policy reaction curve and so affect the slope of the aggregate demand curve, and in this way monetary policymakers can choose the extent to which inflation shocks translate into changes in output or changes in inflation
- By reacting aggressively to inflation shocks, policymakers force current inflation back to target quickly, but at a cost of substantial decreases in output

Stabilization Policy



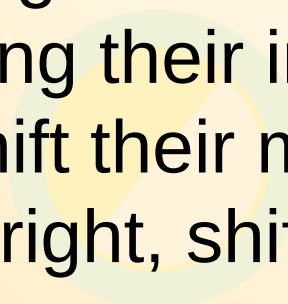
The central bank can respond Cautiously to Minimize Deviations of Current Output from Potential Output

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- When choosing how aggressively to respond to inflation shocks, central bankers decide how to conduct stabilization policy; they can stabilize output or inflation, but not both



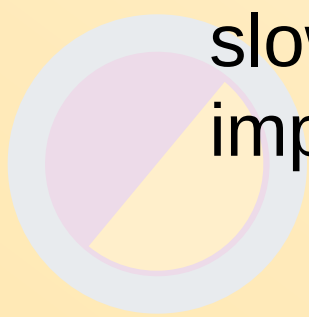
• Opportunities Created by Increased Productivity

- When productivity rises, potential output increases. This shifts the long-run aggregate supply curve to the right, eventually creating a recessionary gap, which exerts downward pressure on inflation.
- This gives policymakers the opportunity to guide the economy to a new, lower inflation target without inducing a recession

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- Since the increase in potential output lowers the long-run real interest rate, rather than reducing their inflation target, policymakers can shift their monetary policy reaction curve to the right, shifting aggregate demand to the right.
 - This will increase current output quickly, leaving inflation unchanged at the target level
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Fiscal Policy

- The people who control the government's tax and expenditure policies can stabilize output and inflation too.
 - Fiscal policy can be used just like monetary policy to neutralize shocks to aggregate demand and stabilize output and inflation.
 - Fiscal policy has two defects: it works slowly and it is almost impossible to implement effectively
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- Most recessions are short, data is available only with a lag, and it takes time for Congress to pass legislation.
- Economics collides with politics where fiscal stimulus is concerned as politicians design stimulus packages based more on political calculation than economic logic.
- Under most circumstances, then, stabilization policy should be left to the central bankers; fiscal policy does have a role but only after monetary policy has run its course

Summary

- Money
- Financial Markets
- Financial Institutions
- Financial Instruments
- Central Bank
- Monetary Policy